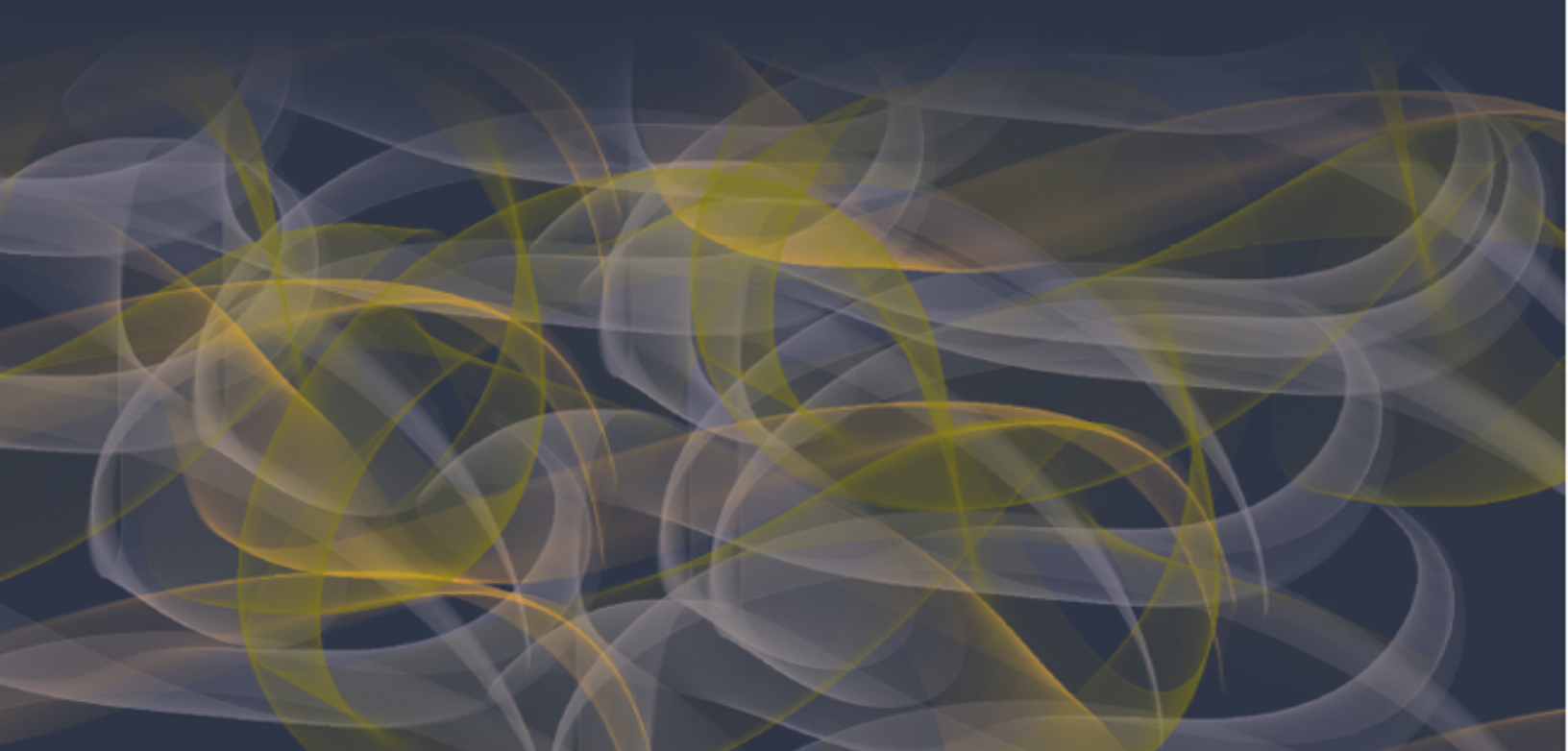


SECOND EDITION



Introduction to Information Science



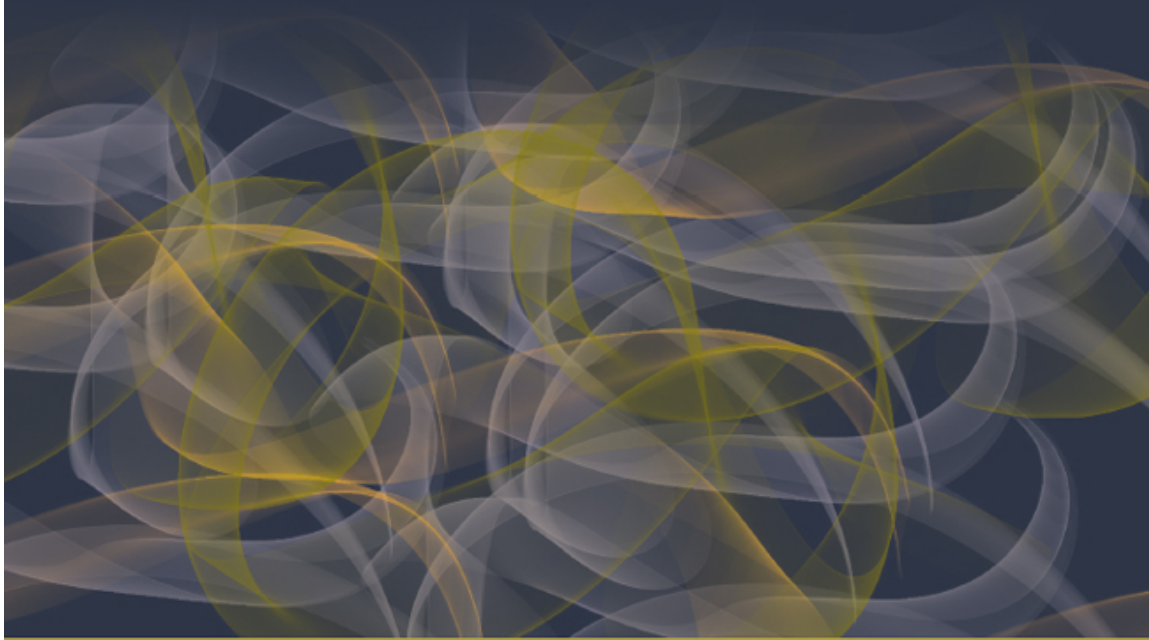
Foreword by Luciano Floridi

DAVID BAWDEN and LYN ROBINSON

SECOND EDITION



Introduction to Information Science



Foreword by Luciano Floridi

DAVID BAWDEN and LYN ROBINSON

Praise for *Introduction to Information Science, Second Edition*

'The publication of this second edition of David Bawden and Lyn Robinson's textbook is a major event in information science. The first edition of 2012 established the highest standards; ten years later, the second edition comfortably exceeds all expectations. Quite simply, it is the clearest, most comprehensive and most useful introduction to the field yet produced, and should receive the strongest possible recommendation – equally for those just starting on their studies and for older hands. The vitality and precision with which the authors organize their vast and complex material make the work highly readable, and of course it is bang up-to-date, with chapters analyzing the latest ideas, practices and technologies. *Introduction to Information Science* has no competition: it is in a class of one. Read it!'

Jonathan Furner, Professor, School of Education & Information Studies, University of California, Los Angeles

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Jutta Haider, Professor of Information Studies, Swedish School of Library and Information Science, University of Borås, Sweden

‘Just as information technologies have come to pervade every aspect of social life, information science has become a required component of all disciplines and research endeavours, with the opportunity and the responsibility to operate in dialogue with other forms of knowledge. This wonderful textbook takes such pervasiveness and cross-contamination as its starting point. It provides essential tools to understand not just what information science is, but how it can best operate within a world transformed by digitalisation and novel forms of information governance. Clear, engaging and replete with useful examples, this is a definitive reference for what an information scientist should know in order to act effectively, reliably and responsibly.’

**Sabina Leonelli, Professor of Philosophy and History of Science,
University of Exeter**

‘This is a welcome update to the excellent first edition, and the breadth of coverage is admirable. This new edition will satisfy those looking for the practical as well as the theoretical concepts necessary to truly understand information science. As a textbook it is an invaluable source that will enhance degree courses across the information science domain, from both soft and hard approaches. As a student text it would support every facet of an information science curriculum, giving excellent overviews of key areas, supported by up-to-date references to explore concepts further. From two of the world’s leading information science academics, this is an absolutely essential purchase for any student or practitioner in the information sciences.’

**Dr David McMenemy, Senior Lecturer in Information Studies,
University of Glasgow**

Introduction to Information Science

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Introduction
to Information
Science

SECOND EDITION

David Bawden and Lyn Robinson



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Preface

The aim of this second edition remains that of the first edition: to provide a clear and succinct synthesis of the information science discipline, in as holistic a way as possible; to give the reader a grasp of the basics and a platform to go into all aspects in more depth. It is intended as an overview, showing the scope and the landscape of the subject. We have tried to examine, in reasonably consistent if brief way, all the aspects of the subject and their interrelations, while not allowing details to obscure the whole picture.

As with the first edition, the book is rooted in the scholarly and professional literature, with a substantial number of references. We hope that it may act as a useful sourcebook for the subject and a reminder that no textbook can be an alternative for engagement with the subject's literature. The conceptual and historical dimensions have been retained, since we feel that it is vital that students and practitioners alike have an appreciation of these, to complement more practice-focused materials.

The book has been substantially rewritten to reflect the great changes in the information environment. Compared to the first edition, there is a greater emphasis on theory, philosophy and ethics, on data and algorithms and on documents and documentation. The increasingly important links between information science and the newer disciplines of data science and digital humanities also receive more attention, as does the avoidance of misinformation and disinformation. [Chapter 3](#) of the first edition (Philosophies and paradigms of information) has been divided into two chapters, dealing with philosophies of information and with paradigms and

theories in the information sciences, respectively. Similarly, the first edition's fourth chapter, which dealt with the basic concepts of the information sciences, has been converted into two, focusing on concepts of data, information and knowledge and on the concepts of documents and documentation. The chapter on information technology has also been divided, into one chapter on data handling and one on information systems. A new chapter on information law and ethics has been spun out from the first edition's chapter on information management and policy. The digital literacy chapter has been largely rewritten, and retitled, to reflect the variety of digital literacies needed in an onlife environment. 'Onlife', a term coined by Luciano Floridi to denote the seamless blending of our online and physical activities, due to the ever-increasing pervasiveness of digital information and communication technologies into all aspects of our lives, is a fundamental underpinning of much of this book. All the other chapters have been extensively revised.

To keep the length of the book within reasonable bounds, and to help readability, content and references have been replaced rather than just added, so that the first edition will still be worth consulting, particularly for those interested in older material. Similarly, new images have been used, with only one retained from the first edition.

We thank Luciano Floridi for providing a foreword to the book, Peter Baker and Michelle Lau of Facet Publishing for their assistance throughout the process, Vanda Broughton for advice on the content of [Chapter 8](#), the Royal Society of Chemistry for use of screenshots from their Chemspider software, and Erica Mosner of the Shelby White and Leon Levy Archives Center, Institute for Advanced Study,

Andrew Robson, Facet Publishing, Mustafa Tupev of the German Archaeological Institute, Cairo and Laurel Bestock and Jess Porter of the Joukowsky Institute of Archaeology at Brown University for assistance with permission to reproduce images. We are grateful to colleagues, and particularly to our students on the CityLIS courses at City, University of London, for their positive reaction to the first edition and for the comments and suggestions which we have, so far as possible, incorporated into this second edition.

Foreword – Curators of Semantic Capital

Sometimes, a task (from baking a cake to studying the evolution of stars, planets, galaxies, and other objects in the universe) requires only one specific agent to be performed. In this case the agent in question, say *A* (Alice or astrophysics, in our example), is supposed to be both necessary and sufficient to deliver the required result. *A* can be *the* agent and the *only* agent for that task, and hence her unique identity as the deliverer is uncontroversial. Because she is both necessary and sufficient, she may find her role ethically demanding in two senses. Not only must she deliver but she can also deliver by herself. When things develop, this twofold responsibility may become less clear. *A* may no longer be sufficient, perhaps she needs to coordinate with other agents, yet she is still necessary, sometimes even more so than in the past.

The debate on the future of Westphalian nation-states seems to suffer from such confusion sometimes. Given the global and very pressing challenges we face, nation-states are far from obsolete. On the contrary, they are even more necessary than in the past. However, they are no longer sufficient, since nation-states now need to coordinate their efforts with many more agents required to make things happen, from companies to NGOs and supranational organisations (think of climate change efforts).

The digital revolution seems to have brought a similar change in information science too. We need information science to curate our expanding space of information (*infosphere*) and all the growing *semantic capital* that we inherit, accumulate, enrich, refine and bestow to future generations. It is a crucial role because semantic capital is

the most precious outcome of all our activities, since it is what enables us to understand, make sense, and shape ourselves and our realities. And yet, information science today is no longer sufficient to take care of it, but must interact, coordinate and collaborate with many other disciplines (think of computer science) and agents (think of Google and its scholarly services) to fulfil its task. This 'insufficient necessity' of information science is challenging for anyone who still thinks of its definition in terms of 'necessary and sufficient' conditions, and hence seeks to identify a role, a field, a subject, an approach, a methodology, a tradition, a canon or any other combination of variables that may establish (for some people, restore) such twofold conditions. But it is an intellectual opportunity for anyone who can embrace a collaborative spirit. From this perspective, information science is one of the necessary curators of semantic capital, has all the responsibility of being increasingly necessary and has the added responsibility of no longer being sufficient but in need of being seriously multidisciplinary in its cooperation with a larger group of stakeholders. This means that the epistemological status of information science has evolved.

Perhaps there was a time when information science was necessary and sufficient and did not *have* to be but could *afford* to be self-reliant and unrelated. But now that it has become even more important and yet insufficient, the solution is to join forces, create relations of trust with other disciplines and agents and become a crucial node in a network of curators of semantic capital. The uniqueness of information science is not (or no longer) provided by some necessary and sufficient status as a scientific discipline, but by the specific

contributions that it must provide to the larger context of taking care of our semantic capital and the infosphere. It is a collaborative and ecological task that readers of this textbook will be able to appreciate.

Luciano Floridi, University of Oxford – University of Bologna

List of Acronyms

AACR	Anglo-American Cataloguing Rules
ACM	Association for Computing Machinery
ACRL	Association of College and Research Libraries (US)
ACTA	Anti-Counterfeiting Trade Agreement
ADI	American Documentation Institute
AI	artificial intelligence
AI4SG	Artificial Intelligence for Social Good
ALA	American Library Association
ANCIL	A New Curriculum in Information Literacy
ANSI	American National Standards Institute
AO3	Archive of our own [repository]
API	Application Programming Interface
AR	augmented reality
ASCII	American Standard Code for Information Interchange
ASI	Academic Standards Institute
ASIS	American Society for Information Science (now ASIST)
ASIST	Association for Information Science and Technology (formerly the American

	Society for Information Science and Technology)
ASK	Anomalous State of Knowledge
ASLIB	Association of Special Libraries and Information Bureaux (UK)
BARTOC	Basic register of thesauri, ontologies and classification
BASH	Bourne Again Shell
BC	Bliss Classification
BC2	Bliss Classification (second edition)
BCC	Basic Concepts Classification
BCE	Before Common Era (equivalent to BC)
BIOS	Basic Input Output System
BL	British Library
BOAI	Budapest Open Access Initiative
BSI	British Standards Institute
CC	Colon Classification
CCO	Cataloging Cultural Objects (metadata standard)
CDWA	Categories for the Description of Works of Art (metadata standard)
CI	Competitor, or Competitive, Intelligence
CIDOC CRM	CIDOC Cultural Reference Model (metadata standard)

CILIP	Chartered Institute of Library and Information Professionals (UK)
CLEF	Cross Language Evaluation Forum
CMS	collection management system
CLI	Command Line Interface
CLIR	Cross Language Information Retrieval
CPU	Central Processing Unit
CRAAP	Currency, Relevance, Authority, Accuracy, Purpose (evaluation checklist)
CRG	Classification Research Group
CSCW	Computer Supported Collaborative Work
CSV	Comma Separated Values (file format)
DACS	Describing Archives: a Content Standard (metadata standard)
DAMS	Digital Asset Management System
DBMS	database management system
DC	Dublin Core
DCMI	Dublin Core Metadata Initiative
DDC	Dewey Decimal Classification
DESI	Digital Economy and Society Index (European Union)
DH	digital humanities
DOI	Digital Object Identifier

DORA	San Francisco Declaration on Responsible Assessment
EAD	Encoded Archival Description (metadata standard)
ECM	enterprise content management
EDRMS	Electronic document and records management system
EFI	Extensible Firmware Interface
eGMS	electronic Government Metadata Standard (UK)
ELIS	Everyday Life Information Seeking
FAIR	findability, accessibility, interoperability, reuse (data sharing principles)
FID	International Federation for Information and Documentation
FOSS	Free and Open Source Software
FRAD	Functional Requirements for Authority Data
FRSAD	Functional Requirements for Subject Authority Data
FRBR	Functional Requirements for Bibliographic Records
FWCI	Field Weighted Citation Impact
GDI	General Definition of Information

GIS	Geographic Information Systems
GLAM	Galleries, Libraries, Archives and Museums
GOFAI	Good Old Fashioned Artificial Intelligence
HCI	human–computer interaction
HIB	Human information behaviour
HMD	Head Mounted Display
HR	Human Relations
HTML	Hypertext Markup Language
HTTP	Hypertext Transfer Protocol
IA	Information Architecture
ICP	International Cataloguing Principles
ICT	information and communication technology
IDI	ICT Development Index (of the ITU)
IGUS	Information Gathering and Utilizing System
IFLA	International Federation of Library Associations
IIB	Institut International de Bibliographie (later FID)
IIS	Institute of Information Scientists (UK – now CILIP)

ILC	Integrative Levels Classification
ILS	Integrated Library System
IM	information management
IO	Information Overload OR Information Organisation
IP	intellectual property
IR	information retrieval
IRM	Information Resource Management
ISAD(G)	International Standard for Archival Description (General)
ISBD	International Standard Bibliographic Description
ISBN	International Standard Book Number
ISCM	Information Seeking and Communication Model
ISI	Institute for Scientific Information
ISIC	Information Seeking in Context (conference series)
ISKO	International Society for Knowledge Organization
ISO	International Standards Organization
ISSN	International Standard Serial Number
IT	Information Technology
ITU	International Telecommunication Union

JIF	Journal Impact Factor
JISC	Joint Information Systems Committee (UK)
KM	knowledge management
KO	Knowledge Organisation
KOS	knowledge organisation system
LCC	Library of Congress Classification
LCSH	Library of Congress Subject Headings
LIS	library and information science
LISA	Library and Information Science Abstracts
LISTA	Library and Information Science and Technology Abstracts
LMS	Library Management System
LOM	Learning Object Metadata
LRM	Library Reference Model
LSP	Library Service Platform
MARC	MAchine Readable Cataloguing
METS	Metadata Encoding and Transmission Standard
MILA	Media and Information Literacy Alliance
MODS	Metadata Object Description Schema
MTC	Mathematic Theory of Communication (Shannon/Weaver)

NGO	Nongovernmental organization
NISO	National Information Standards Organization (USA)
NIST	National Institute of Standards and Technology (USA)
NLM	National Library of Medicine
OA	open access
OAI	Open Access Initiative
OAIPMH	Open Access Initiative Protocol for Metadata Harvesting
OCLC	Online Computer Library Centre
OCR	Optical Character Recognition
OD	Open Data
OJS	Open Journal Systems
ONIX	Online Information Exchange [metadata standard]
OPAC	Online Public Access Catalogue
OpenDOAR	Open Directory of Open Access Repositories
OSI	Open Society Institute
OSS	open source software
OWL	Web Ontology Language
PDA	Personal Digital Assistant

PESTLE	Political Economic Social Technological Legal Environmental
PI	Philosophy of Information
PMEST	Personality, Matter, Energy, Space and Time (facet analysis)
PREMIS	Preservation Metadata: Implementation Strategies (metadata standard)
QR	Quick Response (codes)
RDA	Resource Description and Access
RDBMS	Relational Database Management System
RDF	Resource Description Framework
RDM	Research Data Management
REF	Research Excellence Framework (UK)
RFID	Radio Frequency Identification
ROAR	Registry of Open Access Repositories
ROI	return on investment
SE	Social Epistemology
SCONUL	Society of College, National and University Libraries (UK)
SKOS	Simple Knowledge Organisation System
SLA	Special Libraries Association
SNA	social network analysis
SQL	Structured Query Language
SSCI	Social Science Citation Index

STEM	science, technology, engineering and medicine
SWOT	Strengths Weaknesses Opportunities Threats
TAGS	Twitter Archiving Google Sheet
TCP/IP	Transmission Control Protocol / Internet Protocol
TEI	Text Encoding Initiative
TMI	Too Much Information
TREC	Text Retrieval Conference
UDC	Universal Decimal Classification
UDK	see UDC
UEFI	Unified Extensible Firmware Interface
UK	United Kingdom
UKMARC	obsolescent national version of MARC
UMLS	Unified Medical Language System
UNESCO	United Nations Educational, Scientific and Cultural Organization
UNIMARC	obsolescent international variant of MARC
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
USMARC	obsolescent national variant of MARC
US(A)	United States (of America)

UX	User Experience
VLE	Virtual Learning Environment
VLSI	Very Large Scale Integration
VRA	Visual Resources Association (metadata standard)
VR	Virtual Reality
VRE	Virtual Research Environment
W3C	World Wide Web Consortium
WHO	World Health Organization
WIPO	World Intellectual Property Organization
WWW	World Wide Web
XML	Extensible Markup Language

1 The Information Science Discipline

Information science is, or should be, involved with the whole concept of knowledge in whatever form its manifestations may take.

Jesse Shera (1973, 286)

Let us not restrict ourselves to grubbing around in the garden patch of a limited, little information science, restricted to the relationship between information and machine. Instead, let us expand, reach out, embrace and explore the wider world of information, to develop a vision of information science as a central synthesising discipline in understanding not simply information, but the world we live in. Because the world we live in is surely a world of information.

Tom Wilson (2010)

Introduction

The subject of this book is information science. We begin by asking what information science is, as an academic discipline and profession. Obviously, and simplistically, it is the science of information. But this is not sufficient, since the multiple meanings and implications of information, which will be discussed in [Chapter 5](#), have given rise to different conceptions of information science (Buckland, 2012; Limberg, 2017). One such conception is of information science as being concerned with computing, algorithms and data science, a second with information and communication technologies and a third with information as an entity with physical and biological science. A fourth conception sees information science

as concerned with information recorded in documents, with meaning and knowledge, and hence as growing from the older disciplines of librarianship and documentation. We will focus on the last of these in this book, although we will mention aspects of the others at appropriate points.

We will therefore be following the kind of definition which goes back at least as far as Borko (1968) and is expressed by Saracevic (2010, 2570) as:

Information science is the science and practice dealing with the effective collection, storage, retrieval and use of information. It is concerned with recordable information and knowledge, and the technologies and related services that facilitate their management and use.

This gives us a general idea of the nature of the subject, but there is still scope for much difference in viewpoint as to exactly what the subject comprises and how it should be understood; see Bates (1999), Hjørland (2000), Robinson (2009), Ibeque-SanJuan et al. (2014) and Bawden and Robinson (2016).

We now examine the nature of the subject in more detail.

The nature of information science

Information science is an academic discipline which supports areas of professional practice. We will think first about the discipline, although we should note that there have always been some doubts as to what extent it is a real discipline, still less a 'science' (Robinson, 2009; Buckland, 2012; Furner, 2015).

One way to accommodate the wide range of views about, and diverse approaches to, the subject within a coherent framework is to regard information science as a *field of study*, using this phrase in the specific sense of Paul Hirst, the philosopher of education (Hirst, 1974). A field of study is an alternative to ‘disciplines’ based on a unique form of knowledge, such as mathematics or the physical sciences, and to ‘practical disciplines’ based on one of the forms of knowledge but oriented to solving practical problems, such as engineering or medicine. For Hirst, a field of study is focused on a topic or subject of interest, using any of the forms of knowledge – sociological, mathematical, philosophical, etc. – which may be helpful in studying it. Bawden (2007, 320) argues that it may be appropriate to regard information science as such a field of study, focused on the topic of information, understood as information recorded in documents produced and used by individuals and societies:

a multidisciplinary field of study, involving several forms of knowledge, given coherence by a focus on the central concept of meaningful recorded information.

This is reminiscent of the insistence of Machlup and Mansfield (1983) that the field should be described as the information sciences, emphasising the plural, to show the breadth of approach needed. Ibeque-SanJuan et al. (2014) reinforce this, referring to the ‘pluri-, multi-, trans-, meta- and interdisciplinary’ nature of the discipline; see also Bates (2015), Limberg (2017) and Hjørland (2018). Evidence for the interdisciplinary is also given by the extent to which authors

from the information sciences publish in journals outside the subject, and vice versa; see, for example, Chang (2018a; 2018b).

We can give some more precision to this general idea by arguing that the focus on recorded information can be expressed specifically as a focus on the *communication chain* of recorded information: from its creation, through dissemination, indexing and retrieval, use and archiving or disposal (Duff, 1997; Robinson, 2009). This is implied in earlier formulations, but noting that it explicitly helps to clarify what are the concerns of information science. Details of the chain, and the ways in which it is being changed by new technologies, are discussed in [Chapter 13](#). Information is recorded in *documents*, and information science has a close affinity with the documentation movement and document theory, as will be covered in [Chapter 6](#).

We can also explain more precisely what information science does, in terms of both research and scholarly study and of practice, through the components of *domain analysis*, as propounded by Birger Hjørland (2002). This will be discussed fully in [Chapter 7](#). For now, we will just note that there are a number of aspects which represent both the activities of the information practitioner and the ways in which research and study are carried out. Examples are user studies, historical studies, studies of terminology, research on indexing and retrieval and so on.

Considering information science as an academic discipline, comprising the study of the components of the communication chain of recorded information through the perspective of domain analysis, provides the understanding of information science which we shall use throughout this book.

Information science is a field of study, with recorded information and documentation as its concern, focusing on the components of the information communication chain, studied through the perspective of domain analysis.

Information science is a vocational discipline, underlying a number of professional activities, including: data, information, and knowledge management; librarianship; metadata and taxonomy management; records management and archiving; and documentation in museums, galleries and heritage institutions.

What kind of discipline?

There is little agreement as to how information science should be categorised as an academic discipline. It has been called, among other things, a meta-science, an inter-science, a postmodern science, an interface science, a superior science, a rhetorical science, a nomad science, an interdisciplinary subject which should be renamed knowledge science, a liberal art, a form of cultural engagement, a subject which may assume the role once played by philosophy in mediating science and humanism, and the applied philosophy of information. There have also been national variations in the way the subject is regarded in, for example, the United Kingdom (Robinson and Bawden, 2013), France (Ibeque-SanJuan, 2012; Hudon, 2018), the Nordic countries (Limberg, 2017) and the United States (Buckland, 1996).

University departments teaching and researching the subject may be found in faculties of arts and humanities, social sciences, computing and technology, education, science, and business and

management. This scatter may be a natural consequence of the status of information science as a field of study, but it has been held to be a weakness of the discipline. The related question of the extent to which information science has its own body of concepts and theories, as opposed to simply borrowing them from other disciplines, will be considered in [Chapters 3 and 4](#); see Dillon (2007) for some concerns on this aspect.

It is clear that many other professions are interested in components of the communication chain: journalists are concerned with creation, publishers with dissemination, computer scientists with information retrieval and so on. Even accepting that information science is a meta-discipline, surely it must have some 'academic turf' of its own? We suggest that the uniqueness of information science lies in its concern for *all* components of the communication chain and the ways in which they interact with, and impact upon, each other. Other disciplines are interested in specific aspects, but only the information sciences see their concern as being the totality. We might also name some aspects of information organisation and information behaviour, seemingly invariant to technology or to context, which are the particular concern of information science. But our main claim to a unique area is the totality of the communication chain; for more discussion, see Robinson and Bawden (2012), Robinson (2009) and Robinson and Karamuftuoglu (2010).

Constituents and core

There is no canonical list of topics which fall within the remit of information science, despite much debate, supported by bibliometric

mapping, about such a list, and consequently about what should be in the curriculum for information science education: see, for example, Robinson and Karamuftuoglu (2010) and Milojevic et al. (2011). Commonly mentioned topics are: nature and conceptions of information, documents and collections; information theory and philosophy; information history; information organisation and metadata; information behaviour and practices; information and digital literacies; information seeking and retrieval; information architecture, human–computer interaction and user experience; data, information, and knowledge management; information policies and strategies; document and collection management; records management and archiving; information law and ethics; information research methodology; information society; information systems; publishing and scholarly communication; and information services of all kinds.

A useful model for organising this broad collection of topics is that of the *communication chain* of recorded information (Robinson, 2009). There is no canonical set of constituents of the chain, but it typically comprises stages of creation, dissemination, collection, storage, organisation, description, retrieval, use, and preservation. The elements of the chain form the core of most academic courses in the subject (Robinson and Bawden, 2010).

The subject is delineated by national educational authorities. In the United Kingdom there is a detailed Subject Benchmark statement for the rather broader topic of *Librarianship, Information, Knowledge, Records and Archive Management* (Quality Assurance Agency, 2019). Professional bodies also list components of the subject, typically for

supporting the accreditation of courses and individuals, and generally taking a strongly vocational perspective. In the United Kingdom, the Professional Knowledge and Skills Base (PKSB) of CILIP (the Library and Information Association) fulfils this purpose, with a main PKSB for library and information management generally and specific variants for health information and for knowledge management (CILIP, 2021). The Association for Information Science and Technology (ASIST) also has a set of educational guidelines, specifying broad topics to be included in curricula.

Overlaps with other information disciplines

Information science as understood here is the central discipline focused on recorded information and documents, but it has overlaps with other information-related disciplines. Some of these, particularly with collection disciplines such as librarianship and archival science, are of long standing. Others are more recent, such as those with the newer disciplines of data science and digital humanities (Casarosa et al., 2020). We can distinguish eight such overlaps with different kinds of adjacent disciplines. These overlaps result in the linking of information science with other disciplines in academic departments and faculties, professional bodies, research management structures, journal titles, etc., and the frequency with which those with information science qualifications find employment in these adjacent sectors.

Collection disciplines

Given its focus on information recorded in documents, and its origins in the special library sector, it is not surprising that there is a

particularly significant overlap between information science and the collection disciplines, particularly those concerned with libraries and archives. Bates (2007) distinguishes these as the ‘information sciences’ and the ‘disciplines of the cultural record’. There is frequently a conflation into a wider discipline of *library and information science* (LIS) or *library and information management* (Dillon, 2007; Bawden and Robinson, 2016; Limberg, 2017; Hjørland, 2018), evident in the titles of academic departments, professional associations, academic journals and textbooks, etc. Many people with information science qualifications find employment in the GLAM (Galleries, Libraries, Archives and Museums) sector, further emphasising the overlap with collection issues. This overlap will be discussed later in the book, in sections on digital libraries, collections management, records management and archiving, and others; see Rasmussen and Hjørland (2021) for a discussion of the overlap and convergence of collection institutions.

Digital humanities

Arising in the area of overlap with the collection disciplines is the relatively new discipline of digital humanities (DH). In essence the application of digital computing to the academic study of the humanities DH grew out of humanities computing to become a discipline in its own right (Terras, Nyhan and Vanhoutte, 2013; Warwick, Terras and Nyhan, 2012). DH has always had strong links and overlaps with LIS, and the two disciplines have been seen as natural partners (Koltay, 2016; Clement and Carter, 2017; Alvite-Díez and Barrionuevo, 2020; Casarosa et al., 2020; Engerer, 2021; Gooding, 2021). Both are concerned with recorded information encapsulated

in documents (in the broadest sense), including datasets, and both are concerned with the creation, curation and use of compilations of data and information. The digital tools developed by DH are relevant to, and being used in, LIS activities. DH activities have often been located in libraries, or in academic LIS departments, and librarians have been active participants in DH projects (Fay and Nyhan, 2015; Poole and Garwood, 2018; Wilson, 2020; Zhang, Xue and Xue, 2021). DH will be discussed further in [Chapter 10](#).

Technology disciplines

As already noted, information science originated partly as a response to new information technologies, and the overlap with technology is still very significant, as evidenced by the many academic courses in the overlap area, particularly in institutions presenting as iSchools, and by societies such as ASIST. A number of topics span the area between computer science and information science, such as information retrieval, human–computer interaction, information visualisation and information architecture; these will be considered in [Chapter 10](#).

Data science

The relatively new discipline of data science, concerned with the extraction of knowledge and meaning from large datasets, is a discipline evidently adjacent to information science (Kelleher and Tierney, 2018; Du and Khan, 2020; Stuart, 2020). In order to support data science's activities of data and text mining, data and text visualisation and statistical analysis, it is essential to have data curation, management and analytics, including collecting, cleaning and storing data and making it accessible through good metadata;

all with concern for the data lifecycle. This is closely analogous to, if not identical with, the information management role (Benzing, 2020; as an example, see Darch et al., 2020). There has been considerable interest in introducing data science and data management issues into information science curricula (Robinson and Bawden, 2017; Wang, 2018; Casarosa et al., 2020; Loni et al., 2020). These issues will be discussed further in Chapter 9.

Social science disciplines

Though there is debate as to what kind of discipline information science is, there is no doubt that many of the phenomena it studies are social phenomena, that many of its theories and concepts are social in nature and that much of its research is conducted using social research methods (Roberts, 1976; Hjørland, 2018). There is therefore an overlap with the social sciences, including psychology and law. This is seen particularly with information behaviour and practices ([Chapter 12](#)) and the information society and social informatics ([Chapter 16](#)), but also in national and international information policies ([Chapter 14](#)) and information law ([Chapter 15](#)).

Communication and media disciplines

Technical communication, the writing of abstracts, translation, information design and so on have always been regarded as an area of interest to information science. There is an evident overlap with journalism, publishing and media studies in the *creation* and *dissemination* sections of the communication chain, as will be discussed in [Chapter 13](#). This is reflected in the structure of academic departments and faculties and in research assessment; for example, the conflation of these disciplines in panel D34 (Communication,

Cultural and Media Studies, Library and Information Management) in the UK Research Excellence Framework for 2021.

Management and policy disciplines

Information management and information governance, and to an extent business intelligence, are regarded as important aspects of information science, but equally of relevance to business studies and management science; this is the rationale, no doubt, for some information science departments being located within business schools. Knowledge management also straddles the information science/business divide. National and international information policies, while primarily the concern of politics and social policy, also have significant information science input. These aspects are covered in [Chapter 14](#).

Domain disciplines – subject specialism

Many contributions to the development of information science have been made by those whose expertise is in a discipline where new specialised information systems are needed, particularly in the STEM subjects and within those, particularly in chemistry. Developments in information science continue to be driven by those from such fields, particularly for information resources and informatics systems for chemistry, biology, and medicine, as will be covered in [Chapter 7](#).

History of information science

Recorded information in the form of documents, with associated activities of record-keeping and librarianship, has been important for millennia, as will be discussed in [Chapter 2](#), but it was only in the

19th century that a codified body of professional knowledge emerged, to be developed into an academic discipline (Hjørland, 2018). The Bavarian librarian Martin Schrettinger was the first to argue for formal training for librarians, and used the term *Bibliothekswissenschaft*, which may be reasonably translated 'library science', in 1808 to encompass the tasks of cataloguing, classification, shelf arrangement and library management. Later treatises by Christian Molbech and Frederick Ebert amplified the idea that the skills and knowledge needed to manage document collections were distinct from those of the scholar. The first professional journal for the information disciplines was published in Germany from 1840: Robert Naumann's *Serapeum. Zeitschrift für Bibliothekswissenschaft, Handschriftenkunde und Ältere Litteratur* (*Journal of Library Science, Manuscript Information and Older Literature*). The first formalised educational programmes for the information sciences came with the establishment of courses in librarianship at the University of Göttingen in 1886 and the University of Columbia in 1887, and at Leipzig and Barcelona, both in 1915, followed by the creation of London University's School of Librarianship (later attached to University College London) in 1919. Graduate studies in 'library science' began at the University of Chicago in 1926 and in London a few years later.

The history of information science, as distinct from librarianship and archival record keeping, is more recent; for overviews see Bawden (2008), Buckland (1996; 2017), Buckland and Liu (1995), Hahn and Buckland (1998), Hjørland (2018), Meadows (1987) and Shera and Cleveland (1977). Its origins are in the communications

revolution of the 19th century, which will be discussed in Chapter 2, and with the simultaneous emergence of scientific and technical disciplines (Meadows, 2004), with a consequent need to deal with large volumes of documents, scientific and technical literature in particular. Its specific antecedents were the advent of the documentation movement and the development of the special library sector, both at the beginning of the 20th century; see Williams (1997) for an account of these in the USA. The documentation movement, pioneered by Paul Otlet, espoused a 'scientific' approach to the storage and retrieval of recorded information, with new schemes of classification and the use of the new retrieval technology of the card index (Wright, 2014). The increased awareness of technical information as a resource for science-based industries led to the establishment of special libraries, and to the idea of 'information work' as distinct from librarianship. These, compared with traditional libraries, had a much more proactive role, a strong subject focus and an interest in all forms of information, not just formally published materials (Ditmas, 1950). The need to deal with the 'information explosion', the very rapid expansion in publications of all kinds dealing particularly with scientific and technical information during and after the 1939–45 war, was discussed at the influential 1948 Royal Society conference on scientific information (Line, 1998).

The growing application of new technologies to information handling – initially mechanised documentation techniques and then the digital computer – provided the technological background for the new science (Buckland, 1996; Black, 2007). Vannevar Bush's

influential Memex concept of personal information management with access to the world's information, combined with Shannon and Weaver's Mathematical Theory of Communication and the new 'informetric' laws, such as Bradford's law of scattering (all to be discussed in later chapters), held out the prospect of a genuinely scientific approach to information management.

The establishment of a new discipline or profession has typically been recognised, since the 19th century, by the setting up of a professional body to represent it. The first such body in the area that was to spawn information science was an international body for the co-ordination of the activities of the documentation movement. The Institut International de Bibliographie, later renamed as the International Federation for Information and Documentation, was established by the two Belgian pioneers of documentation, Paul Otlet and Henri La Fontaine, in 1895. In existence for over one hundred years, it was the first recognisable information science association (Rayward, 1997).

On a smaller scale, several associations were set up for special librarians. The Special Libraries Association, founded in the USA in 1909, is still thriving as a worldwide body today. In the UK, the Association of Special Libraries and Information Bureaux (ASLIB), was formed in 1924 with the aim of coordinating the activities of specialist information services in the UK and a role as a national intelligence service for science, commerce and industry (Bawden and Robinson, 2017). National groups of special librarians were formed in many other countries. The UK was rather slow off the mark, perhaps because of the existence of two other bodies catering for a

relatively small sector; the Industrial Group of the Library Association was formed only in 1971 (Mason, 1991). It is still active, though now combined into a group also catering for commercial, legal and science librarians.

The two main bodies representing information science per se formed later, and had shorter lives in their 'pure' form. The American Documentation Institute, created in 1937, became the American Society for Information Science in 1968. It renamed itself as the American Society for Information Science and Technology in 2000 (Sonnenwald, Davis and Shaw, 2017). In the UK, the Institute of Information Science (IIS) was formed in 1958, and joined with the Library Association (itself formed in 1877) to form the Chartered Institute of Library and Information Professionals (CILIP) in 2002, whose Royal Charter states that it will 'work for the benefit of the public to promote education and knowledge through the establishment and development of libraries and information services and to advance information science'. While each association had its particular reasons for the change, the fact that both felt that it was sensible for information science to enter a wider grouping – with technology or with the collection disciplines – illustrates what was said earlier about its multidisciplinary nature.

Specialised journals are also a feature of a mature discipline. Of the major information science journals today, the longest-established is *Journal of Documentation*, the name reflecting its origins in the field of documentation when it was founded in 1945. Both the main information science associations founded journals: *Journal of the American Society for Information Science and Technology* (formerly

Journal of the American Society for Information Science) in 1950, and the IIS's *Journal of Information Science* (formerly *The Information Scientist*) in 1967. Information science material is, of course, published in a much wider range of journals; some of the more significant of these are listed in the additional resources at the end of the book.

The first educational provision for information science came with a course specifically for science graduates in London in 1929, though this closed for lack of interest in 1935. Modern information science education began in 1961, when Jason Farradane set up an evening course in 'collecting and communicating scientific knowledge' at Northampton College of Advanced Technology. Farradane, a British scientist born in Hampstead, London to Polish parents, was originally named Levkowitz and adopted his new name as a tribute to his scientific heroes, Michael Faraday and J. B. S. Haldane. He was the first to use the terms 'information science' and 'information scientist' (Shapiro, 1995). Although his initial concept of an information scientist was a specialist in the handling of scientific and technical information, Farradane pioneered the teaching of information science as a distinct subject and was among those who argued for a 'true science of information', along the lines of the natural sciences (Farradane, 1976).

A Masters course in information science was established at Farradane's Centre for Information Science when the Northampton College became City University London in 1965 (Robinson and Bawden, 2010). Other courses in the subject were developed worldwide in the following years, typically conjoint LIS departments (Um and Feather, 2007; Mezick and Koenig, 2008). Recent years have

seen a restructuring of many such departments, some joining with cultural and media departments, others amalgamating with technical and computing departments. Many of the latter have, since 2005, been associated with the iSchools movement (Golub, Hansson and Selden, 2016).

One recurrent question has been to what extent education in the information sciences should be focused on theories and principles, as against training in practical techniques. We strongly advocate the former, believing that it is much more valuable for students to gain an understanding of principles and concepts on which they can build throughout their professional lives, rather than ephemeral points of practice, and this has been the basis of our courses at City, University of London (Robinson and Bawden, 2010). We are encouraged in this viewpoint by the views of two eminent figures from the past, who both had experience as practitioners and teachers: Jesse Shera from the USA and Brian Vickery from Britain:

Librarianship can be an intellectual discipline in its own right, and education is not a substitute for experience, but a preparation for it. Librarianship ... must abandon the practice of putting its students through 'little fake experiences in the classroom'. We must teach pupils theory, not techniques: principles, not practice.

(Shera, 1973, 335)

Only in a very static profession can one be trained to slot in immediately to an available job, and our profession is far from static. It is more beneficial for the students to give them a generalized grounding in a wide variety of professional activities and concerns, so that they will have some background knowledge

for no matter what job is first available. For those who seek it, our subject also has its cultural value, which can contribute to a general education.

(Vickery, 2004, 29)

This is our rationale for focusing on principles and concepts in this book.

Summary

Information science is a multidisciplinary field of study, focused on meaningful information recorded in documents, and requiring a variety of perspectives and methods. Growing out of special librarianship and documentation, it has strong links with information technology (IT) and computer science and with the collection disciplines, and overlaps with a variety of other academic subjects. It is a coherent academic discipline in its own right, and educational programmes based on concepts and principles are the most valuable.

- The focus of information science is the human record, instantiated in documents and processed through the communication chain of recorded information.
- The uniqueness of information science is in its concern for all components of the communication chain.
- The discipline overlaps with several others which focus on particular aspects of the communication of information.

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2 History of Information: the Story of Documents

The written word – the persistent word – was a prerequisite for conscious thought as we understand it.

James Gleick (2011, 37)

We describe ourselves as living in an information age as if this were something completely new. In fact, many of our current ways of thinking about and handling information descend from patterns of thought and practices that extend back for centuries.

Ann M. Blair (2010, 1)

Introduction

In this chapter we will give some historical context for the rest of the book by looking at the development of recorded information, and the documents which carry it, through time. We will do this only very briefly and informally; there are many detailed treatments of the history of information provision and dissemination, and we will not try to replicate these in any way. We will give some historical perspective on specific topics in several of the chapters which follow.

It is sometimes suggested that, since information science is a relatively young discipline, there is no need to think about the history of anything which came before, say, 1950. We strongly disagree. Some appreciation of the history of information and documents, as well as being of interest in its own right, can help give a perspective on current problems and solutions. Although contexts and technologies may change, many information issues remain constant over time, and a historical perspective can be of practical, as well as academic, value.

Historical studies come in many forms and flavours (Black, 2006). For the information sciences, historical accounts have fallen into a number of categories, including:

- history of libraries, archives and information services
- history of information policies and infrastructures
- history of publishing, books and reading
- history of information technologies (not just digital ones)
- history of the information disciplines and professions
- origins of the information society
- studies of the cultural and social contexts of information communication and use.

The last of these has developed into a new sub-discipline of *information history*, concerned with all aspects of information in societies of the past; how it was understood, used, organised, managed, collected, disseminated, etc. See, for example, Weller (2008), Skouvig and Andersen (2015) and Cortada (2019). It may be regarded as a branch of history focused on an intangible central object of study; analogous to histories of such things as humour or fear. Very often, however, it will necessarily be studied through a focus on the documents which convey information and how people have produced and used these.

We will not follow any of these approaches. We focus on the historical development of documents, and the systems and services set up to deal with them; hence the title of the chapter. The treatment will be brief and informal, and will direct the reader to other works

for details of all the topics mentioned, giving just sufficient material to set the context for what follows in the rest of the book.

Information ages?

It is tempting, in trying to present a short and simple account of a complex and long-term development, to write of distinct 'information ages' based on 'information revolutions'. And there is no doubt that a series of developments in the communication of information have largely dictated the information environment which followed. Language, recorded signs and symbols, writing and scripts, printing, mass communications, the digital computer, and the internet are typically taken as defining developments. However, this is too simplistic a viewpoint. No form of history falls into such neat periods, and developments happen at different points in different parts of the world. In outlining the history of documents, therefore, we will follow the timeline without invoking such distinct categories. For convenience, and acknowledging that this involves gross over-simplification, the timeline is divided into five periods: prehistory and the ancient world; the medieval world; the age of print, including the Renaissance and the Enlightenment; the age of mass communication, roughly approximating to the long 19th century; and the documentation age, to the present day.

Luciano Floridi (2014, chapter 1) has distinguished *prehistory*, when there are no information systems, *history*, when information systems support society, and *hyperhistory*, when society is dependent upon, and defined by, information systems. In our timeline, the first period sees the move from prehistory to history, and the last the move from history to hyperhistory.

A thorough and detailed overview of information in society across all periods is given by Blair et al. (2021), and a shorter account by Gleick (2011). For the history of reading and writing, see Manguel (1997), Robinson (2009; 2013) and Clayton (2013). For book history, see Raven (2020), Duncan and Smyth (2019), Suarez and Woudhuysen (2013) and Pearson (2011), and for the history of printing, see Twyman (1998) and Mosley (2013).

Prehistory and the ancient world

Information communication began with spoken language, but we can only speculate how information may have been transmitted through oral communication. It is only with the origins of recorded information that we can begin a history of information, based on documents. We take this first period as lasting from the first records to the fall of the Western Roman Empire in 476. For overviews of libraries in the period, see Ryholt and Barjamovic (2020), Nicholls (2017), König, Oikonomopoulou and Woolf (2013), Too (2010) and Casson (2001); for archives, see Brosius (2003) and Delsalle (2018); and for writing, reading and literacy, see Houston (2004), Kallendorf (2013) and Kolb (2018).

The earliest artefacts which we might reasonably describe as documents are cave paintings and other forms of rock art, the oldest believed to be more than 40,000 years old; the famous cave paintings of Lascaux in France are 20,000 years old; see <https://archeologie.culture.fr/lascaux> for video images. The purpose of these drawings of animals and people, and seemingly abstract designs, cannot be known, though this has not stopped a lot of speculation. Whether it is reasonable to call them ‘documents’ is a

moot point; but they show that the people of that time had the means to record signs and symbols, and thereby to communicate across time, whatever their purpose may have been (Bahn, 2010; 2016; David, 2017).

Inscriptions believed to be forms of *proto-writing*, with symbols giving a permanent record but of which the meaning is unknown, and dated to 7,000 years ago, appear on portable stone artefacts; messages could now be sent across space as well as time. One example is the Tartaria tablets from Romania, believed by some experts to be the earliest form of writing yet found, though this is still a matter of controversy. More recent examples are the bone tags found in tomb U-J at Abydos, with a precursor of the Egyptian hieroglyphic script c. 3320 BCE ([Figure 2.1](#)).

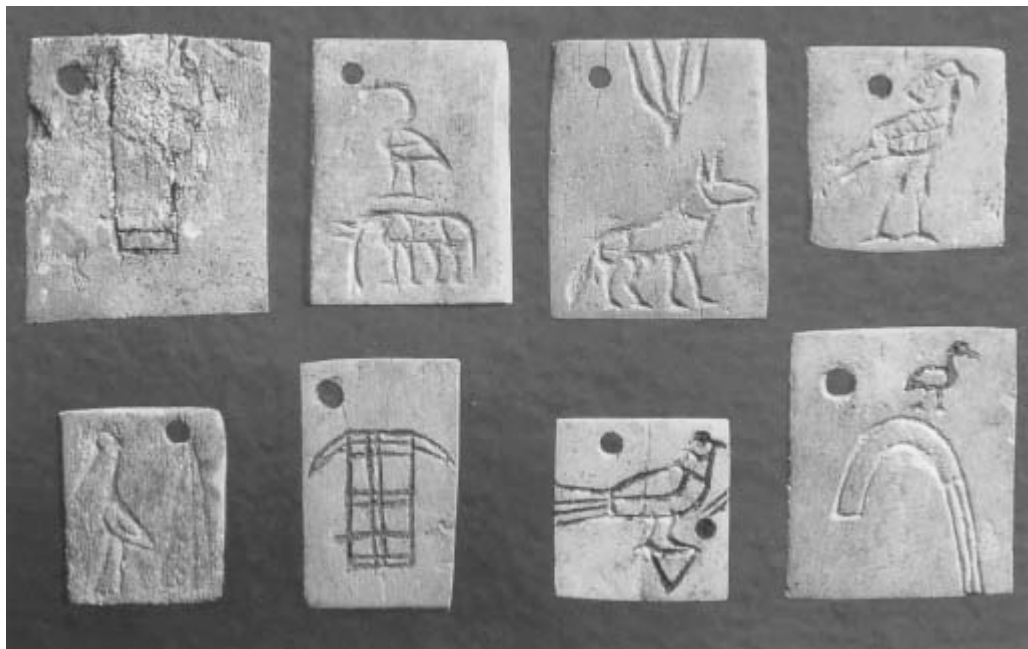


Figure 2.1 *Proto-writing on bone tags from tomb U-J at Abydos, Egypt, c. 3320 BCE* (© DAI Cairo, photographer Dieter Johannes)

The term proto-writing can also describe contemporary sign systems such as computer icons, electrical circuit diagrams, mathematical notation and musical scores. The 'proto' prefix refers not to historical but to functional development.

True writing, the agreed repertoire of signs and symbols that is used to reproduce any of the thoughts and feelings that the writer wishes to express, appeared over 5,000 years ago. It is now believed that writing was invented independently at least four times: in Mesopotamia, Egypt, China and Mesoamerica, although some older texts consider that writing systems may have spread by a mechanism of diffusion. The earliest accurately dated writings are from Uruk in Mesopotamia (3300 BCE).

Mesopotamian writings used the cuneiform script, written on clay tablets using sharpened reeds as writing implements. In Egypt, the hieroglyphic, hieratic, and, later, demotic scripts were written in ink on a variety of materials: the stereotypical papyrus, but also wood, stone and pottery. Because of the nature of the materials used, and the durability of clay tablets, we know much more of collections in Mesopotamia than their equivalents in Egypt. [Figure 2.2](#) shows an example of such a tablet.



Figure 2.2 *Cuneiform tablet* (courtesy of Los Angeles County Museum of Art, public domain)

The earliest writings from Mesopotamia were accounting records. The earliest collections were attached to palaces and temples, with little extant evidence of organisation or intent. It is not clear if any distinction was made between library and archives for these earliest collections. Moving forward in time, there is evidence of different kinds of collections, encompassing religious material, government records, business and trade records, family documents such as genealogical tables, property and inheritance matters, astrological predictions, scientific and medical texts, literary works, and correspondence of all kinds. Three examples are given below, showing the size and sophistication of document collections in ancient Mesopotamia; for more detail, see Postgate (2013), Radner and Robson (2011) and Robson (2019).

The library/archive at the palace of Ebla, in modern Syria, active between 2600 and 2300 BCE, comprised tablets stored on wooden shelves, with different types of materials held in separate rooms. Two 'archive rooms' held over 17,000 tablets of commercial, legal and administrative records. Another room held collections of dictionaries, proverbs, epic narratives, myths, hymns, incantations and rituals. Others contained material on botany, zoology, mineralogy and mathematics. There was even what might be regarded as an early form of reference collection. Many tablets were devoted to lists of persons and items, often in order of precedence and importance: gods, kings, professions, animals, stones, etc. The ordering of items in the rooms was maintained by rudimentary written classification tables.

The temple at Sippar, in modern Iraq, active over a very long period, had a library of tablets stored in brick pigeonholes. Over 40,000 items have been found, divided into records of the temple's activities, including business records and literary texts.

Finally, from the later part of this period, the palace of Assurbanipal at Nineveh, again in modern Iraq, built around 650 BCE, had a large, well-organised and catalogued collection, with around 10,000 distinguishable works on 30,000 tablets, arranged by subject in different rooms. They were stored in earthenware jars on wooden shelves, related items kept together in a basket, with a front tablet describing the set; each tablet had an identification tag, noting its location 'jar-shelf-room'. An early form of catalogue, also written on clay tablets, was provided near the door of each room, giving for each work present its title, opening words, the number of tablets

comprising the work, and its location symbol. A display of tablets from this library in the British Museum is shown in [Figure 2.3](#) opposite.

Cuneiform developed over time from pictograms, through ideograms, to a complex logo-syllabic script. Its use diminished following the absorption of Mesopotamia into the Persian empire with the coming of Cyrus the Great in 539 BCE. Cuneiform was replaced by the Aramaic script, based on the Phoenician alphabet, as Aramaic was widely spoken throughout the Persian empire.

The last known dated Akkadian cuneiform text, from Babylon, is 75 CE, although the script was probably used through later centuries. Cuneiform was used to write a variety of languages, from Sumerian, Akkadian, Hittite and Elamite to old Persian.

Less is known about libraries and archives in ancient Egypt, although it is clear from surviving writings that they existed. Their materials – papyrus and, later, parchment – have for the most part not survived, and the physical structures have yet to be found. It is known that there were institutional libraries, which may have had a state function, at the temples of Tod, Edfu, Esna and Philae, and also groups of papyri in domestic settings which may have been a kind of private library. The only surviving example is from the Temple of Edfu, in Upper Egypt, where there is evidence of a small number of scrolls being kept in a recess, a list of titles of the scrolls being carved on the wall nearby; there is, however, doubt as to whether this can be considered as a true library. There are references to scribes having priestly duties and it is thought that religious leaders may have had a role as librarian/archivist.



Figure 2.3 *Library of Ashibanurpal, British Museum* (photograph Dr Gary Todd; public domain, via Wikimedia Commons)

Evidence from inscriptions within tombs, temples and on stelae, etc. indicate several institutions associated with storing documents, possibly dating back to the earliest dynastic periods of the Old Kingdom (2686–2160 BCE):

- per ankh ‘house of life’
- per medjat ‘house of papyrus rolls’
- hall of written documentation
- storage places for annals (from 2600 BCE).

For more details of libraries and archives in ancient Egypt, see Webb (2013), and for the use of documents and recorded information in general, see Eyre (2013).

The earliest European scripts were originally found in Crete, although examples have now been found around the Aegean. Three

scripts have been identified: Cretan hieroglyphic, dating from 2100–1700 BCE; Linear A, 1750–1450 BCE; and Linear B, somewhat later; although the scripts are thought to have overlapped in time, with no obvious direct line of progression. The latest Linear B inscriptions, found at Pylos, date from 1200 BCE.

Linear B was deciphered in the 1950s and shown to be ancient Greek (Chadwick, 2014). Linear A has been partially deciphered, but appears to be an unknown language, and Cretan hieroglyphic is regarded as a form of proto-writing.

A gap in time occurs between 1200 BCE and 800 BCE, during which no evidence of writing dated to this period has been found in this region. This time is referred to as a ‘dark age’, and thought to evidence a decline in civilisation, possibly due to war or other disaster. The next examples of writings discovered from Greece use the Greek alphabet, entirely different from the Linear B script.

Until further examples of Cretan hieroglyphic or Linear A are found, we are unlikely to move forward in our understanding of the content embedded in these scripts.

Although details have not been established, it is clear that document collections were established in China during this period (1200 BCE), with writings on bamboo, wood or silk – a system including libraries for royal palaces, central government and local administrations, with scholars enlisted to construct catalogues and classification schemes (Allen et al., 2010; Zheng, 2014; Diao, 2019).

In the ancient world we can see already some of the elements of an information infrastructure; physical spaces, with different types of

material arranged by location; descriptions of work and parts of works; shelf lists, locator tags, and early forms of catalogue; subject arrangement; lists and proto-classifications; reference works; and multiple copies of texts, with master copies of important works held centrally.

The classical world

The emergence of the alphabet had a significant impact on writing, and on the intellectual tools used to provide access to the writings. An alphabetical script was more readily understood than the complex logo-syllabic cuneiform, or the numerous Egyptian hieroglyphs, and provided a ready-made solution for arranging materials.

The origins of the alphabet are unclear (Robinson, 2013). It is believed that the Phoenician alphabet, first seen around 1000 BCE, used a small number of symbols to replace the much larger symbol sets of cuneiform and hieroglyphic writing and could be used to write different languages, greatly increasing the flexibility of representation. It influenced the Aramaic/Hebrew and Greek alphabets, leading to the first substantial bodies of alphabetic writings (Steele, 2019) and to the first ordering of books by alphabetical order.

The derivation of the Phoenician alphabet from earlier scripts remains an incomplete story, although current thinking is that the script developed from Egyptian hieroglyphs via a proto-Sinaitic or proto-Canaanite script.

This is based on the discovery in 1904–5, by Flinders Petrie, of inscriptions which he believed to be alphabetic at Serabit el-Khadim (Petrie, 1912). The script became known as Proto-Sinaitic. In 1916, Sir Alan Gardiner concluded that the Sinaitic signs were created by reforming Egyptian hieroglyphic signs (Gardiner, 1916).

The earliest known alphabetic inscription in Greek dates from around 730 BCE, but the date of the invention of the Greek alphabet remains uncertain, placed between 1100 and 800 BCE (Robinson, 2009).

The Greeks wrote on stone, papyrus, pottery and slate covered in wax. Libraries and record centres became widespread throughout the Hellenistic world, with evidence for significant libraries at Pergamum, Antioch, Pella and Alexandria. There were recognisable royal, private and public libraries, supported by scriptoria and a book trade. Texts were written on papyrus rolls and stored on shelves, in pigeonholes or chests, with tags or smaller scrolls to identify title and author. Strabo tells us that Aristotle was the first to teach the Hellenistic kings of Egypt how to arrange a library.

The best-known library of this period is, of course, the Great Library of Alexandria in Egypt, founded by Alexander the Great in the fourth century BCE and developed by the Ptolomaic kings over the following centuries (El-Abbadi and Fathallah, 2008). A centre for scholarship, it was believed at its height to contain over 400,000 Greek papyrus manuscripts: essentially all of human knowledge at that time. Its first director, Xenodotus, is said to have created a shelving system, with a subject division derived from that of Aristotle and items arranged within these divisions by alphabetical

author names. The famous *Pinakes* may have been a catalogue of this library, or a bibliography of literature; presented on many scrolls, it was divided according to the classification of its creator, Callimachus (c. 310 BCE–c. 240 BCE). This had sections for philosophy, law, history, medicine, natural history and miscellanea, with literature divided into rhetoric, lyric, comedy, tragedy and epic works. Including a general guide to the literature of the day, it is regarded as the first recognisably modern form of bibliographic tool.

The Roman world adopted the documentary infrastructure of the Greeks, with many public and private libraries and a flourishing book trade (Nicholls, 2018). The first Roman public libraries were initiated by the military commander Asinius Pollio (75 BCE–4 CE) around 39 BCE, and by the emperor Augustus some years later. The *codex*, the forerunner of the modern book, was developed by the Romans from wooden writing tablets tied or hinged together, later replaced by parchment pages. Considerably more convenient to store, arrange and use than scrolls, codices replaced scrolls for most purposes over the period 100–500 CE. The first reference to literary works in the form of a parchment codex is made by Martial during the years 84–86 CE (Reynolds and Wilson, 2013, 35). The impetus for change from the scroll to the codex is thought to have come from early Christians in biblical texts; pagan texts in codex form were rare in the second century CE. The transfer of classical texts from scroll to codex happened gradually and may have resulted in the loss of texts considered less interesting or important at the time. All modern books are technically codices, but the term is generally reserved for handwritten books produced from late antiquity through the Middle

Ages. Some of the earliest surviving works are parchment codices from the fourth century CE. The *Codex Sinaiticus*, a manuscript of the Christian Bible written in Greek in the fourth century, is a well-known example.

The medieval world

Developments begun in the classical period continued throughout medieval times, which is conventionally taken as lasting from 477 to the fall of Constantinople in 1453, although the establishment of Gutenberg's printing press a few years earlier might be a more appropriate end marker for our purposes.

The earlier part of the medieval period used to be referred to as the 'Dark Ages', but this gives a quite misleading impression of a loss of all culture and documentation. In fact, there was a rich and evolving book culture throughout the period, based on the manuscript codex, which began with the transmission of classical works but moved on beyond it (Kwakkel, 2018; Reynolds and Wilson, 2013). Initially, the documentary legacy of the classical world was maintained in the Eastern Roman (Byzantine) empire, with Constantinople a centre for scholarship and the route through which many classical writings were preserved (Wilson, 2013). There were three major libraries, which can be categorised as royal, religious and academic.

Subsequently, innovations in information communication continued, in Europe and beyond. Parchment, able to be written on both sides and better able to accept illuminations, became the norm, with the sheets folded and sewn together in codex form.

This was the period of the illuminated manuscript, now the treasures of many national libraries. Other forms of information representation – maps, musical notations and mathematical symbolism – were developed, the last using the so-called Arabic numerals, originally an Indian innovation. The Islamic world was at that time much ahead of the West in document production, having embarked on a major programme of translating Greek, Persian and Indian texts and extending them by original scholarship (Al-Khalili, 2010; Lyons, 2009). This process was pioneered by the ‘House of Wisdom’ (Bayt al-Hikma) established in ninth-century Baghdad by the Abbasid caliph Al-Ma’mun, who wished to collect all the world’s books under one roof. Nothing remains of this institution, and its nature and purpose remain speculative. The first known star charts appeared at this period, in China (Figure 2.4).

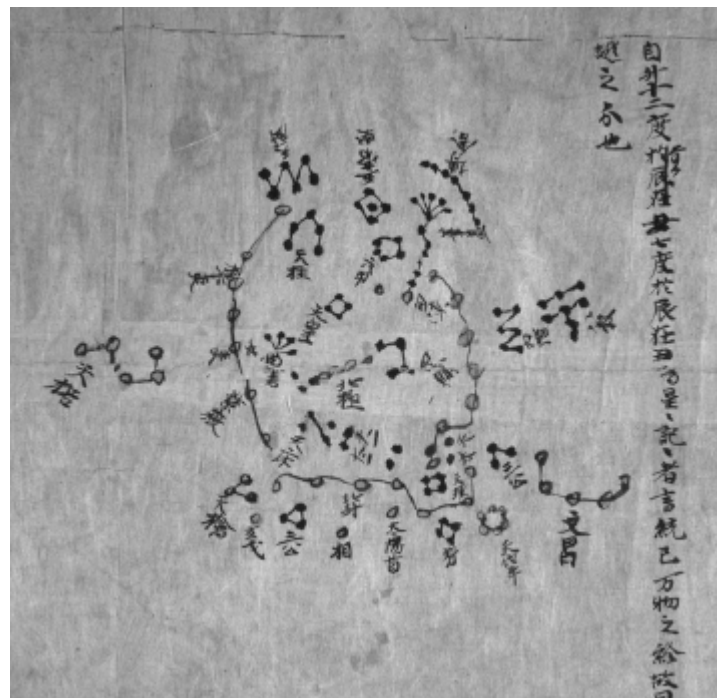


Figure 2.4 *Chinese star chart, earliest-known manuscript atlas of the night sky, produced in central China around 700 (public domain, via the British Library [Or.8210/S.3326])*

For much of this period in the West, writing, and to a large degree reading, was focused in religious institutions; see, for example, Dyson (2019) on the books used by priests in Anglo-Saxon England. Monasteries and cathedrals had substantial collections of parchment codices and were the major producers of such works. Notable examples include:

- St Augustine Gospels (late sixth century, Latin)
- Bede's Ecclesiastical History of the English People (completed in 731) ([Figure 2.5](#))
- St Cuthbert Gospel (early eighth century, Latin) ([Figure 2.6](#))
- Lindisfarne Gospels (around 700, Latin/Old English)
- Anglo-Saxon Chronicle (compilation of manuscripts produced around 890)
- Beowulf (undated, c. 700–1000, Old English, held at the British Library)

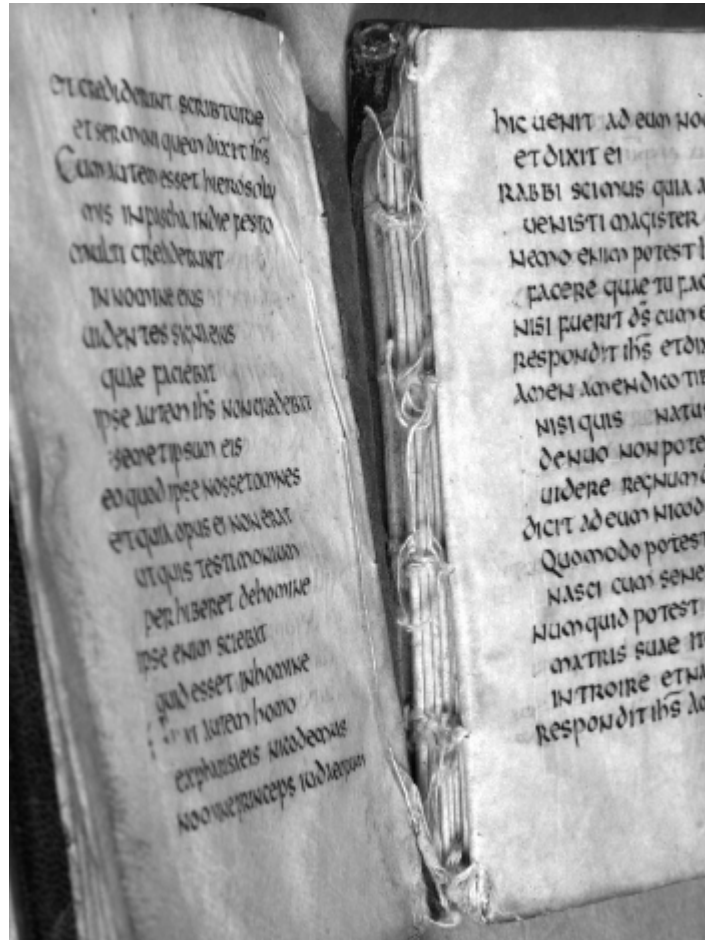


Figure 2.6 *St Cuthbert Gospel of St John (Stonyhurst Gospel), the oldest intact European book, early 8th century* (public domain, via Wikimedia Commons)

Following the end of the Anglo-Saxon period in 1066, the Norman rule of England saw the production of the Domesday Book in 1086 (Latin, Middle English, held at the National Archives), giving documentary evidence of the established English state ([Figure 2.7](#) on the next page).

By the end of the 12th century secular scribes were banding together into guilds and workshops. Works in local languages – as opposed to Latin or Greek – were becoming more common, and

treatises on mathematics, logic, philosophy and astronomy began to appear, often translated from Arabic. It is not unreasonable to suggest that, two hundred years before the introduction of printing, a publishing industry had been established.

This was promoted by the rise of secular universities throughout Western Europe, stimulating a need for multiple authoritative copies of texts. This was coupled with a demand for popular and professional books, including medical texts, educational books, novels and cookery books. The production of manuscripts attained a professional status, with scribal fraternities and apprenticeship schemes. Paper had reached Europe in the 12th century, a Chinese innovation from the second century or earlier (Allen et al., 2010; Zheng, 2014), passed on via the Arabic world after its introduction to Samarkand from China in 751. However, it was initially used only for formal documents, and not for books. The first paper mill in England was established at the end of the 15th century, in order to meet the demands of printers.



Figure 2.7 A page from the Domesday Book covering modern Oxfordshire (public domain, via archive.org)

Toward the end of the period, several European monarchies, for example, those of England, France, Austria and Hungary, established royal collections, which provided the foundations of later national libraries; see Tanner (2008) for a detailed account of the Hungarian example. University libraries appeared, some of which survive to the present day, at Oxford, Cambridge, Paris, Padua and elsewhere.

Catalogues of this period were accession lists of authors and titles, and served to support an active programme of document loans between libraries and scholars (Reed-Leal, 2020). The first union catalogues were created; in 14th-century Paris, for example, the Sorbonne university library had a union catalogue of several Parisian libraries. Secular classification was typically based on the classical

seven liberal arts: the *trivium* of grammar, rhetoric and dialectic (or logic) and the *quadrivium* of arithmetic, geometry, music and astronomy. This form of classification was first devised in the sixth century by Flavius Magnus Aurelius Cassiodorus (c. 490–c. 585), a philosopher and theologian living in the last days of the Roman world. Classifications arranged by the ecclesiastical calendar and the names of saints met the needs of religious institutions, such that material appropriate for reading on each day could easily be located.

The age of print

This period from the introduction of printing to the conventional beginning of the ‘long 19th century’ in 1789 saw the introduction of an information infrastructure based on printed documents.

Printing with moveable characters had been established in China – where characters of baked clay were used initially – for several centuries before it was introduced into Europe in the middle of the 15th century (Zheng, 2014). The earliest dated printed book is the Diamond Sutra, held by the British Library, printed in Chinese translation in north-west China in 868 ([Figure 2.8](#)).



Figure 2.8 *Frontispiece of the Diamond Sutra* (public domain, International Dunhuang Project, via Wikimedia Commons)

Credit for the introduction of the printing press into Europe is usually given to Johannes Gutenberg, who established a press in Mainz around 1447, his colleague Peter Schoeffer having discovered a viable method of casting the letters using an alloy of tin and lead. The first printing press was set up in England in 1476, by William Caxton; the first product was *The Dictes or Sayings of the Philosophers*. For more details of early printing and printed books, see Twyman (1998), Wagner and Reed (2010), de Hamel (2013), Dondi (2013) and Mosley (2013).

Printing houses spread throughout Europe during the next century, combining the talents of compositors, engravers and type-casters to give a recognisably modern printing industry, able to

produce both text and images, though by a rather clumsy process with images reproduced separately, initially on wood-block carvings and later on metal plates. Printing had two general effects. One was the availability of multiple identical copies of works, calling for bibliographic tools – catalogues, bibliographies, tables of contents and indexes – in which there was a single description for a printed work, rather than for each unique handwritten manuscript. The other was that printing led to the original ‘information explosion’, with many more copies of more titles, of an increasingly diverse nature (Eisenstein, 2005; Barber, 2017).

The first bibliographies in a modern sense appeared at this stage; notably Conrad Gesner’s *Bibliotheca Universalis* of 1545, which pioneered ideas of selection, arrangement and indexing. Gesner (1516–65), who has been described as the first would-be universal bibliographer and the last to have had any chance of succeeding, managed to include 15,000 titles in Hebrew, Greek and Latin, estimated at 20–25% of the total available at the time. The publisher, bookseller and bibliographer Andrew Maunsell (d. 1595) published a *Catalogue of English Printed Bookes* in 1595, the first attempt at a British national bibliography. Classification schemes, attempting to reconcile a philosophical organisation of knowledge with the practical need to arrange books in libraries, were developed by several scholars, notably Francis Bacon (1561–1626) and Gottfried von Leibnitz (1646–1716) (Schulte-Albert, 1971). A variety of reference tools and techniques were developed for organising and managing information (Blair, 2010; Krajewski, 2011).

In the 17th century advances in printing technology led to both a more rapid production process and the ability – through lithography – to print text and images at the same time. This led to the increased numbers and readership of newspapers, magazines and scientific and professional journals. The first scientific journals were established in 1665, the Paris-based *Journal des Savants*, first published on 5 January, just claiming precedence over the *Philosophical Transactions of the Royal Society* of London, which was first published on 6 March.

The so-called Enlightenment period of the 18th century saw a further expansion of the information infrastructure. Major reference tools, such as Chambers' *Cyclopedia* (1728), Diderot's *Encyclopédie* (1761), Samuel Johnson's *Dictionary* (1755) and the *Encyclopedia Britannica* (1768), appeared for the first time, accompanied by national and subject bibliographies and new ideas of classification and bibliographic principles. National libraries and 'memory institutions' began to appear: the British Museum and its associated library in 1753 and the Bibliothèque Nationale in 1789. Headrick (2000) shows that many of the conceptual developments in the organisation, presentation and communication of information made during this period enabled the technical advances of the following century.

The age of mass communication

The advent of mass communication, due to the adoption of the products of the Industrial Revolution for the purposes of information dissemination, laid the foundations of the modern information environment, and the long 19th century (1789–1914) can

be regarded as the mass communication age. The first steam-powered presses, introduced early in the 19th century, increased the rate of printing considerably, but this soon went up by orders of magnitude; one of the first such machines, Koenig's steam press of 1814, is shown in [Figure 2.9](#) on the next page. Whereas a hand press typically printed 250 pages in an hour, by the 1820s steam-powered presses could produce 4,000, and by the 1850s newspaper rotary presses were achieving 20,000 sheets an hour. Steamships and steam trains provided rapid and reliable transport for this new volume of printed material. An example of the way in which steam-powered printing and transport affected the creation and dissemination of information is given in Fyfe's (2010) account of the Edinburgh-based publishers W. & R. Chambers.

A much greater range of books and serial publications was produced, both popular and academic; in particular, many more specialised scientific and professional journals, monographs and textbooks to meet the needs of the rapidly professionalising and specialising scientific and technical activities (Meadows, 2004). Communication in general improved, with the introduction of the telegraph in the 1830s, national postal services in the 1840s and the telephone in the 1870s. Photography was introduced in the 1840s, and three decades later photographs were commonly being reproduced in printed documents. Charles Babbage's analogue computers of the 1830s, although their concept proved to be beyond what the technology of the time could deliver, gave intimations of things to come.

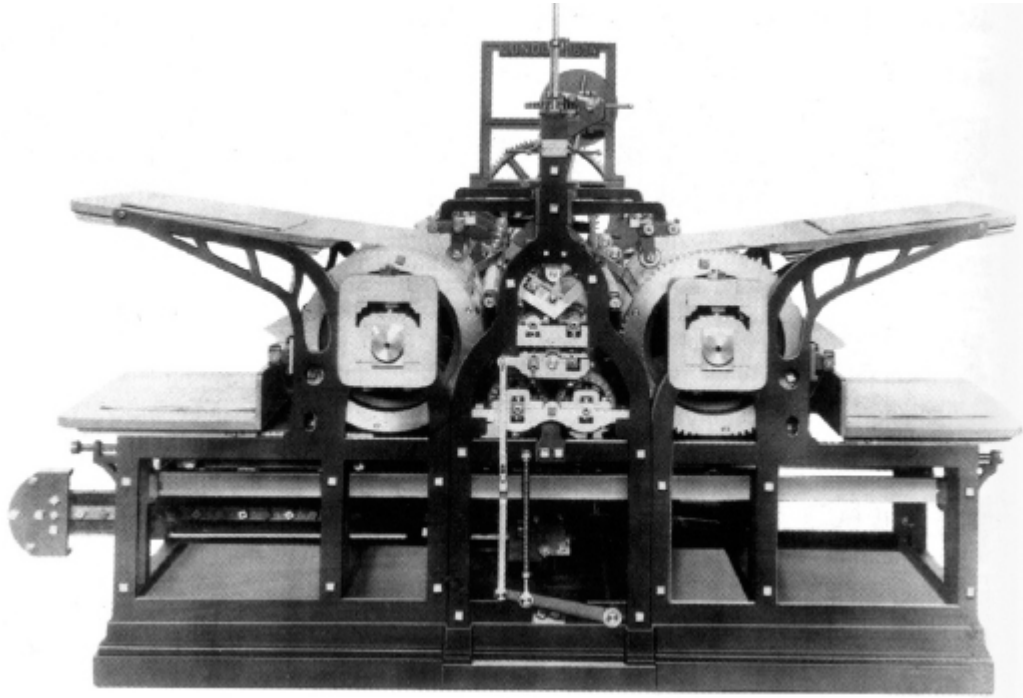


Figure 2.9 *Koenig's steam-powered printing press, 1814* (Wikimedia Commons, PD-Art)

A range of library and information services was created during the 19th century: national libraries, such as the Library of Congress (1800); specialised national information institutions, such as the French Académie Nationale de Médecine and the US National Library of Medicine (1836); memory institutions such as the National Gallery (1824) and the UK Public Record Office (1838); many academic and professional societies, typically with their own library and/or archive; modern university libraries; the first abstracting and indexing services – *Chemisches Zentralblatt* (1830) for chemistry and *Index Medicus* (1879) for medicine; and public library services, both subscription and free.

The intellectual tools for dealing with information also made major advances. Anthony Panizzi's 91 rules (expanded from the

original 73) for the British Museum library catalogue of 1841 were a major development (Carpenter, 2002), paralleled by the principles developed by Charles Jewett for the library of the Smithsonian Institute in the USA. Dewey's decimal classification first appeared in 1876, and the first modern subject headings lists appeared towards the end of the century, notably the Library of Congress Subject Headings in 1898.

Such familiar information-handling tools as filing cabinets, standard-sized envelopes, pencils with erasers, paper clips and typewriters were also developed during this period, the card index and catalogue became the major tool for information management (Figure 2.10) and a variety of new tools and networks for information exchange were established (Krajewski, 2011; Rayward, 2014). In many ways, despite technical advances, the information environment showed its Victorian origins until well into the 20th century.



Figure 2.10 *Subject card catalogue of the library of the University of Graz*
(Dr Marcus Grossler, Wikimedia Commons, CC BY-SA)

The documentation age

The 20th century saw the introduction of new forms of analogue document; photography, cinematography, sound recording, microforms and, later, audio and video tape. Mechanisation was introduced to documentation, in the form of punched cards, edge-notched cards, optical co-incidence cards and microforms, which allowed complex searches using Boolean logic, based on detailed indexing. For example, Lloyd's Register of Shipping introduced a co-ordinate system of edge-punched cards to index surveyors' reports in the late 1940s, converting this to punched cards in the late 1950s (Boyd, 1963).

The advent of the digital computer at the mid-point of the 20th century led to convergence of formats; text, images, sounds, and video all became digital. New intellectual tools, such as faceted classifications and thesauri (discussed in [Chapter 8](#)), were introduced, in an attempt to deal with the increasing volumes of materials in all formats. Many discipline-specific information services were introduced; see Powell (2000) for the development of one important example, *Chemical Abstracts*. Information and knowledge management became established disciplines, and the concept of an 'information society' (discussed in [Chapter 16](#)) was conceived. Many print-based information services moved to digital networked form and the concept of *online searching* was established (Bourne and Hahn, 2003). For overviews of developments in this period, see Rayward (2008), Black and Brunt (1999) and Black, Muddiman and Plant (2007).

At the end of the century the introduction of the internet and World Wide Web, and the advent of social media, brought about new forms of document and of communication, leading to repeatedly voiced suggestions that this is a change in the information environment as dramatic as, and potentially even more far-reaching than, that due to printing, or even writing. These 20th-century developments are covered in the relevant chapters.

The centuries-old desire for universal bibliographic control was pursued by the new 'documentation movement', pioneered by the Belgian scholars Paul Otlet and Henri La Fontaine ([Figures 2.11](#) and [2.12](#) opposite). Making use of new intellectual and physical tools for information organisation, Otlet – particularly through his

Mundaneum documentation centre and his influential book *Traité de documentation* (1934) – provided the foundations for the modern sciences of information and documentation (Rayward, 1994; Wright, 2014).

Summary

This overview of the history of information has necessarily been brief, and somewhat simplistic. Economic, social, technical and academic developments have gone alongside the growth of knowledge and the technologies and systems which have tried to control the resulting increases in information, records and documents. These developments have never been as orderly as they may seem, as each influences, and is influenced by, the others. We can, nonetheless, learn lessons from the information past, and use them to understand the information present and to imagine the information future.



Figure 2.11 *Paul Otlet in his home office, 1937* (public domain, via Wikimedia Commons)



Figure 2.12 *Henri La Fontaine* (photographie Geruzet Freres, Bruxelles) (public domain, via Wikimedia Commons)

- Understanding the history of information and documents is not only intrinsically interesting, but is helpful in understanding the present and in seeing analogies with past problems and issues.
- There are several different approaches and perspectives in the study of the history of information and documents.
- The developments which have, each in turn, drastically changed the information environment have been language, writing, printing, mass communications and networked digital information.
- We live in what is still, to an extent, a Victorian information environment.

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3 Philosophies of Information

Today, philosophy faces the challenge of providing a foundational treatment of the phenomena and the ideas underlying the information revolution, in order to foster our understanding and guide both the responsible construction of our society and the sustainable management of our natural and synthetic environments. In short, we need to develop a *philosophy of information*.

Luciano Floridi (2013, xii)

The philosophy of information is congruent with library tradition. From Bliss and Danton's discussion in the 1930s, through Egan (without Shera), Nitecki, and Floridi, there stands a very definite genealogical line.

Ariel A. Morán-Reyes (2015, 587)

Introduction

Topics at the core of the information sciences, such as the nature of knowledge and what we can know, how meaning and truth are ascertained, and how things can be distinguished, named, categorised, and classified, have been the concern of philosophy for many centuries. Yet information science has no agreed philosophy of its own; rather, a wide variety of philosophies and philosophers have been pressed into service to analyse its concerns. In this chapter we will look at some ways in which philosophy and information science overlap and interact, and how philosophical ideas may directly affect the theory and practice of the information disciplines. This is an extensive, and in some aspects technical, topic, and only a

selective and informal overview will be given, with references to the literature for those wishing to take it further.

Philosophy and its relevance to information science

There are usually held to be three main branches of philosophy: *ontology* and *metaphysics* – what is it to ‘be’, and what kinds of things can exist; *epistemology* – what is it to ‘know’, what can we know, and how can we be sure of our knowledge; and *ethics* – how can actions be ‘right’, and how should we act. All of these have relevance to the information sciences; see Furner (2017a) for more detailed discussion. Metaphysics (the study of what can exist) and ontology (the study of fundamental categories and kinds of things) address the existence and nature of entities such as information, documents, relevance, aboutness and many more. Epistemology questions what is meant by knowledge, with very practical consequences for knowledge management. Ethics questions what is the right use of information, and touches on issues such as censorship and privacy. This is treated in detail in [Chapter 15](#), and will not be repeated here. For accessible short introductions to philosophy itself, see Craig (2020), Grayling (2020) and Williamson (2020); for ontology and metaphysics, see Ney (2014) and Rea (2020); and for epistemology, see Audi (2011), Pritchard (2018) and Nagel (2014).

Furner (2010) gives a review of the nature of philosophy and its relation to the information disciplines, reminding us that there are two main aspects: analysing the nature of the information disciplines from a philosophical viewpoint, and analysing the nature of entities which are important for the subject. The first has led to consideration of the value for the information sciences of general philosophical

positions such as realism, positivism and constructionism. The second has led, in particular, to analyses of the nature of data, information and knowledge. Hardiman (2014) gives a wide-ranging review of philosophical positions, showing their relevance to archives and records management.

There has been limited examination of the influence of philosophical positions on information science, despite the fact that they may have a significant, if implicit, influence on practice. A *realist* perspective, for example, asserting that an objective external world exists, and we can understand it to an extent, underlies the systems paradigm in information science, and research methods such as experiment and quantitative surveys. A *constructivist* or *constructionist* perspective, asserting that reality is, at least to a degree, subjective, and created by individuals and groups, underlies the cognitive paradigm, and a preference for holistic and qualitative research, focusing on individuals' knowledge and experience. See Hjørland (2004; 2005) for realist and positivist perspectives in information science, and Talja, Tuominen and Savolainen (2005) for constructivist/constructionist perspectives. Other philosophical positions examined from an information science perspective are *pragmatism* and *neo-pragmatism* (Sundin and Johannisson, 2005; Buschman, 2017), and *empiricism* (Hjørland, 2005; 2011).

There has been no consensus that the information disciplines need a foundational philosophy. For example, Zwaldo (1997), noting that, in their respective literatures, librarians rarely discussed philosophy and philosophers never discussed library science, suggested that it was unnecessary, and rather self-defeating, to seek for a single

philosophy, or indeed any philosophy of LIS. For a rejoinder, see Radford and Budd (1997), who argue that a lack of philosophical discussion in the theory and practice of LIS does not mean that philosophy was absent; rather, that philosophical positions, typically rather simplistic, have been, and are, assumed and taken for granted.

The idea that the information disciplines have not been interested in philosophical issues is far from the truth; as Fallis and Whitcomb (2009, 176) write, with numerous examples, 'quite a lot of applied epistemology has been done by information scientists'. Many commentators have argued that it would be to the advantage of the information disciplines to find a more secure theoretical foundation; see Bawden and Robinson (2018) for a review. Most notably, Luciano Floridi (2002) argued that LIS needs a foundational philosophy and that the discipline's identity crisis, originating in the 1930s, has been due to a search for such a foundation. LIS and philosophy, in Floridi's argument, are closely associated, because of the topics in which they are interested and because of the level and scope of their research.

There have been criticisms of the rather superficial way in which some of those who advocate for the value of philosophy in information science have applied it. Floridi (2002, 47) writes that information researchers 'have been lured by a variety of friendly but pre-established philosophies instead of fighting for their own place in the philosophical field'. Zwaldo (1997) likens it to borrowing a book, which is never ours, and which we will soon return and replace with another. Vickery (1997, 458) expressed this tendency as 'examining the ideas of a ... philosopher, extracting principles and

offering them as presuppositions upon which information science may be based’, while Frohmann (1992, 366) noted eloquently that ‘high-flying LIS researchers swoop indiscriminately down upon the theoretical terrain, colonising Popperian worlds, or cannibalizing hermeneutics, phenomenology, general systems theory, symbolic interactionism, decision theory, existentialism, structural-functionalism, cognitive science, or philosophy of language, to name just a few . . .’.

There have, however, been three strands of thought which have developed a deep relation between philosophy and the information disciplines, with the opportunity for information science to influence and add to the philosophy, rather than just drawing from it. We can categorise these as philosophies of information, philosophies of knowledge and critical theory as related to documents and media, and we now consider each in turn.

Philosophies of information

Most philosophies of information are ontological or metaphysical, that is to say they focus on analysing what information is. This aspect is discussed in detail in [Chapter 5](#), and only the general philosophical issues are covered here; for an overview from a variety of perspectives, see Floridi (2016), and for a shorter review, see Compton (2015).

The attempt to build a philosophy around information and data, rather than knowledge, which has been a concern of philosophers since the earliest times, began with Shannon’s mathematical theory of communication. This was the first successful attempt to give a rigorous quantitative theory of information, at the cost of neglecting

the meaning of an information-bearing message. Several theorists attempted to extend Shannon's information theory to include the meaning of, and knowledge in, the communicated information, the most notable being the American philosopher Fred Dretske (1932–2013), with his concept that knowledge is information-produced belief (Dretske, 1981). Floridi (2019a) reviews these attempts, concluding that none has been fully successful.

Karl Popper

The Austrian-born philosopher Karl Popper (1902–94), who spent most of his career at the London School of Economics, developed his 'Three Worlds' ontology, in part as a response to the limitations of the problematic traditional philosophical *justified, true belief* view of knowledge: World 1 is the physical world, World 2 is the subjective mental world and World 3 consists of objective, communicable knowledge, 'knowledge without a knowing subject', which Popper denoted rather vaguely as 'the contents of books and libraries', which includes errors and grows and develops in an evolutionary way (Popper, 1992; 1979; Notturmo, 2002). Popper's idea, and particularly his World 3, were heralded by Brookes (1980) to be a philosophical foundation for information science, and the task of information science to be the exploration and organisation of World 3. However, its impact has been limited. Although it has been popular as a pedagogical model, and used on occasions for analysing the information environment (see, for example, Bawden, 2002 and Gnoli, 2018), it has generated little in the way of either detailed conceptual insight or practical guidance. Popper's conception did, however, influence Luciano Floridi's development of

his philosophy of information (PI) (Gorichanaz et al., 2020, 593 and 608–9).

Luciano Floridi

Floridi's PI has been developed since the 1990s and is currently the most thorough and sophisticated approach of its kind; for detailed treatment, see Floridi (2011; 2013; 2019b), and for accessible overviews, see Floridi (2010; 2014a). Rather than being the philosophy of a restricted subject area, it is an all-encompassing philosophy for the information age: 'philosophy of our time, for our time' (Floridi, 2019b, xii).

It is rooted in an ontology which sees reality as ultimately informational, a form of realism termed *Informational Structural Realism* (ISR). Data is defined as any difference or lack of uniformity, and information as well-formed, meaningful, truthful data, as will be discussed fully in [Chapter 5](#). This is extended into a fully developed philosophical system, including epistemological, ethical and political facets, which is able to address many current issues, including information governance, information society, good use of AI and algorithmic decision making, digital well-being and privacy. Ess (2008) notes that it was remarkable that PI's ontological analysis of information could lead directly into a system of ethics, and its subsequent developments emphasise the point. No other information philosophy foundation has attempted such a broad scope.

Six concepts within PI are of particular relevance to the information sciences. The *fourth revolution*, following the earlier revolutions of Copernicus, Darwin and Freud, is the shift which

places information and communication technologies at the heart of our society. Following prehistory, in which there was no recorded information, and history, in which recorded information is important, we are now at the stage of *hyperhistory*, in which our society is entirely dependent on information and communication technologies (ICTs). This is characterised by the *infosphere*, an information environment blending physical and digital, in which we live *onlife*, again seamlessly blending physical and digital. Communication and interaction in the infosphere is determined by *informational frictions*: whatever impedes the free flow of information. The transition to onlife happens only once in the history of humanity. These concepts give us a new perspective for understanding the radical, rapidly developing and often confusing changes in the information environment which have caused continual reassessment of the nature and purpose of the information disciplines.

Floridi (2002, 2004) put forward the idea that not merely was PI a suitable foundation for LIS, but that LIS should be viewed as applied PI, and hence research and reflective practice in LIS could contribute to the development of philosophy. This idea has been debated since then, and, although by no means universally agreed, the idea is gaining support; for reviews and commentaries, see Dinneen and Brauner (2015), Fyffe (2015), Van der Veer Martens (2017), Bawden and Robinson (2018) and Gorichanaz et al. (2020). In summary, we can give a number of reasons why PI may be a suitable philosophical foundation for the information sciences. The definition of information within PI is more appropriate than that in numerous

other theories of information (Dinneen and Brauner, 2015). Although PI's definition of information includes 'truthful', this does not ignore the fact that 'libraries are full of "knowledge"' (Floridi, 2004, 662); in the PI perspective, LIS deals with *semantic content* recorded in documents. Semantic content includes information (true), misinformation (false) and disinformation (deliberately false). It also includes fiction and other forms of material of relevance to LIS (Van der Veer Martens, 2015). Information science is not concerned with the truth of information per se, but with the quality of semantic content. PI gives special importance, and ethical value, to documents because of their status as rich informational entities; although because every entity is assigned some moral (informational) value to begin with, this does not imply that they will continue to have the same value after any thought has been applied: 'the point at stake is not some daft idea about the intrinsic value of Shakespeare versus Dan Brown' (Floridi, 2008, 193).

The information sciences can also contribute to PI as well as take from it; a response to the criticisms about their taking from a variety of 'foreign' philosophies: 'By contributing to the development of PI, LIS can carry on the task of developing its own theoretical foundation from within' (Floridi, 2002, 48); 'LIS may just be as important to PI as PI is to LIS in terms of deepening our mutual understanding of information ontologies, the dynamics of informational domains, and the variety of evolving relationships among information organisms and information objects' (Van der Veer Martens, 2015, 348).

This leads to the idea that the task of the information sciences is the stewardship of the semantic environment, or curation of the infosphere (Bawden and Robinson, 2018): a stewardship based on the PI concept of *ontic trust*, by which all objects with some informational nature are worthy of respect (Fyffe, 2015), and on the notion that the ultimate ethical good is the flourishing of the infosphere.

Although Floridi's is the best-known philosophy of information and with the greatest reach, there have been other approaches. These are typically focused on the nature of information and have not, as yet at least, developed any interactions with the sciences of recorded information and documentation. Examples, very much in the analytical philosophical tradition, are the mathematical/logical theory of information developed by Burgin and Feistel (2017) and the metaphilosophy, combining philosophies of science and of information, of Wu and Brenner (2017). These are discussed further in [Chapter 5](#).

Though all these approaches are centred on the concept of information, they also typically include data; for perspectives on the philosophy of data per se, see Furner (2017b) and Leonelli (2016).

Philosophies of knowledge

Knowledge has been a subject for philosophical enquiry for very many years (Nagel, 2014; Pritchard, 2018). The standard philosophical understanding of knowledge is *justified, true belief*. This is problematic for information science, since it requires someone to be actively believing something, that what they are believing is true and that their belief be justified, typically by the senses – the believer

saw or heard something. This does not seem helpful for disciplines which deal with knowledge recorded in documents, which may well be in error, or may be fictional. This form of knowledge is justified by its being evidenced by what is written in other documents. In philosophical terms, this is *testimony*, a social source and justification of knowledge, additional to individual perception, memory and reason (Lackey, 2010; Audi, 2011, chapter 7; Audi, 2013; Floridi, 2014b). Analysis of the accuracy, authority and truth of testimony – that is to say, with the quality of semantic content and the identification of misinformation and disinformation – is of evident importance to the evaluation of information discussed in [Chapter 17](#) (Fallis, 2004; 2016). For a review of aspects of epistemology of particular relevance to information science, see Kelly (2021).

Perspectives from the tradition of moral philosophy, debating the right use of knowledge, have also had influence within the information sciences. Most notable have been the writings of Mary Midgley (1989); for an assessment, see McElwain (2020). These assert that the point of gaining knowledge cannot be just to store it; there must be an emphasis on its use for developing understanding to good ends.

Michael Polanyi

A distinct approach to knowledge is that of Michael Polanyi (1891–1976), Hungarian-born but who spent most of his career in Britain. Originally a physical chemist, who later contributed to philosophy and social science, he developed an epistemology based on *tacit knowledge* or *personal knowledge* (Polanyi, 1962; 1966). Things are known which cannot be expressed fully in words, diagrams,

formulae or any means of objective communication. This is summed up in his dictum that 'we may know more than we can tell'. Tacit knowledge is often regarded as a 'knowing how' rather than 'knowing that', and Polanyi argued that all knowledge is either tacit, or is rooted in tacit knowledge. The concept has been developed by others; for example, to distinguish between knowledge which is tacit because no one has gone to the effort of making it explicit and things our bodies can do that we cannot describe even in principle, and between individual and social tacit knowledge (Collins, 2010). Its significance for the information sciences is the impetus it gave to the development of knowledge management, to be covered in [Chapter 14](#).

Social epistemology

One significant philosophical movement based on knowledge was, remarkably, originated by librarians, Margaret Egan (1905–59) and Jesse Shera (1903–82) (Egan and Shera, 1952; Furner, 2004; Fyffe, 2015). Although Egan and Shera were by no means the first to reflect on the social nature of knowledge, they were the first to use a specific term, *social epistemology*, to denote it. This is the only example so far of the information sciences contributing a new concept to philosophy. Social epistemology has at its root the collaborative study and communication of knowledge, in order to discover truth and create new knowledge, and how knowledge can be possessed by societies and by groups; it stands in contrast to traditional epistemology, which focuses on the knowledge of individual persons. This is not fully congruent with the interests of the information disciplines, which are mainly concerned with

sources and records of knowledge, rather than knowledge per se. Nonetheless, Shera (1970, 108) asserted that social epistemology 'can give librarianship its intellectual foundation for which we have been searching for so long', while Fyffe (2015) suggested that LIS has been treated as a branch of social epistemology since the mid-20th century.

The nature of social epistemology has changed since its inception; see, for example, Zandonade (2004), Fallis (2006), Fuller (2012) and Dragos (2019). Its focus has always been on knowledge in society, the communication of ideas throughout the whole social fabric, and the ways in which this is affected by social factors, norms and institutions. This is a broad concept, covering a variety of approaches and foci. For reviews of the current state of social epistemology, see Goldman and O'Connor (2021) and Fricker et al. (2019).

Egan and Shera held that a library's purpose is to be the centre for a community's knowledge creation and communication, and that libraries and similar institutions have a central role in social epistemology. More recent formulations do not give information institutions such a central place. Nonetheless, there remains a strong relation between social epistemology and the information sciences, whose nature is still a subject for debate.

Floridi (2002) argued that social epistemology cannot be a basis for LIS, since it has as its basis the study of knowledge in order to discover truth, while LIS has to deal with a much wider range of sources than those which provide 'true knowledge', and its remit is not knowledge itself, but the documents and records which convey

knowledge. The two are certainly related, but both are in need of a common foundation, which Floridi identifies as PI. Fyffe (2015) supports this viewpoint; Furner (2015) makes a similar point: that 'knowledge studies', which social epistemology would equate to, would be far wider than LIS could reasonably encompass. Fyffe (2015) supports this argument; social epistemology is intimately connected with the work of the information professions, but is not appropriate or sufficient as a foundation for the information disciplines. Fallis (2006) suggests that, even if we accept PI as the foundation for LIS, social epistemology (SE) may be valuable as a framework for some of the interests of the information sciences, specifically those focused on knowledge acquisition. This seems a reasonable viewpoint which might be expanded, in view of the increasing interest of social epistemology in knowledge creation and sharing, and the role of testimony, all of direct relevance to information science.

Critical theory and media theory

Critical theory is a broad term, including structuralist, post-structuralist and post-modernist theory, phenomenology, hermeneutics and media and communication theory. Its style of philosophy is typically qualitative, interpretivist and literary, rather than formal, analytical and logical.

It is generally associated with continental European philosophers, such as Adorno, Barthes, Baudrillard, Benjamin, Bourdieu, Deleuze, Derrida, Foucault, Gadamer, Guattari, Habermas, Heidegger and Lacan, and also with the Toronto school of communication theory, epitomised by McLuhan and Innis (Bronner, 2017; Bohman, 2019).

Although it has been suggested that there are common interests in the individual's need for knowledge and in the creation of meaning from sources (Benoit, 2007), the use of such philosophical theory in information science has been limited, and this has been explicitly criticised by Cronin and Meho (2009) and Dewey (2016). Nonetheless, a variety of such philosophies and philosophers have been cited in relation to a variety of issues in information and documentation, including archives, intellectual freedom, information behaviour and practices, bibliometrics, classification and knowledge organisation, production and use of media, and professionalism. For overviews, see Day (2005), Leckie, Given and Buschman (2010) and, in the specific context of knowledge organisation, Martinez-Ávila, Semidão and Ferreira (2016).

It is difficult to choose representative examples from so wide and diverse an area, when it appears that there is little systematic use of the thought of any particular philosophers or styles of critical theory, but we can give some examples to convey a flavour of the way this kind of philosophy has been applied.

The work of Michel Foucault (1929–64) – in particular, his *Archaeology of Knowledge* and *Order of Things* – on knowledge, classification, archives, documents, texts, discourse, and information spaces has been applied in the information sciences, though arguably not as widely as it deserves (Dewey, 2016; 2020; Radford, Radford and Lingel, 2015; Olsson and Heizmann, 2015).

Pierre Bourdieu (1930–2002) developed analyses of disciplines and domains which have been applied in information science, particularly in respect of domain analysis, as discussed in [Chapter 7](#)

(Nascimento and Marteleto, 2008; Marteleto and dos Santos Carvalho, 2015). Bourdieu's work in cultural production, social capital, and practice theory has also been used in information settings (Knox, 2014; Desrochers et al., 2018; Wojciechowska, 2021).

Concepts put forward by the Gilles Deleuze (1925–70) and Félix Guattari (1930–62) have also been applied in information science. These include the *rhizome*, an abstract non-hierarchical structure with multiple entry and exit points and multiple paths, an alternative to the Aristotelian hierarchical tree structure. This has been applied to analyses of knowledge organisation, social networks, document collections, and educational issues (Cronjé, 2021; Burnett and Bonnici, 2013; Duff and Haskell, 2015; Robinson and Maguire, 2010). The concept of *assemblage*, an analysis of the way in which entities come together through dynamic relationships, has been applied to contexts as diverse as information institutions and data management infrastructures (Gerolami, 2015; Bates et al., 2019).

A branch of philosophy focusing on examining the internal *life world* of individuals, studying conscious experience from the first-person viewpoint, was developed in the early 20th century by Edmund Husserl (1859–1938), Martin Heidegger (1889–1976) and others. *Phenomenology*, which facilitates the study and understanding of activities and experiences of individuals from their own perspective, has been applied within the information sciences to studies of information behaviour and practices, information experience, and information literacy (VanScoy and Evenstad, 2015; Suorsa, 2015; Gorichanaz and Latham, 2016; Dabengwa, Raju and Matingwina, 2020).

Closely associated and overlapping with critical theory is the work of the communication and media theorists of the Frankfurt and Toronto schools, notably Walter Benjamin (1892–1940), Theodor Adorno (1903–69), Harold Innis (1894–1952), Marshall McLuhan (1911–80) and Jurgen Habermas (b. 1891). Their analyses of the role of media in shaping culture and society, and of the influence of communication technologies on what is communicated, typified by McLuhan's well-known dictum 'the medium is the message', have been generally associated with the disciplines of cultural studies, media studies and publishing. They have had some influence within the information sciences; see, for example, Turner (2015) and Larsen (2021). However, as with critical theory in general, this influence is less than might have been expected.

Summary

Despite continuing and increasing interest in the intersection of philosophy and the sciences of information, there is no agreement on any single philosophy, or philosophical approach, as a foundation for information science. It may be that there never will be such in a diverse and multidisciplinary field.

Nonetheless, the search for satisfactory foundations for the discipline is a vital task which will bring benefits to practice as much as to academia. We currently have a variety of philosophies competing with, and to a degree complementing, each other. It seems that Floridi's PI offers much as the most generally appropriate foundation.

- There is no single, generally accepted philosophical or theoretical foundation for the information sciences; rather a variety of alternatives have been suggested.
- Floridi's PI gives the promise of a viable foundational basis.
- Further development in foundational theory and philosophy of the information sciences is important both for the academic subjects and for the practical disciplines.

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4 Paradigms, Turns and Theories in the Information Sciences

It can scarcely be denied that the supreme goal of all theory is to make the irreducible basic elements as simple and as few as possible without having to surrender the adequate representation of a single datum of experience.

Albert Einstein (Herbert Spencer Lecture, Oxford, 10 June 1933)

You can work out all the theories you like, but the real world is more complex than your theories.

Claude Shannon (quoted in Soni, 2019)

Introduction

In this chapter we consider paradigms and turns, approaches to the study of the information disciplines and the practice of the information professions, which are generally regarded as more specific and contextual than the philosophies discussed in the previous chapter. We then consider the role of theory in information science. A theory is regarded as more specific and contextualised than a paradigm, although types of theory are closely associated with particular paradigms. However, the terms are not used consistently, and the philosophy–paradigm–theory hierarchy is not strictly observed in writings on information science.

Paradigms and turns

The Oxford English dictionary (December 2020) defines a *paradigm*, in the sense in which the term is used here, as ‘a conceptual or methodological model underlying the theories and practices of a science or discipline at a particular time; hence a generally accepted world view’. The term, brought into common use by the philosopher

of science Thomas Kuhn (1922–96) in his work on scientific revolutions, describes a collection of beliefs about the subject matter of a subject, and a set of agreements about how the subject should be studied. Acceptance of a paradigm is therefore mainly applicable to academic study and research, but it also influences practice, since it encapsulates a view of what a subject is like on the whole. It is important to realise that the emergence of a new paradigm does not mean the cessation of work based in older paradigms; the process is more one of addition, and perhaps gradual replacement. Information science has certainly seen several paradigms operating at the same time. This is somewhat different to Kuhn's original concept, by which there would usually be a single paradigm for a science, and a new paradigm would replace the old quite rapidly.

A turn, in this context, implies a fairly rapid change of direction within a field of study, affecting the topics thought to be relevant and important and the appropriate ways of studying them. Nolin (2007, 2) characterises a turn as 'some sort of cognitive interruption with a research tradition', and Hartel (2019, 2) as 'a recognizable intellectual project in which a group of scholars enthusiastically embrace a new set of theoretical, methodological, or substantive commitments [and] typically critique the status quo and focus attention on a different research agenda or frontier'. A turn can be seen as roughly equivalent to the adoption of a new paradigm, and we will treat paradigms and turns as synonymous in this chapter. They have also been referred in the information science literature as perspectives, viewpoints and traditions, and these are also subsumed here.

There have been numerous named paradigms and turns in the information sciences. They include the system paradigm, the user paradigm, the behavioural paradigm, the information paradigm, the object paradigm, the bibliometric paradigm, the cognitive paradigm, the communication paradigm, the physical paradigm and the socio-cognitive paradigm. Some of these have an overlap, and others have not been influential. For discussion of some of these, and analysis of the idea of paradigms in information science, see Ellis (1992), Hjørland (2002; 2018) and Tang et al. (2021).

Similarly, there have been numerous named turns. Cronin (2008) notes the user turn and the sociological turn, and also the linguistic turn, of wider scope but impacting on the information disciplines. Nolin and Åström (2010) identify the cognitive turn, the informational turn, the user-centred turn, the epistemological turn and the pragmatic turn. They note that these often have wider scope than just the information sciences; the cognitive turn, for example, originated in psychology. Hartel (2019) describes seven turns in the information sciences: cognitive, affective, neo-documentary, socio-cognitive, everyday life, social constructionist and embodied, and also mentions three turns in more general scholarship which have been influential: linguistic, cultural and material.

We will outline four of the more influential paradigms/terms, in order to make the nature of information paradigms clearer: systems, cognitive, socio-cognitive and neo-documentary.

The systems paradigm

This is also termed the physical paradigm. Recognisable from the 1950s onwards, it is the original paradigm of the information

sciences, used as the basis for the evaluation and improvement for information systems and services. A realist perspective, following the methods of the natural sciences, it emphasised quantitative and experimental methods for evaluating the performance of systems, particularly retrieval systems; see, for example, Sparck Jones (1981) and Robertson (2008). Quantitative user surveys, bibliometric analyses and performance metrics for information services also fit within this paradigm. It is sometimes referred to as the Cranfield paradigm, after the well-known Cranfield information retrieval experiments, which were the first major studies carried out in this way, with test queries, test collections and metrics for comparison; in the case of Cranfield, comparing alternative indexing and classification systems. This experimental style of research is still popular in studies of retrieval algorithms and system interfaces (see Chapter 18) (Harman and Voorhees, 2006; Voorhees et al., 2020) and is a good illustration of the way in which paradigms continue, even if complemented, and to an extent over-shadowed, by new paradigms.

The cognitive paradigm

Starting in the 1970s and gaining impact in the 1980s, the cognitive paradigm, was a move away from the systems paradigm and towards a focus on the internal thought world of the individual information user. Based on Popper's philosophy, it was particularly influenced by the 'fundamental equation of information science' proposed by Brookes (1980), in the form:

$$K(S) + \Delta I = K(S + \Delta S)$$

A knowledge structure, $K(S)$ is changed into an altered knowledge structure $K(S + \Delta S)$ by an input of information, ΔI . ΔS is an indicator of the effect of the modification. The same information may have different effects on different knowledge structures. It is a description of the information communication process as it affects one individual's knowledge, and hence is regarded as a foundation of the cognitive paradigm in information science, although it does not function as a true equation (Bawden, 2011).

A number of theories and models were advanced, based on cognitive aspects of information use, often considering mental models of users, and applied in information retrieval and information behaviour contexts (Ingwersen, 1999). The best-known example, the idea of an Anomalous State of Knowledge (ASK) (Belkin, Oddy and Brooks, 1982), assumes that a user recognises some anomaly or inadequacy in their knowledge of some issue, which they translate first into an information need and then into queries to some kind of information system. The answers go some way to resolving the anomaly in their knowledge, but may generate more information needs. This kind of analysis led to ideas for retrieval systems built on different principles to those generally available, typically with a much greater degree of interaction. Nolin (2007, 5), admittedly writing before some of the more recent turns, considered that 'without doubt, the most influential turn in the history of information science has been the cognitive turn', and its influence is still significant.

The socio-cognitive paradigm

The socio-cognitive paradigm originated in the 1990s and is based on the idea that information behaviour and practices are embedded in a specific context: social, professional, educational, or disciplinary (Hjørland, 1995; 2002). It focuses on the construction and use of knowledge within communities, and in that sense is strongly related to social epistemology. It is strongly related to Birger Hjørland's domain analysis (discussed in [Chapter 7](#)), which offers a framework for research, for disciplinary development and for professional practice. Although it focuses on the group and the community, it does not neglect the individual: rather, it allows 'an interplay between domain structures and individual knowledge, an interaction between the individual and the social level' (Hjørland, 1995, 409). Although it deals with knowledge construction, Hjørland identifies it as a basically realist approach: 'trying to find the basis for information science in factors that are external to the individualistic-subjective perceptions of the users as opposed to . . . the behavioural and cognitive paradigms' (Hjørland, 1995, 400).

The neo-documentary paradigm

This paradigm also originated in the 1990s, with a focus on recorded information instantiated in documents of all kinds at the heart of information science. It refers back to the pioneers of the documentation movement, Otlet and Briet in particular, and was inspired by Michael Buckland's concept of information-as-thing (Skare, Lund and Vårheim, 2007; Buckland, 1991). This perspective has developed around the new document theory of authors such as Buckland, Skare, Latham and Gorichanaz (Lund, 2011; Lund, Gorichanaz and Latham, 2016). The interest of this viewpoint is in

the nature and types of documents, in their significance and use within society, and in the processes of documentation (authorship, annotation, preservation, retrieval, etc.), as discussed in [Chapter 6](#).

Theories

Theories are usually taken to be more specific than paradigms, addressing one topic or one aspect of a subject. We first consider what we mean by theory, then examine the types of theory there can be, evaluate the role of theory in the information sciences and finally look at the relevance of theory to practitioners. For fuller discussion of theory in general see Pinfield et al. (2020, chapter 3), and for a detailed analysis of theory in the humanities and social sciences, the kind of theory most relevant to the information sciences, see Corvellec (2013).

What is theory?

‘Theory’ is a much-used term in science, philosophy and everyday life language. It is therefore strange that it is relatively seldom examined.

(Hjørland, 2015, 113)

The term ‘theory’ is indeed much used, and in different ways. Perhaps the best and simplest explanation of the term is that of the philosopher Theodore Schatzki (2001), who suggests that a theory is simply a general and abstract account of something.

Comparison of dictionary definitions suggests that ‘theory’ has three generally agreed connotations (Pinfield et al., 2020, chapter 3):

- something which provides an abstraction of reality into general principles

- something that explains, or aids understanding
- something that acts as a guide to practice.

All three meanings can be found in the theories of information science. The ‘understanding’ aspect is of particular importance: ‘Understanding is what theory is about: those other attributes of theory, prediction and application, are side-effects only, secondary to the main purpose’ (Robertson, 2000, 1).

The meaning of ‘theory’ varies across different disciplines, and types of discipline, and not in a simplistic way; see Pinfield et al. (2020, chapter 3) for an analysis. It is often thought that the physical and biological sciences use rigorous quantitative ‘laws’ and ‘theories’, while the social and human sciences rely on ‘models’ and ‘frameworks’, but matters are not so straightforward. Qualitative and classificatory models are widely used even in the natural sciences, while the social sciences make use of quantitative and formal theories. In the humanities and social sciences, theory may also mean critical theory or literary theory or art theory – loosely defined qualitative, hermeneutic forms of theory building, which would not be recognised as theory at all in the sciences. Given the multidisciplinary nature of information science, it is not surprising to find a wide variety of conceptions of ‘theory’ being used in the discipline.

This idea that no single idea of theory is adequate for information science was emphasised in the well-known study of use of theory in the discipline by Pettigrew and McKechnie (2001), whose analysis of over a thousand information science articles found that explicit theories were mentioned in only one third of the articles, a majority

being theories brought in from other disciplines. Where theories from within information science itself were present, these were typically from a wide range of rather specific theories, such as models of information seeking, rather than broader theoretical perspectives. They noted that 'there was no singular definition that would encompass all the varied uses of the term in the articles . . . For example, articles about information retrieval differed vastly in their use of the term from those that focused on information behavior, history, or information policy' (Pettigrew and McKechnie, 2001, 65).

Types of theory

Several typologies or categorisations have been devised to clarify the various forms of theory presented. Two are widely known, and relevant to information science: those due to Reynolds and to Gregor respectively.

For the social sciences generally, the sociologist Paul Reynolds identified four forms of theory (Reynolds, 1971):

- a set of laws, i.e., well-supported empirical generalisations
- an interrelated set of definitions, axioms and propositions
- descriptions of causal processes
- vague concepts, untested hypotheses, prescriptions for good behaviour.

Shirley Gregor, an information systems scholar, proposed five types of theory for that discipline (Gregor, 2006). This typology has gained wide use beyond the information systems context. Its five categories are:

- Type 1 – theory for analysing, typically initial descriptive attempts at theory
- Type 2 – theory for explaining
- Type 3 – theory for predicting
- Type 4 – theory for both explaining and predicting, typically described as ‘grand theories’ or meta-theories
- Type 5 – theory for design and action.

Both these typologies are quite broad and hospitable, in that they allow the inclusion of theoretical entities which may be termed differently: frameworks, models and so on. Both have been used to analyse the creation and use of theory in the information sciences.

Reynolds’ first category (laws, meaning well-supported empirical generalisations) is unlikely to be of value for information science. More appropriate are the second, encompassing models for conceptual understanding and, particularly, the third, encompassing most of the process models for information retrieval, information behaviour, and information literacy. Arguably, all too often it is the fourth which comprises much that has been labelled theory for the information disciplines.

Gregor’s typology (2006) has been more quoted than any other in an information research context. All of Gregor’s five types of theory may be found in the literature, although types 1 and 2 predominate (Sonnenwald, 2016b). Gregor’s typology has been suggested as a useful framework for understanding theory in the information disciplines by, for example, Sonnenwald (2016b) and Thelwall (2016), although Hjørland (2017) finds it to be superficial and ahistorical.

We can also categorise theories as to their scope, from a large-scale 'grand theory', 'Theory of Everything' or metatheory, to very local small-theories, explaining specific and limited points. A kind of theory of particular interest to information science is the 'mid-range' or 'middle-range' theory, defined by the sociologist Robert Merton as 'theories that lie between the minor but necessary working hypotheses that evolve in abundance during day-to-day research and the all-exclusive systematic efforts to develop a unified theory that will explain all the observed uniformities of social behaviour, social organization and social change' (Merton, 1968, 51). This idea seems to match very well the numerous conceptual models and frameworks developed in information science, and the importance of mid-range theories for LIS has been noted by Lor (2014), by Hjørland (2015) and by Meadows (2016).

The descriptions of the creation of theory in information science in the collection edited by Sonnenwald (2016a) serve to reinforce this point. There is some mention of methods of 'grand theory' construction, but the examples of theory construction presented are mostly personal and contextual, with little in the way of an overarching and directing metatheoretical approach. This seems a fair reflection of the ways in which theory has been created in information science: bottom-up, and as a pragmatic response to contextual needs.

Theory in the information sciences

One of the concerns that has long been expressed about information science as a discipline is that it lacks a theory base of its own, and indeed makes relatively little use of theory per se. For example, Lor

(2014, 30) states that 'although some use is made of theory from other fields such as psychology, sociology or management, LIS has produced very little theory of any significance'. Even when theory is used in information science, it may go under another name. For example, Thelwall (2016) notes that, although his webometric studies may be fitted within Gregor's typology, he did not consider that he was creating theory, and that, although he rarely cited explicitly named theories, he often used 'theoretical insights'. Similarly, Sugimoto (2016, 1) writes that 'one difficulty in identifying theories of informetrics and scholarly communication is the diversity of terminology around theories'. She notes that contributors to a volume on theory in that aspect of the information sciences used 'theories', 'models', 'taxonomies', 'typologies', 'frameworks', 'indices', 'hypotheses' and 'principles', sometimes synonymously. And, to complete the set of problems, theoretical contributions do not necessarily build upon one another. As Hjørland (2017, 1797) puts it: 'The overall situation in information science today is a chaos of theoretical contributions, each paying no or much too little interest in the existing ones.' Overviews of theory in the area of information behaviour, a particularly fruitful area within information science for the development of models and theories, are given by Bates (2005) and other authors in Fisher, Erdelez and McKechnie (2005), by Case and Given (2016, chapter 8) and by Wilson (2020a).

With the exception of Shannon's mathematical information theory and its derivatives (discussed in [Chapter 5](#)), models have typically been the way into theory building in information science. These are

typically qualitative conceptual models, expressed as text or diagrams. Hjørland (2015) has emphasised the importance of concepts in LIS theory building, and Savolainen (2019) has analysed in detail the kind of models used in information behaviour research. Use of metatheory, and of more formal or general approaches, has been limited.

We now give some examples of theory creation and use within the information sciences under four headings: use of metatheory; developing models *de nova*; elaborating and extending existing models; and bringing in 'pre-prepared' theory from other disciplines. Fuller discussion, and more examples, are given by Pinfield et al. (2020, chapter 3).

Using metatheory

Metatheory, the analysis of theory itself, and the application of broad general theories to particular topics, has been little used in information science. Since Bates noted its lack of application in 2005, there have been only a few proponents and examples of these approaches.

Hjørland has argued consistently that information science theories should be developed, and existing theories analysed, on the basis of metatheory (Hjørland, 1998; 2011a), specifically using broad theories of knowledge and of science. He has given examples of the application of metatheory to, for example, browsing (Hjørland, 2011a), indexing and retrieval (Hjørland, 2011b), and informetrics (Hjørland, 2016). Vakkari and Kuokkanen (1997) applied a sociological metatheory approach due to Wagner and Berger (1985) to an example of theory growth in information-seeking research.

This was a 'reconstruction', showing how the actual theory development via models could be interpreted in light of the metatheory. We may also consider the applications, albeit limited as yet, of Floridi's PI within information science as metatheory.

Developing theory de novo

This approach aims at the creation of qualitative theory, often termed a model or a framework, by identification of concepts and their relationships. This is done using one of a variety of methods for developing theory from qualitative data without a prior guiding framework. These include the various flavours of grounded theory, or methods such as thematic analysis. It is certainly the most common means by which new theory is created in the information sciences. Examples are:

- grounded theory used to derive a model of concepts and relations for the search process in a multilingual retrieval system (Vassilakaki and Johnson, 2015)
- grounded theory used in a qualitative study of the information practices of welfare workers, developing a series of conceptual models (French and Williamson, 2016)
- thematic analysis used to generate a model for evidence-based practice in librarianship (Gillespie, 2014).

Elaborating existing models

This approach involves theory building by the elaboration and incremental improvement of conceptual models, rather than by continually creating new models from scratch, without aiming for an over-ambitious, all-embracing model or theory. Robertson has

argued specifically for this kind of theory building: 'I believe that the models we have at present can indeed be extended, by theoretical argument as well as by [pragmatic considerations], to cover more ground than they do at present and to be more useful as tools. But when I read a paper (as one does occasionally) which seems to make a claim to represent a Grand Theory, then I shall continue to take it with a pinch of salt' (Robertson, 2000, 10).

The most evident example of this has been a variety of extensions to Tom Wilson's original information behaviour model, by Wilson himself and by others, as discussed in [Chapter 12](#).

Wilson (2016) has argued that the incremental extension and improvement of his series of conceptual models of information behaviour now amount to a general theory of that topic, although the term 'theory' had never been used by Wilson himself.

Other examples of model extension and combination include:

- the combination of models for information seeking and for communication to create a new model, of broader scope (Robson and Robinson, 2013)
- successive improvement and elaboration of conceptual models of information literacy, leading to the introduction of a new theoretical concept, information discernment (Walton, 2017)
- development of a theory of digital media, by comparing three existing theories, identifying deficiencies in each, and developing a new theory to compensate for these deficiencies (Schroeder, 2018).

Bringing in theory from other disciplines

It has often been noted, sometimes as a criticism, that the information disciplines tend to adopt theory from other disciplines, rather than to create their own. As Dillon (2016, 230) puts it, ‘the information sciences have no theoretical core, but derive theoretical structures from outside disciplines . . . by its very nature, information science is an interdisciplinary field and therefore it must draw theory from outside. This is certainly plausible, but my concern is that if we only derive our theoretical analyses from outside, it is even more difficult to demonstrate the process of theory building . . . in the information sciences’.

The final method for introducing theory in information science is therefore to bring in a ‘pre-prepared’ theory from another discipline and apply it in the information context with little or no adaption. Some examples of this are:

- activity theory applied to the study of information practices of medical researchers (Roos, 2012)
- innovation diffusion theory applied to the study of funding of open access publication (Pinfield and Middleton, 2016)
- rational choice theory used to study unethical scholarly publishing (Xia, 2018)
- Lewin’s gatekeeping theory used to examine censorship within library collections (Steele, 2018).

Other examples are discussed by Pinfield et al. (2020, chapter 3). Imported theory may sometimes not be readily understandable by an audience from the information disciplines, because it is based in unfamiliar concepts and perspectives. Examples of this problem,

identified in a study of theory of open access issues, are the use of quantum game theory and of aspects of critical theory (Pinfield et al., 2020, chapter 3).

While it is generally true to say that information science has been an importer rather than an exporter of theory, there are some indications that this may be changing. Wilson (2020b) has demonstrated how other disciplines are making use of concepts of information behaviour developed within information science, while Willson (2019) gives the example of information behaviour studies contributing to the development of transition theory, originally developed in nursing studies and increasingly widely applied in the social sciences generally. As noted in Chapter 3, Floridi has suggested that LIS may contribute to PI.

In summary, we can say that these four general methods encapsulate the main ways in which theory has been developed within information science. The most common method has been the generation of new models by elaboration of existing models; least common has been use of any form of metatheory.

Theory and the practitioner

Concerns have been expressed over the years about the limited use of theory by practitioners in the information professions (Pinfield et al., 2020, chapter 4); the same is said of many other professions, at least outside technology and medicine, such as education. There is a parallel, and indeed an overlap, with the supposed non-use of research results by information practitioners, discussed in [Chapter 18](#).

Floridi (2004) reminds us that there are three levels at which theory may be applied, for information science as for other disciplines: the level of day-to-day routine practice; the level of the delineation of the discipline and its knowledge base, as given in academic courses; and the foundational, and necessarily abstract, level. Theory most clearly operates at the last of these, though it may certainly inform the first two. It is not necessarily a concern that theory is not routinely invoked by practitioners, as it may have served to form their professional knowledge. It may also, together with research results, inform evidence-based practice (Koufogiannakis and Brettle, 2016). It is certainly the case that theory may be unconsciously used by practitioners: anyone using the management terminology of 'quick wins' or 'low-hanging fruit', for example, is using, consciously or otherwise, the language of theories of change.

Furthermore, the non-use of theory by practitioners may be exaggerated. A detailed study of the theory/practice relation in the context of open access (Pinfield et al., 2020) showed that open access practitioners, including publishers and librarians, did indeed create and use theory; conversely, researchers in the area were not necessarily consistent and confident users of theory. VanScoy, Thomson and Hartel (2020) give an example of how theory may be used directly in practice, applying the serious leisure perspective to programming in public libraries.

Summary

Paradigms and theories in the information sciences are numerous, ever increasing and applied inconsistently. Even when theory is

used, in research or practice, it may not be called 'theory', and its creators and users may not regard it as such. A variety of paradigms or turns have been identified in information science, generally adding to those already there rather than replacing them. Theory in information science is likely to be 'mid-range', and to be Reynolds' 2 or 3 (conceptual understanding and conceptual process models), and Gregor's 1 or 2 (description and explanation) or possibly 5 (action research). It largely comprises either conceptual frameworks for understanding or process models. It will most usually be created by incremental extension of existing models; less commonly by creating a model *de novo* or by bringing in a pre-packaged theory from another discipline.

- Paradigms are the general way in which a subject is regarded, in research and in practice.
- Turns are the introduction of a new paradigm, which generally adds to, rather than replaces, existing paradigms.
- There are many individual theories in information science, typically, qualitative conceptual models and frameworks.
- There is no generally accepted overarching grand theory for the information sciences.

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5 Information

It is hardly to be expected that a single concept of information would satisfactorily account for the numerous possible applications of this general field.

Claude Shannon (1993, 180)

Almost every scientific discipline uses the concept of information within its own context and with regard to specific phenomena ... There are many concepts of information, and they are embedded in more or less explicit theoretical structures

Rafael Capurro and Birger Hjørland (2003, 356 and 396)

Information is in everything. It has become the foremost concept in biology, physics, economics, engineering, philosophy, decision-making, methodology of science, artificial intelligence, computer science, etc. Naturally, in different domains, information appeared in different forms and shapes, which depend on the context and environment . . . Due to such a diversity of information forms and shapes, researchers have built many specialized information theories trying to reflect important aspects of information

Mark Burgin and Wolfgang Hofkirchner (2017a, 1)

Introduction

Information is a complex and contested concept. It is central to information science, but it is also important in many other disciplines, in the physical sciences, in the biological sciences, in computing and communication technology and in the social sciences. It is reasonable to ask whether the concept of information is

in some sense ‘the same’ in all of these, or if it is just a coincidence that the same English word is used for different things.

The word has had a number of different meanings over the years. Its entry in the Oxford English Dictionary (March 2021 edition) gives its current meanings as the imparting of knowledge, or the giving of form or character to something. The former is what is usually understood in the information sciences, the latter is relevant to wider uses of the term. As the quotation from Shannon, the originator of information theory, reminds us, we should not expect to find a single, simple idea of information which will do in all circumstances.

One reason for this complexity is that information may be discussed and analysed at a number of different levels. For example, Liebenau and Backhouse (1990) show that any communication of information can be understood simultaneously at four levels:

- empiric: the physical transmission
- syntactic: the language or coding used
- semantic: the meaning of the message
- pragmatic: the significance of the message to a recipient in a particular context.

We will begin with an account of information in the computing and communications disciplines, and with Shannon’s information theory and its successors, as this has provided the only concept of information which has been applied in all other domains. We will then look at information in the physical and the biological sciences, and then in the social world; this last will examine the relations between information and the related concepts of *data* and *knowledge*.

Finally, we consider approaches to providing a unified view of information, bridging the gaps between its different manifestations.

For overviews of the scope of the concept of information, see Machlup and Mansfield (1983), Davies and Gregersen (2010), Gleick (2011) and Floridi (2010; 2016a).

Information in computing and communications: Shannon theory

Shannon and Wiener and I

Have found it confusing to try

To measure sagacity

And channel capacity

By $\sum p_i \log p_i$

Anonymous, *Behavioural Science*, 1962, 7(July), 395

Until the mid-20th century, information was regarded as the communication of meaningful facts or knowledge, without any attempt being made to provide any more formal definition. The first formal quantitative definition of information was provided by Claude Shannon (1916–2001) in his *Mathematical Theory of Communication* (Shannon and Weaver, 1949). Shannon, a mathematician working for the Bell Laboratories, was a polymath, who made contributions to the design of digital computers, such that electronic circuits could implement Boolean algebra, and to robotics (Shannon, 1993; Gleick, 2011; Nahin, 2017; Soni and Goodman, 2017).

Shannon was concerned with the efficiency of transmission of information in communication by telegraph and telephone, and hence not at all concerned with the meaning of the messages sent.

His work built on that of his colleagues, Harry Nyquist (1889–1976), who in 1924 showed how to estimate the amount of information which could be transmitted in a channel of given bandwidth, and Ralph Hartley (1889–1970), who in 1928 established a quantitative measure of information so as to compare the transmission capacities of different systems.

Shannon's theory was a purely statistical formalism, relating the information content of a message to the probability of occurrence of its symbols, regardless of its meaning. Indeed, he termed it a theory *of* communication, rather than *on* information. High information content in Shannon theory is related to low probability combinations of symbols, such that information is equated to surprise, lack of predictability and randomness. This has caused problems for those who have tried to apply Shannon theory to meaningful information. Nonetheless Shannon's formalism is universally accepted as the basis for any theory of information.



Figure 5.1 *Claude Shannon* (Konrad Jacobs, Wikimedia Commons, CC BY-SA)

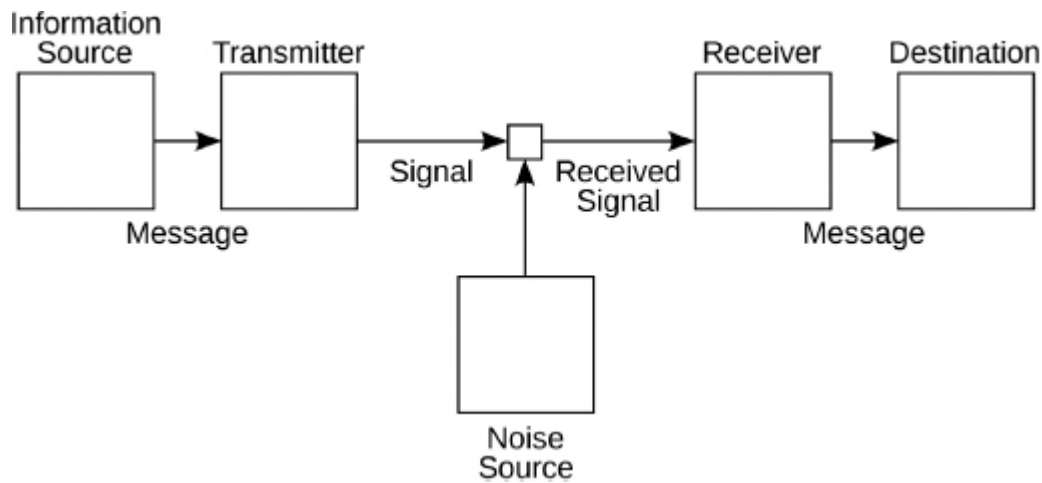


Figure 5.2 *Information communication in Shannon theory* (public domain, via Wikimedia Commons)

Shannon's formalism was introduced in an article in 1948, later republished in book form (Shannon and Weaver, 1949). Warren Weaver (1894–1978), a distinguished mathematician, scientific administrator and proponent of the public understanding of science, contributed an essay – philosophical, non-mathematical, wide ranging and considerably easier to read than Shannon's analysis – which has arguably had greater influence in spreading the ideas of information theory than Shannon's own writing.

Shannon defined the fundamental problem of communication as the accurate reproduction at one point of a message selected from another point. Meaning is ignored: 'these semantic aspects of communication are irrelevant to the engineering problem' (Shannon and Weaver, 1949, 3). What matters is that the actual message in each case is one which is selected from the set of possible messages, and,

if the number of possible messages is finite, then the information associated with any message is a function of the number of possible messages.

Shannon derived his well-known formula for H , the measure of information:

$$H = -K \sum p_i \log p_i$$

where p_i is the probability of each symbol and K is a constant defining the units. (The minus sign is included to make the quantity of information, H , positive; necessarily a probability will always be less than 1, and the log of such a number is always negative.)

Shannon pointed out that formulae of the general form $H = - \sum p_i \log p_i$ appear very often in information theory as measures of information, choice and uncertainty; the three concepts seem almost synonymous for his purposes.

Shannon's was not the only attempt to derive a mathematical theory of information based on ideas of probability and uncertainty. The British statistician R. A. Fisher (1892–1962) derived such a measure, as did the American mathematician Norbert Wiener (1894–1964), the originator of cybernetics. The latter's mathematical formalism was the same as Shannon's but, confusingly, he treated information as the negative of physical entropy, associating it with structure and order, whereas Shannon equated information with entropy. Shannon's information is, in effect, the opposite of Wiener's, which has caused confusion ever since for those who seek to understand the meaning of the mathematics (Qvortrup, 1993; Bawden and Robinson, 2015a). The relations between information,

order, structure, and complexity are subtle, and far from fully understood (Bawden and Robinson, 2015b).

Shannon theory gives the only convincing quantitative measure of information yet derived, and a great deal of work has been done to try to apply it in a variety of disciplines, from the sciences through the social sciences to the humanities (Floridi, 2010; Gleick, 2011; Bawden and Robinson, 2015a; 2015b). Numerous attempts have been made to extend it to deal with meaningful semantic information, and to develop logico-mathematical models for information flow, by authors such as Bar-Hillel, Dretske, Barwise and Seligman, and Devlin. None has been fully successful; for a review, see Floridi (2019). Nonetheless, Shannon theory is the basis of all current formal conceptions of information.

Information in the physical world

In the *physical* world, information has been introduced in the last few decades as an explicit quantity in physical theories, a physical concept like energy or matter (Davies and Gregersen, 2010; Bawden and Robinson, 2013; Bawden & Robinson, 2020; Durham and Rickles, 2017; Bawden, 2019). This was dramatically stated by Nobel-winning physicist Frank Wilczek (2015, 17) in a prediction about physics in 100 years: ‘Fundamental action principles, and thus the laws of physics, will be interpreted as statements about information and its transformations.’

In computing and communications technology, information is processed by physical devices, so that it seems natural to think of information in this domain as a physical quantity. This was clearly established by Rolf Landauer, a computer scientist at IBM, who

coined the phrase 'Information is physical'; that is to say, information must always be instantiated in some physical system (Landauer, 1991). Landauer made calculations of the equivalence between energy and the processing of information in digital computers, emphasising the nature of information as a physical quantity.

At the heart of this issue is the relation between information and entropy, roughly a measure of the disorder in physical systems; the mathematical formalism of physical entropy is essentially the same as Shannon's information content. The idea of information as the opposite of entropy was popularised, with the label of 'negentropy', by Leon Brillouin (1956), who first advocated the wide use of information theory in science. The greater the entropy of the system, the more information it contains. This may seem paradoxical for those who intuitively associate information with organisation and understanding, but think of the fact that it takes many more words to describe in detail a disordered system than an ordered, simpler one; the extra words needed are due to the extra entropy/information in the system. The relation between information and entropy is subtle, and still not fully understood; for discussion and a literature review, see Bawden and Robinson (2015a), Parrondo, Horowitz and Sagawa (2015) and Hemmo and Shenker (2016).

There are other ways in which information is entering the physical sciences. Formulations of quantum mechanics are being rewritten in information terms (D'Ariano, Chiribella and Perinotti, 2019). Some theoreticians regard information as the most fundamental aspect of the physical universe from which everything else is built, following

American physicist John Wheeler's dictum 'It from Bit' (Barrow, Davies and Harper, 2004; Durham and Rickles, 2017). And there is the question of whether information about the state of the physical world can ever be destroyed, or whether it is always conserved. This is Stephen Hawking's famous 'black hole information paradox'. Hawking initially believed that information was irrevocably destroyed when matter fell into a black hole; the matter is not yet fully solved, but it seems likely that information can re-emerge from a black hole.

The informational approach to physics is increasing in influence, although there are those who argue that neither the Shannon formalism nor the ideas of negentropy are able to provide a satisfactory explanation of physical phenomena; see, for example, Anta (2021).

Information in the biological world

A similar idea to Wiener's conception of information as negative entropy was proposed by the German physicist Erwin Schrödinger, one of the pioneers of quantum mechanics, who suggested that living organisms feed upon negative entropy (Schrödinger, 1944; Moberg, 2020). Since then information has become a central concept in the *biological* sciences, with the ability to process and transmit information being taken as a part of the definition of life; see Davies (2019) for an accessible overview.

There are a number of specific instances for applying information concepts in biology (Robinson and Bawden, 2013; Bawden and Robinson, 2020). The encoding of genetic information in DNA is perhaps most obvious (Yockey, 2011). The communication of

information between living cells, plants and animals, and the idea that non-human species may use external memory devices, such as spiders' webs, have been studied for many years (Stegmann, 2018; Karban, 2015). More recent are the ideas of *code biology*, the analysis of informational codes in living systems at various levels from the genetic code to codes of culture (Barbieri, 2015), and various *integrated information theories*, frameworks for explaining consciousness as associated with the integration of information through complex processing (Tononi et al., 2016; Brogaard, Chomanski and Gatzia, 2021).

Information in the social world

Where is the life we have lost in living?

Where is the wisdom we have lost in knowledge?

Where is the knowledge we have lost in information?

(T. S. Eliot, *Choruses from the rock*)

In the social world, information is generally regarded in line with the dictionary definitions mentioned above, as associated with meaning and with the communication of facts and knowledge.

In line with the philosophical view, discussed in [Chapter 3](#), that knowledge must be true, it is usual to regard information as something that is truthful, i.e. correct, and to distinguish it from *misinformation* (incorrect information) and *disinformation* (deliberately incorrect information). We may also distinguish *malinformation*, information which is true, but communicated from bad motives. All these varieties of information are subsumed under the heading of *semantic content*, and it is semantic content which

makes up the recorded information with which information science deals. For more detailed analysis of the varieties of information, see Floridi (2010, chapter 4), Fallis (2016) and S   (2019; 2021).

Another quality of information in the social realm is the extent to which it is *relevant* to its potential users. The provision of relevant information is a central task for all information services, and the proportion of information provided which is relevant is an important metric for judging the success of such services. Saracevic (2007, 1915) suggests that relevance is ‘a, if not even *the*, key notion in information science in general’. However, the concept of relevance is complex and contested, and a variety of slightly different related concepts have been proposed, such as *utility*, *usefulness*, *topicality*, *pertinence* and *resonance*. For example, while one document may be judged relevant, another ten similar documents, while equally relevant, may be seen as being of decreasing interest and usefulness. Again, a document may be relevant in terms of subject, but if it is old, or in a language the user cannot read, or written for professional experts while the user is a lay person, it may be of limited usefulness. For reviews of the concept of relevance, see Saracevic (2007; 2016) and Ruthven (2020).

A further quality of information is that it may be regarded as *embodied*, that is to say that human bodies may be sources and repositories of information. Such a view has implications for how information behaviour and practices are understood; see, for example, Olsson and Lloyd (2017) and Leug (2020).

In the social world, information implies meaning, and communication between people. There are various conceptions of

this kind of information, even within the information sciences; see Capurro and Hjørland (2003), Bates (2017) and Mingers and Standing (2018) for overviews.

We may note that, although it is usually taken for granted that information is a central concept for information science, Bates (2017, 2060) commenting that ‘the much-debated concept of information remains at the lively heart of information science’, there are contrarian views. Furner (2015), for example, makes a case that information science is not wholly, or even primarily, concerned with information per se.

Buckland (1991; 2017, chapter 2) distinguishes three uses of the term ‘information’: information-as-thing, where the information is associated with a document, i.e. a digital or physical entity; information-as-process, where the information is that which changes a person’s knowledge state, in the process of their becoming informed; and information-as-knowledge, where the information is equated with the knowledge which it imparts.

Information-as-thing regards information as physical and objective, or at least as being ‘contained within’ physical documents, and essentially equivalent to them. The other two meanings treat information as abstract and intangible. Buckland gives arguments in favour of the information-as-thing approach, as being very directly relevant to information science, inasmuch as it deals primarily with information in the form of documents. Information-as-process underlies theories of information behaviour with a focus on the experience of individuals, while information-as-knowledge invokes

the ideas that information and knowledge are closely related, without specifying an exact relation.

One well-known model places information on a spectrum with data and knowledge, and sometimes other entities, implying that they are 'the same sort of thing', and one may be transformed into another. This model is commonly represented as a pyramid, with the lower layers containing a large amount of material, which is 'refined' into a smaller amount of material at higher levels. The refining is done by value-added processes within information systems, such as compilation, comparison, evaluation, contextualisation, and summarisation (Taylor, 1986). Although this model, often referred to as the DIKW hierarchy, has been criticised for its over-simplicity, it has remained influential (Rowley, 2011; Frické, 2019). The original presentation of this model by Ackoff (1989) had five layers: data, information, knowledge, understanding, and wisdom (Figure 5.3). *Understanding* was rapidly dropped from the model, though it has been suggested that it would be desirable for it to be reinstated (Bawden and Robinson, 2016). An additional layer is sometimes added between data and information: *capta* are data of interest which are collected and stored.

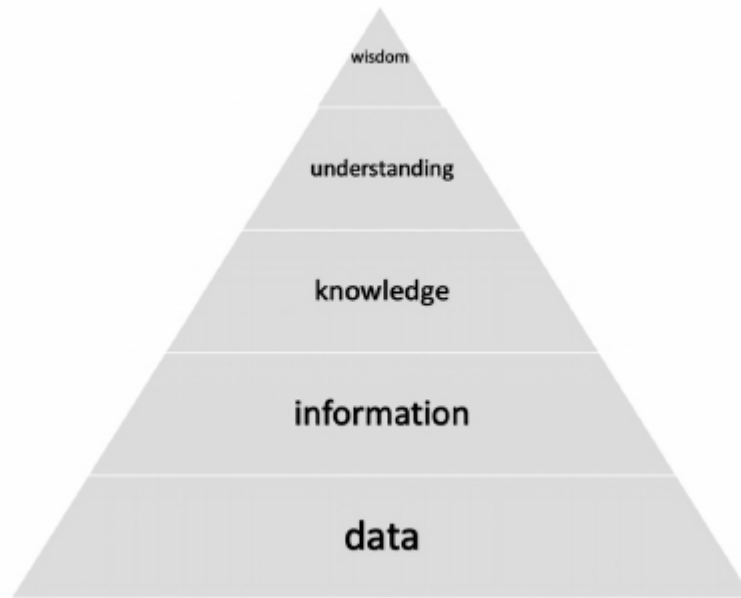


Figure 5.3 *The DIKUW hierarchy*

An alternative viewpoint treats information and knowledge in a different way. In this perspective, knowledge is subjective and personal, something known by a person, while information is the objective form in which it is communicated and recorded, through speech, writing, diagrams, etc. This is a conception associated with Polanyi, and his idea of personal knowledge, and with Popper, and his ‘Three Worlds’ epistemology. Ma (2012) compares this model with the DIKW hierarchy, and with Shannon theory. The nature of information in each is different; it is not reasonable to say that any meaning is ‘the right one’, but which is chosen will have significant implications in practice for the information disciplines.

There are other approaches to the understanding of information in the social world, which have had a lesser influence in information science, arguably less than they deserve. One is Rafael Capurro’s *angeletics*, from the Greek *angelos*, messenger, which places the message at the heart of information theory. Rooted in the continental

European tradition of hermeneutic philosophy, this is concerned broadly with the interpretation and understanding of communicated information (Capurro, 2009; Capurro, 2019; Kelly and Bielby, 2016). Related approaches stem from *semiotics*, with information instantiated in the use of signs and symbols – physical forms which stand for something other than themselves – to communicate meaning. Examples for the information sciences are given by Liebenau and Backhouse (1990), Mingers and Willcocks (2017) and Cárdenas-García and Ireland (2019). These ideas are the basis for Brier's *cybersemiotics*, discussed later. Other approaches based on constructivist philosophy are reviewed by Bosancic and Matijevic (2020).

Mind the gap: unified theories of information

There have been attempts over the years to provide general theories of information, able to explain the concept in all domains. We will discuss three which have specific relevance to LIS, those due to Stonier, Bates and Floridi, mentioning others briefly.

Tom Stonier (1927–99), a scholar of science and society, proposed a conception of information integrated across the physical, biological and mental worlds (Stonier, 1990; Stonier, 1992; Stonier, 1997). In this conception, information is a physical entity analogous to matter and to energy, being the capacity to create and maintain organisation. This organising function of information is expressed in the biological world and then in human intelligence, and in meaning, finally expressed in recorded information. Although it has been suggested that a framework for LIS could be developed based on Stonier's theory of information (Morán-Reyes, 2015), there has been little

explicit development of his ideas, and his work is regarded as an early example of a unified gap-bridging theory of information.

Marcia Bates, a library science academic, has presented an all-encompassing model, which she characterises as 'evolutionary' (Bates, 2005; 2006). It distinguished a number of 'information-like' entities:

- Information 1 – the pattern of organisation of matter and energy
- Information 2 – some pattern of organisation of matter and energy given meaning by a living being
- Data 1 – that portion of the entire information environment available to a sensing organism that is taken in, or processed, by that organism
- Data 2 – information selected or generated by human beings for social purposes
- Knowledge – information given meaning and integrated with other contents of understanding.

An ambitious attempt at integrating information in all its contexts, this remains somewhat vague and has not been further developed.

Luciano Floridi (2010; 2011) has presented a General Definition of Information (GDI) as part of his Philosophy of Information (PI), defining it as 'well-formed, meaningful and truthful data'. Data is understood here as simply a lack of uniformity, a noticeable difference or distinction in something. To count as information, individual data elements must be compiled into a collection which must be well formed (put together correctly according to relevant syntax), meaningful (complying with relevant semantics) and

truthful, as distinct from misinformation (false) and disinformation (deliberately false). Taken together, information, misinformation and disinformation amount to semantic content.

Floridi's analysis also includes biological information in detail; noting that it is complex and multifaceted, he treats, for example, genetic and neural information separately. Meaningful information and knowledge are part of the same conceptual family; information is converted to knowledge by being interrelated; knowledge is true information set in context. Meaningful information must also be relevant in order to qualify as knowledge, and this aspect may be formally modelled, as also the distinction between 'knowing', 'believing', and 'being informed'. In this way, Floridi avoids the problems of the traditional philosophical definition of knowledge as 'justified true belief', which is too focused on the individual knower to be helpful in analysing information issues.

Floridi emphasises that PI is not intended to provide a single, unified theory of information in all domains but, rather, to provide a formal framework for linking and relating the various aspects and manifestations of information, and hence for integrating, though not in a reductionist way, all information-centred disciplines (Floridi, 2016b). His formal treatment of information seems to be most promising for useful applications in LIS of those currently available, particularly since it supports a full philosophy and ethics of information, as detailed in [Chapters 3](#) and [15](#).

There are other general, or unified, theories of information, looking at the meaning of concepts of information in different domains. They are generally mathematical and logical in nature, and

have not yet been applied in LIS; see Burgin and Hofkirchner (2017b) and Bawden and Robinson (2020) for reviews. We will give a brief mention of four.

Burgin has developed a mathematical General Theory of Information over several decades, incorporating the insights of Shannon, Bar-Hillel, Dretske and others, and aiming to explain relations between the meanings of the word 'information'. It treats information as a form of energy which has a capacity to cause changes in a system, creating a mathematical framework which may, in principle, encompass all forms of information including the physical, biological, mental and social (Burgin, 2013; Burgin and Feistel, 2017).

Wu and Brenner have devised a metaphilosophy with information at its centre (Brenner, 2015; Wu and Brenner, 2017). This examines the role of information, with both its physical and non-physical components, in complex processes across various domains, using a non-standard logic, Logic in Reality (LIR), for its analyses. LIR is a physical theory, and, through it, information may be directly related to aspects of the physical and biological worlds.

Hofkirchner's Unified Theory of Information devised a 'superconcept', which would encompass all the different manifestations of information, including the physical and the nonphysical, through a framework based on complex systems theory and emergent properties (Hofkirchner, 2009; 2010). This is arguably the most elaborate and sophisticated account of information in terms of complexity and emergence, but has not been further developed.

Finally, Brier's influential idea of *cybersemiotics* provides a transdisciplinary framework for information communication, focusing on signs, meaning and interpretation in biological systems, including the human and the social, and eschewing a purely logical and mathematical foundation (Brier, 2008; 2017).

Summary

Despite some suggestions to the contrary, information is generally seen as the central concept for information science. It is also an increasingly important entity within all disciplines, including the physical and biological sciences, but whether we are dealing with the 'same thing', in any sense, in these different domains remains unclear. It seems worthwhile to seek for bridging, if not unifying, conceptions of information, and information science should have a role to play in this.

- Information is generally regarded as a core concept, if not the core concept for information science.
- Information is a complex and multifaceted concept, and no single understanding of the idea will be effective in all contexts.
- Shannon theory is commonly accepted as a necessary basis for all general theories of information.
- The approach to information in Floridi's PI seems to be the most appropriate for the information sciences.

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6 Documents and Documentation

Is a star a document? Is a pebble rolled by a torrent a document? Is a living animal a document? No. But the photographs and the catalogues of stars, the stones in a museum of mineralogy, and the animals that are catalogued and shown in a zoo, are documents.

Suzanne Briet (1951, 10)

An object is to be considered a document when there is an assertion or a perception of evidence for some belief.

Michael Buckland (2017)

Introduction

Documents, as carriers of information, play a central role in the information disciplines and professions, and the documentation movement of the first half of the 20th century was one of the predecessors of information science. However, although documents carry information and data, they are not synonymous with either. The complex relation between these concepts, and the significance of that relation for the information disciplines, has been discussed for many years, gaining new impetus in recent years. For diverse and detailed perspectives on the concept of document, and its relationship to information and data, see Hjørland (2000), Ørom (2007), Koscijew and Richard (2015), Furner (2016) and Schöpfel et al. (2020).

We will look first at the document theory, introduced by the first documentation movement, and further developed by the neo-documentalists from around 1990, and then consider some current and future issues relating to document categorisation and

organisation. Finally, we will examine the changing nature of collections of documents.

Document theory

The English word 'document' comes from the Latin *docere*, to teach or to inform, with '-ment' implying the means. The Oxford English Dictionary defines a document as 'something written, inscribed, etc., which furnishes evidence or information upon any subject', and this is how documents are generally understood.

It might therefore be thought that the meaning of 'document' is straightforward and that the only issues that can arise relate to the differences between printed and electronic documents. This is far from the case. It can be argued that if a 'document' is some physical thing which records thoughts or ideas – information – then we should include paintings, sculpture, and perhaps even any artefact, as documents. But then, what about geological specimens in museums, carefully labelled and collocated with similar minerals; might we not go to look at them to find out about the mineral as an alternative to reading about it in a book? An antelope in the wild may be just an antelope; but in a zoo, placed where it can be studied, in surroundings indicative of its natural habitat, does it not convey information? And is an information-bearing 'thing' not a document? If a photograph of the lunar surface is a document, why not the real thing seen through a telescope? And so on. These issues were addressed by European documentalists in the first half of the 20th century, particularly Paul Otlet (1868–1944), Henri La Fontaine (1854–1943), Frits Donker Duyvis (1894–1961) and Suzanne Briet (1894–1989), whose 1951 text sums up the ideas of documentation in

this period. For appreciations of their work, see Wright (2014), Maack (2004) and Lancaster-Jones (1961); their classic texts are also still worth rereading (Otlet, 1934; Briet, 1951).

The work of these early documentalists was in response to the greatly increased volume of published scholarly information, particularly in science and technology, from the middle of the 19th century, and to the consequent need for better ways to control it. It built on the earlier study of bibliography, which, since the 18th century, had studied the selection, organisation and dissemination of printed works. The documentalists regarded bibliography as an aspect of a wider subject of documentation, which also encompassed authorship, printing and publishing and library services. They raised for the first time the questions of what were the limits of what a document is and what were the processes of documentation, coming to the view that any object should be regarded as a document if it provided information and evidence.

More recently, a number of scholars, particularly those associated with the Document Academy and with the Tromsø Institute for Documentation Studies, have espoused the centrality of the document idea for information science, looking backwards to its origins in the documentation movement and forwards to the newer ideas of document theory. See, for example, Skare, Lund and Vårheim (2007), Document Academy (n.d.), Lund and Buckland (2008), Lund, Gorichanaz and Latham (2016) and Buckland (2018).

Michael Buckland brought the issues to attention in his influential article on 'What is a document?' (Buckland, 1997). His views stem from his advocacy of 'information-as-thing', as against 'information-

as-knowledge' or 'information-as-process' (Buckland, 1991), and hence the idea that a document is a physical entity. Reviewing and critiquing the views of the early documentation movement, he showed how they extended the idea of a document beyond printed texts, and indeed beyond items created with the intention of communicating information. The emphasis moved from communication to evidence: a document was some physical item which gave evidence, and was in that way informational. However, there had to be something else: to count as a document, the item had to be placed in relation to other evidence-bearing items, by being put in a collection, indexed, cross-referenced, etc. This is shown here using examples given by Briet.

Object	Document
star in the sky	No
photo of star	Yes
stone in river	No
stone in museum	Yes
animal in wild	No
animal in zoo	Yes

There are practical consequences of accepting this wider view of document: whole sub-disciplines such as museum documentation make sense only if it is accepted. The practice of some botanical gardens of arranging library books, dried herbarium specimens and living plants according to the same classification scheme is another example. The emphasis is on whether some item functions as a document rather than what its physical form or nature may be.

Buckland's presentation has been reviewed and updated by several authors, including Buckland himself (1998), Frohmann (2009), Kallinikos, Aaltonen and Marton (2010), Robinson (2015), and Gorichanaz and Latham (2016), particularly with respect to the specific issues of digital documents. These cannot be defined by form, medium or location in the way that physical documents can, being less 'solid' and permanent; hence the pragmatic 'documentation' approach, defining a document by function, is particularly appropriate.

Document theory has allowed analyses of the 'document' nature of a wide range of entities. Some might have been considered as documents previously: for example, religious icons (Walsh, 2012), museum objects (Latham, 2012), art works (Gorichanaz, 2017; Berryman, 2018), cave paintings and sculptures (Narayan, 2015). Others are not likely to have been regarded as documents without a document-theoretic perspective: for example, clothing and fashion items (Peirson-Smith and Peirson-Smith, 2020), bodycam footage (Blanchette and Becker, 2018), living people (Nord, 2020), human remains (Berryman, 2019), landforms (Grenersen, 2012; Grenersen, Kemi and Nilsen, 2016), oral utterances (Turner, 2012), library community services (Turner, 2015), Tai Chi (Hansson, 2017) and performances (Sant, 2017). Document theory may also provide guidance to understand new emerging forms of document, such as multisensory, immersive experiences (Robinson, 2015).

Modern document theory, sometimes termed *neo-document theory*, examines the concept of 'document', and how it can be used in conjunction with other concepts to better understand the

communication of information and knowledge in an infosphere where many documents take digital form (Lund, 2010; Lund, 2011; Lund and Skare, 2017; Document Academy, n.d.; Buckland, 2014; Buckland, 2017). Alongside the nature and attributes of documents, document theory studies the processes of documentation, by which documents are created, organised and used. These are essentially the components of the information communication chain (Robinson, 2009).

Documents may take very many diverse forms, but there is a common distinction between two main forms:

- documents by intention: deliberately created as a document, such as books and photographs
- documents by attribution: objects not created as documents, but which become documents later, such as plant specimens in a herbarium.

Buckland makes a threefold distinction between: items created as documents; those made into or presented as documents; and those which are regarded as documents by someone, whether or not they were created with a documentary intention. Under this last, semiotic viewpoint, any object may be a document.

Although most documents will be recognised as such by everyone as able to convey information and meaning across space and/or time, some objects may also be regarded as a document by a person at a particular time for a particular purpose. This means that many things may be documents, so document theory focuses on a set of

attributes which documents, of whatever nature, possess, at least to some extent.

Indexicality: a document must be about something, or point to something (like an index finger, hence the name). It has an 'aboutness', and may act as a symbol.

Documentality: the special power of a document, by which it fulfils a social function in a particular context, at a time and place. It has an agency such that it can affect its users and bring change in the world. Because of this power, people will generally need training so as to effectively create and use documents.

Complementarity: from the principle of complementarity in physics, which dictates that objects have properties that cannot be observed at the same time. A document has three forms of attribute: *material*, its physical or technical make-up; *informational*, relating to the mental, cognitive activity of its creators and users; and *social*, its economic, political or cultural significance and connotations.

Fixity: documents must stay largely stable over time to fulfil their function. Books can last for hundreds of years, essentially unaltered; digital documents are more mutable, while remaining useful, and hence neo-document theory recognises the associated notion of fluidity.

Productivity: new documents can be created from an original. This was regarded by the original document theorists as a hierarchical process, with a primary document leading to second and tertiary documents. An example would be an article leading to an entry in

an abstracting and indexing tool or in a bibliography; or a geological specimen generating photographs, descriptive notes, etc. Neo-document theory also allows for non-hierarchical, or rhizomic, productivity, without making an indication of any particular document as primary.

The processes of documentation, in document theory, include: authoring or otherwise creating; selection and collection; organisation, arrangement, classification and indexing; copying, reproduction and dissemination; preservation, conservation and digitisation; and destruction. These are, of course, the processes of the communication chain of recorded information, the basis for the operation of libraries, archives and other document institutions. However, the practical value of document theory is not limited to documentation in such institutions, nor is it limited to the kinds of documents with which they are typically concerned. Documentary processes are also important in the everyday life of individuals, both in recording the past and in planning for the future (McKenzie and Davies, 2021). Regarding other kinds of entities as documents is also valuable. For example, treating landscapes as documents allowed them to be used as evidence in court cases relating to land and water rights (Grenersen, 2012; Grenersen, Kemi and Nilsen, 2016), while considering museum objects as documents will help curators to better understand and improve the experience of museum visitors (Latham, 2012).

Categorising and organising documents

Documentation, as noted above, has developed from the earlier study of bibliography, which, since the 18th century, had studied the

description, selection for particular purposes, organisation and dissemination of printed work (Buckland, 2018). This led also to systems for resource description and classification (discussed in [Chapter 8](#)). We can make some points here about documents as information resources.

Even if we restrict ourselves to the usual viewpoint of the information sciences, regarding documents as being discrete entities deliberately created for the purpose of conveying information, we still find some difficulties. In particular, we may have a problem deciding when two documents are 'the same'. If I have a copy of a textbook and you have a copy of the same edition of the same book, then these are clearly different objects. But in a library catalogue they will be treated as two examples of the same document. What if the book is translated, word for word, as precisely as possible, into another language: is it the same document, or different? At what point do an author's ideas, on their way to becoming a published book, become a 'document'?

These are quite difficult issues conceptually and pose real practical problems for resource description, in particular for library cataloguing codes. They led to the development of the Functional Requirements for Bibliographic Records (FRBR) conceptual model and its successor, the Library Reference Model (LRM) (Žumer, 2018). These models are based on the distinction of four levels:

- work: the intellectual or artistic content of a distinct creation
- expression: a distinct combination of signs conveying intellectual or artistic content

- manifestation: a set of all carriers that are assumed to share the
- same characteristics as to intellectual or artistic content and aspects of physical form. That set is defined by both the overall content and the production plan for its carrier or carriers, e.g. a specific edition of Shakespeare's *Hamlet*
 - item: an object carrying signs intended to convey intellectual or artistic content; 'a single exemplar of a manifestation'.

This might be made clearer by a specific example. A work is a distinct creative output, say Shakespeare's *Hamlet*. An expression is the realisation of a work – the English text of *Hamlet*. A manifestation is the physical embodiment of an expression – a specific edition of *Hamlet* in English. And an item is a single example of a manifestation – a particular copy of that edition of *Hamlet*.

There has been scope for disagreement on how these four levels should be understood. Are adaptations or abridgements of an original to be considered as different expressions of the work, or as new works? When do small changes in a text, perhaps just the correction of errors when a book is reprinted, make a new expression, rather than just variations of the original expression? Do we need the idea of a 'superwork', to include commentaries, criticisms, different formats – e.g. a film based on a book, or fanworks based on an original? This last is very similar to the document theory concept of an 'unbounded' document; for example, the music, programmes, booklets, newspaper supplements, photographs, decorations in theatres and nearby stores and exhibition items associated with an early film (Skare, 2016). It is related to the idea of *paratext*, elements apart from a main text which

make a document usable, including metadata, formatting elements, front matter of books, supplementary materials, recommendations to other documents, citations to a document and many more (Skare, 2020; Skare, 2021).

Much of this ambiguity stems from the basic question of what is a 'document' and, similarly, what is a 'work'. Although FRBR and LRM have formed the basis for the Resource Description and Access (RDA) standard for bibliographic description (discussed in [Chapter 8](#)), it is not clear that the understanding of work and document is as yet sufficiently developed to give a fully satisfactory solution for description of all forms of documents. For discussion of these issues, see Žumer (2018), Smiraglia (2019) and Holden (2021).

A simple categorisation of documents by resource level has been a useful, if simplistic, organising principle in the information disciplines. The levels are:

- *zeroth*: raw information before it enters the communication chain, for example the output from laboratory instruments, search interface/API (Application Programming Interface) output
- *primary*: the 'original information', e.g. journal articles, blog posts, tweets
- *secondary*: value-added information, from some organisation of primary material, e.g. textbooks, reviews, abstracts
- *tertiary*: materials pointing to, and aiding the use of, primary and secondary materials, e.g., bibliographies, directories
- *quaternary*: giving access to resource listings at a higher level, essentially 'lists of lists'

- *quintenary*: lists of lists of lists.

The two highest levels have proliferated in the web environment, whereas in the days of print a few quaternary ‘bibliographies of bibliographies’ were all that could be found.

Categorisations of this sort are necessarily subjective, according to how the resource is used and viewed by its user; a bibliography of medical articles, for example, would normally be taken as a tertiary resource, but to a future historian of healthcare information it would be a primary document. There are also disciplinary differences; a historian, for example, would think of original archival materials as primary, and journal articles based on them as secondary; a lawyer would think of legislation and case reports as primary, and articles, again, as secondary.

Collections

Collections, in the sense of those of interest to the information sciences, and documents go naturally together. As we have seen, one of the things which makes a document a document is that it is in an organised collection of related documents; and it is difficult to envisage a collection of information without documents. We discuss the management of collections in [Chapter 14](#), and focus here on the concept of collection.

The Oxford English Dictionary defines collection as ‘a number of objects collected or gathered together, viewed as a whole; a group of things collected and arranged’. The collection, an organised set of information-bearing items chosen for a particular purpose in a particular context or environment, and usually unique to that

situation, is fundamental to the information sciences. The library is the archetypical form of information collection: as Corral (2012, 3) puts it, 'the whole notion of a library is fundamentally associated with the idea of a collection, to the extent that the words "library" and "collection" are almost synonymous'. However, the same principles apply to collections in records centres, archives, museums, galleries and all forms of information centre.

A useful distinction, though not an absolute one, is between two forms of collection:

- collections of *ideas*, embodied in documents
- collections of *objects*, which may provoke ideas in the viewer.

The former is typified by the library, the latter by the gallery.

It is the selection and organisation of the collection, with its documents having some rationally assessed similarity, which distinguishes the typical information collection from the 'cabinet of curiosities' style of seemingly anarchic or random objects, intended to provoke surprise, awe or humour. For thoughts on the relevance of the cabinet of curiosities to information science, see Frohmann (2009). With different forms of documents, and different purposes, the collection forms the basis for the activities of information centres, libraries, records centres and archives, museums (of all kinds, including natural history) and galleries (although the last might not consider the information-bearing nature of the items to be the most important). The disciplines which underlie these professional activities are often referred to as the 'collection sciences'. The 'collection' in each of these cases may refer to the totality of their

information resources, i.e. documents, or to a subset; for example, a special collection within a library devoted to some topic, or the records of a particular organisation within an archive centre.

The move towards a digital information environment has implications for the nature of the collection itself, as well as for the practicalities of managing it. Some collections are truly digital, and these fully digital collections may be assembled very rapidly, in a way in which physical collections cannot be. Conversely, very few collections now avoid any digital dimension. Managing 'hybrid' or 'complex' collections presents particular problems of maintaining consistency of approach and presenting the collection in a unified way.

The move to digital collections generally means a move from 'ownership' and 'location' to 'access', as has been recognised for many years. The collection takes on a dynamic and impermanent nature, with varied and unpredictable lifecycles, and to some degree no longer under local control.

Any digital resource, perceived as a single entity by its users, may in fact be a composite or aggregate of distributed documents: a collection, composed of such entities, is even more an aggregate. Given that the resources of which it is composed will often link to other resources, it becomes very difficult to place a boundary around the collection clearly distinguishing what is inside it. Furthermore, a single digital document may be simultaneously in several digital collections, which could not happen in the pre-digital world.

This means that such a collection can no longer be thought of as a set of physical items in a particular location, or even as a collection of

digital items on a particular computer, but, rather, as a means of selecting documents from the universe of knowledge. Such a collection may be defined as 'a set of criteria for selecting resources from the broader information space', with collection membership defined 'through criteria rather than containment' (Lagoze and Fielding, 1998).

Summary

Documents are at the heart of the information sciences, generally organised in collections. The number of entities recognised as documents is increasing, both because of the recognition of the range of entities with documentary properties and because of new forms of digital documents being created. Document theory provides a means of understanding documents and documentation processes.

- A document is any item which conveys information or provides evidence.
- Documents may be created as such, or made into documents, or may be any entities regarded as documents.
- Documents are denoted as such by possessing indexicality, documentality, complementarity, fixity and productivity.
- The processes of documentation are those associated with the communication chain of recorded information.

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7 Domain Analysis

Domain analysis suggests that the specific competencies that information specialists have, or should have, are information about information infrastructures and information retrieval . . .

Domain analysis further suggests that subject qualifications and LIS qualifications are not independent of each other.

Birger Hjørland (2017, 452)

Introduction

The nature of information and knowledge, and the ways in which information services are provided, differ considerably in different contexts. This is an important theoretical and practical issue for the information sciences. In this chapter we consider what is meant by an information context or domain, and how domains may be characterised or understood, with a note on the importance of terminology in defining and representing domains. We then examine domain analysis, the approach developed for this topic. Some brief examples of particular domains are given to show the distinctions which may be made between them, and their effects in practice. Finally, we consider the changing nature of information work in specific domains.

Information domains

The contexts of information communication and use may be described in various ways: by geographic area, e.g. Europe; by type of institution, e.g. universities; or by the demographics of the information users, e.g. children and young people. The most helpful distinction, particularly for understanding information resources

and information practices, is the information domain, which often aligns with an academic discipline, e.g. mathematics or history; a professional or vocational activity, e.g. healthcare or finance; or a leisure interest, e.g. book collecting; or with a more loosely defined area, e.g. everyday information. Hjørland (2010, 1650) suggests that a domain may be a scientific discipline or a scholarly field. It may also be a discourse community connected to a political party, a religion, a trade, or a hobby. Domains are defined and explained by three dimensions: ontological, epistemological and sociological. The ontological dimension defines the domain by its main object of interest: botany by plants, history by the past, theology by the divine, etc.; this is the most usual way of defining a domain. The epistemological dimension relates to the kind of knowledge in the domain, or perhaps different kinds of knowledge associated with different paradigms or ways of understanding. The sociological dimension relates to the kind of people and groups involved in the domain.

It is clear that there are distinct differences in the way information is regarded and used in different domains, affecting information literacy, searching behaviour and information-sharing behaviour, among other things; see, for example, Pinto and Sales (2015), Tamine and Chouquet (2017), Fry et al. (2016), Madden et al. (2018), Russell-Rose, Chamberlain and Azzopardi (2018), Gregory et al. (2019), and Dreisiebner and Schlögl (2019). Hence the need for an awareness of domain and for approaches like domain analysis.

A framework to give a clear idea of what a domain is, and how domains differ one from another, will be helpful in understanding

their information aspects. Categorising the domain with which one is dealing in this way helps to show what its 'informational nature' may be, and also helps to identify similarities and analogies with other domains which may be matched by similarities in the most appropriate kind of information provision.

There are several models and frameworks for understanding subjects and domains. Three of these have been found useful for the information sciences: those of Hirst, Becher and Whitley, discussed below. Domain analysis has also been used in conjunction with the social theories of Pierre Bourdieu (Morado Nascimento and Marteleto, 2008). Other understandings of 'domain' are noted by Hjørland (2017) and, under the heading of 'disciplinarity', by Sugimoto and Weingart (2015).

The educational philosopher Paul Hirst (1946–2003) argued that, since disciplines are closely associated with their knowledge base, we can understand a discipline by understanding its 'form of knowledge' (Hirst, 1974; Walsh, 1993). There are seven main domains or forms of knowledge, defined by the fundamental nature of the knowledge and concepts with which they deal: mathematics; physical sciences; human sciences; literature and the fine arts; morality; religion; and philosophy. Where a discipline equates to one of these forms, it is what would be regarded as a 'pure' academic subject. Hirst also recognises 'practical disciplines', based on one of the forms but oriented toward solving practical problems. Engineering, for example would be a practical discipline based on the form of the physical sciences. Some academic subjects, however, do not align neatly with any form. Rather, they are focused on a

topic or subject of interest, using any of the forms which are useful in studying and understanding it. Hirst refers to these as *fields of study*; they are typically, though not necessarily, multidisciplinary. Examples are peace studies, women's studies and media studies; it was suggested in [Chapter 1](#) that the information disciplines might also be of this nature.

Tony Becher (1930–2009), a British higher education researcher, analysed the social and cognitive nature of academic disciplines along two rather traditional dimensions: pure/applied and hard/soft (Becher and Trowler, 2001). Clearly, these have strong implications for information practices (Fry, 2006); for example, contrasting disciplines whose knowledge structures are hard/pure (e.g. physics) with soft/pure (e.g. anthropology) or hard/applied (e.g. mechanical engineering).

Richard Whitley, a sociologist from Manchester Business School, analysed sciences as systems for the production of different kinds of knowledge by focusing on their internal structure, and identified seven different kinds of scientific disciplines (Whitley, 2000). Examples are 'professional adhocracies' (e.g. biomedical sciences) and 'conceptually integrated bureaucracies' (e.g. physics). In his model, disciplines are distinguished primarily by the degree of mutual dependence between researchers in creating new knowledge and the degree of task uncertainty in agreeing on research problems, methods and validity of knowledge acquired. Whitley's model has been used by Fry (2006) and Roos and Hedlund (2016) to contrast the differences in information practices between different disciplines.

Domain terminologies

Terminology, the structured naming of things, is one of the most basic and familiar tools for organising information and communicating it accurately and unambiguously. Many domains, particularly in scientific, medical and technical areas, have their own specialised terminologies, which define and shape the domain and are vital to the communication of information within it. All domains have their own terminology; the terminology of everyday information, for example, is the dictionary; the standard dictionary of the language, and perhaps dictionaries of regional accents and of slang terms. An understanding of the terminology of a domain is a necessity for effective information provision.

Terminology resources encompass a number of overlapping categories: general dictionaries, which may be multilingual; dictionaries and glossaries for specific subjects; special-purpose dictionaries (e.g. rhyming, quotations, crossword clues); dictionaries of the proper names of persons (biographical dictionaries) and of places (gazetteers); thesauri (in the sense of Roget's word finder); lists of specific names (for products, medicines, chemicals, etc.); and tools intended primarily for retrieval but which may function as terminologies (classifications, taxonomies, ontologies, subject headings and retrieval thesauri). For the nature and history of some terminology tools, see Hitchens (2005), Hüllen (2004), Muggleston (2011), Tackabery (2005) and Stirling (2007).

There are also special and detailed terminologies within the sciences for chemical structures and reactions, for living things, and for many aspects of biomedicine and healthcare. The notations of subjects such as mathematics, chemistry, music, and dance may also

be regarded as forms of terminology, or at least as things which must be understood to make sense of terminology.

Domain analysis

Providing a theoretical basis for the study of information domains is the concept of domain analysis, derived by the Danish academic Birger Hjørland (2002; 2010; 2017). It is an approach within the socio-cognitive paradigm of information science (discussed in [Chapter 4](#)). As Hartel (2019, 5) summarises it, 'domain analysis refutes the notion of universal information phenomena, and focuses upon informational patterns within social worlds. [It] directs the [information worker] towards mastery of their subject by producing literature guides, special classifications, and user studies, and to a strategic understanding of the history, culture, epidemiology, and institutional dynamics of their fields'. Domain analysis is a realist approach in philosophical terms; it seeks a basis for information science in factors external to the individual which are objective rather than subjective, and which may be located in the expertise and practices of subject specialists.

Hjørland sees domain analysis as central to the work of the information scientist and as comprising eleven distinct ways in which information science may approach a domain. These approaches are:

- the production of literature guides and subject gateways
- the production of special classifications and thesauri
- researching indexing and retrieval in specialist subjects
- empirical user studies

- bibliometric studies
- historical studies
- studies of documents and 'genres'
- epistemological and critical studies
- studies of terminology and special languages, and discourse analysis
- studies of structures and organisations in the communication of information
- studies in cognition, computing and artificial intelligence.

Some alternative ways have been suggested, including discourse analysis, database semantics, and archival provenance (Hjørland, 2017). For an example, see Robinson (2010) for an analysis of the healthcare domain, which has a particularly rich and diverse set of resources and users, using all eleven aspects.

Domain analysis is an approach for both research and practice, and provides a bridge between the two. An information professional working in a particular subject area need not carry out such domain analysis studies, though they might well do so, but they would certainly need to be aware of the results of such studies. For example, a healthcare information specialist would need to be aware of special terminologies used in collections of healthcare information (for example, the International Classification of Diseases), though they need not themselves have carried out comparative studies. Similarly, a law librarian should be aware of what is known about the information behaviour of lawyers, law students, etc., though they might not carry out such studies themselves.

Examples of domains

In order to illustrate the ideas of information domains more fully, we will discuss aspects of information in three domains: chemistry, as an example of a scientific discipline; history, a humanities and social science discipline; and serious leisure, a domain outside the academic realm. Briefer remarks will be made about some other domains.

Chemistry

Chemistry is the study of substances, their properties and reactions, is, in Hirst's terms, an academic discipline, with the form of knowledge of physical science, although its theories are often conceptual or semi-qualitative. It is the basis of numerous practical disciplines. It has no popular or public dimension as an activity in its own right, but there is a strong interest in chemically related information, for example to do with the environment or with foods.

Chemistry is traditionally divided into three branches: *organic*, dealing with carbon compounds; *inorganic*, dealing with all other elements and compounds; and *physical*, the application of the concepts and methods of physics to chemical problems. There are many subfields and crossover disciplines, such as biochemistry, polymer chemistry, medicinal chemistry, forensic chemistry, and environmental chemistry. This natural subdivision strongly influences information resources.

Chemistry is important scientifically, socially and economically; is fundamental to the life sciences and to many engineering and technology areas; and underlies the chemical and pharmaceutical

industries. Its information base is extensive. It is a 'lending science', hence the prevalence of resources in 'chemistry for . . .'. The subject has often led the development of information products and services: learned journals, textbooks and monographs, and abstracting and indexing services in the 19th century; and applications of bibliometrics, mechanised documentation, online databases, and use of the web in the 20th century (Willett, 2008).

Major resources, in addition to textbooks, monographs and journals, include bibliographic databases and property databanks. The best-known examples are the information sources of the Chemical Abstracts Service, part of the American Chemical Society, founded in 1907, and the Beilstein data compilations, dating from 1881. Older chemical information is often of value. Chemical information systems are used not only for information retrieval but also for molecular modelling and property estimation, and for correlation of structures with properties, for example for drug design. This links chemical information with the related subject of chemoinformatics. These resources, organised by structure and reaction type, are unique to chemistry and define it as an information domain (Solla, Currano and Roth, 2013).

Chemistry has a wide range of knowledge organisation tools: nomenclatures, terminologies, classifications and taxonomies, and chemistry-specific ways of organising information, based on the unique attributes of the main entities within chemical information – chemical substances and chemical reactions (Bawden, 2015). For inorganic chemistry, the periodic table of the elements ([Figure 7.1](#)) has been a vital tool for information organisation, with many

variants over time and for particular purposes, showing different information and relations for the elements, e.g. for geology or metallurgy (Hjørland, 2011; Scerri, 2019); as an example of its use, see the Royal Society of Chemistry's interactive periodic table (www.rsc.org/periodic-table, as of July 2021). Special classifications have also been developed for bulk forms of substances, such as minerals, and for special purposes such as the hazardous nature of substances.

Group Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	* 71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	* 103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
			* 57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb		
			* 89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No		

Figure 7.1 *The periodic table of the elements* (Wikimedia Commons, CC BY-SA)

Chemical substances may be represented in a variety of ways: trivial names; systematic names; registry numbers; alphanumeric notations; and 2D and 3D structure diagrams and their computer representations. The structure diagram, devised around 1860, has been a very powerful tool for representing chemical information and facilitating written communication. Reactions may similarly be represented in a variety of ways: by trivial names; by originators' names, giving an insight into the history of the subject; by systematic

names; by classification of structural change; and by diagrammatic representation of structural change and its computer representation. [Figure 7.2](#) on the next page shows the variety of representations for the anti-leprosy drug dapsone, from the Royal Society of Chemistry's Chemspider software.

Chemistry has always been known as an information-intensive science, and chemists as particularly information-conscious users, as has been borne out by numerous studies of information behaviour in the domain; see, for example, Gordon et al. (2018) and Rose-Wiles and Marzabadi (2018). There has always been a strong emphasis on context-specific information literacy in the domain (Bawden and Robinson, 2017).

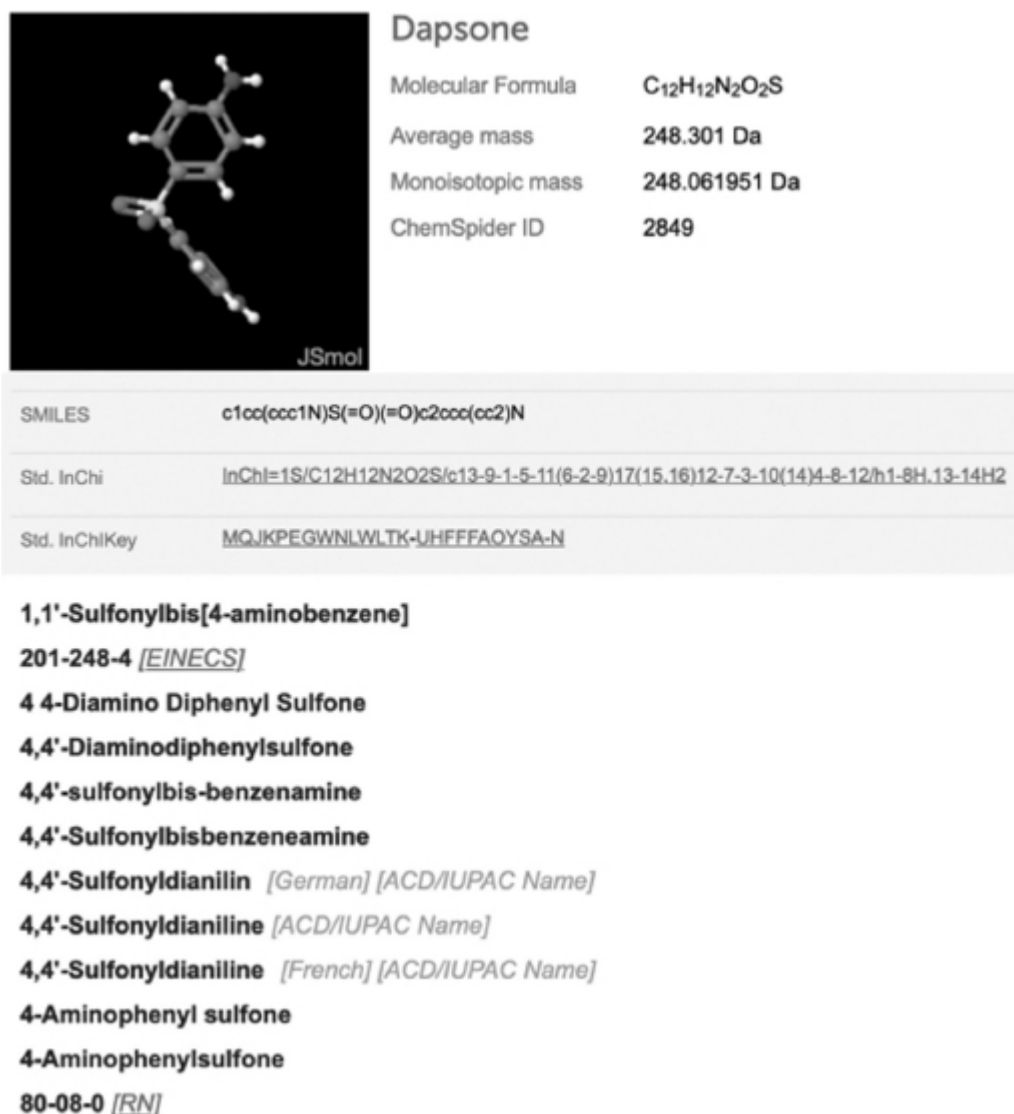


Figure 7.2 Chemical structure representations (Royal Society of Chemistry, CSID:2849, www.chemspider.com/Chemical-Structure.2849.html)

History

History is one of the longest-established and accepted academic disciplines, with a literature extending over thousands of years (Burrows, 2009). It is an academic subject, but without a practical discipline directly associated with it. Professional historians tend to

be teachers, researchers, museum and archive professionals, or authors/presenters of popular history. There is a great public interest in the subject, as attested by the many books, magazines, TV and radio programmes and web resources devoted to it. Users of history information therefore span the spectrum from a relatively small group of academics and professionals to the much larger number of the interested general public.

There has always been debate as to what kind of a subject it is; this has significance for what kind of information resources will be regarded as most important. Those who see it as located within the social sciences will look for general trends and explanations of events from social data compilations, seeking to establish general historical 'laws'. Those who locate it in the humanities, seeing history as the interpretation of unique events, will focus more on individual accounts.

The discipline is traditionally divided into sub-disciplines: social history, economic history, political history, military history and so on. Another common division is history by country or region: English history, European history, etc., and a third by time period. This gives a natural structuring to the discipline's information resources. Most, if not all, other subjects have a historical dimension, and several of these have formed interdisciplinary subjects in their own right, e.g. history of science, history of medicine and history of art. All of these sub-disciplines and overlap areas have their own information resources.

History has a particularly wide range of information sources. As noted in Chapter 6, their categorisation is different from the usual;

historians regard journal articles and research reports as secondary, primary documents being original archival materials. History's primary resources include archival documents of many kinds, but also include physical (museum) artefacts of almost unlimited nature. Digitisation of all these materials is making them available to a wider audience. In the academic sector, the traditional journal, scholarly monograph and monograph series and textbook are still predominant. Older material of this kind is still valued. In the popular sphere, books, magazines, radio and TV programmes (even whole channels devoted to history programming), and a variety of multimedia and web sources are all widely used.

There are a limited number of subject-specific abstracting and indexing services, while many general sources include historical material. There is an increasing number of historically focused digital libraries, which include documents of very varied kinds.

The very diverse nature of historical materials, and the particular needs of museum documentation, have led to the development of a variety of information organisation tools, thesauri in particular. Examples are those maintained by Historic England, including thesauri for maritime place names, maritime cargoes, building materials, archaeological sciences and historic aircraft types, and by the British Museum, including thesauri for objects and for materials.

Dissemination of history information involves a range of institutions, notably universities, learned societies, publishers, broadcasters and central and local government. Academic libraries, public libraries (particularly local studies centres), special libraries (particularly those of learned societies and research institutions),

archives and record centres, and museum libraries and documentation centres are all important contributors.

Studies of the information practices of historians, both amateur and professional, suggest that the traditional behaviours associated with humanities and social science scholars predominate. There has been an emphasis on printed materials, browsing as a means of identifying information and enthusiasm for sources such as book reviews, though digital systems are having an increasing impact; see, for example, Vilar et al. (2016), Martin and Quan-Haase (2016), Dalton and Charnigo (2004), Rhee (2012) and Sinn and Soares (2014). Amateurs, particularly genealogists and family historians, often have sophisticated information practices; see, for example, Darby and Clough (2013).

Serious leisure

Serious leisure is an area which, though not academic or professional, may certainly be regarded as an information domain. Since about 2010 there has been an increased interest in the information practices and information resources associated with serious leisure. Before that, the focus of information for the general public was on education, employment and (to a lesser extent) citizenship. However, it has been realised that the information resources and behaviours for leisure pursuits – where the emphasis is on play, pleasure and enjoyment, rather than on solving problems and carrying out tasks – are just as sophisticated and significant as those for work and study.

The idea of ‘serious leisure’ was devised by the Canada-based sociologist Robert Stebbins (2009; 2020). One of the main contentions

is that those involved develop a very deep and rich knowledge of their leisure topic and develop sophisticated practices in creating, organising, sharing and using information. The study of the information dimension of serious leisure was pioneered by Jenna Hartel (2010) and has generated a considerable literature of its own.

There are various categorisations of serious leisure activities. A typical one divides the area into four (often overlapping) aspects:

- hobbyist (e.g. collecting things, crafts)
- volunteer
- amateur (e.g. music or athletics)
- fandom (sports, music, cult media . . .).

All of these aspects have been studied from the information point of view, typically using ethnographic study methods (although a wide variety of methods have been applied), and applying a number of information behaviour models, including the everyday life information models of Savolainen and of Hektor, and the small-world model of Chatman. They have found extensive, sophisticated and sometimes idiosyncratic information resources and practices, including creating, sharing, organising, tagging and archiving (Hartel, Cox and Griggin, 2016; Case and Given, 2016, 330–3; de Kosnik, 2016; Mansourian, 2020). These are now becoming of increasing interest to library and information services, most obviously with an interest in providing access to materials produced by serious leisure protagonists, such as fanworks (Price and Robinson, 2017), and in programming and providing facilities such as makerspaces (VanScoy, Thomson and Hartel, 2020). There is also

the question of whether some of the methods devised in the serious leisure domain, such as tagging methods, may be more generally applicable.

Other domains

Other domains have informational features of interest; we briefly mention four here.

Mathematics

Mathematics is a ubiquitous subject, studied by every person during their formal education, and applied in some ways by most people in their everyday life. It has a uniquely wide set of users, from professional mathematicians and students of all levels, through workers in the many occupations which use mathematics, to the enthusiasts for recreational mathematics. It therefore has a particularly wide range of types of resources. The information practices of those using mathematical information lie between those of scientific research and of applied science and engineering, reflecting the nature of mathematics as both a discipline in its own right and a tool for application in many other contexts, with the relatively slow and incremental process of mathematical research having a particular influence on information behaviour (Sapa, Krakowska and Janiak, 2014; Barsky, 2012; Gordon et al., 2020). Mathematical knowledge is one of Paul Hirst's unique forms of knowledge, and the subject has a uniquely abstract knowledge base, with a variety of classifications (Fraser, 2020) and an extensive and sophisticated language with a variety of notations developed over hundreds of years (Mazur, 2014). For this reason, it has an unusually wide range of 'logic and language' resources, including guides to

how to read and write mathematics, as well as how to understand its terminology and notation.

Law

Law is an example of a subject which might be thought to form a homogeneous academic and professional domain, but which is very diverse geographically, including in informational terms. Not merely are laws different in different countries, but in those jurisdictions, typically in Anglophone countries, which have a common law system, legislation is interpreted and nuanced by judicial decisions in a way that does not happen in systems of civil law, as in many continental European countries. The consequence is that in countries such as the UK and the USA there is a wide range of legal resources providing information on case reports, precedents and citations to cases, which has no equivalent elsewhere. Terminology may pose problems, with legal use of words and phrases differing from the usual dictionary definition. In the UK there is much use of Latin terms and obscure abbreviations, with texts and resources known by the proper names of their original authors or editors.

Art, design and fashion

These comprise a domain which has perhaps the widest range of document forms, and correspondingly diverse collections, including images of all kinds, architectural models, materials and fashion pieces (Glassman and Dyki, 2017; Peirson-Smith and Peirson-Smith, 2020). Information practices are focused on assisting inspiration and creativity, and include a need for printed materials as well as digital, and a preference for browsing and informal access.

Pharmaceuticals

Pharmaceuticals, although rooted in the academic disciplines of pharmacology, medicinal chemistry and pharmacy, is a domain closely involved with one of the most information-intensive industries and with the equally information-intensive clinical use of medicines. Because of this background, and because of its economic and social importance, this area has perhaps the widest range of resources of any domain, from chemistry and biology to pharmaceutical technologies to medicine, from pure science to commercial and patent information, and from highly specialised resources to those aimed at the general public (Bawden and Robinson, 2011).

Domain-specialist information work

One of the main origins of information science is domain-specific information work, in special libraries, information services and documentation centres. The traditional roles of the subject librarian and the subject-specialist information officer have changed, particularly as users are now well able to do much of their own searching online, doing away to an extent with the intermediary role. Nonetheless, subject-specialist roles remain of considerable importance, particularly in areas such as law and medicine.

What kind of subject knowledge, if any, is needed for such roles is a very long-standing question. It is generally agreed that a domain-specialist information professional does not need the same degree of subject knowledge as their users: medical librarians are not usually qualified doctors or nurses. What is needed is some subject

knowledge, plus knowledge of subject-specific sources, user needs, etc.: the subject background necessary to allow insight into needs and domain-specific sources, and to enable the interpretation and evaluation of information. This is often described as an understanding of the *logic and language* of a subject area; an appreciation that, for example, *Phyllostachys aurea* is a plant rather than a Greek philosopher (Morris-Knowler, 2001). How such domain understanding is best gained, and whether information skills or subject knowledge are more important, are questions which recur frequently. Moreau (2019) gives a detailed account of how a librarian new to the health science area gained such understanding, and the surveys of Sterner (2020) and Koos and Scheinfeld (2020) show that, while a subject background is valuable to science and health librarians, subject expertise can be acquired in a variety of ways.

A variety of activities have been noted as appropriate for the subject specialist information worker, most commonly:

- creating user guides
- creating terminologies and taxonomies
- carrying out 'difficult' searches and reference queries
- assisting, or leading, systematic reviewing
- evaluating and interpreting information
- providing current awareness
- suggesting new resources and aiding collection development
- promoting specialist resources
- cataloguing and resource description

- research data management
- coordinating expertise and facilitating knowledge communities
- acting as instructor, trainer, consultant and advisor.

There is a clear resonance here with aspects of domain analysis.

As examples, see an article on the building of a collection of material on the composer Debussy, requiring strong subject knowledge (Glaeser, 2013), and on new roles for medical reference librarians (Sullo and Gomes, 2016) and agricultural librarians (Bracke, 2017). Kranich et al. (2020) give an account of the changing role of the liaison librarian, with emphasis on building relationships and partnerships; Kallaher et al. (2020) describe the role of the library in supporting systematic reviewing; and Darch et al. (2020) describe library involvement in the curation of datasets, in their case astronomy data. Day and Novak (2019) point out new roles for the subject specialist in the 'inside out' collection model (discussed in Chapter 10).

Summary

Much of information science can be understood only in a disciplinary, or similar, context. This includes: the nature of documents and resources; terminology and information organisation; information behaviour, practices and literacies; bibliometric patterns; and the role of institutions in the communication chain. The concept of domain, and the approach of domain analysis, allows us to understand and deal with these differences, in both research and practice. The linking of domain analysis with the communication chain of recorded information

provides a unique character for information science (Robinson, 2009). The subject-specialist role, dealing with information within a particular domain, is of importance, particularly in areas such as law, business, healthcare and the sciences.

- Domain analysis is a framework for studying information communication within subject areas and user groups, and for the provision of information services to such groups.
- It is a socio-cognitive approach, based on examining the nature of knowledge within social groups and implications for information provision.
- It provides a basis for the work of the subject-specialist information practitioner.
- It provides a bridge between research and practice in the information sciences.

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8 Information Organisation

The first step in wisdom is to know the things themselves: this notion consists in having a true idea of the objects: objects are distinguished and known by classifying them methodically and giving them appropriate names. Therefore, classification and name-giving will be the foundation of our science.

Carl Linnaeus

Obviously there is no classification of the universe that is not arbitrary and conjectural. The reason is very simple; we do not know what the universe is.

Jorge Luis Borges, *The Analytical Language of John Wilkins*

Introduction

The organisation of information, and of documents which convey information, is one of the fundamental aspects of information science. In many respects, this amounts to classifying and name giving: as essential in our science as Linnaeus proclaimed it to be in his, though for rather different reasons. These reasons are the needs to describe the nature of a document (its title, creator, date of creation, physical form, publishing details, etc.), fully and accurately, through cataloguing and metadata creation, and to denote its subject, what it is about, through classifying, indexing and summarising.

The subject may be referred to as either *information organisation* (IO) or *knowledge organisation* (KO). Often these terms are treated as synonymous. KO strictly refers to the organising and structure of knowledge itself, a rather abstract and philosophical activity, while

IO is usually taken as the seeking of pragmatic solutions to the organisation of information-bearing documents, physical or digital, so that they can be accessed and retrieved. Hjørland (2012) notes that the terms are typically used in different contexts, but that the underlying theoretical principles are identical; he suggests that KO is the more appropriate term, because it relates to the social context of information rather than to information theory and computing. We will use the term IO throughout this chapter, for consistency with the title of the book and other chapters.

One of the main aspects of the subject is the way in which long-standing concepts and techniques are being adapted to the modern digital environment. Some fundamentals of the topic, particular aspects of the theory of classification, go back many centuries, and some of the main tools used today have their origins in the 19th century. They were developed as part of the attempt to provide bibliographic control of printed materials; to record, identify and make accessible all the intellectual output of humanity, as expressed in recorded knowledge. They, and newer equivalents, are now being used to provide access to the rapidly expanding stock of digital material. The processes of IO, themselves part of the wider set of processes of documentation, have always been intellectual activities, carried out by expert people. However, automated, or at least machine-aided, systems for metadata creation, including automatic classification and automated indexing, are making a significant impact.

IO is an extensive and complex subject in its own right, but fortunately it is covered by a wide range of textbooks, articles and

web resources. Wide coverage of the whole topic is given by Gnoli (2020a), Joudrey and Taylor (2017) and Glushko (2013); also valuable are the articles in the growing online encyclopedia of knowledge organisation created by the International Society for Knowledge Organization (ISKO)

We will outline the main topics and issues, pointing to where more detailed treatments can be found. We will look first at some of the fundamental issues of information organisation, before examining the main tools, issues and processes: metadata, resource description, classification, taxonomy and ontology, and alphabetic vocabularies.

Fundamentals of information organisation

The questions raised by IO have been asked since the earliest days of philosophy. Is there a single ‘natural’ classification of everything in the world? On what basis do we assign things to categories? How do our mental concepts relate to physical things? How distinct can two things be and still be given the same name? And so on. There is a great deal of theory underlying all organisations of information and knowledge, whether classification schemes for documents or scientific taxonomies, of plants, animals, rocks, stars, etc. For detailed treatment of the theory of IO, see Svenonius (2000), Hjørland (2016a; 2021) and Gnoli (2020a); only an outline is given here.

IO has been approached from a number of perspectives. Following Gnoli (2020a), we can identify five *dimensions* in which IO has been developed, in theory and in practice.

Phenomenon: rooted in ontology, this leads to organisation by the

- phenomena of reality, the things that exist in the world, rather than by how we think of them. Such considerations have led to the development of classification schemes such as Brown and Integrative Levels. IO systems built on this basis include all subjects which may be envisaged, whether or not any documents mention them; the opposite of the principle of literary warrant, noted below.
- *Perspective*: rooted in epistemology this leads to organisation by how our knowledge is structured, typically by academic discipline, and leads to discipline-based classification such as Dewey. It may lead to IO being influenced by philosophical perspectives such as realism or pragmatism.
- *Document*: a focus on information-bearing documents as the units to be organised leads to metadata formats for different kinds of documents and to the inclusion of document- or publication-type indicators in classification schemes, thesauri, etc. It may also lead to IO tools being developed on the basis of *literary warrant*, such that only subjects and topics which appear in existing documents should be included in a vocabulary (Barité, 2018).
- *Collection*: IO is often applied to a collection of some kinds of document, its purpose being to structure the collection to make it understandable and usable. Although the principles of IO remain the same, the details will differ with the kind of collection. Hjørland (2016a), for example, notes the ways that IO is applied in libraries, archives, museums, bibliographic databases and the internet.

- *User*: IO systems need to respect the needs, preferences and capabilities of their users. For example, many classifications and taxonomies have between five and ten main headings, reflecting the number of entities people can hold in mind at one time (Bawden and Robinson, 2016). A user-centred focus is also seen when small or specialised libraries use a local scheme for arranging physical materials, in preference to a general classification scheme.

Hjørland (2017) proposes that there are four basic approaches to classification, which are applicable to all IO:

- *rationalism*, based on the logical application of rules based on clearly stated fundamental principles
- *empiricism*, based on assessment of similarity according to neutral comparison of all available criteria, without any assessment of which criteria are important
- *historicism*, based on analysis of development over time, bringing together entities which have a common ancestry, or which are related in the development of theory
- *pragmatism*, based on a critical and subjective analysis of the purpose of the organisation to bring out relationships meaningful for that purpose.

Examples of all these approaches may be seen in practical IO systems.

Several fundamental ideas within IO can be identified, which will recur as different aspects are considered. We can summarise them

under three headings: concepts and categories; hierarchy and facets; and controlled vocabulary.

Concepts and categories

A concept is a basic unit of knowledge, an idea, a building block of thought. The exact nature of concepts, and in particular the relation between a concept as a representation of external reality, as an internal subject of individual thought and as a culturally agreed concept, has been the subject of much debate; see, for example, Van Loocke (1999), Hjørland (2017) and Margolis and Laurence (2021). Concepts are of significance to IO in two main ways. Analysis of the conceptual nature of any document determines its *aboutness*, by which it may be classified or indexed (Holley and Joudrey, 2021). And the tools of IO, such as classification schemes and thesauri, are constructed of concepts, albeit necessarily expressed as words or notations.

Concepts are divided into categories, groups of generally similar concepts, which may be overlapping (a concept may be in more than one category) or mutually exclusive (a concept fits into one and only one category). The idea of categories and how they are formed, which is at the heart of much IO (Hjørland, 2017; Glushko et al., 2013), goes back to the philosophers of classical Greece, and to Aristotle in particular, and has again been the subject of much philosophical debate (Thomasson, 2019).

Hierarchy and facets

Division into categories is typically by hierarchy, again an idea going back to classical Greece (Bergman, 2021). It is usually taken that

hierarchies may be formed in three ways, with *classes* divided into *subclasses*:

- by *type* or *kind*, e.g.

PLANTS > FLOWERING PLANTS > ROSES

- by *part* of a whole, e.g.

HUMAN BODY > ARM > ELBOW

- by *instance* or *example*, e.g.

OCEANS > ATLANTIC.

Although there are examples of non-hierarchical structures in IO (see, for example, Robinson and Maguire, 2010), hierarchy is generally fundamental.

Hierarchy gives a way of organising any knowledge structure or collection of information, but it is rigid and one dimensional and does not allow any ways of showing other relationships between the concepts. To deal with this, the idea of facet and of facet analysis was originated. Facets are non-hierarchical aspects or characteristics of a subject area, used to form classes, and facet analysis involves dividing the concepts within a subject domain into consistent sections, based on facets (La Barre, 2010; Hjørland, 2013; Hudon, 2020). The concept of facet analysis was originated by the Indian librarian and documentalist S. R. Ranganathan (1892–1972), who proposed a set of five ‘fundamental facets’, applicable to any subject: personality, matter, energy, space and time (PMEST). Extensions to this, and alternative sets of fundamental facets, have been proposed subsequently, but facets are usually derived to suit each case. For example, the subject of ‘historic buildings’ might have the facets

PURPOSE (house, church, school . . .), STYLE (Gothic, classical, Arts and Crafts . . .), CONSTRUCTION (stone, brick, timber . . .), AGE (Victorian, medieval . . .), etc. For an example of detailed analysis of the facet structure of a subject, music, see Lee, Bawden and Robinson (2021). Facet analysis is a fundamental structuring principle for IO, information architecture and information retrieval, and is applied in the construction of classifications and thesauri, the design of interfaces and the construction of complex search logics (Broughton, 2017a).

Controlled vocabulary

A controlled vocabulary, another foundational concept within KO, is a set of agreed and standardised terms to be used for some IO purpose, including indexing, retrieval and classificatory arrangements (Joudrey and Taylor, 2017, chapter 10; Smith, 2021). They vary from the very simple, such as keyword lists and synonym rings, to which others are more complex, including inter-term relations and explanations of terms, such as subject-heading lists, classifications, taxonomies, and thesauri. They control the variability and redundancy of natural language. The opposite is an uncontrolled vocabulary: full text, freely chosen keywords and tags, etc. The term *knowledge organisation system* (KOS) is used as the widest descriptor for IO tools, including all controlled vocabularies, lists of uncontrolled terms and also scientific taxonomies, such as those for living things (Gnoli, 2020a, chapter 7). An argument about whether controlled or uncontrolled vocabularies are ‘best’ has continued for many years. The obvious answer is that a combination of both is desirable: a controlled vocabulary for consistency and for

its term relations, with uncontrolled terms for precision and for new terms which have not been included in the controlled vocabulary.

KOSs are usually described as either *alphabetic*, with the terms arranged in alphabetical order according to the language of the vocabulary, or *systematic*, with the terms arranged according to the system of a classification or taxonomy. Some simple KOSs are restricted to one form; for example, a list of terms without any inter-term relations can only be presented either alphabetically or in an arbitrary order. KOSs with some complexity in term information can generally be viewed in different ways; for example, classifications like Dewey have alphabetic indexes, and taxonomies and thesauri can be interconverted.

Finally, we may note that most KOSs have terms representing concepts and subjects, rather than proper names of people, places, etc. The latter will be found in separate authority files, although some special-purpose vocabularies, particularly taxonomies, will include them.

Metadata

The Oxford English Dictionary defines metadata as ‘data whose purpose is to describe and give information about other data’. In the context of IO, it is generally understood as data which describes the nature and subject of a document; short, structured and standardised records, which can act as surrogates for the original document for search and display. It may be encountered in files of metadata, the classic example being a library catalogue where each record gives the metadata for a book or other item in the document

itself, or it may be encoded within digital documents, for example embedded in HTML or XML coding.

Metadata is crucial to IO, giving the information necessary for organising, managing, searching and accessing documents. Metadata creators seek to provide answers to questions such as these for a document: 'what is this?', 'how does it differ from similar things?', 'how can it best be described?', 'how can it best be made retrievable?', and 'how can it clearly be shown how, and by whom, it can be used?' For relatively short introductions to metadata, see Pomerantz (2015), Gartner (2016) and Hider (2018), and for more detailed treatment see Haynes (2018), Joudrey and Taylor (2017) and Zeng and Qin (2016), the last giving many examples of metadata formats and schemas.

The creation and use of metadata have been regarded as intellectual processes carried out by expert people. Increasingly, however, metadata, even of the kinds of relevance to IO, is being created and applied by machine. There are some metadata standards intended primarily for automated use, such as the ONIX (Online Information Exchange) family of standards, used to transmit bibliographic and licensing data about new publications from publishers to purchasing institutions.

There have been two separate approaches to metadata. One is the longstanding *bibliographic* approach, library related and typified by the MACHine Readable Cataloguing (MARC) standard; the other is the internet and digital resource approach, typified by Dublin Core. The two approaches are becoming integrated in more recent metadata standards and schemas.

Bibliographic metadata has traditionally been thought of as having two components: *descriptive* and *subject*. These were traditionally instantiated in different physical forms; for example, the distinct 'author/title catalogue' and 'subject catalogue', in libraries using drawers of catalogue cards, but both are now usually subsumed within one metadata format, and described as descriptive metadata. This includes a description of the item; its title, author, date of publication or creation, physical form, etc., and an indication of its subject, or aboutness. In terms of Popper's '3 Worlds', this describes both the World 1 information objects and their World 3 content. Descriptive metadata answers questions such as 'what is this called?', 'who wrote it?', 'how old is it?', 'how big is it?', 'what kind of thing is it?', and 'where is it?' The subject elements answer the question 'what is it about?' The details of description will vary according to the physical nature of the item: e.g. 'how big is it?' will be answered in terms of pages and centimetres for a book, and megabytes for a digital resource. Subject description is invariant to physical form, since it denotes the intrinsic content, and the same tools are therefore used regardless of form; for example, the Dewey Decimal Classification, best known for shelf arrangement of books in libraries, has been used in internet directories.

Metadata serves numerous purposes. Haynes (2018), for example, distinguishes six: identifying and describing information resources; retrieving information; managing information resources; managing intellectual property rights; supporting e-government and e-commerce; and enabling information governance. Supporting cultural heritage is sometimes identified as a further distinct

purpose. As a result, different types of metadata can be distinguished. Descriptive metadata is best known, since this is what enables access to, and choice of, resources; for this reason, it is sometimes referred to as *finding metadata*. Other kinds of metadata which have been distinguished include:

- *administrative metadata*: gives information on factors such as acquisition, appraisal, location, responsibility for an item and dates by which an item should be reviewed or discarded
- *rights metadata*: sets out intellectual property rights, access restrictions and instructions on reproduction and dissemination
- *preservation metadata*: describes physical condition of materials and actions taken to preserve them
- *technical metadata*: gives information on technical issues such as file formats, digitisation quality and software requirements
- *structural metadata*: provides links between data elements to form them into complex digital documents, for example linking pages into chapters in an e-book.

Metadata standards

Metadata of all kinds is prescribed by metadata standards, sometimes also referred to as metadata formats or schemas; there are various formal definitions, but a standard is generally regarded as a set of fields, or elements, with rules for their use, designed for a particular context. Because of the varied purposes and contexts for metadata, there are 'a jungle of standards. They appear to be everywhere, proliferating wildly, constantly evolving and giving birth to offshoots which take on a life of their own' (Gartner, 2016,

53). Space prevents us discussing standards in any detail; we will make some general remarks and give some examples, referring the reader to the metadata texts, and in particular to Joudrey and Taylor (2017) and Zeng and Qin (2016), for details and further examples.

All metadata is generally taken to have three components (Gartner, 2016):

- *semantics*: the meanings of the fields or elements of the standard, for example, exactly what is meant by a 'title'. Does it have to be a formal title, as on the title page of a book? Or can it be an 'unofficial' name by which a document is known?
- *content rules*: which govern the content of the metadata itself, for example, exactly how we state the name of an author. Family name first or last? Given names or initials?
- *syntax*: the way in which the metadata is encoded, for example, coding in XML.

Zeng and Qin (2016, 23) distinguish four categories of metadata standard:

- *standards for data structures*: the best-known form of data standards, sometimes referred to as *metadata vocabularies* or *element sets*, these govern metadata semantics, specifying the elements or fields to be used. These include the Dublin Core standard for web materials, and bibliographic metadata standards such as the MARC family, MODS (Metadata Object Description Schema), BIBFRAME. There are standards for specific contexts: examples include EAD (Encoded Archival Description) for archives, Categories for the Description of Works of Art

(CDWA) for museums and galleries, CIDOC CRM (Cultural Reference Model) for cultural heritage and VRA Core (Visual Resources Association) for visual materials.

- *standards for data contents*: these govern the content rules, specifying how the metadata content is to be created. Examples are the bibliographic standards, AACR (Anglo American Cataloguing Rules) and its successor RDA (Resource Description and Access), the archive standards ISAD(G) (International Standard for Archival Description (General) and DACS (Describing Archives: a Content Standard) and CCO (Cataloging Cultural Objects) for galleries and museums.
- *standards for data exchange*: these govern the metadata syntax. Examples are XML, which is widely used to express element sets and create metadata schemas with a *namespace* denoting the location of components and definitions (Zeng and Qin, 2016, chapter 5), and the Resource Description Framework (RDF), increasingly used to provide metadata in linked data form. RDF is a general metadata model, based on the description of resources as a series of subject-relation-object expressions termed *triples*. RDF and XML are used to build more specific metadata models, such as the Web Ontology Language (OWL), designed to represent ontologies, and the Simple Knowledge Organization System (SKOS), a model for representing controlled vocabularies such as classification schemes, taxonomies, subject heading lists and thesauri (van Hooland and Verborgh, 2017). As an example of a metadata standard encoded in SKOS, see the treatment of the Integrative Levels Classification by Binding et al. (2020).

- *standards for data values*: these are the KOS used to ensure consistency in metadata creation. Examples are classification schemes, thesauri and authority lists for personal names.

There are also:

- *standards for structural data*: ‘packaging standards’, defining how digital documents are created from smaller units. An example is the Metadata Encoding and Transmission Standard (METS), originally developed by the Library of Congress for archiving digital items.
- *standards for preservation metadata*: an example being PREMIS (Preservation Metadata: Implementation Strategies).
- *standards for document markup*: these provide a consistent way of labelling parts of a document. Examples are XML, and Text Encoding Initiative (TEI), a specialised mark-up and metadata standard for digital humanities.

Resource description

This is the provision of descriptive metadata for documents, following content rules. It has developed from the *cataloguing* of bibliographic and library materials, and that term is still used for the creation of bibliographic metadata; resource description is more general. In practice, resource description for a document is usually taken to have four stages: describing the document’s form; describing its subject; checking authority files for correct forms of proper names, etc. (Wiederhold and Reeve, 2021); and encoding the record in an appropriate format. For detailed accounts, see Welsh

and Batley (2012), Joudrey, Taylor and Miller (2015) and Dobreski (2021).

Charles Ammi Cutter (1837–1903), an American librarian who was one of the 19th-century originators of modern ideas of resource description, argued that a catalogue should fulfil the following functions (phrased in modern idiom):

- to allow a user to *find* a resource, for which they know one or more of author, title or subject
- to *show* what resources are available written by particular authors or on specified subjects
- to *help* the user choose the best resource for their needs, by *edition* (date, publisher, etc.) and by *character* (style, level, etc.).

These are still relevant aims. The Library Reference Model (LRM) of the International Federation of Library Associations (IFLA) (Riva, LeBoeuf and Žumer, 2017) ([Figure 8.1](#) on the next page) states rather similar ideas as a series of five tasks which resource description should allow the user to achieve:

- *find*: to bring together information about one or more resources of interest by searching on any relevant criteria
- *identify*: to clearly understand the nature of the resources found and to distinguish between similar resources
- *select*: to determine the suitability of the resource found, and to be enabled to either accept or reject specific resources
- *obtain*: to access the content of the resource

- *explore*: to discover resources using the relationships between them and thus place the resources in context.

Although there is evidence of some form of bibliographic description in the libraries and archives of the ancient world, with continuing development since then, it is from the 19th century that resource description as currently understood has developed (Dobreski, 2021; Miksa, 2021). The aim has been to have consistent rules, applied as widely as possible and based on principles, rather than tradition and pragmatism. The first major step was taken by Sir Anthony Pannizzi (1797–1879) with his *Rules for the Compilation of the Catalogue* (1841), giving 91 rules for the cataloguing of books in the British Museum library. This approach was developed in the USA by Charles Jewett (1816–68) with *On the Construction of Catalogues of Libraries* (1852) and Cutter's *Rules for a Dictionary Catalogue* (1876). The desire for international standards saw the *Prussian Instructions* developed for German-speaking countries in the 1890s and then the first Anglo-American Cataloguing Rules (AACR) in 1908. The next major step was the development of the Paris Principles (1961), a set of guiding principles to help make national and regional codes consistent. British and US practices, which had diverged over the years, were reconciled in the 1967 revision of the AACR.

The desire for a more rational set of principles led to the International Standard Bibliographic Description (ISBD) (1969), which prescribed at a general level what was needed when describing a bibliographic item. This required the following elements to be specified: title and statement of responsibility ['author']; edition; material, or type of publication ['physical form'];

publication, distribution, etc. ['publisher']; physical description [size, etc.]; series; note; standard number; and terms of availability.

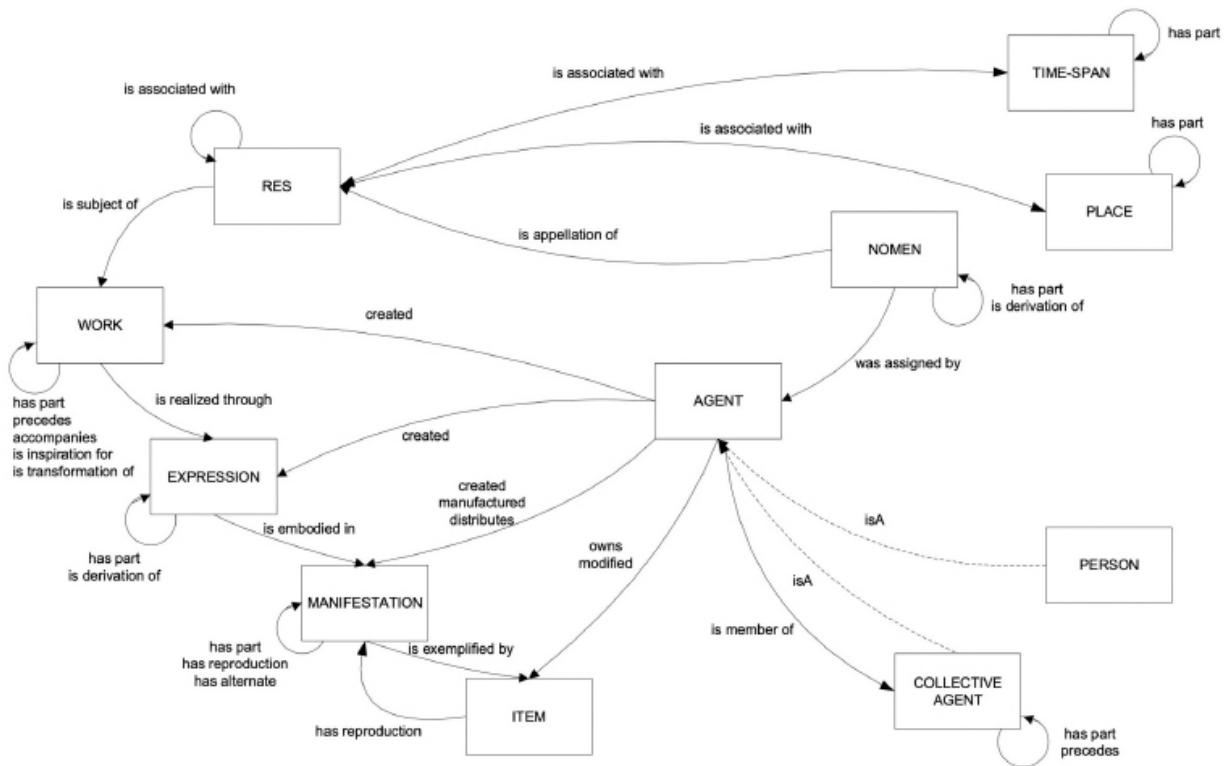


Figure 8.1 IFLA LRM: overview of relationships (reproduced from Riva, LeBoeuf and Žumer, 2017)

The *standard number* required by ISBD is a unique identifier for a published work. Examples are the International Standard Book Number (ISBN) for a single work and International Standard Serial Number (ISSN) for periodicals, dating from the 1960s and 1970s respectively, and Digital Object Identifier (DOI) for digital objects of greater granularity, such as journal articles, introduced in 2000.

A revision of the AACR in 1978 incorporated these principles, but at the cost of considerable detail for each type of document, with separate chapters for books, maps, music scores, etc. The perceived limitations of such codes led to the recognition of the advantage of a

formal data model for bibliographic description, avoiding the complexity of separate rules for describing different kinds of material, and also minimising ad hoc rules. The first such model, the Functional Requirements for Bibliographic Records (FRBR) was introduced in 1998. FRBR is based on an entity-relation model, instantiating the relations between different kinds of documents and their attributes, and the people and organisations which create and disseminate them. It was joined by two related models, the Functional Requirements for Authority Data (FRAD), which provides standard forms for the names of people, organisations, works, etc., and the Functional Requirements for Subject Authority Data (FRSAD) for subject description. Subsequently, the FRBR family of models were subsumed into IFLA's Library Reference Model (LRM) (Riva, LeBoeuf and Žumer, 2017; Žumer, 2018). This is regarded as a complete model of the bibliographic universe, able to deal with materials in any collection institution, not just libraries, and also compatible with all web materials. As noted in Chapter 6, FRBR and LRM are based on the concepts of work, expression, manifestation and item, whose definition is not fully defined or agreed. Nonetheless, there is some evidence that this is a natural way of looking at bibliographic entities (Pisanski and Žumer, 2010).

These models are the basis of the latest standard for metadata contents, Resource Description and Access (RDA) (Brenndorfer, 2016; Oliver, 2021). The original version, introduced in 2013, is based on FRBR; later versions are based on LRM. Library cataloguing has typically allowed searching at the 'manifestation' level, and display of results at the 'item' level. Systems based on the LRM should allow

for more flexible search and display, to allow for greater precision; e.g. 'retrieve records for this edition of this book, where there is a copy available for loan in a library to which I have access'. It should also allow broader searching, retrieving, for example, everything on Shakespeare's *The Tempest*: the play itself, in its various editions, novels based on it, translations, commentaries, audiobook versions and film versions, including related films, such as *Forbidden Planet* and *Prospero's Books*; the 'superwork' mentioned in Chapter 6.

The discussion above has referred mainly to the library context, in which resource description has been most complex and has received most attention. The archive sector, for example, has taken a different and somewhat more straightforward approach, through standards such as ISAD(G) and DACS (Cook, 1999, chapter 6; Meehan, 2014; Millar, 2017, chapter 11). General web resources have, of necessity because of their number, generally had only basic descriptions, as for instance in the original Dublin Core. The hope is that universally applicable models, such as LRM, if widely accepted, may allow a consistent level of description across all document forms, and strike a balance between convenience and precision.

We now turn to forms of KOS for subject description, beginning with classification and following with taxonomies and ontologies, and then alphabetical vocabularies.

Classification and categorisation

Classifications and categorisations (and also taxonomies and ontologies, considered in the next section) are forms of IO which aim to show the relationships between concepts, generally although not always hierarchical relationships, by bringing together terms

representing concepts' similar meanings. They are therefore referred to as *concept-based* or *systematic* KOSs: constructed of concepts, according to a 'system'. Classification is often thought of as a system for arranging collections of printed materials by subject, but it is also useful for visualising and browsing digital materials. In all contexts, a classification can offer a map of a subject or collection, showing the interrelation of all its parts. Dealing with concepts and classes as opposed to terms and phrases, classifications show the relations between concepts more clearly and explicitly than alphabetic vocabularies, to which they are complementary. There is a considerable body of classification theory, and principles of classification also provide the rationale for the more complex alphabetic vocabularies, such as thesauri. We will first consider general principles of classification and then look at examples of some important schemes.

Dividing the world into distinct aspects appears to be a deep-rooted human impulse. See Bowker and Star (2000) and Hjørland (2017) for wide-ranging discussions of classification in general, including but not limited to IO schemes.

Much classification theory comes not from information science but from other disciplines, especially from philosophy and from the taxonomic sciences, biology in particular. The information sciences have been relatively late on the scene, with all the main classification schemes in current use being developed in the late 19th and early 20th centuries, and theoretical examination of classification for IO purposes beginning only in the 20th century. The philosophical roots of classification lie in Aristotle's hierarchical division of the world

according to the attributes of its entities, with non-overlapping categories created by applying one criterion of division at a time, and alternatively on Wittgenstein's ideas of overlapping categories based on 'family resemblances' with multiple similarities. The two approaches, which are as applicable to scientific taxonomy as to classification of knowledge and documents, are termed *monothetic* and *polythetic*, respectively (Leazer, Montoya and Furner, 2020).

Because classifications, in the IO sense, are created on the basis of concepts and their meaning, they are necessarily subjective, since the many ways concepts can be categorised and ordered all have validity. There is no claim, as there is with some ontologies and scientific taxonomies, that they represent a truth about the world. It follows that there cannot be such a thing as a 'correct' classification; though this does not mean that classifications cannot be judged as being more or less consistent, complete or useful. George Perec (1936–82), the French novelist and essayist, noted a list of ways in which a library might be arranged: alphabetically; by continent or country; by colour; by date of purchase; by date of publication; by format; by genre; by literary period; by language; according to our reading priorities; according to the binding of the materials; and by series. His gloomy conclusion was that, although all are valid to an extent, none is 'satisfactory in itself' (Perec, 2020, 60–9). The same, alas, is true of any classification; none can be perfect for all purposes. (We thank Vanda Broughton for reminding us of Perec's ideas.)

An issue faced by all IO classification is that subjects, and documents, rarely deal with just one concept but, rather, with multiple concepts in combination, so that we cannot say that a

document deals with just one topic and hence has one unique place in a classification scheme. For example, should a document dealing with *scientific research in Victorian London* be classified with materials on science, on the Victorian period, or on London? There is no evident solution; the best that can be achieved is to specify a consistent ordering of the aspects, so-called *citation order*, and ideally to have a system allowing access by any aspect.

Types of classification

A number of different types of classification in the IO sense have been distinguished (Broughton, 2015, chapter 6; Joudrey, Taylor and Miller, 2015, part IV; Gnoli, 2020a). One distinction is between *general* classifications, covering all of knowledge, such as the major bibliographic classifications, including Dewey, Universal Decimal Classification (UDC) and Library of Congress, and *special* classifications, for restricted subject areas. Examples of the latter are mathematics (the Mathematics Subject Classification), law (Moys classification), medicine (the National Library of Medicine classification, and the Wessex scheme, a UK version). Two other major distinctions are between aspect and entity classification, and between enumerative, analytico-synthetic and faceted classification.

Aspect classification

An aspect or disciplinary classification is organised by disciplines or areas of study. All major bibliographic classifications are of this nature; the Dewey Decimal Classification, for example, with its ten main classes for social sciences, science, technology, arts and recreation and so on. This is convenient for finding materials of the general nature and perspective required, but scatters any particular

concept or entity. In a general classification of this kind, for example, the concept of 'horse' will appear in multiple places: zoology; veterinary medicine; use of horses in agriculture, in transportation, and in warfare; cookery, for those cultures who eat horsemeat; horseracing and equine sports; keeping and care of horses; laws relating to horses; horses in art and literature; etc. They will have to be identified from an alphabetical index to the classification, or from scanning the schedules.

Entity classification

An entity or phenomenon classification keeps entities together, but scatters the contexts in which they are found. Everything to do with horses will be in one place, but there is no place for the subject of zoology, which is scattered over all the animals mentioned in the classification. There are few examples of such classifications, and they have been developed largely for research into classification issues. An early example was Brown's Subject Classification; current examples under development are the Integrative Levels Classification (ILO) and the Basic Concepts Classification (BCC).

Enumerative classification

Enumerative classifications are designed to list completely, to enumerate, all aspects of knowledge within their scope; they are invariably hierarchically arranged, dividing and subdividing knowledge. The Dewey Decimal Classification, for example, is divided into ten main classes, each class into ten divisions, and each division into ten sections, with further subdivision as needed, as illustrated in [Figures 8.2, 8.3 and 8.4](#) Classifications of this kind are

well suited for arranging large volumes of material, especially when a physical arrangement, with a place for each item, is required, and hence are widely used for library classification. Many of their classes are pre-combined linkages of topics, but they have no flexibility in creating new topics. Since they are static, they are problematic for rapidly changing subject fields, since any new topic must be included by a revision of the scheme.

0	Generalities
1	Philosophy, including psychology
2	Religion
3	Social sciences
4	Language
5	Mathematics and natural sciences
6	Applied sciences, technology
7	The arts
8	Literature
9	Geography, history, and auxiliary subjects

Figure 8.2 *Main classes in the Dewey Decimal Classification* (reproduced from Broughton, 2015)

700	The arts
710	Civic and landscape arts
720	Architecture
730	Sculpture and the plastic arts
740	Drawing and decorative arts
750	Painting
760	Graphic arts
770	Photography
780	Music
790	Recreation, sport and performing arts

Figure 8.3 *Dewey first hundred subdivisions – the 700s* (reproduced from Broughton, 2015)

780	Music
781	General principles
782	Vocal music
783	Music for single voices
784	Instruments and instrumental ensembles
785	Ensembles with one instrument per part
786	Keyboard and other instruments
787	Stringed instruments
788	Wind instruments
789	(Optional number)

Figure 8.4 *Dewey first thousand subdivision – the 780s* (reproduced from Broughton, 2015)

Faceted classifications

Faceted and analytico-synthetic classifications are designed to classify complex material, at a high level of subject specification, and to be flexible in dealing with new topics. An analytico-synthetic classification allows notations to be combined to create new topics and typically has sets of commonly combined concepts such as place and time. However, such classifications also typically have many pre-combined complex classes. Several classifications originally created as purely enumerative are now of this form, with a synthetic, *number-building* capability for complex topics. The UDC, for example, is largely of this nature, Dewey to a more limited extent. An example of the complexity of number building possible is shown by the UDC example in [Figure 8.5](#).

*A conference on wading birds and their brackish water
habitats held in London in 1999*

waders	598.24
habitat	574.2
brackish water (289)	
conference	(063)
London	(410.111)
1999	"1999"

598.24:574.2(289)(063)(410.111)"1999"

Figure 8.5 *Number building in UDC* (Vanda Broughton)

Fully faceted classifications take this principle of combination more radically. Terminology is grouped into related concepts by facet analysis, from which the classification for any item can be

constructed, so that codes can be created for new topics by combination, with a minimum of predefined combinations. Faceted classification, along with facet analysis, was devised by Ranganathan in his Colon Classification, named for the punctuation mark which characterises its notation. Many faceted classification schemes in specific subject areas, particularly in science, technology and social sciences, were created during the 1950s and 1960s for classifying specialised or technical material, particularly in systems using techniques of mechanised documentation and, later, early applications of computers. The concept was researched and promoted, in particular, by the UK-based Classification Research Group from 1948 (McIlwaine and Broughton, 2000; Broughton, 2017a; Miksa, 2017). Faceted classification lost popularity, however, because of its perceived complexity and difficulty of use, and fully faceted schemes are little used in practice, although schemes such as the Integrative Levels Classification and the second edition of the Bliss Classification continue to be developed. For an assessment of a variety of issues of faceted classification in a digital environment see the chapters in Slavic and Gnoli (2017), in particular Broughton (2017a).

General schemes of classification

Space does not permit a detailed account of any schemes, but we will briefly mention some of the more significant general schemes, to illustrate the breadth of classification types, with a note of where to find further information. Other schemes are described by articles in the *ISKO Encyclopedia of Knowledge Organization*, including national

and regional schemes from China, Japan, Korea, and Russia, and special schemes for various subjects.

- The Dewey Decimal Classification (Broughton, 2015, chapter 17) is the most widely used classification, applied in hundreds of thousands of libraries worldwide, as well as in digital collections. Devised by the American librarian Melville Dewey and first published in 1876, it is now in its 23rd edition. Originally a purely enumerative scheme, with its well-known ten major disciplinary classes, divided and subdivided by ten to allow its convenient decimal notation, its structure reflects the 19th-century Western world and the 'liberal arts' setting in which it was created, and its theoretical basis is limited. It is an example of an IO tool whose success is due to its reach, convenience and good support from its parent organisation.
- The Universal Decimal Classification (Broughton, 2015, chapters 18 and 19) was created in 1905 by the documentalists Paul Otlet and Henri La Fontaine, as an extension of Dewey's scheme, to allow retrieval of documents rather than just shelf arrangement of books. The first scheme with an analytico-synthetic nature, it was more flexible in dealing with detailed and complex topics than previous schemes. Subsequent editions have introduced more faceted elements (Slavic and Davis, 2017).
- The Library of Congress Classification (Chan, Intner and Weihs, 2016) was devised by that library, and begun in 1901. It remains the purest example of a large enumerative scheme which has limited synthetic capabilities and provides very detailed,

sometimes idiosyncratic, subject listings, based upon enumeration within 21 major disciplinary classes.

- The Colon Classification (Satija, 2017) is a fully faceted disciplinary classification, the first of its kind, published by Ranganathan in 1933, developed through two major revisions and numerous editions. Organised by the PMEST facet analysis principle, it is intended for use with any kinds of documents in a library context. It has had little application in practice, even in the Indian subcontinent, but has been very influential on the development of later schemes.
- The first edition of the Bliss Bibliographic Classification was published in the 1930s by Henry Evelyn Bliss (1870–1955), one of the early theorists of IO. An enumerative disciplinary scheme, though with a synthetic number-building capability, it was regarded as the most scholarly and philosophically sound of the enumerative schemes. A second edition of this scheme, still under development, is fully faceted and is regarded as particularly theoretically sound (Broughton, 2018).
- The Brown Subject Classification (Beghtol, 2004), developed by British librarian James Duff Brown (1862–1914) and published in 1906, is an early example of a phenomenon classification, with an analytico-synthetic nature. Although no longer used in practice, it has a place in classification history and has been an inspiration for current developments.
- The Integrative Levels Classification (Gnoli, 2020b) is a fully faceted phenomenon classification based on the idea of levels of reality in general systems theories and applicable to any kind of

document. It is being developed primarily to instantiate ideas of classification theory, though it has been applied to a number of small bibliographic databases.

- The Basic Concept Classification (Szostak, 2020) is a fully faceted phenomenon classification under development, designed to connect users to any document, object or idea and to provide a general classification of particular value for interdisciplinary issues. It is constructed so far as possible from *basic concepts*, i.e. those concepts which are so fundamental that they are understood in the same way in all contexts and societies.

So far, we have described formal classifications, published and intended to be widely applied. There are, of course, very many simpler classifications and categorisations in use in contexts including personal collections, bookshops, small libraries and leisure and fan communities; some of these are gaining wide visibility through their use on social media.

Taxonomy and ontology

Taxonomy

The word *taxonomy* comes from the Greek *taxis*, meaning an order or arrangement. It has most commonly been used to mean a scientific classification of some aspect of the natural world, the best-known being biological ‘tree of life’ taxonomies. These classifications are purely hierarchical and, like the ontologies mentioned later, are intended to be realistic, to represent an aspect of the world as it really is. When such taxonomies are amended it is supposed to be done on the basis of objective facts, not subjective opinion or

convenience. They are usually associated with a systematic nomenclature for naming the entities in the taxonomy.

In the information sciences a taxonomy is regarded as a classification devised for a particular environment or a particular set of information; it does not necessarily cover a subject area comprehensively, as a library classification would. Taxonomies usually reflect the local conditions closely, are adapted to the local 'culture' and may be modified and extended more frequently than other information organisation tools. See Hedden (2016) for a thorough discussion of this kind of taxonomy. Taxonomies are usually intellectually constructed, but may be automatically generated and maintained; for a review of methods for automatic taxonomy generation, see Irfan et al. (2019). They are a popular tool for organising digital information resources and supporting browsing and searching, and may be linked to a thesaurus as a way of viewing the taxonomy terms alphabetically.

Taxonomies may differ from the norms of classification, for example by allowing the same concept to appear more than once, at different levels, if this is helpful to the user's browsing. They may be a high-level description of subject matter; for example, a 'corporate taxonomy' is a way of expressing the interests of an organisation so as to, for example, organise in-house data. Alternatively, taxonomies may include many specific examples – names of places, people or departments, for example, as well as general concept headings – and may have more similarity with a thesaurus than a library classification. They may also include much descriptive information about the concepts and items, acting as a kind of glossary.

Taxonomies are typically used to give access to diverse forms of materials – databases, documents, e-mails, people – and hence can support knowledge-management programmes.

Ontology

Ontology, coming from the Greek *ontos*, ‘that which exists’, has been used and still is used within philosophy to mean the study of what kinds of things can exist and how they can be described. It is sometimes used to describe, informally, a general set of things of concern; the ontology of an area in this sense would be the major concepts within it.

In the information sciences the term is now used to mean a formal description of a domain of knowledge, in terms of the entities within it and their relationships, which may be processed directly by computer for deduction, question answering, etc. For the changing meaning of the word, see Almeida (2013) and Machado, Almeida and Souza (2020); for ontology use in the biological sciences see Hoehndorf, Schofield and Gkoutos (2015) and Zheng et al. (2018); and for discussion of ontologies in the LIS context, see Stuart (2016) and Biagetti (2020).

Ontologies are widely used in the natural sciences, generally for organising and codifying the knowledge base of a subject, for standardising term usage and for aiding interoperability between systems and databases, and have sometimes been used directly for indexing and retrieval; see, for example, the evaluation of the Gene Ontology by Mayor and Robinson (2014). Such ontologies:

- are intended to represent reality directly, to represent the truth about the items in the ontology, and so to show one correct classification of entities: so-called *ontological reality*
- include whatever kind of inter-entity relations are necessary to represent this truth; these are often specific and detailed relations, going far beyond the hierarchy and association relations of LIS vocabularies. They may also include components such as axioms and rules, to assist machine processing
- are designed to be represented in formal languages, such as OWL (Web Ontology Language – an extension of RDF) for processing by computer (Antoniou et al., 2012).

It is often suggested that LIS vocabularies, such as classifications, taxonomies, and thesauri, are simple ontologies, limited in the range of inter-entity relations allowed, while others argue that ontologies should be regarded as distinct from other forms of vocabulary, due to the way that entities are defined and relations created; see, for example, Kless et al. (2015) and Iliadis (2019). One evident difference is that LIS vocabularies acknowledge subjectivity and pragmatism in their creation, as distinct from the claim of ontological realism. LIS vocabularies have generally relied on human intervention for their creation and maintenance, while it is common for automated tools to be used in the creation, expansion and quality control of ontologies (Agrawal and Cui, 2020).

Ontologies are of importance in LIS for three reasons:

- they are being increasingly used for information organisation and retrieval, as an alternative to a taxonomy or thesaurus (see, for

example, Gödert, 2016 and Padmavathi, 2020)

- there is an increasing overlap between the nature and use of ontologies and other vocabularies, particularly when taxonomies and thesauri are commonly represented on ontology languages such as OWL
- LIS input to ontology creation and maintenance can be valuable, particularly since the design of an ontology may encounter issues long familiar within LIS (Mayor and Robinson, 2014).

Alphabetic vocabularies

There are several kinds of alphabetic controlled vocabularies in which the terms – words or phrases – are arranged in alphabetical order. There are no well-defined distinctions between them; in general, the distinction lies in how much information is provided about each term, including term definitions and interrelations.

Keyword lists are the simplest form of alphabetic controlled vocabulary. Often they consist of nothing more than a list of ‘approved terms’; sometimes they include synonyms. They are a simple and cheap form of terminology to develop and use, but are very limited in their usefulness and applicable only to small files and simple searches.

Subject headings are lists of terms – often quite lengthy to represent complex concepts – which are used for indexing for retrieval, and sometimes for browsing. They will generally include synonyms, and sometimes hierarchical and ‘SEE ALSO’ relations. The Library of Congress Subject Headings (LCSH) is the most widely known and widely used terminology of this sort (Broughton, 2017b). A very

large vocabulary, first published in 1914, its 42nd edition was published in 2020. It has over 250,000 terms, covering all subject areas, with approximately 4,000 added each year. USE/USE FOR, BROADER TERM/NARROWER TERM and RELATED TERM relations are included. The large number of terms reflects the need for this kind of vocabulary to provide detailed description for precision in retrieval. This is partly achieved by including many proper names of people, places, living things, vehicles, diseases, etc., partly by creating pre-coordinated terms including more than one concept and partly by subdividing headings by topical subdivisions (aspects of a topic, listed for each heading), by geographical subdivisions (where it is allowed for a heading) and by free-floating subdivisions, including subjects, form and time period.

Thesauri in the information sciences should not be confused with reference works such as *Roget's Thesaurus*. Both are concerned with precision in terminology. Roget-style thesauri provide alternatives for a given term, since, as Roget himself put it, 'specious phraseology' spreads 'prejudice and error' (Kendall, 2008, 8). Retrieval thesauri generally control the vocabulary used in indexing and/or retrieval. Such thesauri are defined by international standards (ISO, 2011; ISO, 2013), although not all vocabularies called thesauri observe their prescriptions. For detailed coverage of thesauri, albeit dated in detail, see Aitchison, Gilchrist and Bawden (2000) and Broughton (2006); for a more recent review see Dextre Clarke (2019); and for commentary on their continuing relevance see Shiri (2012), Hjørland (2016b), Bawden (2016) and Dextre Clarke (2016).

The ISO 25964 standard defines a thesaurus as a 'controlled and structured vocabulary in which concepts are represented by terms, organised so that relationships between the concepts are made explicit, and preferred terms are accompanied by lead-in entries for synonyms or quasi-synonyms'. The standard recommends, for consistency, that thesaurus terms should be nouns or noun phrases, but other parts of speech can be found in many thesauri. The usual set of thesaurus inter-term relationships is equivalence, hierarchy and associative.

The *equivalence* relation (USE/USE FOR) denotes terms which are *synonyms* or *quasi-synonyms*. The allowance for *quasi-synonyms*, terms which do not mean exactly the same but which are sufficiently similar to be regarded as synonymous for the purposes of the thesaurus, shows that the guiding principle for this kind of vocabulary is pragmatic convenience, rather than ontological realism. The requirement for selection of one term as *preferred*, rather than just collecting synonyms, reflects the origin of the thesaurus in the age of printed indexes, when it was necessary to have a single entry point for a concept: there can be only one preferred term for a concept, but any number of non-preferred terms or none. Preferred terms have a full thesaurus entry; non-preferred terms usually just a single-line USE relation, pointing to the preferred term.

The *hierarchy* relation (BROADER TERM/NARROWER TERM) is used when one term falls entirely within the scope of the other. It establishes the structure of a thesaurus and allows it to be inter-converted with a taxonomy. The standard recognises three forms of hierarchy: *generic* (e.g. METALS – IRON), *part-whole* (e.g. HANDS –

FINGERS) and *instance* (e.g. OCEANS – ATLANTIC). A term generally has one broader term (none for terms at the top of the hierarchy); some thesauri are *multi-hierarchy* and allow more than one broader term. A term may have any number of narrower terms; none for terms at the bottom of the hierarchy. The relation is symmetrical (except in some special circumstances, such as some multilingual thesauri): if A is the broader term of B, then B must necessarily be a narrower term of A.

The *associative* relation (RELATED TERM, or SEE ALSO) denotes relations which are neither equivalence nor hierarchy, but point to additional terms which might be of use to indexers or searchers. The standard gives examples such as terms with overlapping meaning (e.g. SHIPS – BOATS), activities and places where they are carried out (e.g. NURSING – HOSPITALS) and disciplines and their object of study (e.g. ASTRONOMY – STARS). Inclusion of this relation is typically pragmatic, according to perceived user needs. The relation is not necessarily symmetrical: if A is a related term to B, then B may or may not be a related term of A.

Thesauri in areas with a strong hierarchical structure, typically in science and medicine, are likely to have fewer associative terms than those in areas of humanities, social subjects and everyday life.

A further relation used in some thesauri is 'Top Term' or 'Heading Parent', identifying the hierarchy in which the term occurs. A few thesauri have used other forms of inter-term relation but, in general, equivalence, hierarchy and associative are regarded as sufficient for retrieval.

ISO25964 also recommends *scope notes*, giving definitions or explanations of terms, and/or prescribing their use in indexing and retrieval, although these need not be present for all terms.

An example of a thesaurus term showing all these relations is shown in [Figure 8.6](#) on the next page.

Thesauri were originally designed to be used by both indexer and searcher, to ensure that both were using the same term for the same concept. In the digital environment, thesauri may still be used in this way, or may be used only for indexing or only for searching by, for example, assisting automatic indexing; automatically expanding searches by adding additional terms; and suggesting alternative terms to a searcher. See Shiri (2012) and Dextre Clarke (2019) for examples.

SEAL MATRIX	
UF	Amulet Seal
SN	The object used to make impressions in wax as seals.
CL	TOOLS AND EQUIPMENT
BT	MARKING EQUIPMENT
NT	INTAGLIO
	OCULISTS STAMP
	POTTERS STAMP
	SCARAB SEAL
RT	SEALING WAX

Figure 8.6 Example of a thesaurus term, from the Archaeological Object

Type thesaurus (www.heritage-standards.org.uk/wp-content/uploads/2021/02/ArchObj_alpha.pdf)

Creating subject metadata

Adding subject information to metadata records involves classifying or indexing, generally using the one or more of the KOS noted above. It may also including abstracting or summarising.

Classifying implies the addition of one or more classification codes from whatever classification scheme is in use. The ethos of most classification schemes is that one unique code should be assigned, although in digital environments there is no reason why more than one code should not be added.

Indexing is usually taken to mean the assignment of a number of terms from an alphabetic vocabulary; if no controlled vocabulary is used, the process is called *free term indexing* or *tagging*. If the tagging is carried out by individual users, rather than indexers, who share metadata production, it may be known as *social tagging* and the resultant set of terms as a *folksonomy* (Rafferty, 2018; Price and Robinson, 2021). Indexing is usually assumed to be applicable to text documents, but it may be necessary to index other forms of material, such as images, datasets and maps. The process of indexing is governed by international standards (ISO, 1985; ISO, 1996). For overviews of the principles and practice of indexing, see Lancaster (2003), Hjørland (2018) and Duncan (2021).

The indexing of any particular document, according to the ISO 5963 standard is a three-stage process: understanding of the content – the indexer decides what the document is ‘about’; analysis of the content – the indexer decides which concepts are to be indexed, and to what depth; translation of concepts into indexing terms – the indexer chooses the most appropriate terms from the vocabulary being used. There are two main criteria for how indexing is carried out: *exhaustivity*, the extent to which all possible concepts are included; and *depth*, the degree of specificity with which concepts are described.

An abstract or summary of a document may also be a part of subject metadata, acting as a surrogate for the document for retrieval or display, and is a long-established part of IO (Borko and Bernier, 1975; Lancaster, 2003; Chowdhury, 2017, chapter 8). According to the relevant international standard (ISO, 1976), an abstract is ‘an abbreviated, accurate representation of the contents of a document, without added interpretation or criticism and without distinction as to who wrote the abstract’. Abstracting has a long history; summaries of articles have been used as means of keeping up with the scientific, medical and professional literature in particular, for many years. Borko and Bernier (1975) suggest that its origins can be traced back to classical times, with the first recognisable abstracting journals, and sections for abstracts in general journals, dating from the 17th century, their use expanding greatly in the 19th century, together with the creation of subject-specialist abstracting and indexing services.

Abstracts may vary in a number of respects, and so may be categorised differently. A fundamental difference is whether they are *informative*, intended to give sufficient information to be a replacement for the original, or *indicative*, giving just enough information for a reader to decide whether the item is of interest. They are increasingly likely to be *structured* (Jamar, Šauperl and Bawden, 2014; Hartley and Cabanac, 2017) or *graphical* (Stahl-Timmins, Black and Simpson, 2019).

Systems for automatic, or at least machine-aided, classification, categorisation and indexing are of increasing importance, with a variety of statistical and machine-learning approaches being applied

(Golub, 2019). Automatic abstracting has been a goal for many years, since the first research on the topic, carried out by H. P. Luhn in the late 1950s. Much effort has been expended since then on the design of systems, using a variety of approaches to automatic abstracting and summarising, including sentence extraction, text mining and machine learning (Chowdhury, 2017, chapter 8).

Summary

Intellectual tools for IO have changed surprisingly slowly; some of those used today, more than a century after they were first introduced, would be recognisable to Melville Dewey and to Anthony Panizzi. The dramatic changes in the technical means by which documents are created and disseminated has not been fully matched by the means by which they are managed. The influence of automated means for metadata creation, particularly those based on AI, is increasing, and may have dramatic effects in the coming years.

- IO remains at the heart of information science, its importance enhanced in new information environments.
- New forms of descriptive metadata have emerged to deal with new forms of document, while subject description remains relatively little altered.
- Long-established theories and concepts remain important, despite technical advances.
- The full effect of automated IO systems is yet to be realised.

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310-18.

9 Digital Technologies and Data Systems

While computing technologies are young by comparison with other efforts of human ingenuity, their origins go back to many early chapters in the human quest for the understanding and realization of mechanical aids to knowledge.

Giuseppi Primiero (2020, 7)

Computing is symbol processing. Any automaton capable of processing symbol structures is a computer . . . We may choose to call such symbol structures information, data, or knowledge, depending in the particular ‘culture’ within computer science to which we belong.

Subrata Dasgupta (2016, 121)

Data science is the process by which the power of data is realised, it is how we find actionable insights from among the swathes of data that are available.

David Stuart (2020, xvi)

Introduction

Technology, from the Greek *techné*, meaning art, skill or craft, is usually taken to mean the understanding of how to use tools, in the broadest sense of the word. The term *information technology* was first used in the 1950s to describe the application of mechanised documentation and then-new digital computers, and became widely used in the 1980s to describe the wider-spread use of digital technology (Zorkoczy, 1982).

Information technology is usually associated with computers and networks. But, in a wider sense stemming from the original meaning

of the word, the technologies of information include all the tools and machines which have been used to assist the creation and dissemination of information throughout history, as discussed in [Chapter 2](#); from ink and paper, through printing to microforms, document reproduction and photocopying, and mechanised documentation technologies such as card indexes, punched cards, edge-notched cards, and optical coincidence cards. Krajewski (2011) examines the idea of card index files as a ‘universal paper machine’, in a sense the forerunner of the computer, between the 16th and 20th centuries.

Our focus here is on digital technologies, echoing the view expressed by Gilster (1997) that all information today is digital, has been digital or may be digital. This chapter covers the basics of digital technologies and the handling of data; the following chapter deals with information systems.

Digital technologies

We will describe these aspects only in outline; see Ince (2011) and Dasgupta (2016) for more detailed but accessible accounts, and Primiero (2020) for a more advanced summary; for accounts of the historical development of the computer, see Ceruzzi (2012) and Haigh and Ceruzzi (2021).

Any digital device represents data in the form of *binary digits* or *bits*. Patterns of bits, *bit strings* with bits conventionally represented by 0 and 1, may represent data or instructions. A collection of eight bits is known as a *byte*. Quantities of data are represented as multiples of bytes, for example: a kilobyte, defined as 2^{10} bytes, i.e. 1024 bytes.

Any character or symbol may be represented by a bit string, but this requires an agreed coding. The most widely used code since the beginning of the computer age was the American National Standards Institute's ASCII (American Standard Code for Information Interchange) 7-bit code, but it is limited to the Latin alphabet, Arabic numerals and a few other symbols. It has been supplanted by Unicode, which can handle a wider variety of symbols and scripts. It can do this by using a 32-bit code; the more bits in the code, the more different symbols can be coded. Codes provide arbitrary representations of characters; for example, the ASCII code for the letter L is the bit string 1001100.

The idea that a general-purpose computing machine might be possible stems from the theoretical studies of the British mathematician Alan Turing (1912–54), whose concept of the *Turing machine*, developed in the 1930s, showed that a single machine limited to a few simple operations (read, write, delete, compare) could undertake any task if appropriately programmed.

The basic architecture for a practical digital computer was set out by John von Neumann (1903–57), the Hungarian-American mathematician, in the 1940s ([Figure 9.1](#)). His design, which gave the first formal description of a single-memory stored-program computer, is shown in [Figure 9.2](#).

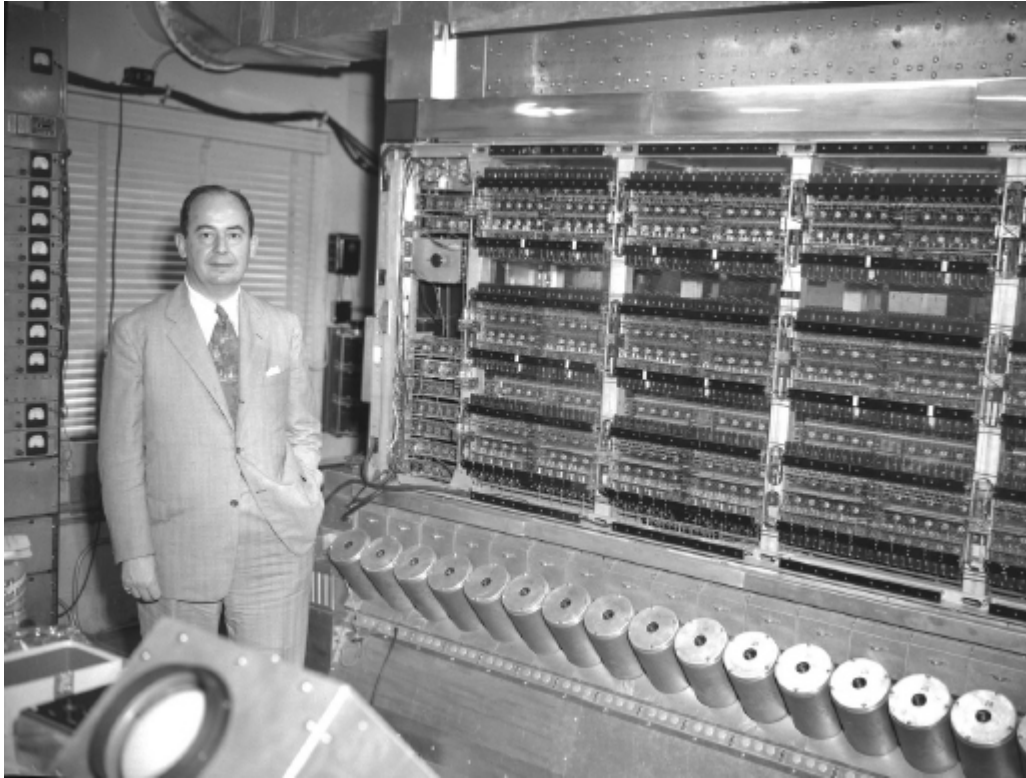


Figure 9.1 *John von Neumann* (Alan Richards photographer. From the Shelby White and Leon Levy Archives Center, Institute for Advanced Study, Princeton, NJ, USA)

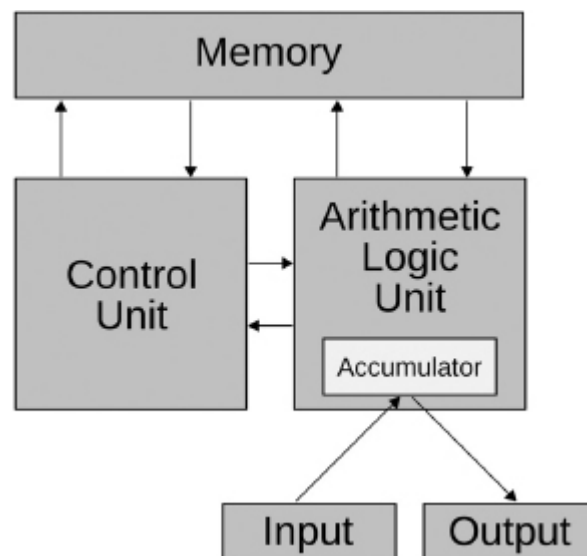


Figure 9.2 *Von Neumann computer architecture* (Wikimedia Commons, CC BY-SA)

This architecture is general purpose, in the sense that it can run a variety of programs. This distinguishes it from special-purpose digital computers which carry out only one task, as in digital cameras, kitchen appliances, cars, etc. A von Neumann machine loads and runs programs as necessary, to accomplish very different tasks.

The heart of the computer, the *processor*, often referred to as the *central processing unit* (CPU), carries out a set of very basic arithmetical and logical operations with instructions and data pulled in from the *memory*, also referred to as *main* or *working* memory. This sequence is referred to as the *fetch–execute cycle*. Two components of the processor are sometimes distinguished: an *arithmetic and logic unit*, which carries out the operations, and a *control unit*, which governs the operations of the cycle.

While programs and data are being used they are kept in the memory. While not being used they are stored long-term in *file storage*. Items in memory are accessible much more rapidly than items in file storage, but memory is more expensive, so computers have much more file storage than memory.

Data comes into the computer through its *input devices* and is sent into the outside world through the *output devices*. The components are linked together through circuits usually denoted as a *bus*, sometimes referred to more specifically as a *data bus* or *address bus*.

All of these components have undergone considerable change since the first computers were designed. Processor design has gone through three main technological stages. The *first generation* of computers used valves as processor components, and the *second*

generation used transistors. Computers of the *third generation*, including all present-day computers, use circuits on 'silicon chips', by very-large-scale integration (VLSI), which allows millions of components to be placed on a single computer chip, thus increasing processing speed.

The storage elements of the computer's memory comprise regular arrays of silicon-based units, each holding a bit of data, and created using the same VLSI methods as processors. Earlier generations of computers used *core storage*, with data held on tiny, magnetised elements arranged in a three-dimensional lattice.

File storage, holding data and programs that are not needed immediately, has always used magnetic media, which can hold large volumes of data cheaply, provided that quick access is not required. Earlier generations of computers used a variety of tapes, drums and disks; current computers use *hard disks*, rapidly spinning magnetisable disks with moveable arms with *read/write heads*, to read data from, or write data to, any area of the disk. Regardless of the technology, each area holds one bit of information, according to its magnetic state.

Input devices fall into three categories: those which take input interactively from the user; those which accept data from other digital sources; and those which convert paper data into digital form. The first category comprises the venerable 'QWERTY' keyboard, originally developed for typewriters in the 1870s, together with more recently developed devices: the mouse and other pointing devices and the touch-screen. The second category comprises the silicon-memory data stick, replacing the various forms of portable

magnetic *floppy disks* used previously, and the ports and circuits by which the computer communicates with networked resources. The third category is that of the scanner, digitising print and images from paper sources.

Output devices are similarly categorised; display screens, which allow user interaction through visual and sound output; devices which output digital data, such as data sticks, network circuits and ports; and devices which print paper output, typically laser or inkjet printers.

The most recent development is the trend towards mixed reality and immersive environments, which may require specific input/output devices (Greengard, 2019; Pangilinan, Lukos and Mohan, 2019). *Virtual reality* (VR) provides immersive experiences in a computer-generated world, often using head-mounted display devices, such as Oculus Rift. *Augmented reality* (AR) overlays computer-generated information, often visual but sometimes multisensory, on the real world. *Mixed reality* is the combination of real and virtual environments, a spectrum with AR at one end and VR at the other. These are of increasing relevance to the provision of information services (Robinson, 2015; Varnum, 2019; Dahya et al., 2021; Kroski, 2021).

Networks

Virtually all computers are connected via some form of network to others, to enable communication and information access and sharing. Since the 1990s the internet and the World Wide Web have become ubiquitous, such that it has become difficult to think about digital technologies without these at centre stage.

The growth of networked computing has been driven by three factors: communications technology, software and standards. In terms of technology, older forms of wired network, originating in the twisted copper-pair cables used for telegraph systems from the mid-19th century, have been succeeded by fibre-optic cables and various forms of wireless transmission. These allow much faster transmission speeds and greater information-carrying capacity; such systems are described loosely as *broadband*. Software systems have improved the efficiency and reliability of transmission greatly: an important example is *packet switching*, by which messages may be split up and their constituents sent by the fastest available route, being recombined before delivery.

Standards are fundamentally important, so that different forms of network, and different kinds of computers connected to them, can communicate effectively. The internet, originating in 1960s networks built for defence research in the USA, is a worldwide network of networks, integrated by common standards: the internet control protocols, commonly known as TCP/IP (Transmission Control Protocol/Internet Protocol).

The World Wide Web, often spoken of as if it were synonymous with the internet, is in fact a major internet application. A system for allowing access to interlinked hypertext documents stored on networked computers, it originated in the work of Sir Tim Berners-Lee at CERN, the European nuclear research establishment. Berners-Lee introduced the idea in 1989 in an internal memorandum with the modest title 'Information management: a proposal'.

The web is based on *client–server* architecture: a web browser, the client, on the user’s computer accesses the website on the remote server computer, through the internet, and downloads the required web page. This relies on a number of standards. For example, web pages must be created using a mark-up language, typically HTML, sometimes referred to as the *lingua franca* of the internet, and be identified by a Uniform Resource Identifier, commonly referred to as a Uniform Resource Locator or URL; a Hypertext Transfer Protocol (HTTP) enables communication between client and server.

The *Internet of Things* (Greengard, 2021) is a generic term for the aggregation of network-enabled ‘smart’ devices of all kinds, able to collect and share data.

The internet and web, despite their great reach and influence, are by no means the only significant computer networks. Many private, local and regional networks exist, generally using the standards of the internet and web for compatibility.

Software

Software, in its various forms, provides the instructions for digital computers to carry out their tasks and can be categorised into *systems software* (firmware and operating systems) and *application software*; for short overviews, see Ince (2011), White and Downs (2015) and Dasgupta (2016).

Firmware, such as BIOS (Basic Input Output System), EFI (Extensible Firmware Interface) and UEFI (Unified Extensible Firmware Interface), comes pre-installed on all computers and is

activated when the computer is powered up, to initialise the hardware and start the operating system.

The *operating system*, specific to a particular type of computer, controls the hardware, allocates system resources, such as memory, and runs applications. Users interact with the operating system to manage files, use input and output devices, and run applications. Examples of operating systems are Windows, Unix and several variants of Unix, including Linux and Mac OS.

Users interact with the operating system via a program termed a *shell* because it forms an outer layer around an operating system; examples of shells are Bash, zsh and Powershell. Most users will rely on a graphical user interface to the shell for convenience. An alternative form of interface uses a *command line*, with the user typing instructions (such interfaces are commonly referred to as shells, although strictly the shell is the program to which they are the interface). This type of interface offers a more direct control over computer operations than with a graphical interface, but requires the user to understand the set of commands and the syntax, and to have some understanding of the workings of the operating system.

Applications software, discrete programs for carrying out specific tasks, for a specific purpose, may either be installed on individual computers, as, for example, word processors, spreadsheets, and web browsers typically are, or may be accessed on the web. The latter is usual for search engines, library management systems, and social media platforms. From the users' perspectives, application programs cause the computer to carry out tasks defined at a fairly high level: search the bibliographic database for authors with this name;

calculate the mean of this column of figures; insert this image into this blog post; and so on.

All software must be written in a programming language, though this will be invisible to the user, who will have no reason to know what language any particular application is written in. At the risk of over-simplification, we can say there are four kinds of software language: low-level and high-level programming languages, scripting languages and mark-up languages.

Low-level programming languages, also referred to as *assembly* languages or *machine code*, encode instructions at the very detailed level of processor operations; these languages are therefore necessarily specific to a particular type of computer. This kind of programming is complex and difficult, but results in very efficient operation; it is reserved for situations where reliably fast processing is essential.

High-level programming languages express instructions in terms closer to user intentions, and are converted by other software systems, *compilers* and *interpreters* into processor instructions in machine code for a particular hardware. Programs are therefore much easier to write, understand and modify than those in machine code. Such languages can be used on different types of computer. There have been many such languages, some general purpose and some aimed at a particular kind of application. The first examples were developed in the 1950s, and two of the earliest are still in use: FORTRAN, still a language of choice for scientific and engineering applications, and COBOL, still used in many legacy systems for business applications. Currently popular languages are Java, C++, R

and Python. Compilers convert a whole program into machine code, whereas interpreters convert one line at time. Programming languages are described as either *compiled*, for example C++, or *interpreted*, for example Python.

Scripting languages are a particularly significant form of high-level language, designed to support interaction, particularly with web resources; for example, to update a web page in response to a user's input rather than reloading the page each time, or to create a *mashup*, an integration of data from several web sources. Examples are JavaScript, PHP and Perl.

Mark-up languages are somewhat different in nature, as they are designed to annotate text so as to denote either structural elements ('this is an author name'), presentation ('print this in italic') or both. The first such languages were designed for formatting documents for printing; an example is LaTeX. More recent examples control the format and structure of web resources; examples are HTML and XML.

Regardless of the language used, software is created in essentially the same way. The abstract process to be followed, the *algorithm*, is determined and its steps are converted into statements in an appropriate language; this is the process of *coding* or *programming*. For small-scale software development, perhaps for a piece of code to be used temporarily or just by one person, writing the code and testing that it works correctly is sufficient. For a major piece of software, other stages will include design, testing and maintenance; these stages, together with the coding stages, make up *system*

development or *software engineering*. For accessible introductions to these topics, see Louridas (2020) and Montfort (2021).

One influence on the way software is created is the availability of libraries of *subroutines*, packages of code in a particular language to undertake specific tasks which can be included in programs by a single call. If a *subroutine library* (also called a *code library*) is available for a particular topic, it means that a coder can create a program very quickly and economically by a series of calls to subroutines, without having to write the code in detail; see Wintjen (2020) for detailed examples. There are an increasing number of subroutine libraries available for topics of relevance to the information disciplines. For example, the Python language has libraries of subroutines for text processing, for extracting data from URLs, for extracting data from mark-up languages such as HTML and XML and for processing data from Wikipedia.

An innovation in the way software is provided has come from the *open source* movement. Commercial software is usually provided as a 'black box'; the user has no access to the program code and therefore cannot modify the system at all, nor even know exactly how it works. With open source software, the user is given the full source code – the original programs – which they are free to modify. This makes it easy to customise software to meet local needs and preferences. It also allows users to collaborate in extending and improving the systems, correcting errors, etc. Most such software is free, leading it to be known as FOSS (Free and Open Source Software). Well-known examples of full FOSS systems relevant to library and information management are the Koha and FOLIO-ERM

library management systems, the EPrints repository system, the Drupal content management system and the Open Journal Systems (OJS) e-journal publishing system; see, for example, Breeding (2017) and Choo and Pruett (2019). Many smaller-scale FOSS software products are available; the GitHub repository is the most commonly used means for making these available (Beer, 2017). Github is commonly integrated with the Jupyter notebook environment, which combines simple word processing with the ability to create and run programs; this integration allows for testing, sharing and distribution of open source software and is widely used for data-wrangling functionality, including library/information applications (Wintjen, 2020).

There are a number of reasons why information professionals may wish to do their own coding, particularly now that relevant code libraries make it so much quicker and easier to do so. The main reasons would be to customise a procedure to a specific situation, to merge data from different sources, to create tidy, consistent data or to link systems and data together in a particular context.

Artificial intelligence

Artificial intelligence (AI) encompasses a complex array of concepts and processes, directed to making it possible for digital computers to do the kind of things that minds can do; see Boden (2016), Cantwell Smith (2019) and Mitchell (2019) for accessible overviews, Floridi (2019) for ideas of its future and Mitchell (2021) for caveats about how much we can expect from AI, especially with regard to endowing machines with common knowledge and common sense. It is 'a growing resource of interactive, autonomous, self-learning

agency, which enables computational artefacts to perform tasks that otherwise would require human intelligence to be carried out successfully' (Taddeo and Floridi, 2018, 751).

Although AI as such has been possible only since the middle of the 20th century, it has been preceded by several centuries of interest in mechanical automata, game-playing machines, calculating machines, automated reasoning and the like; see Pickover (2019) for an informal history. Although Ada Lovelace mused on a 'calculus of the nervous system', and the possibility of mechanical devices composing music, AI became a prospect only with the development of the digital computer.

Alan Turing's 1950 article 'Computing Machinery and Intelligence' was the first serious philosophical investigation of the question 'can machines think?' In order to avoid the problem that a machine might be doing what might be regarded as thinking but in a very different way from a human, Turing proposed that the question be replaced by the question as to whether machines could pass what is now known as the *Turing Test*. Turing proposed an 'Imitation Game', in which a judge would interact remotely with a human and a computer by typed questions and answers, and would decide which was the computer; if the judge could not reliably decide which was the computer, then the machine passed the test, as Turing believed that digital computers would be able to do. Turing's article also proposed for the first time how a machine might learn, and noted a series of objections to the idea of AI which are still raised today.

The term 'artificial intelligence' is attributed to John McCarthy (1927–2011) and appears in a proposal for a workshop at Dartmouth

College, in which ‘the artificial intelligence problem is taken to be that of making a machine behave in ways that would be called intelligent if a human were so behaving’ (McCarthy et al., 1955). The workshop, held in 1956, established the topic as a field for realistic academic study (Nilsson, 2012). A fundamental insight was provided by another founder of AI, Marvin Minsky (1927–2016), who argued that intelligence was not the result of a single sophisticated agent or principle but, rather, the result of the interaction of many simple processes (Minsky, 1987). This suggested that AI might be achieved by the combination of relatively simple programs.

The first approaches to practical AI were centred on attempting to capture in a *logic-based* program the knowledge and reasoning processes of a human expert: a doctor diagnosing disease, or a chemist interpreting analytical data or planning the synthesis of new compounds. This symbolic approach to AI, now sometimes referred to as Good Old Fashioned AI (GOFAI), had some successes, but proved to be limited by the difficulty of entering the large amounts of knowledge, particularly general ‘common sense’ knowledge, required for even seemingly simple tasks.

A newer generation, or ‘second wave’, of AI systems relies on programs which enable machines to learn for themselves, given very large datasets and massive computing power. (Big Data is not necessarily a *sine qua non* for AI; Floridi (2019) notes that in some cases small datasets of high quality may be more valuable.) For example, in 2011, IBM’s Watson AI program beat a human champion at the game Jeopardy, which requires general knowledge questions,

including puns, jokes and cultural references, to be solved quickly, without access to any external resources. Watson's success was due to massive computing power using probabilistic reasoning, plus extensive resources, such as the whole of Wikipedia and many other reference works, held in computer memory. In 2017 the program AlphaZero was given only the rules of chess as a starting point, and trained its neural networks by playing against itself many millions of times. Within hours it was at a level to compete with the best chess programs. This 'brute force' approach has led to greater success, but at the cost that it may be impossible to understand how and why the system is working; as Turing foresaw, machines may be 'thinking' but they do it differently. In particular, we should not attribute any element of 'understanding' to an AI because it can equal, or better, human performance at tasks which would require a human to understand a situation.

At the core of most modern AI systems is *machine learning*, a way of programming computers to learn from experience, using large sets of data for training (Alpaydin, 2016). This comes in three forms: *supervised* learning, in which the algorithms are given large sets of labelled examples; *unsupervised* learning, in which the algorithms discover patterns *de novo* in sets of unlabelled data; and *reinforcement* learning, in which the system receives feedback as to whether the change it made was good or bad. This approach encompasses the *neural net* data structure, a complex linkage of many simple processing units and *evolutionary* or *genetic* algorithms, which allows programs to develop themselves to improve their performance. So-called *deep learning* uses a neural net with many layers by which

machines train themselves to perform tasks (Kelleher, 2019). This can lead to very high performance, at the cost of creating models of such complexity that it may be difficult to understand why the artificial agent is making its judgements. All machine-learning systems face such problems of explanation, and also problems of algorithmic bias if the original data is incorrect or biased in any way.

Although somewhat displaced by machine learning and statistical algorithms, research continues on knowledge representation, using data structures such as *frames* or *semantic nets*, so as to allow artificial agents to simulate human reasoning, applying logical deduction, application of rules, or probabilistic reasoning. This is incorporated in *expert systems*, dealing with one specific task, typically of diagnosis, classification, recommendation or prediction, in scientific, medical, legal and financial domains. The two approaches, symbolic and statistical, are beginning to be used together in systems combining the power of both (Boden, 2016).

AI has raised ethical concerns, initially because of fears about ‘killer robots’ and ‘superhuman intelligences’, more recently and realistically because of dangers of algorithmic bias (Coeckelbergh, 2020); a full discussion is given in [Chapter 15](#).

Robotics, the design of machines to undertake tasks without direct human control, is linked with AI, since the machines must be programmed to undertake activities which would otherwise require the direction of an intelligent person. Indeed, a robot may be understood as an AI given autonomous agency, albeit usually of very limited scope, in the physical world. Agents on the internet which emulate a human user are termed *bots*, by analogy with robots

in the physical world; those which are able to simulate human conversation may be termed *chatbots*. The term ‘robot’ was first used in English in Karel Capek’s play *R. U. R. (Rossum’s Universal Robots)* of 1921, from Czech ‘robata’, meaning ‘forced labour’. The use of robots is well established in many areas, including libraries and archives, typically undertaking repetitive tasks; the number of robots in use depends on the definition used, but is certainly in the millions.

Application of AI in libraries, archives and similar institutions is, for the most part, at an initial exploratory stage, but the possibilities are becoming evident; see Griffey (2019) and Rolan et al. (2019). They include metadata creation, information extraction and text and data mining.

Interfaces, interaction and information architecture

The ways in which people interact with computers, generally described under the headings of *human–computer interaction* (HCI), *human–information interaction*, *interaction design* or *usability*, is an area of evident importance to the information sciences. It is particularly concerned with the design and evaluation of interfaces to information systems of all kinds (Sharp, Preece and Rogers, 2019).

Information architecture is the organising and structuring of complex digital information spaces, typically on websites. It draws from principles of information design on the printed page, of HCI and of categorisation and taxonomy. The term was coined in the 1970s by Saul Wurman, originally a ‘real’ architect, who came to believe that the principles behind the design of buildings and physical places could be applicable to information spaces. Information architecture came into its own from the late 1990s, with

the expansion of the web and the need to bring order and structure to web-based information. For overviews, see Rosenfeld, Morville and Arango (2015) and Arango (2018).

Data systems

The growth of data science, and the increased emphasis on managing and curating data, as distinct from information, has been a major influence on information science since 2010. For an accessible general introduction, see Kelleher and Tierney (2018), for an overview aimed at library/information applications, see Stuart (2020), and for a more detailed and practical treatment, see Shah (2020) and Wintjen (2020). For an analysis of the idea of data itself, see Hjørland (2020), and for examples of data practices in science, including the issues discussed later in this chapter, see Leonelli and Tempini (2020). We discuss this under two main headings: data wrangling, the processes of acquiring, cleaning and storing data; and techniques for finding meaning in data.

Data wrangling

The term *data wrangling* is sometimes used to refer to all activities in the handling of data. We will consider them, and the digital tools which support them, under four headings: collecting; storing; cleaning; and combining and reusing. A useful overview from an information science perspective is given in Stuart (2020, chapter 4).

Collecting

Collecting data is most simply achieved by typing it in from scratch. If it already exists in printed form, with a recognisable structure, then optical character recognition (OCR) may be used, though this is

rarely error-free, and entry must be followed by cleaning. If the data exists in some digital form, then it may be selected and copied; a tool such as OpenRefine may be useful in collecting and parsing data, and in converting between formats.

If the required data is to be found on one or more web pages, then a process of *web scraping* can be used to obtain it. This is the automatic extraction of information from websites, taking elements of unstructured website data and creating structured data from it. Web scraping can be done with code written for the purpose, or by using a browser extension such as Crawly or Parsehub. This is easier if an Applications Programming Interface (API) can be accessed. Described as ‘an interface for software rather than people’, an API in general is a part of a software system designed to be accessed and manipulated by other programs rather than by human users. Web APIs allow programs to extract structured data from a web page using a formal syntax for queries and a standard structure for output; see Lynch, Gibson and Han (2020) for a metadata aggregation example.

Storing

Data may be stored in a wide variety of formats and systems. These include database systems such as MySQL, Oracle and Microsoft Access, and data-analysis packages such as SAS or SPSS. Most common is the spreadsheet, such as Microsoft Excel and Google Sheets, widely used for data entry and storage, with some facilities for analysis and visualisation of data. All kinds of structured data and text are stored in a two-dimensional row–column arrangement of cells, each cell containing one data element. Consistency is of

great importance, in the codes and names used for variables, the format for dates, etc., and can be supported by a *data dictionary*, a separate file stating the name and type of each data element (Broman and Woo, 2018).

Exchange formats are relatively simple, used to move data between more complex formats or systems. One widely used example is the CSV (Comma Separated Values) format. It holds data in plain text form as a series of values separated by commas in a series of lines, the values and lines being equivalent to the cells and rows of a spreadsheet. JSON (Javascript Object Notation) is a more complex XML-like format used for the same purposes; it can represent diverse data structures and relationships, rather than CSV's two-dimensional flat files.

Data collections are typically bedevilled by factors such as missing data, duplicate data records, multiple values in a cell, meaningless values and inconsistent data presentation.

Cleaning

Data-cleaning processes are applied to create *tidy data* as opposed to *messy data*. For a data sheet, tidy data requires that: each item is in a row; each variable is in a column; each value has its own cell; there are no empty cells; and the values of all variables are consistent.

To achieve this, data may be scanned by eye, and errors corrected manually. An automated process is more efficient and less error prone. Examples are the use of Python scripts for identification of place names in collections of historical documents (Won, Murrieta-Flores and Martins, 2018) and for the normalisation of Latin place

names in a catalogue of rare books (Davis, 2020); see Walsh (2021) for a variety of examples. Data-handling packages, such SAS and SPSS, have their own data-cleaning functions.

One specific data-cleaning tool has been much used by library/information practitioners, as well as by scholars in the digital humanities. OpenRefine, a tool designed for improving messy data, provides a consistent approach to automatically identifying and correcting errors, grouping similar items so inconsistencies can be readily spotted by eye and identifying outliers, and hence possible errors, in numeric values. Examples of its use include the cleaning of metadata for museum collections (van Hooland et al., 2013) and extraction in a consistent form of proper names from descriptive metadata (van Hooland et al., 2015).

The most precise means of algorithmic data cleaning uses Regular Expressions (RegEx), a subset of a formal language for describing patterns in text, which ensures consistency. They were devised originally within theoretical computer science by the American mathematician Stephen Kleen in the 1950s, using the mathematics of set theory as a way of formally describing all the operations that can be performed by computers in handling text.

In practice, regular expressions are search strings with special characters to denote exactly what is wanted. Examples of these special characters are:

- . matches any character ('wildcard')
- [] matches any of the characters within the brackets
- ^ matches beginning of a line of text

These expressions are used primarily for searching, parsing, cleaning and processing text, and also for the formal modelling of documents and databases; see applications to inconsistencies in bibliographic records (Monaco, 2020), errors in DOIs (Xu et al., 2019), metadata aggregation (Lynch, Gibson and Han, 2020) and identification of locations in the text of journal articles (Karl, 2019).

Combining and reusing

Data in digital form offers the possibilities of combining and integrating different formats and media. The results are sometimes described as *mashups*, usually meaning data from more than one source displayed in a single view. Data may also be reused for different purposes. The term *remix* is used for the amending or repurposing of data to create something novel; for examples, see Engard (2014). These processes may offer a reimagining of a dataset or collection of datasets so as to show different perspectives, and have the potential of creating new knowledge.

Finding meaning in data

One important strand of data handling in the information sciences is to analyse datasets so as to find meaning in them. We will outline the methods used in four broad and over-lapping themes: data mining and statistical analysis; data visualisation; text visualisation; and text mining. For more detailed treatment of this area, see Shah (2020) and Wintjen (2020).

Data mining and statistical analysis

Data mining and statistical analysis identifies nuggets of value in large volumes of unhelpful data (hence the mining analogy) and

tests the significance of what is found by statistical methods. This is most commonly carried out using specialist subroutine libraries with the Python and R programming languages. A variety of techniques may be used, including: association, correlation and regression; classification and clustering; analysis of variance and model building; and time series. When dealing with large sets of data, and typically investigating many possible correlations and interactions, it is essential to have a thorough understanding of statistical significance, ways to guard against misleading and chance findings.

Data visualisation

Data visualisation is now easier than ever before, but the results are not always pleasing or informative. There are many ways of achieving this, including routines in coding libraries, functions in spreadsheet and database software and specialist software packages. The ease of creating a visualisation means that the emphasis is now on representing the data in an understandable way, free from bias and distortion. See Tufte's classic 1990 book, Healy (2019), Kirk (2019) and Dick (2020).

One useful form of visualisation of particular value for information science is the network diagram, displaying nodes, with their size, and links, with their strength. There are numerous software packages available for this purpose, such as the NodeXL extension to Microsoft Excel (Ahmed and Lugovic, 2019), and packages such as VOSviewer and CitNetExplorer, designed for analysis and display of citation networks in published literature (Eck and Waltman, 2017; Williams, 2020).

Text visualisation

There are many ways of visualising textual data; a survey by Linnaeus University identified 440 distinct techniques (Kucher and Kerren, 2015). Few, however, have gained wide use.

The most common way of visualising text documents or collections of documents is the *word cloud*, a relatively simple tool for producing images comprising the words of the text, with greater prominence given to the words that appear most frequently. They are popular for giving quickly assimilated summaries and comparisons of the contents of lengthy documents, but have been criticised for giving overly simplistic views of complex data. The same is true of other simple text-visualisation methods, such as the *termsberry* and the *streamgraph*. Some suggestions for improvement by manipulating the layout of word clouds to make them more meaningful are given by Hearst et al. (2020).

Text mining

Text mining involves analysing and displaying the structure of texts. There are a variety of means to achieve this. Coding with specialised libraries, such as the Python language's TextBlob, gives greatest flexibility. General-purpose qualitative analysis packages, such as nVivo and MAXQDA, can be used for text mining, but are complex solutions to what may be only a simple need. Web-based environments for analysis, visualisation and distant reading of texts, such as Voyant Tools, provide a variety of analyses, including word clouds, word-frequency lists, common words shown in context, and occurrence of terms in different sections of a web document.

Network analysis software such as VOSviewer can be used with bodies of text, producing visual displays of the frequency and co-occurrence of words and phrases. For library/information applications of this type of software, see Tokarz (2017), Miller (2018) and Hendrigan (2019). A more specialised tool, the *n-gram viewer*, allows the comparison of frequency of occurrence of words or phrases between datasets, or over time in a single dataset. The best-known example is Google Books Ngram Viewer, which provides a picture of how word, phrase and name occurrences have changed over time in Google's corpus of digitised books.

Summary

Digital technologies and data systems form the bedrock of the infosphere. An understanding of their nature and significance, and an ability to make practical use of relevant aspects, is essential for all information professionals.

- The technologies of information include all the tools and machines which have been used to assist the creation and dissemination of information throughout history.
- An understanding of the basics of computer and network architecture, coding, AI and HCI and the principles of data science is essential for all information professionals.

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10 Information Systems

Our dependence on increasingly pervasive technology has served to make it a significant determinant of the modern human condition. To comprehend this condition, one must therefore understand the human–machine relationship.

Sachi Arafat and Elham Ashoori (2019, 1)

Introduction

In [Chapter 9](#) we gave an overview of digital technologies and some of their applications. In this chapter we examine their use in information systems, focusing on the kinds of systems of particular interest to information science. After an initial look at the concept of information itself, we cover four general kinds of system: for retrieval of data and information; for managing documents in libraries, repositories and other collections; for facilitating the processes of digital scholarship; and for supporting creativity and innovation.

Information systems

An information system is a formal socio-technical system to collect, process and share information of any kind. It is a means to link people, information and technology to carry out particular tasks in a defined manner. We might think of pre-digital systems, such as a collection of records with access via a card index as an information system, and such a system would fit our general description. However, information systems are generally understood to be digital systems, comprising hardware and software, interfaces for data entry and information access, roles for the people involved and

clearly specified processes governing their operation. They may be general purpose, or specific in nature. General-purpose systems are described by terms such as business information systems, management information systems, decision-support systems, enterprise systems, group-support or cooperative-working systems, and transaction-processing systems. Domain-specific examples are health information systems, and geographic information systems. For overviews of the current information systems, see Whiteley (2013) and Valacich and Schneider (2018), for reflections on the nature and significance of such systems, see Checkland and Holwell (1998) and Swanson (2020) and for a perspective on the broader concept of *system*, see Siskin (2016).

The ways in which information systems are created and modified are generally termed systems analysis and design (Dennis, Wixom and Roth, 2019). These methods have changed greatly since their inception, with a variety of formal models being paralleled by 'softer' approaches, variously termed *user-centred*, *participatory*, *responsive*, *persuasive* and *interactive*, which emphasise an incremental, interactive approach, changing the system to suit the needs of the user, rather than forcing the user to adapt to the system. Information practitioners are often well positioned to be involved in the softer forms of design process; see, for example, Urquhart et al. (2017) and Makri (2020).

It is worth remembering that the introduction and operation of information systems is not a purely technical matter; human, organisational and ethical issues are of importance. In the context of library management systems, but more generally applicable,

Breeding (2021) lists ten principles, or concepts, for the assessment, introduction and use of systems: strategy, privacy, security, accessibility, efficiency, reliability, value, durability, transparency, and innovation. Such considerations help to ensure that technology and systems support the organisation and its people, rather than allowing technology, as Breeding puts it, to set the agenda. Ethical issues are discussed in [Chapter 15](#); as examples of the specific ethical problems that may arise with these systems, see the analysis of ethical problems of library analytics by Doty (2020), where the advantages of a detailed understanding of how library resources are used, provided by the system, must be balanced against the potential harms of surveillance of library users, and the study of the privacy of users of the arXiv repository by Menkov, Ginsparg and Kantor (2020).

Information retrieval, databases and discovery tools

Database is a rather overused and general term for any collection of digital data. It is often used to refer to systems which handle structured data, typically in the form of numbers and short pieces of text: the *database management system* (DBMS) or, in an even more restricted sense, the *relational database management system* (RDBMS).

In the library/information world, the term is generally used in the context of *information retrieval* (IR) systems. These handle data which is less structured than the facts and figures dealt with by DBMSs, but more structured than web search engines and ‘enterprise search’ systems, both of which combine data in diverse formats into a single index.

At one end of the spectrum, highly structured data is handled by RDBMS, or the similarly highly structured *graph databases*. At the other, a variety of general web search engines are available, including Google, Bing and DuckDuckGo, as well as specialist search engines for particular forms of material (Bradley, 2017). They are popular because of their wide coverage, ease of use and ability almost always to present some results, but have some drawbacks for academic use, in particular, lack of clarity as to what is covered and how searches are conducted: see, for example, Bates et al. (2017). For a perspective on the significance of search of this nature in society, see Haider and Sundin (2019). Within an organisation, information in a variety of formats, such as databases, documents and e-mails, may be searched by this kind of search engine, or by an enterprise search system such as Apache Solr, IDOL or Sinequa.

The ‘classic’ online bibliographic database typically has the following form of hierarchical structure.

- *Host system*: a system which gives access to a variety of databases through a common search interface. Examples are Proquest and Ovid.
- *Database*: a collection of material of a common subject or genre, produced by a single organisation to a common format. Initially these were often the digital equivalents of printed abstracting and indexing services, bibliographies, etc., or the printed catalogues and bibliographies. Examples are Web of Science, Medline/Pubmed and Library and Information Science and Technology Abstracts.

- *File*: a searchable form of a database. Database and file are usually the same, but a file may be a subset of a database, e.g. a date range.
- *Record*: a surrogate document description. Examples are bibliographic references, sometimes with abstracts and indexing added.
- *Field*: the smallest unit of the record. Examples are: title, author, abstract, journal name.

Typically, one would connect to a host system, choose one or more files/databases, search on fields and view records.

Such systems were initially developed in the 1960s and 1970s, almost always based on an original printed index, and typically beginning by making searchable the files used for printed index production; for accounts of the history of these systems, see Bourne and Hahn (2003) and Hall (2011). A much wider range is available today, with traditional subject- and genre-based databases being joined by searchable bundles of digital material offered by publishers, such as ScienceDirect (Elsevier) and EmeraldInsight, by web search engines, including those focusing on academic materials such as Google Scholar, and by repositories and open access directories. These all have their value, but differ markedly in how useful they are for any particular query; see, for example, Gusenbauer and Haddaway (2020; 2021).

Components of IR systems

IR systems are usually analysed in terms of system components; however, it seems that each writer has their own preferred set of

components. We will take a simple view, considering just four main components or subsystems: input; indexing; search; and interface. For fuller discussions, see Chowdhury (2017) and Arafat and Ashoori (2019), and for a focus on internet search engines, see Bradley (2017) and Russell (2019). For commentary on the relation between IR, the organisation of information and knowledge and philosophical perspectives, see Hjørland (2021).

Input

The input subsystem, depending on the purpose of the overall system, may have to include materials of widely different kinds, or may handle a more restricted range. What is entered may be a whole document, for a full-text retrieval system, or a document surrogate, a shorter summary record such as a bibliographic reference. Virtually all IR systems use an 'inverted file' structure: a file of index terms, created by the indexing subsystem, with pointers from each term to the documents to which it applies, themselves held in another file in sequential order. This allows rapid look-up of relevant documents, without having to scan the whole file.

Indexing

The index file is created by the indexing component. Indexing may be full-text, with every word in the document indexed, or more selective; if the latter, it may be automatic, with terms chosen statistically according to their frequency of occurrence, or it may be done by an indexer. A controlled vocabulary may or may not be used, as well as, or in addition to, the terms in the original document and/or added free terms.

Search

The heart of an IR system is its search component, which does the work of retrieval. Search is carried out in different ways, reflecting the nature of algorithms used in the search process.

- *Boolean*: this works by manipulating sets of documents indexed with particular terms, using the Boolean operators AND, OR, NOT. It is clear and straightforward: a search for '(CATS OR DOGS) NOT HORSES' produces all those documents indexed with the term CATS or the term DOGS provided they are not indexed with the term HORSES. A drawback is the binary all-or-nothing metric. There is no way to say that one document is all about cats, while another barely mentions them in passing; documents are indexed CATS or not. Also, users often find it difficult to express their information need in the exact way required by a Boolean search.
- *Vector*: this approach calculates the similarity between the query and each document according to the terms in common. It is more subtle than Boolean, as it allows for different degrees of matching, rather than just yes/no, and it allows results to be ranked in order of similarity to the query.
- *Probabilistic*: this approach includes a number of methods which attempt to calculate the probability that a document will be relevant to a query, on the basis of statistical analysis of the occurrence of terms in the query and in the set of documents. Some of the methods use 'relevance feedback', allowing the user to assess retrieved documents as relevant or not, amending

probabilities according to these judgements and searching again; this process continues until the user is satisfied with the result.

See Aly, Demeester and Robertson (2014) for more detail.

Van Rijsbergen (2004) introduced a formalism based on the Hilbert spaces used in quantum mechanics for representing IR models, in effect combining the probabilistic, logical and vector space approaches to IR in a single mathematical framework; see Melucci (2015) for an updated treatment.

Sometimes the Boolean model is described as ‘exact’ retrieval, while the other two models are denoted as ‘best match’; the latter will always return some result as the closest to the query, whereas the former will return exactly what was specified, which may be nothing. Some retrieval systems use information other than apparent relevance to rank the output; an obvious alternative is to present newer material first. The Google search engine’s PageRank algorithm orders items according to how many other web pages link to them, a surrogate measure of the interest value of the site. The closed nature of some search algorithms, including that of Google, makes it difficult to ascertain exactly how the ordering of search results is determined.

User interface

The interface component allows the user to put their query to the system and to receive the results. Interfaces have changed greatly over the years, although the operations which they evoke behind the scenes have not. There have been, and still are, a variety of interface styles; see Chowdhury (2017) and Morville and Callender (2010) for

examples. We will mention three important general styles, which have developed to match the typical computer interfaces of the time.

The first was the *command line* interface, stemming from the 1960s and developed for the first generation of online bibliographic search systems. At least one of these, for the Dialog host system, was still in use almost unchanged after nearly 50 years. These require the user to know a limited set of commands: typically, to search for documents indexed with a term, browse the index, combine sets with Boolean operators and display results. They also allowed for searching for phrases, truncated terms beginning or ending in a certain way, terms within a specified proximity of one another, terms in particular fields of the record and so on. For instance, in the 'classic' Dialog systems, the command:

s digital(w)library/ti AND (online OR retrieval OR search*) AND
au=jones ?

will retrieve documents with the phrase 'digital libraries' in the title, and any of the terms 'online', 'retrieval' or anything beginning 'search', for example search, searches, searching and so on, anywhere in the subject parts of the record, and also an author with the name 'Jones' with any forenames or initials.

This kind of interface gives the user complete control over a Boolean search, and allows very precise specification of detailed searches. However, it is not easy to use, and the user must take some time to learn the system commands. Although these types of interfaces are still available in some host systems, their functions have largely been absorbed within other styles of interface.

At the opposite extreme is the simplicity of the 'search box', pioneered by Google; a single box, into which the user types whatever they wish. This may be a Boolean statement, if the system works in this way, but will more usually be just a few terms or phrases; most searches on Google are single words. This simplicity is so attractive to most users, particularly if combined with a search function which gives ranked output, that many IR systems have adopted it, accepting that it loses the control and precision possible with the command line.

An intermediate stage, often provided as an 'advanced search' function, attempts to combine the simplicity of the search box with the power of the command line by providing a matrix of rows and columns for the user to fill in, assisted by prompts and drop-down menus, giving a limited choice of Boolean operators and field specification.

All these forms of interface, and variants of them, are to be found in IR systems. System designers seek to strike a balance between simplicity and precision of search specification; most users prefer simplicity most of the time.

Other search systems and strategies

So far, we have discussed IR systems in terms of text retrieval, by contrast with the facts-and-figures searching of DBMS. Although this is certainly the most widely encountered form of search system, there are systems for specific types of information, such as images, music and chemical structures. Another alternative is *citation searching*, a form of retrieval which relies on ready-made indicators of document similarity, which do not necessitate any indexing – the

links made when one document references or mentions another. A source organised in this way is referred to as a citation index and its use as citation searching (Belter, 2016; McLaughlin and Tucker, 2017; Carina de Araújo, Castanha and Hjørland, 2021). These are used to find newer material on a topic, which has cited the original. While bibliographic items, such as articles and books, are the usual cited items searched for, a wider range is possible; paintings and other art works, and pieces of music, for example, may be cited items. Citation databases include the Web of Science/Web of Knowledge, created by the Institute of Scientific Information and now provided by Clarivate Analytics, Elsevier's Scopus, Microsoft Academic, Digital Science's Dimensions, Crossref and Google Scholar; for a comparison of coverage, see Martin-Martin et al. (2021).

Libraries are increasingly employing 'discovery tools', IR systems giving one-stop access to all the library's resources, usually including open access sources and search engines outside the conventional idea of the collection defined by ownership and storage; what Dempsey (2016) terms the 'facilitated collection'. They will also contain metadata for printed resources. A well-known example is Serials Solutions' Summon. There is a potential conflict between providing the simplest and quickest results, via the search box approach, and the best control of the search, with understanding of the approach being used, of more complex interfaces. For analyses of the impact of such systems, see McLeish (2020) and Haggerty and Scott (2019).

Studies of the use of IR systems have identified a small number of search strategies and tactics, general approaches to retrieval, used

frequently; these can be put to best effect with those interfaces which give full control over Boolean searching in structured records with subject indexing. These strategies were first studied by the American library science academic Marcia Bates (1979); for more recent examples, see Russell-Rose, Chamberlain and Azzopardi (2018) and Stapleton, Carter and Bredahl (2020).

Some examples of these strategies are shown in Box 10.1: they are mainly used informally and in combination, but it may be helpful to adopt them if no useful material, or too much, is being found.

Box 10.1 Examples of search strategies for information retrieval

Berrypicking: carry out the search using a small number of search terms; choose the most interesting two or three items; read these, and then search again modifying the search terms as necessary; repeat until enough items have been found (the topic of the search may have changed in the process).

Building blocks: divide the search into distinct topics, represent each topic by a set of synonyms linked by OR and then link all the topics with AND. Begin with the topic likely to have the smallest number of documents in the collection, and stop when the number of retrieved documents is small enough to scan.

Pearl growing: start with a known relevant document, the 'seed', and look it up in the collection. Examine the index terms and use these as starting points for a search. Examine the retrieved documents, choose relevant ones and repeat the process. Continue until no new relevant documents are found.

Successive fractions: do an initial search, using just one term, or a few terms linked with AND. Examine some of the documents found and find terms occurring only in those which are relevant or those which are not relevant. Repeat the search, adding 'relevant' terms with AND, and 'not relevant' terms with NOT. Continue until the retrieved set has been reduced to a small enough number to be scanned.

Quicksearch: divide the search into distinct topics, represent each with one index term and search, linking the terms with AND. Any relevant documents found can then be used as the 'seed' for pearl growing.

Library, repository and collection management systems

The term *digital library* has been extensively used over past decades, though with lack of precision as to its meaning. It has sometimes been used in a very general way to imply 'future libraries', and it is generally accepted that few, if any libraries, described in this way are exclusively digital, and even they may also include physical items. An early analysis by Borgman (1999, 2000) is still helpful in reminding us that the concept of digital library may relate both to technical architecture and to services: it may be seen as a collection of multimedia information resources, datasets and databases, and equally as a set of tools and capabilities for locating, accessing and using available data and information.

There are two main forms of digital library management system. The older *integrated library system* (ILS), comprises a set of components hosted within an institution. Examples are SirsiDynix's Symphony and the Koha open source system. The new style of

library service platforms (LSP), sometimes referred to as *second generation ILS*, are cloud-based systems, offering resource management, web services and a discovery layer, and interoperable with systems outside the library, such as links to a virtual learning environment (VLE) in an academic library. Examples are Ex Libris' Alma, with the Primo discovery layer, OCLC's Worldshare, with the WorldCat discovery layer, and the Folio open source system. Both ILSs and LSPs offer components, or modules, for activities such as: *acquisitions*, selecting and purchasing materials, and managing the budget; *cataloguing*, creating, importing and editing metadata records; *circulation*, managing loans of physical items; *interlending* of physical and digital materials between institutions; *reporting*, providing analytics and management information; *searching*, whether by an Online Public Access Catalogue (OPAC) or by a discovery layer; and *digital engagement* with both operators and users of the system. Detailed discussion of all aspects of library management systems is given by Matthews and Block (2019) and Banerjee and Reese (2018), and practical advice on the use of library technologies by Smith (2021). Provision of analytics, allowing decisions to be made on the basis of data, is one of the main advantages of such systems over their predecessors (Showers, 2017).

Institutional and subject repositories, typically operated by libraries and similar institutions, require software for maintaining digital archives, allowing storage and preservation of and access to their open access materials. Because of their purpose, and the nature of their content, they do not require many of the features of library management systems, and their acquisition and cataloguing

functions are relatively simple. Open source software systems, using the Open Access Initiative Protocol for Metadata Harvesting (OAI-PMH) to ensure interoperability through consistent metadata standards, are generally used for such repositories. Examples are EPrints and DSpace. Repositories using systems of this kind are listed in two main directories: Directory of Open Access Repositories and Registry of Open Access Repositories. For details of the operation of such repositories, see Finlay (2020). Such systems enable Lorcan Dempsey's concept of the 'inside-out' collection, by which libraries make their internally generated resources, such as research materials in institutional repositories, digitised images from a library's materials and open educational resources developed at the institution, available externally as well as internally (Dempsey, 2016). This complements the traditional 'outside-in' collection model, by which external materials are acquired for internal use. As examples of systems supporting the inside-out approach, see Gwynn, Henry and Craft (2019) and Day and Novak (2019).

Collection institutions, including archives, museums and galleries, make use of two general types of system: *collection management systems* (CMS) and *digital asset management systems* (DAMS). The former generally deal with physical objects, the latter with digital items, though some systems may deal with both. CMS, sometimes referred to as *collections information systems*, allow the recording and editing of all relevant information about items in the collection, and may also control collection workflows. The exact functionality will depend on the context. A system for museum documentation will deal, *inter alia* with acquisition, catalogue description, object location,

provenance, curatorial and conservation information, and exhibition history. An archival system will deal with aspects such as accessioning, archival description and metadata, authority control, preservation, and reading room management. Examples of such systems are ArchivesSpace, Minisis M2A, and Axiell Collections. DAMS offer similar functionality, but specifically for digital resources; an example is Axiell DAMS.

Collections of documents and records in other kinds of organisations may be managed using DAMS, or by systems variously termed *electronic document and records management systems*, *document management systems*, or *content management systems*. These are designed to support the processes of the document lifecycle, as discussed in [Chapter 13](#), emphasising features such as document capture, organisation and metadata, version control, and workflow control. They may also be used to support knowledge-management programmes. Examples are Drupal, ECM Documentum, and Alfresco.

Systems for digital scholarship

By this we mean systems to support what is sometimes termed *digital scholarship* or *e-research*. This covers a wide area of academic endeavour, spanning science, social science and the humanities. There are, strictly speaking, no systems devoted solely to such studies. Rather, they all rely on systems of the sort discussed in this chapter for the collection, storage, access and preservation of relevant data collections, which are then analysed by the sort of techniques outlined in [Chapter 9](#), including data cleaning and organising, visualisation, data and text mining, and machine

learning. There is therefore a clear link between the information sciences and this kind of research.

The link is particularly close in the case of the *digital humanities* (DH), which grew out of humanities computing to become a discipline in its own right, with its own theoretical foundations as well as practical activities of curation and analysis of humanities data. Much of its activity has focused on the creation of digital resources for this purpose, and on development of techniques of data analysis, and text mining and visualisation. In particular, there has been a focus on *distant reading*, developing an understanding of texts by analysis of their large-scale composition and structure (Underwood, 2017).

The nature and scope of DH has been extensively debated since the discipline was first recognised. Various perspectives are given by Terras, Nyhan and Vanhoutte (2013), Warwick, Terras and Nyhan (2012), Gardiner and Musto (2015), Svensson and Goldberg (2015) and Edmond (2020).

DH has always had strong links and overlaps with the library/information disciplines, and the two have been seen as natural partners; see, for example, Poole and Garwood (2018), Clement and Carter (2017), Wilson (2020), Alvite-Díez and Barrionuevo (2020) and Chambers (2021). Both are concerned with recorded information encapsulated in documents (in the broadest sense), including datasets, and both are concerned with the creation, curation and use of compilations of data and information. The digital tools developed by DH, particularly for the digitisation and

cleaning of data, are relevant to, and being used in, the activities of the information sciences, and vice versa.

E-science, while its overlap with the information sciences has not been discussed to the same extent as that of DH, is also reliant upon curated open data, and an infrastructure for the communication, sharing and collaborative processing of information and data (Koschtial, Köhler and Felden, 2021). The increasingly important role of research data management (Cox and Verbaan, 2018) is currently information science's main contribution to e-science. The same is true of the newer processes of e-social science, which is increasingly adopting methods such as big data analysis and data mining (Fielding, Lee and Blank, 2016).

Information systems for creativity and innovation

Information systems may in some cases be required to support and promote innovation and creativity (Bawden, 1986; 2011). But here we encounter the paradox put forward by the poet Lord Byron: 'To be perfectly original one should think much and read little and this is impossible, for one must have read before one has learnt to think.'

Innovation and creativity are said to depend on the 'prepared mind', which will generally rely on the input from information systems. But there may be some specific, perhaps unusual, features of information which may support creativity, for example: inclusion of peripheral and speculative (even incorrect) material; provision of interdisciplinary information; representations of information to bring out analogies, patterns, exceptions, etc.; emphasis on browsing facilities; encouragement of informal channels; and provision of information geared to individual preferences and requirements.

There are also some examples of information systems specifically geared to *literature discovery*, by identifying patterns and relationships in information collections; for example, common title words, index terms or citations, an approach pioneered by the American information scientist Donald Swanson (1924–2012): see, for example, Swanson and Smalheiser (1999) and Kostoff et al. (2009), and for recent developments in literature discovery, see Henry and McInnes (2017). A browsing approach to information has also often been associated with creativity, and we may speculate that the more that resources and interfaces are provided to support browsing and information encountering, the more effective information systems may be at supporting innovation (Bawden, 2011; Björneborn, 2017). In general, we may hope that new ways of using information systems for the direct support of creativity and innovation may be found.

Summary

The information systems of primary interest to information science are those for storing and retrieving data and information. These take different forms according to the context, systems for libraries, archives, museums, etc. being necessarily different in detail, though their functions and capabilities are increasingly overlapping. Increasingly such systems are cloud based rather than locally hosted, and interoperable with systems which are not strictly ‘information systems’, such as pedagogic and administrative systems. In the information retrieval area, traditionally the core of information science, the balance between the convenience and ubiquity of web search engines and the capability of special-purpose interfaces and

retrieval algorithms is an important issue. The increasing importance of digital research and scholarship offers new roles and opportunities for information science.

- The information systems of particular relevance to information science are those for storing and retrieving data, information and documents in a variety of contexts.
- Information science can draw from, and contribute to, new trends in the design of information systems and in the infrastructures of digital research and scholarship.
- Information systems should be able to make a greater contribution to innovation and creativity.

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11 Informetrics

To measure is to know. If you cannot measure it, you cannot improve it.

Attributed, in various forms, to William Thomson, Lord Kelvin

Introduction

This chapter is about measurement; specifically, measurement of the quantitative aspects of the creation, communication and use of information. As Lord Kelvin reminds us (above), measurement is vital; but only if the right things are measured and the measurements are meaningful and up to date.

In this chapter we will first examine the nature of informetrics and its components, before looking at how the subject has developed since its origins. We will consider the very basic question of how much information there is before looking, without extensive mathematics and with a focus on qualitative significance, at the main informetric laws – those of Lotka, Bradford and Zipf. Finally, we will look at how informetrics techniques may be applied in information research and practice.

Informetrics is the study of the quantitative aspects of information resources and of the communication of information. The term ‘informetrie’ was introduced by Otto Nacke (1979), a German documentalist and medical information specialist, and popularised as ‘informetrics’ by the British information scientist Bertie Brookes. Informetrics is usually taken to include several more specific subjects: *bibliometrics*, the study of quantitative aspects of published documentation; *webometrics* (also termed *webliometrics* or

cybermetrics), the study of quantitative aspects of web resources; *scientometrics*, the quantitative study of the growth and change of academic disciplines, and *altmetrics*, assessing significance by measures not reliant on formal publication.

Ball (2021), Abrizah et al. (2014), Gingras (2016), Cronin and Sugimoto (2014), Sugimoto (2016), Bar-Ilan (2008; 2017), Hood and Wilson (2001), Wilson (1999) and Tague-Sutcliffe (1992) all give detailed accounts of the origins, meanings and interrelations of these terms and of others which have been used for this subject, and review the subject in general; see also Bawden and Robinson (2016) for an account of some numbers which are significant for the information sciences.

The major bibliometric laws – actually approximate statistical relations, rather than true laws describing patterns and regularities in quantitative aspects of documentation – are those of Lotka, Bradford and Zipf. They describe general patterns, typically rank frequency distributions, observed in social situations including in the world of bibliography and documentation. They show, for example, that for any subject much of its content will be found in a relatively small number of sources (the ‘core’), while the rest is ‘scattered’ throughout a wide range of sources.

We will also include here within the scope of informetrics three other topics: the fundamental question of how much information there is; the quantitative analysis of information networks; and the controversial application of these methods to the evaluation of research impact and productivity. Some authors have proposed still wider scope: Tague-Sutcliffe (1992), for example, argued for

inclusion of topics such as the quantitative assessment of information itself, by Shannon formula and similar measures; quantitative measures of retrieval system effectiveness, such as recall and precision; and measures of use of resources, such as library performance metrics. We find it more helpful to treat these topics in their appropriate chapters, focusing here on analysis of quantitative aspects of documents and collections.

Informetrics has both a strong theoretical base and a number of highly practical applications. It is generally regarded as one of the central components of the information science discipline and is also one of the aspects of domain analysis. An understanding of the principles and practice of aspects of informetrics is therefore important for all information professionals.

Historical developments

Only a brief outline of the development of the subject will be given here; for more detailed accounts, see Danesh and Mardani-Nejad (2021a), Rousseau (2010), De Bellis (2014), Abrizah et al. (2014) and Wilson (1999).

Informetrics may be said to have been inaugurated by Charles Babbage, who in 1830 suggested that the counting of authorship would be a way to identify scientific excellence. The Royal Society Catalogue of Scientific Papers of 1868 gave the first example of such paper counting. The first example of unease about the misuse of these metrics came in that same year, when Wilhelm von Haidinger, an Austrian mineralogist, complained in a newspaper article that mere comparison of numbers of publications is no basis to make judgements of value; he rather compromised his position two years

later, by publishing a chart comparing prolific authors in each region of Europe, and lamenting the low position of Austria in the tables.

The earliest detailed informetric study was carried out by Francis Cole, a zoology professor at the University of Reading, England, and Nellie Eales, the curator of the University's zoology museum and rare book collection. They studied the growth of the discipline of comparative anatomy between 1550 and 1860, using what they termed 'statistical analysis' of the literature, to analyse, for instance, contributions from different countries (Cole and Eales, 1917). They presented graphically an analysis of 6,436 documents, showing changes in the subjects of anatomical study and relating these to external influences.

In 1923, a study with wider scope was carried out by Edward Wyndham Hulme, librarian at the London Patent Office, who related the growth of scientific literature to economic development and development of civilisation, using the term 'statistical bibliography'. He analysed data from annual issues of the *International Catalogue of Scientific Literature* from 1901 to 1913, counting the entries by subject – and particularly noting when subjects divided, denoting increased specialisation – and by author nationality.

These studies established the nature of the topic as the counting of documents in printed collections: counts of items (books, journals, articles, etc.) and of their attributes (country, language, references, etc.). They were rapidly applied to practical problems; for example, the first use of bibliographic data for collection development, in the case of a chemistry collection in an academic library (Gross and Gross, 1927). The authors assessed how often other journals were

referenced in the *Journal of the American Chemical Society*, chosen as being most representative of the activities and needs of American chemists. They noted not only the most common of the 247 journals cited, but also trends over time, so as to make cost-effective recommendations. Foreshadowing a later main bibliometric finding, they noted that valuable materials were scattered outside the discipline, recommending that journals of physics and of general science were as important as those of chemistry per se, and pointing out that the data showed the need for materials in languages other than English, particularly German and French.

Origins of informetric laws

The first general laws of what would come to be called bibliometrics were also developed from the 1920s onwards. We will focus on those of Lotka, Bradford and Zipf, but there were numerous others, such as the laws of Leimkuhler and Mandelbrot. It was finally shown that many, if all, of the individual bibliometric laws have a common mathematical foundation (Egghe, 2005; 2021; Rousseau, 2005; 2010).

Alfred Lotka, a mathematician best known for his work in theoretical biology, was working as a statistician for an insurance company when he derived the law bearing his name, concerned with the distribution of authors by their productivity (Lotka, 1926). He analysed the number of papers contributed by each first-named author mentioned in certain sections of abstracting journals in physics and chemistry. Ironically, although his law is the first of the important informetric laws, Lotka was never involved in the information disciplines, and the study which derived this law is a small and untypical part of his work.

Samuel Clement (usually known as just S. C.) Bradford, the librarian of the Science Museum in London, established the next development, with his 'law of scattering', showing how the periodical literature of a given subject area is distributed across journal titles (Bradford, 1934). He analysed bibliographies of geophysics and of lubrication engineering, showing the way the entries were spread across a range of sources and deriving from this a practical means for librarians to choose sources for a collection. Bradford's is the only major informetric law derived by someone from the information sciences.

The third of the major bibliographic laws was established by George Kingsley Zipf, an American philosopher and linguist, who derived a distribution for the frequency of occurrence of words in text (Zipf, 1935). Zipf wrote that an unnamed friend had pointed out this idea to him, and indeed ideas of this kind had been mentioned by several authors before Zipf systematised and promoted them (Rousseau, 2010). This formed the basis for his later 'principle of least effort' (Zipf, 1949). There has been some debate about whether Zipf's formulations should count as true bibliometric laws, but their influence has been great. They clearly apply to the concerns of the information sciences, but are of much more general applicability (Chang, 2016).

Informetric studies were given a boost by the appearance of *citation indexes*. The importance of indexes of this type is that they allow identification of which items have cited older items; this in turn leads to the identification of 'impact', when items receive many citations, and to analysis of the structure of disciplines by examining

patterns of citing. Indexes to legal citations had existed for many years, but their wider value for information retrieval was realised and expounded by Eugene Garfield (1955). The main tools which made this possible, the citation indexes of Garfield's Institute for Scientific Information (ISI), now the Clarivate 'Web of Science/Web of Knowledge' services, were introduced between 1961 and 1974 (Pendlebury, 2021).

By this time, the concept of bibliometrics was well established. Derek De Solla Price's book *Little Science, Big Science* (1963) used a variety of quantitative methods to document the growth of scientific and technical literature and to relate this to the history and sociology of science. This classic work formed the basis for *scientometric* study of the research process through bibliometric data (Danesh and Mardani-Nejad, 2021b). The term 'bibliometrics' had itself become established, derived by Alan Pritchard (1969), as an improvement on the 'statistical bibliography' designation still in use. The term had not been used before, although Paul Otlet had used the French equivalent 'bibliométrie' in his *Traité de documentation* of 1934.

Eugene Garfield (1972) was also among the first to emphasise the potential value of citation data in the evaluation of the quality of journals and as a tool for the evaluation of research; see Bensman (2007) for a detailed account. This was not the first time that literature analysis had been recommended, and indeed used, to assess the impact of research. George Banay (1945) was the pioneer in this respect, identifying literature references to a research programme on the treatment of mental illness at Worcester State

Hospital, Massachusetts, USA. However, the availability of citation databases and analysis tools gave a boost to this concept.

At about the same time Henry Small, a colleague of Garfield's at ISI, proposed the method of co-citation analysis (Small, 1973), which examines the joint citing of two or more items by later items; the number of times this occurs shows the strength of the link between the two items. This has proved to be a very valuable method of showing connections between key documents which can be used to map the structure of subjects and disciplines. Use of computer visualisation techniques alongside informetric analysis has led to visually appealing maps and other images; see, for example, Börner (2010).

The term 'scientometrics' for the quantitative analysis of the growth of science, and other academic and professional areas, had been introduced by the Russian scientist Vasily V. Nalimov, as a translation of the Russian 'naukometriya'. It became generally accepted during the 1970s and was used as the title of the main journal of the field, launched in 1978. *Scientometrics* has published many papers with essentially bibliometric content, showing the overlap between these sub-branches of informetrics. Growth of interest in this aspect has led to wide, and sometimes controversial, use of these techniques for productivity assessment and research evaluation, applied to countries, institutions and individuals.

The advent of the internet, and later the World Wide Web, quickly led to quantitative studies of resources and their interrelations in these new environments. A wide variety of terms were devised to describe this new area of informetrics; most commonly used now are

cybermetrics (for quantitative studies of anything on the internet) and *webometrics* (for the more restricted area of the web), although these terms are often used interchangeably (Thelwall, 2010; 2017).

Most recently, the approach of *altmetrics* (alternative metrics) has sought to find measures for the impact and research of scholarly works beyond counts of publication and citations, typically looking at mentions in, for example, social media or news media; usage, for example downloads; or impact, for example mention in patents, public policy reports or educational syllabi (Priem, 2014; Tattersall, 2016; Cave, 2017; Blümel and Gauch, 2021).

How much information is there?

One of the most fundamental questions for informetrics is how much information there is, in various contexts, and how much could there be. These questions can be answered only approximately, but even order of magnitude estimates can be illuminating. We will look at three variants of this question: how much information can the universe hold; how much information, and data, have humans created; and how much information, and data, is being created and transmitted at the present time?

How much information there can be in the universe is a question for physics, answered essentially by assessing the smallest unit of volume that could hold the smallest item of information, a bit, and estimating how many such units there are in the universe; the answer is of the order of 10^{90} bits (Bawden and Robinson, 2016). While this has little to do with the activities of information specialists, it reminds us that information, necessarily associated with a physical substance as a carrier, must be limited in quantity.

On the other hand, it has been suggested that knowledge, unlike information, may be infinite, since it is 'made of' information, which may be combined in infinitely many ways; see, for example, Deutsch (2011).

The question as to how much information has been created has more direct relevance to the information sciences. Human recorded information is estimated at about 300 exabytes, amounting to about 2 million Libraries of Congress, or 60 metres of bookshelf for every person on the planet, and including over 50 million scholarly articles. It is estimated that more data was created in 2012 than in the previous 5,000 years, and the rate has greatly increased since then. However, it is important to distinguish data created by machines to be read by other machines from human-created or- curated datasets, which will be orders of magnitude less. For example, in April 2021 the British Library estimated that its printed collections took up nearly 750 km, growing at 8 km per year, with 1 pentabyte of digital collections.

Current production of information also shows remarkable numbers, with estimates that, for example, every minute of the year 2020 saw an average of 19 million text messages, 4.1 million Google searches, 190 million e-mails and 4.7 million videos viewed on YouTube. (There are a number of versions of the 'Internet Minute'; these data come from the 2020 infographic due to Lori Lewis and Chadd Callahan of Cumulus Media.)

The main informetric laws

We mention here the main laws of informetrics and their implications, making no attempt at a detailed treatment or at

mathematical precision; for those, see Egghe (2005; 2021), Rousseau (2010) and Wilson (1999).

This topic may appear confusing. There are several main ‘laws’, with numerous variants and reformulations, some of which are given different names. There are claims that these laws can be interrelated and reflect one underlying law or principle. They are, in any case, not rigid laws like the laws of science, but statistical approximations which do not hold in all circumstances. And, although all these laws may be applied in library and information contexts, some of them come from outside the field and are of much wider applicability.

We will simplify matters by focusing on the three most important informetric laws, or ‘families of laws’ as they are sometimes termed to emphasise their different variants and formulations. These are the laws of Lotka, Bradford and Zipf.

Lotka’s law

Lotka’s Law of Scientific Productivity, as it was originally known, is an inverse square law relating the total number of documents in some collection and the number which are authored by one person. The number of people who are authors of n documents will be $1/n^2$. More formally, this is described as a power law of form C/n^a , where C is a constant and a is approximately 2. The number of authors with y documents is given by:

$$f(y) = C/y^a$$

So, the number of authors producing two items will be one-quarter the number of those producing just one item; the number producing

three will be one-ninth of those producing one and so on. About 60% of authors in a collection will produce just one item. So, for every 1,000 authors contributing 1 paper, there are 250 authors contributing 2 papers, 111 authors contributing 3 papers and only 60 authors contributing 4 papers, 40 authors contributing 5 papers, etc. This means that for any subject or collection we would expect there to be a small number of very productive people, institutions, countries, etc. and a large number of those making a small contribution. This is an example of what would later come to be called the 'long tail'.

Of course, like all the informetric laws, this is only an approximate statistical law, describing a general regularity in collections of documents; the calculations will never be exact. Lotka derived his result – which he described as a 'formula' or a 'distribution', rather than a law – by analysing first authorship of scientific articles, taking the data from abstracting and indexing services. It is, however, applicable to much wider contexts, whenever we have identifiable producers of documents.

Bradford's law

Bradford's Law of Scattering, as it is generally known, described regularities in the sources in which interesting documents are found. His original analysis examined articles (documents) in scientific journals (sources). In general, if sources relevant to a particular subject are listed in decreasing order of 'productivity' – meaning the number of relevant items included in them – then a pattern emerges. If the list of sources is divided into three parts, each containing an equal number of useful documents, then the ratio of sources in the

three parts is roughly $1:n:n^2$, where n is a constant, varying between subjects and types of material. Again, this is an approximate statistical relationship, rather than anything exact. What this means is that a small number of sources produces a large number of useful items; this is generally referred to as the 'core'. An alternative way of looking at it is that a third of useful material is scattered over a wide range of sources: another example of the 'long tail'.

Subsequent theoretical work has extended Bradford's formula to any number of groups, rather than just three. Leimkuhler (1967) produced an alternative mathematical formalism, which is sometimes referred to as Leimkuhler's Law. This may be expressed as:

$$N(r) = a \log (1 + br)$$

where $N(r)$ is the total number of items in journals of rank equal to or lower than r ; a and b are constants, depending on the nature of the collection.

Zipf's law

Zipf's Law of Word Occurrence, as it was originally known, looks at rank/frequency distributions in word use. If we order the words in some text by the number of times that they occur, then the *rank* of the word – its place in the list – will be determined by its *frequency* – the number of times it occurs. Zipf's law states that if the words are ordered in decreasing frequency, then the rank of the word multiplied by its frequency is a constant. Mathematically, for a word with an occurrence of y and a rank of r , then

$$y = k/r$$

where k is a constant, depending on the context.

Benoit Mandelbrot, best known for his development of fractals, derived an alternative and more general formulation of Zipf's law, so that the number of times a word occurs, n , if it has rank r , is given by

$$N = C_1 / (1 + C_2 \cdot r)^\beta$$

where C_1 , C_2 and β are all constants depending on the context. This is generally referred to as Mandelbrot's Law.

Zipfian distributions of this kind are very general, and are found in many aspects of informetrics.

Are they all the same law?

There was considerable interest, from the 1960s onwards, in proving that the empirical informetric laws, formulae, functions and distributions have a common basis and are essentially equivalent. Some commonality was obvious from the first. They show very similar patterns in various aspects of documents and collections and are based on similar rankings by size or frequency of attributes. They relate in general terms to some kind of source and yield: journals–articles, authors–publications, etc. Indeed, Zipf referred to Lotka's work when first announcing his own findings. They also have obvious similarity to familiar distributions observed outside the library and information world, such as the ubiquitous Pareto principle, often referred to as the '80:20 rule', derived originally from the observation by the Italian economist Vilfredo Pareto that 80% of the wealth of Italy at the beginning of the 20th century was owned by 20% of the population. However, considerable mathematical difficulties were encountered, and it was not until more than 50

years after the first empirical laws were derived that a satisfactory proof of their equivalence was provided. For details of this, readers with a mathematical bent are referred to Rousseau (2010) and to Egghe (2005; 2021).

There has also been interest in seeking an underlying explanation for these laws. The most commonly offered is the principle known variously as 'success breeds success', 'cumulative advantage', 'preferential attachment' and the 'Matthew effect'; the last from the verse in St Matthew's gospel that states: 'For whosoever hath, to him shall be given, and he shall have more abundance: but whosoever hath not, from him shall be taken away even that he hath.' The more productive a source is, in any sense, the more likely it is that it will continue to be productive. An article in a journal that has been cited many times already is much more likely to be cited again than a rarely cited article; an author of many papers is much more likely to publish again than one who is less prolific; and so on.

Other explanations are more mathematical in nature: Mandelbrot argued, for example, that fractal theory was the underlying explanation for these laws. Still others point to power laws, of which the informetric laws are an example, being ubiquitous in both the physical and the social worlds. Accounts of these and other explanations are given by Rousseau (2010) and Newman (2005).

Network analysis

A further kind of quantitative method within informetrics is network theory in its various guises. This may be adapted to study networks of different kinds which are of relevance to the information sciences, from computer networks to social networks. Network analysis is

often used in conjunction with other informetric tools. Its formal structure is very different from them, relying on the mathematics of graph theory rather than that of statistical distributions, and often used semi-quantitatively. Börner, Sanyal and Vespignani (2007) give a clear account of the underlying ideas. Thelwall (2006) identifies four main approaches in network analysis: from statistical physics, aimed at modelling the structure and growth of the web; from computer science, aimed at facilitating information retrieval and data mining; from the information and social sciences, aimed at studying networks of people; and from the information and social sciences, aimed at studying networks of information.

For informetric applications, the main applications are in studying networks of publications and citations, and in social network analysis (SNA), studying the interactions and relationships between people and groups. Many SNA studies are based on the 'small world' or 'six degrees of separation' concept, which implies that large networks are structured so that all parts are connected through a small number of links. For overviews of SNA, see Otte and Rousseau (2002), Scott (2017), Blümel (2021) and Hicks, Cavanagh and VanScoy (2020). Library and information applications include study of citation and co-citation networks, and of networks of collaborating researchers and of people seeking, accessing and sharing information: as examples, see Hellstein and Leyesdorff (2020), Johnson (2019) and da Silva, Barbosa and Gomes (2019).

Applications of informetrics

A variety of indexes, data sources and software for carrying out informetric analyses and visualising the results are now available

(Bauer, 2021). While this is of evident benefit, using different systems and sources will inevitably lead to different results, so that the results must be interpreted with care.

We will consider four main categories for applications of informetrics: understanding the nature and structure of literature and scholarship; collection management; analysing a set of literature; and assessing impact.

Understanding the nature and structure of literature and scholarship

This will generally be done as part of a study of a subject or topic, either to assess its status at a particular date or to track changes over time. Factors examined include: the amount of material published and the amount necessary to get 'good coverage'; the most important sources, authors and institutions; spread across sources, countries and languages; content of journals and monograph series; authorship patterns; and use of citations, footnotes and acknowledgements. An understanding of the structure of literatures, of disciplines and of collaborating groups can be obtained through techniques such as bibliographic coupling and co-citation analysis. The former involves counting the number of references that a given pair of documents have in common; the latter counts the number of times that a given pair of documents (or authors or journals) are co-cited by third parties. Webometrics uses essentially the same techniques, with hyperlinks replacing the reference or citation as the function linking documents. These techniques rely on the assumption that citations are rationally made, so that the more citations in common, the stronger the relation between documents. However, citation is a complex process and not always rational or

truly subject related, so the results must be assessed with care (Nicolaisen, 2007; Ma, 2020). Citation and co-citation analysis have been much used to produce 'maps' of subjects and disciplines, and combined with other methods to identify the underlying social and intellectual relationships.

Many such studies have been carried out. For overviews, see Borgman and Furner (2002) and Franssen and Wouters (2019), and as examples, see Walton and Morris (2013), Lund (2019), Mokhtari et al. (2021) and Tomaszewski (2020).

Collection management

It is of evident value for managing any collection of documents to have an understanding of the size of the literature of an area, its rate of growth, the types of resources and the most important resources. This is seemingly an obvious case for the use of bibliometric data, particularly Bradford distributions and citation ranks for the development of collections, core lists, etc., relying on the intuitively sensible ideas that any information collection has a relatively small core, but with some material scattered over a wide range of sources, and that highly cited items are worth having. However, practical use seems to have been limited, despite the technique being specifically recommended for this purpose; for examples, see Belter and Kaske (2016) and Kohn and Gordon (2014). Nicolaisen and Hjørland (2007) argue that Bradford's distribution is not as objective a tool as may appear, since a first necessity is to specify a subject for analysis, and the way in which the subject is defined will lead to very different results. Sole reliance on informetrics for collection development has been limited, although it is often used in conjunction with other

methods. The same is true of use of the bibliometric phenomena of *obsolescence* and *half-life*, the rate at which literature loses its value over time, for the maintenance of collections by 'weeding'; this is a complex issue, and supposed obsolescence may be affected by many factors, so that uncritical use of informetric data in practice may be unwise.

Analysing a set of literature

On a smaller scale, bibliometric considerations may be used in desk research, to assist with literature analysis. Most simply, given a set of retrieved literature, attention can be focused on those items cited most often as a way of identifying the most notable items, important or influential authors, institutions, journals, etc. If a search system supports simple bibliometrics, a searcher can quickly summarise factors such as year of publication, language, country of publication, source journal or subject area, and thereby provide a useful synopsis of the content of a large amount of material retrieved.

Assessing impact

The most controversial use of bibliometrics has been in assessing the impact (or quality, influence or visibility) of individuals, academic departments, institutions, grant recipients, countries, journals, websites and more. Research managers and policy makers have been attracted by a low-cost and seemingly objective means of quality assessment. For overviews, see Gingras (2016), Cronin and Sugimoto (2014) and Sugimoto and Larivière (2018).

Counts of published works, especially in sources regarded – often on the basis of bibliometric data – as high quality, and counts of

citations, are the main metrics used. There has been much debate as to how well such measures truly capture 'quality', and how much reliance should be placed on them, as evidenced by regular controversy about their place in national research-evaluation programmes. At issue is the number of possible measures, with over one hundred having been put forward (Wildgaard, Schenider and Larsen, 2014; Hammarfelt and Åström, 2021). Citation rates differ greatly between academic disciplines. Metrics have been derived such as the Field Weighted Citation Index, the ratio of the number of citations to the average for the discipline; use of such metrics to assess individual scholars has been particularly controversial.

For journals, the Journal Impact Factor is taken as a measure of quality: essentially, the frequency with which an 'average article' has been cited in a particular time period, thus offering a quantitative tool for ranking and comparing journals (Tunger, 2021). Other metrics for journals include the *diffusion factor*, which measures how many other sources cite it, and *immediacy* and *half-life*, measuring respectively, the rapidity with which the journal's material makes an impact and how long it remains useful.

A popular measure is the 'h-index', applicable to journals, departments and, particularly, to individuals (Fraumann and Mutz, 2021; Barnes, 2017). This is a measure of overall impact: an author has an index of N if at least N of their publications have been cited N times but N+1 of their papers have not been cited N+1 times. It was devised as a way of avoiding the simplicity of 'citation counts', which can be greatly influenced by a single highly cited paper. It is

now commonly, and controversially, used as a single measure of academic 'quality'.

Because of disquiet about misuse of bibliometric indicators, there has been interest in the use of *responsible metrics*, promoting a thoughtful and balanced use of informetric indicators. Initiatives include the San Francisco Declaration on Research Assessment (DORA) and the Leiden Manifesto (Lasda, 2020).

Summary

Informetrics has developed from simple counting of the attributes of printed resources to a subject in its own right with a high degree of mathematical and technical sophistication, applied in information environments very different from its origins. Its adoption in the practice of library and information work remains limited, and its use for research evaluation is controversial. It is a central topic for information science, now and in the future, and it is important that both researchers and practitioners are fully aware of its promise and limitations.

- The laws of informetrics are well understood mathematically, but their origin remains a matter for debate.
- Appropriate application, for practice and for policy, has lagged behind theory.
- Informetrics is a fundamental part of the information sciences.

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[Authoritative multi-authored handbook]

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12 Information Behaviour

One thing we know now ... is that underlying human propensities with regard to information emerge again and again, as each new technology becomes familiar and its use second nature.

Marcia Bates (2010, 2385)

All people are individuals and will seek and use information in different ways ... [information gathering] is an integral part of our personalities, and we all do it differently. There is no such thing as a homogeneous body of information users.

Maurice Line (1998, 223)

It can be seen that information behavior research has developed along multiple lines, and maintained its popularity. The field still retains themes, theories, and methodologies from half a century ago, and a few of these older approaches remain useful. At the same time it has embraced new perspectives, theories, and methods that would have been considered highly unusual.

Donald Case and Lisa M. Given (2016, 365)

Introduction

Information behaviour, the ways in which it is studied, and the significance for practice of the results of such studies, are central topics within information science. Without reliable knowledge of the way in which people find and use information, provision of effective services can be based only on guesswork and assumptions.

This has been a popular area within information research and one of the richest for the creation of concepts, models and frameworks to explain the complex data produced, and this chapter can only

provide an overview. For more in-depth treatment of all aspects, see Case and Given (2016) and Ford (2015); for a broad approach showing the overlap of information behaviour with information literacy and with HCI, see St Jean, Gorham and Bonsignore (2021); and for a review of information behaviour in the specific context of health, see Zimmerman and Shaw (2020). We will first examine the origins and development of information behaviour studies, with a summary of changing research methods. A summary of information behaviour concepts and a selective mention of significant models and theories follows. The chapter concludes with examples of studies of the information behaviour of different groups and the idea of individual styles of information behaviour.

Origins and development of information behaviour studies

Information behaviour studies have a long history (Case, 2006; 2014; Wilson, 1994; 2008). A specific concern for users of information has been evident from the earliest days of the information professions. Bates (2010) reminds us that Samuel Green, one of the founders of the library profession in the USA, who began his 1891 presidential address to the American Library Association with 'The function of the library is to serve its users', was writing as early as 1876 that librarians should 'mingle freely' with their users and 'help them in every way' (Green, 1876, 78). Green pointed out that one benefit of such mingling was that 'you find out what books the actual users of the library need ... what subjects the constituency of the institution are interested in, and what is the degree of simplicity they require in the presentation of knowledge'. Systematic studies of library use and users began in the early 20th century (McDiarmid, 1940). Case (2014)

distinguishes three phases in information behaviour research: studies of use of library collections, 1836–1935; studies of use of documents and resources, 1936–58; and studies of use of information per se, regardless of the source, 1959–present.

Like much else in the information sciences, research into information behaviour was stimulated by a concern for improved methods for handling scientific and technical information, catalysed by the 1948 London conference of the Royal Society, at which use of documents and libraries was discussed. By the time of the follow-up 1958 Washington conference on scientific information, a significant part of the programme was devoted to information needs and sources. As a consequence, the first major programme of research on the topic was carried out during the 1950s and 1960s on the information needs and behaviours of academics and practitioners in STEM (science, technology, engineering and medicine) subjects, mainly using quantitative surveys. These early studies are reviewed by Menzel (1966) and Meadows (1974). From this first phase of information behaviour research activity, many ideas of wide significance emerged, such as the ‘information gatekeeper’ (a person who acts as an information resource for colleagues) and the ‘invisible college’ (a network for communicating academic and professional information outside the formal publication process).

During the 1960s and 1970s there was an expansion of interest in information behaviour relevant to the social sciences, including business and management – initially that of academic and professional groups in the area and then spreading to an interest in ‘ordinary people’ and to information from outside formal published

sources. Project INISS (Information in Social Services Departments), a major study of information needs and behaviour in UK social services, is a good example and was highly influential in the development of concerns and methods (Streatfield and Wilson, 1982). These developments were fuelled by an enthusiasm, particularly in the USA, for 'evaluation' studies of government programmes to address poverty, inequality, poor education, discrimination, and so on: this provided both new areas for information behaviour studies, and new tools – both quantitative and qualitative – to address them (Bawden, 1990).

The term 'information behaviour' began to be used widely during the 1970s; previously such research had been classed as 'user studies', 'information needs and use', 'communication behaviour', and so on. Coincident with this was a new interest in social science methodology, qualitative studies, and the creation of conceptual models of information behaviour (Ellis, 2011).

Only during the 1990s did attention turn to the information behaviours of academics and practitioners in the arts and humanities. This progress of interest in information behaviours, beginning with the STEM subjects, progressing to the social sciences and culminating with the humanities, is a response to the external factors noted and also reflects the general advancement of information services in different domains. The introduction of online information services, for example, and other technological developments followed the same pattern. From the 1980s much attention was paid to the use of digital tools: online and CD-ROM databases, online catalogues and, later, internet resources.

From 1990 onwards there was an increase in academic interest in the area. This manifested itself not just in a larger number of studies but also in an increase in the range of research methods used and of theories, models and frameworks adopted. The broader concepts of information seeking, information practices, and information experience emerged. Everyday information and information for serious leisure became topics for studies of users and practices, expanding the scope of information behaviour far beyond its original remit of the use of formal information resources and systems by academic and professional users (Case, 2014; Savolainen, 2008; Ocepek, 2018; Hartel, Cox and Griffin, 2016).

Methods for studying information behaviour

We will consider information research methods in [Chapter 18](#), but we should note here that a wide variety of research approaches and methods have been used to investigate information behaviour (Case and Given, 2016, part 4; Greifeneder, 2014). In summary, we can say that surveys of one kind or another have been the most common method, but methods have included questionnaires; telephone and web surveys; interviews, focus groups and Delphi studies; observation; diaries, logs, audio journals and photo diaries; HCI methods, such as eye-tracking; think-aloud; ethnography and phenomenography, critical success factors, critical incidents and vignettes; web log analysis; content and discourse analysis; conceptual and thematic analysis; systematic reviews and meta-synthesis of the literature. At the most detailed, and reductionist, level, neuroscience methods have been applied to investigate the

brain function correlates of information behaviour (Sharot and Sunstein, 2020).

In general terms, there has been a move away from largely quantitative survey methods, almost universal in the first decades of 'user studies', and to greater use of in-depth qualitative techniques as interest has moved towards investigating information behaviour and practices in wider contexts. The focus has similarly changed. Initial studies were generally carried out from a systems perspective, investigating which systems and services were used, by whom and to what extent, with limited consideration of the purpose or success of their use. Later studies considered the information behaviour of individuals from a cognitive viewpoint, while a primarily social viewpoint followed, including detailed consideration of information in the wider context of the life and work of groups and communities. Interest has also been focused on collaborative and sharing information behaviours (du Preez, 2018; Ford, 2015, chapter 5; Hertzum and Hansen, 2019), on liminal or transition behaviours as people change status and roles (Willson, 2019) and on broadening the concept of information behaviour to include 'non-users' and those who avoid or ignore information (Manheim, 2014; Case and Given, 2016, 117–19). There has been a recognition of the link between information behaviour and information literacy, even that they are 'two sides of the same coin' (Limberg and Sundin, 2006). An understanding of the former can provide guidance for promoting the latter.

Information behaviour concepts

Information behaviour itself has been a contested concept. Firstly, there is the grammatical objection that information cannot behave, so the correct term should be 'information-related behaviour'. Though this is correct, the grammatically dubious term is ubiquitous, so we use it here. Secondly, and more substantially, 'behaviour' has connotations of the discredited psychological concept of behaviourism and could be taken to be limited to observable, objective actions, which would greatly limit the scope of the idea. However, in general usage, behaviour is here taken to include cognitive behaviour: needs, opinions, motives, knowledge and so on, and that is the sense in which we use the term. Thirdly, it has been suggested that the correct term should be 'human information behaviour' (HIB), as distinct from the information behaviour of animals, extra-terrestrials, machines, etc. This point is reasonable, since it is well established that many animal species have mechanisms for conveying rich information (Stegmann, 2013), and it has been suggested that the behaviour of complex algorithms should be studied by anthropological methods (Rahwan et al., 2019). The best definition would perhaps encompass the information behaviour of any inforgs, that is, any entities in the infosphere which can generate, gather and use information. That is the sense in which we will use the term, recognising that, to date, studies are largely limited to humans.

Information behaviour, originally understood as limited to interaction with formal information sources and systems, is now defined more broadly as, for example, 'the many ways in which human beings interact with information, in particular the ways in

which people seek and utilize information' (Bates, 2010, 2381) and as 'the totality of human behavior in relation to sources and channels of information, including both active and passive information seeking and information use. Thus, it includes face-to-face communication with others, as well as the passive reception of information as in, for example, watching TV advertisements, without any intention to act on the information given' (Wilson, 2000, 49). Some commentators, for example Case (2006), have warned against too wide a broadening of scope, as otherwise there may be nothing in the information sciences which could not be called information behaviour. Nonetheless, the continuing trend is a broadening of the scope of the concept and related ideas (Savolainen, 2007).

Information practice is a term increasingly used alongside information behaviour. The difference is one of emphasis and perspective, the behaviour perspective coming from a psychological background, while the practice perspective comes from sociology (Savolainen, 2007; Wilson et al., 2009). The former reflects a cognitive viewpoint: what individual people think and know, what information needs they have, and what they do as a result. The latter reflects a socio-cognitive and constructivist viewpoint: how information activities are embedded in the wider social practices of the group, and particularly in the cultural factors affecting information sharing, the repeated actions characteristic of a particular community (Orlikowski, 2002). Information practices will therefore be aligned with the ideas of domain analysis, also a socio-cognitive approach (Fry, 2006).

The concept of *information experience* or *document experience* focuses on the subjective experience of an individual's engagement with information, perhaps in the form of documents, with the aim of giving a holistic understanding of how individuals deal with information as part of their experience of the world (Bruce et al., 2014; Gorichanaz, 2019; Savolainen, 2020).

Several terms have been used to denote components within the overall information behaviour/practice concept. One influential pictorial model is due to Wilson (1999). This shows information behaviour as a wide concept, set within the whole of an individual's life world. Within it is *information seeking*, the purposeful activities of looking for information to meet a need, solve a problem or increase understanding, and within that *information retrieval* or *information search*, the seeking for defined, needed information within some kind of information system.

All of these concepts have received detailed analysis. Savolainen (2007, 2016a) gives a conceptual analysis of information seeking in terms of active/passive and directed/undirected aspects. This is sufficiently broad as to include undirected browsing and information encountering, which are sometimes excluded from purposeful information seeking.

Some authors have identified the additional concept of *information gathering*, lying between information seeking and the broadest idea of information behaviour and covering a variety of means of information acquisition in addition to purposeful seeking (Savolainen, 2007; Solomon and Bronstein, 2021); this might also be taken to include the appropriation of known but unused information

(Huvila, 2019). This is represented visually in [Figure 12.1](#) opposite, with the elements of Wilson's model augmented by information gathering.

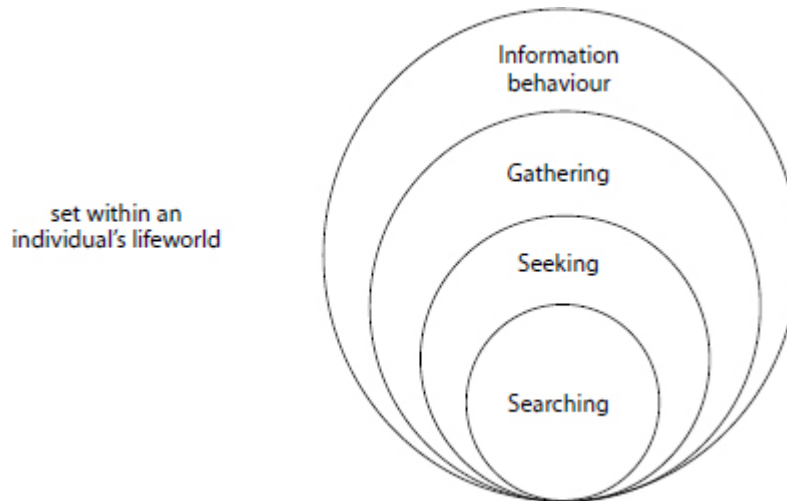


Figure 12.1 *Components of information behaviour*

Information gathering is carried out in response to an information need and is followed by information use. The nature of these two concepts has also been analysed over the years, with no clear consensus.

Ideas of *information need* involve a gap between what one knows and what one wants to know, perhaps associated with a desire to seek answers, reduce uncertainty or make sense. However, this gap may be conscious and explicit or unconscious and implicit; it may be a vague dissatisfaction or a clearly expressed requirement. Nor is the relation between information need and subsequent action to satisfy it well understood. Over four decades ago, Wilson (1981, 5) wrote that 'within the field of user studies the investigation of "information needs" has presented seemingly intractable problems . . . progress towards some theoretical understanding of the concept of

“information need” has been slow’; for summaries over time, see Taylor (1968), Wilson (1981), Case and Given (2016, chapter 5), Savolainen (2017), Borlund and Pharo (2019) and Ruthven (2019).

Information use implies one or both of two things: that the information is used to change an individual’s mental state, closing a gap in knowledge or resolving conflicting ideas; or that the acquisition of information leads to some decision or action, which may include communicating or sharing it. However, this is a relatively little-studied aspect of information behaviour and is far from a simple concept (Savolainen, 2009a; Case and O’Connor, 2016). The context within which information is used, and the information practices associated with that context, will largely determine the nature of information use and the nature of information gathering prior to use.

Information sharing, or *knowledge sharing*, is clearly an important aspect of information behaviour, typically regarded as something which happens, like information use, after information has been acquired. The exact relation between information sharing and other aspects of information behaviour has been regarded in different ways, particularly as to how linear a relation there is between the processes; for analysis, see Savolainen (2019).

Theories and models of information behaviour

There has been a proliferation of theories and models in the information behaviour area, much more so than in any other aspect of information science. A compilation from the early 21st century listed 75 (Fisher, Erdelez and McKechnie, 2005), and more have been added since. Many have been adopted from other disciplines, such

as psychology and sociology, but an increasing proportion have been devised within information science. Only a few have had wide usage, and attempts to combine them have been limited (Savolainen, 2016b). They are, in the terms discussed in [Chapter 4](#), theories for description and explanation, mainly listing the concepts involved and showing their interrelations, often in the form of simple conceptual diagrams or process flow charts. Wilson (2017) categorises them as descriptive, decision based, and causal. We will not attempt to enumerate them, as they have been reviewed fully elsewhere; see Wilson (1999; 2016; 2017), Ford (2015), Robson and Robinson (2013), Savolainen (2016b) and Case and Given (2016, part 3). Rather, we will mention some examples, focusing on those which have been widely recognised, to show their variety.

Descriptive models

Descriptive models list the factors and activities involved in the aspects of information behaviour being considered; they may be presented graphically or just as a list. The earliest of these was a diagrammatic representation of a very general information communication model, showing relations between users and information resources, mediated by technology and tools, due to Tom Wilson (1981). Wilson, a British information science professor who spent most of his career at Sheffield University, has been an innovator in the study of information behaviour, developing a series of influential models (Bawden, 2006; Wilson, 2016), and was instrumental in establishing the series of 'Information Seeking in Context' (ISIC) international conferences.

Ellis's behavioural model is an influential example of this kind (Ellis, 1989; Ellis, Cox and Hall, 1993). This identifies eight features of behaviour within information seeking: starting; chaining (following references, citations, etc.); browsing; differentiating (filtering material by source and quality); monitoring (keeping up to date by checking sources regularly); extracting (systematically working through a source); verifying (checking accuracy and validity of information); and ending. The model does not define the interrelations between the features, nor the order in which they are carried out; that is determined for each situation. This is one of the most tested of all information behaviour models, having been used with studies of groups including scientists and engineers, lawyers, web users and social science researchers, and its features having been embedded in a model for slow information behaviour (Poirier and Robinson, 2014).

Process models

A second class of models, denoted as process or casual models, show the aspects of information behaviour in sequence; typically, these are represented as flow-charts or process diagrams. The first of these is known as Wilson's first model; he acknowledged its limitations, suggesting that it was little more than a map of the area to be explored, with little explanatory power (Wilson, 1981). A similar model was produced by Krikelas (1983).

Other models of this kind focused on occupational groups, tasks and work roles. Byström and Järvelin (1995) focused on the complexity of the task and the expertise of the individual, based on a study of civil servants. The model of Leckie, Pettigrew and Sylvain

(1996) mapped the information behaviour of professionals in broad terms, focusing on role-driven information needs and the characteristics of information sources, and has been used with groups including engineers, lawyers, doctors, nurses and dentists. A very similar model was presented by Freund (2015), based on an analysis of software engineers.

Cognitive models

Later models added cognitive and contextual aspects. The best known of these is Wilson's 'expanded model', which builds on Wilson's first model by including contextual and environmental variables and theoretical considerations, such as stress/coping theory, risk/reward theory and social learning (Wilson, 1997). This model has been adapted for specific contexts, for example the information behaviour of research students and supervisors (Al-Muomen, Morris and Maynard, 2012) and of distance-learning students (Tury, Robinson and Bawden, 2015); for other examples, see Wilson (2016).

Other models involving cognitive aspects were presented by Ingwersen (1996), focusing specifically on the cognitive aspects of information retrieval, and by Choo (2006), taking a wider scope of information behaviour, from needs, through seeking, to use.

Finally, in this category, is the 'information search process' model, derived by Carol Kuhlthau, a library science professor with a long career at Rutgers University. This model, developed between 1980 and 2000, is based on theories of learning and considers six stages of a search process: initiation, selection, exploration, formulation, collection, and presentation. For each of these stages, three aspects

are described: thoughts, feelings, and actions. This is a rather different kind of model from the others: it is focused on the user's subjective experience as the search progresses, has to be understood holistically, and is not usually displayed as a diagram (Kuhlthau, 2003). It has been used in studies of school and college students, and also with occupational groups.

Complex models

A group of more elaborate models introduce a greater degree of context and an increased number of perspectives. These are typically non-linear or multidirectional, rather than having a single sequence of steps. This nonlinear aspect has been most clearly captured in the model devised by Foster (2004) and developed by Foster and Urquhart (2012). This model defines three 'contexts of interaction' for the information seeker: external (e.g. the social or organisational setting); internal (e.g. feeling and thoughts); and their cognitive approach (e.g. openness). They influence three 'core processes': opening (e.g. browsing); orientation (e.g. problem definition); and consolidation (e.g. verifying). Behaviour patterns described in this way are likened to an artist's palette, able to be combined in any way to reflect the dynamic and holistic nature of information seeking.

A model described as 'comprehensive', aiming to include factors not included in previous information behaviour models, was devised to explain the information behaviour of cancer patients, although it has also been used with engineers (Johnson, 1997). Using some of the ideas of Dervin's Sense-Making, it focuses on the factors relating to people and sources and on the reasons for particular forms of information seeking, rather than on the details of

information seeking. It has been used to design interventions to help build information-seeking skills (Johnson, Andrews and Allard, 2001).

Other formulations of this kind include two described as ‘general models’: those of Niedzwiedzka (2003), who developed Wilson’s expanded model by extending it to broader areas of information behaviour and emphasising the character of the activities, and of Godbold (2006), who combined Wilson’s model with Dervin’s Sense-Making concept and with the Fundamental Equation of Brookes (1980). Ingwersen and Järvelin (2005) developed Ingwersen’s original cognitive model in similar ways. The Information Seeking and Communication Model (Robson and Robinson, 2013; 2015) combines elements of a number of information behaviour models with those of models created within the area of communication studies; see [Figure 12.2](#). It is therefore particularly broad in scope, placing equal emphasis on the characteristics of the communicator or provider of information and on the seeker. Savolainen (2016b) evaluates this kind of combination of models.

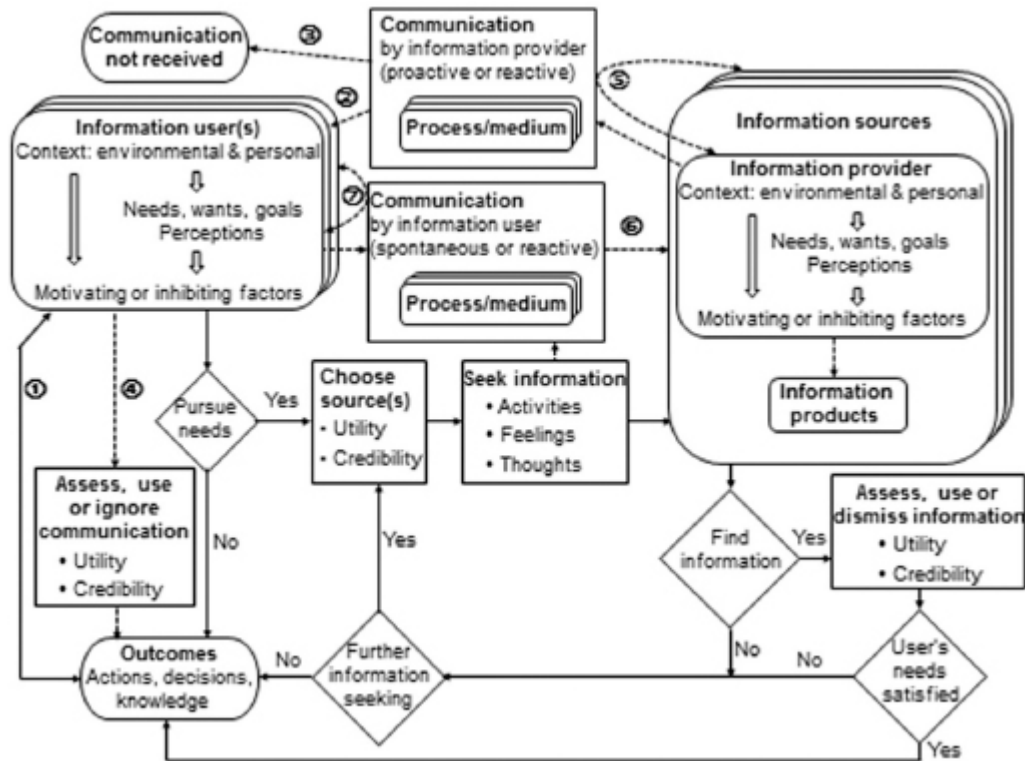


Figure 12.2 *The Information Seeking and Communication Model* (Andrew Robson)

Constructivist models

Another approach to explaining information behaviour has been through theories or frameworks based on a constructivist approach, focusing on the subjective perceptions of individuals as they go about dealing with information. A wide scope is generally taken in what counts as information behaviour; there is certainly no limitation to formal sources and retrieval processes.

Of these theories, the most influential has certainly been Brenda Dervin's Sense-Making methodology. Dervin, a US academic whose work straddles the information and communication sciences, has developed the idea over several decades to provide a general methodology, rather than a specific model or theory, for

understanding how people derive meaning from information, and thereby to understand their information behaviour in the widest sense (Dervin, Foreman-Wernet and Lauterbach, 2003; Savolainen, 2006; Chang and Li, 2020). The ideas have influenced more specific models of information behaviour, some noted above.

There are other examples of this kind of theory, of which we will mention four well known ones. Reijo Savolainen's 'everyday life information seeking' (ELIS) gives a holistic framework for understanding the social and psychological factors underlying the way people use information in everyday settings (Savolainen, 1995, 2008). Elfreda Chatman's 'life in the round' or 'small world' theories similarly seek to explain the everyday information behaviour of people in their social context (Chatman, 1999; Savolainen, 2009b). The related *information grounds* theory, developed by Karen Fisher (formerly Pettigrew), concentrates on the detail of the social contexts in which people find and share information (Fisher, Durrance and Hinton, 2004; Savolainen, 2009b; Rohman, 2020). These are many and varied; as well as libraries and information centres, they include clinics, playgrounds, shops, restaurants and transport settings. Sandra Erdelez's concept of *information encountering* focuses on the ways in which people make unexpected discoveries of interesting information, and the extent to which they actively seek to make this happen (Erdelez, 2004; Erdelez and Makri, 2020).

Finally, we can mention an approach which has arisen by analogy with the biological world. *Information foraging* identifies behaviours in biological terms such as following an *information scent* (Pirolli, 2007; Savolainen, 2018).

Information behaviour of groups

Information research has focused on a wide variety of groups of user, becoming wider as time has passed, from an initial close focus on academic and professional groups. Examples of the kind of groups studied are given below; for more examples see Case and Given (2016, chapter 10).

- By occupation or discipline: scientists, engineers, doctors, nurses, pharmacists, social scientists, humanities scholars, psychologists, industrial managers, journalists, lawyers, farmers, artists, police officers, arts administrators, musicians, historians, architects, teachers, janitors, religious leaders.
- By role: patients, carers, students, researchers, professors, fans, citizens, voters, job-seekers, hobbyists (e.g. cooks, singers, knitters, hikers, runners, genealogists, collectors), library users, shoppers, readers, internet users, consumers.
- By demographic: children, teenagers, women, mothers, transgender people, older people, immigrants, refugees, poor people, homeless people, retired people, temporary employees, inhabitants of particular countries or areas, ethnic minorities.

Individual differences and information styles

It has long been recognised that information users, even with the same subject interests and same education level, cannot be treated as a homogeneous mass. There are individual differences, shown in individual information styles (Line, 1998). However, there is no agreed method for determining such styles reliably.

Several of the models and frameworks mentioned above focus on the individual information user, their actions, thoughts and motivations. An interesting extension of this is to try to determine, and label or categorise, an individual's typical pattern of information behaviour, which we might call their 'information style' (Bawden and Robinson, 2011). Much of the work in this area has analysed the relation between personality factors or thinking styles, and interaction with information systems, such as search engines or e-journals.

There have been fewer examples of studies of individual differences of wider information behaviour; O'Brien, Dickinson and Askin (2017) give a review of research between 2000 and 2015. An early, and typical, example is a study of scientists working at an agricultural research organisation (Palmer, 1991). Cluster analysis of survey data showed the participants falling into five groups: *non-seekers*, for whom information access was not a priority; *lone, wide rangers*, preferring to work alone, reading and scanning widely and relying on serendipitous information discovery; *unsettled, self-conscious seekers*, concerned about missing important information; *confident collectors*, amassing their own information collections, rather than routinely searching for information; and *hunters*, with regular information-gathering routines and a focus on currently relevant information. A more recent example examined the behaviour of hospital patients in respect of information about their treatment; grounded theory analysis of observation and interview data showed five groups: *avoiders*; *receivers*, who accepted information but did not actively seek it; *askers*, who sought information only from their care

providers; *seekers*, who looked for information for various sources; and *verifiers*, who checked information from multiple sources (Costello and Veinot, 2019). Subjective categorisations of this kind may spark insights, and there are a number in the literature; however, they are rarely consistent, and their objective validity may be doubted.

A more systematic approach is to try to determine information behaviour by some kind of survey and then to compare the results with personality as assessed by some psychometric test. An example is the work of Jannica Heinström (2003; 2005), who assessed three patterns of information behaviour – fast surfing, broad scanning and deep diving – which were related to personality factors and learning style. For example, the fast surfing behaviour – associated with a rapid search for a few highly relevant documents, ideally supporting already formed viewpoints – was related to neuroticism, to cautiousness (low openness to experience) and to carelessness (low conscientiousness), as well as to a typically superficial and non-strategic approach to study. These ideas led to identification of five ‘information attitudes’, linked to personality traits, and resulting in typical patterns of information behaviour: invitational; exploring; purposeful; passive; and avoiding (Heinström, 2010).

Summary

Information behaviour has been systematically studied for over 50 years, using a wide variety of research methods, initially largely quantitative (typically surveys) but now largely qualitative (typically interviews and ethnographic observation). Initially, interest was focused on information retrieval, using formal library and

information systems, but interest has broadened to the wider information seeking or gathering, and to the inclusion of all aspects of dealing with information, including creation, sharing and avoidance.

There are many, arguably too many, qualitative theories and conceptual models dealing with information behaviour, but few have been widely applied in research or practice. A few general principles are understood. Choice of sources is determined largely by least effort, familiarity and convenience, and the seeking process by the principle of *satisficing*, getting what is good enough (Connaway, Dickey and Radford, 2011; Warwick et al., 2009; Prabha et al., 2007) and getting it 'just-in-time' (Miller et al., 2021) – arguably, following Zipf's Principle of Least Effort.

Information behaviour has mainly been studied within groups defined by occupation and academic subject, though there has recently been interest in 'everyday life' information behaviour, including 'serious leisure' information practices. Information behaviour is often subtle and difficult to interpret, and there is an interest in considering individuals' information style.

- Information behaviour has been studied for many years, using an increasingly wide range of methods.
- There are numerous conceptual models and qualitative theories accounting for different aspects and contexts of information behaviour, but few general principles.
- Information behaviour is determined primarily by domain and task, but is also affected by individual differences.

Key readings

Case, D. O. and Given, L. M. (2016) *Looking for information: a survey of research on information seeking, needs and behavior* (4th edn), Bingley: Emerald. [The three earlier editions of this book are also useful for older material.]

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13 Communicating Information: Changing Contexts

For millennia libraries have acquired resources, organized them, preserved them, and made them accessible (or not) to patrons. But the forms of those resources have changed – from scrolls and codices; to LPs and LaserDiscs; to e-books, electronic databases and open data refs. Libraries have had to at least comprehend, if not become a key node within, evolving systems of media production and distribution.

Shannon Mattern (2014, 1–2)

We are all patrons of the Library of Babel now, and we are the librarians too.

James Gleick (2011, 426)

Introduction

In this chapter we will consider some of the changes which have occurred, and are still occurring, in the way in which recorded information is communicated, largely as a result of digital technologies but also because of associated economic and sociocultural factors. This amounts to a consideration of the changing infrastructure of the infosphere. It has a wide scope, including publishing, data curation, news media, collection and memory institutions including libraries and archives, and social media. These will mainly be covered in outline, with references to other sources for more detail. The chapter is divided into three sections. The first deals with information chains and lifecycles, simple models which help to understand changes in the communication of information; the second deals with the

communication of knowledge, through publishing in the broadest sense; and the third with information places and spaces, which act as nodes for the communication and sharing of information.

Information chains and lifecycles

The idea of a *communication chain*, or *information chain*, or *information lifecycle* of recorded information is fundamental for the information sciences, and it is also commonly invoked in explanations of information and digital literacies. Such a chain describes the processes of documentation from creation, through dissemination, organisation, storage and use to disposal. They have been applied to information and data, and to journal articles, books, archival records and other forms of document. Communicating information typically falls within ‘dissemination’ or similar steps, but is obviously dependent upon other stages.

There have been many variants of chains and lifecycles used over the years to explain the nature of information dissemination in particular, and the changes in it brought about by technological and other factors (Duff, 1997; Robinson, 2009; Cox and Tam, 2018; Wing, 2019; Yakel, Faniel and Maiorana, 2019). However, the general processes within the chain remain invariant; information resources must be created by someone, disseminated in some way, etc. The communication chain conceptual model, of which a typical example is shown in [Figure 13.1](#), is therefore a simple but effective way of categorising and understanding changes in the information landscape.



Figure 13.1 *The information communication chain*

Borgman (2019) suggests that, in the case of data, linear models imply a single sequence of stages from creation to use to end of usefulness, with the implication that the data can be recreated if needed subsequently, while cyclical models imply the possibility of reuse, and are more applicable for data that may be difficult to recreate, and keep their value indefinitely, such as astronomical observations. This latter implies the active processes of *data stewardship* and *data curation* (also referred to as *digital curation*), which may include elements such as migration between storage media (Oliver and Harvey, 2016; Higgins, 2018). For a detailed specific example, see Darch et al. (2020). This leads to a rather complex lifecycle, such as that of the Digital Curation Centre in [Figure 13.2](#) opposite.

This is a considerable contrast to the original, rather static and linear information chain of printed documents, which typically took the form of:

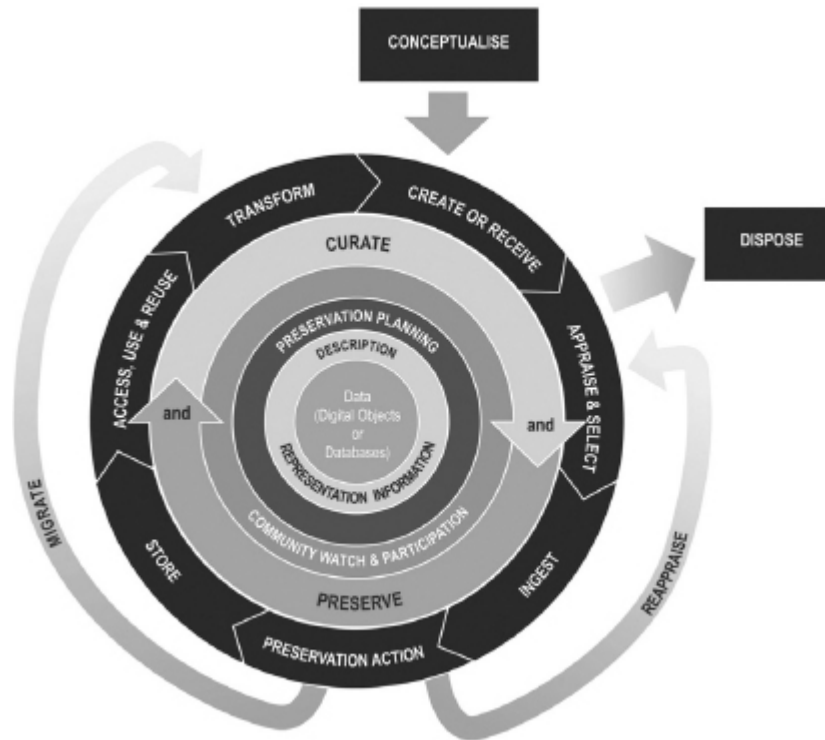


Figure 13.2 *The data curation lifecycle* (Data Curation Centre, CC BY)

- creation (author)
- reviewing and editing (publisher)
- printing, warehousing, and distribution
- providing access (booksellers, libraries)
- conservation and preservation
- discarding or archiving.

Each step in the chain necessarily followed its predecessor, and each was carried out by the appropriate professional group. This kind of simple chain has been considerably subverted by the digital transition. Some stages, such as printing, warehousing and distribution, have been essentially eliminated. Others may be carried out by the same people: an author of a professional or academic

article can now prepare a manuscript, edit it into a suitable form and deposit it, with suitable metadata, in a repository such as arXiv or Humanities Commons. The whole chain is completed by one person, needing only the repository infrastructure to be in place, and access to digital authoring tools. Outside the academic/professional context, the same can be achieved by amateurs; the communication chain has become much more widely accessible. Authors of the fan works mentioned in Chapter 7, for example, can create their own material and deposit it in a repository such as Archive of our Own (AO3) (Hill and Pecoskie, 2017).

One consequence is that the information disciplines are now taking more of an interest in the creation stage of the chain, previously considered to be the province of the domain expert, the journalist, and the creative writer. As examples of information-centred studies of the creation stage, see Gorichanaz (2018), Lee, Ocepek and Makri (2021) and Multas and Hirvonen (2021).

Cox and Tam (2018) have analysed and compared a number of lifecycle models for research data, information and knowledge, pointing out that they necessarily over-simplify a complex process and that other metaphors and models may be helpful in capturing the richness of the ways in which research data and information are created, processed, used, and reused. Other forms of document may have simpler lifecycles; organisational records, for example, are likely to have a relatively clear and predictable cycle.

On the whole, despite their limitations, chains and cycles are helpful models for understanding the disruptive effects of the digital transition as we move to new infrastructures for the infosphere. The

new information infrastructures, and the documents whose communication they facilitate, are digital, mobile, pervasive, immersive, and social. We now outline some of them, distinguishing the communication of data, particularly research data, from the communication of knowledge through publishing in its broadest sense, including the communication of news. There are overlaps between these categories, for instance the relatively recent development of *data journalism*, whereby methods for analysing and visualising data are used to create news stories (Bounegru and Gray, 2021). This is an example of the way in which distinctions between the communication of types of information and of documents are breaking down in the digital environment. In turn, this is an aspect of what Floridi (2017) terms digital's cleaving power, the ability to couple and recouple, to separate and recombine, information and documents in varied ways.

Communicating data

One of the major new roles for information science which has developed since 2000 has been the curation and stewardship of data and datasets. Data is discussed at several points in this book, and here we focus on issues in the communication of data. In the past, it was common for data not to be included in publications; at best it might be noted as being available from the authors on request. However, it is now regarded as good practice to make all data be made available at the time of publication, and this is increasingly required by funders of research and by regulatory bodies. This has required a change in publishing practices, particularly in the scientific literature.

Data may most obviously be published in a traditional way, as a part of a document such as a journal article, or as ancillary data deposited with a document or deposited in a data repository operated by a publisher for all their journal titles. However, this is not always required and many published scientific studies do not present the data supporting their findings (Miyakawa, 2020). There are an increasing number of *data journals*, devoted to the publication and promotion of datasets (Candela et al., 2015; Kim, 2020). Alternatively, datasets may be deposited in an institutional repository or in a subject repository; the latter may be formally constituted and funded, or may be an informal community resource run by researchers themselves.

In order that data may be communicated it must first be collected or created, and then actively curated, in the ways encapsulated in data lifecycles noted earlier. Borgman (2019) makes the point that keeping digital data compilations accessible and usable over time requires active management, such as checking its continued validity, and migrating it as necessary between formats, software versions, and storage media; the ‘benign neglect’ which allows paper documents to remain viable indefinitely in appropriate storage conditions does not apply here. Metadata and data description standards are of obvious importance if datasets are to be combined and reused effectively, but are often lacking (Yakel, Faniel and Maiorana, 2019). An influential set of guidelines is the FAIR (Findability, Accessibility, Interoperability, and Reuse) guiding principles for scientific data stewardship and publication, but adherence to these is variable in scientific publications (Wilkinson et

al., 2016; Wiley, 2020). There are particular issues in the way data and datasets are cited so that they can be subsequently accessed (Yoon et al., 2021).

Communicating knowledge

For several centuries, the dissemination of knowledge in the form of written documents has been the responsibility of the publishing industry. To an extent it still is, although the digital transition has made it possible for authors, and other actors in the communication chain such as libraries, to undertake activities which have heretofore been the sole remit of publishers. This has led to suggestions that a publishing industry, at least as it has been known, is no longer needed. However, publishing, although much changed, is still an important element of the information environment. For a forthright defence of the publisher's role, particularly for academic publishing, see the blog post enumerating 96 things that publishers do (Anderson, 2016).

It is conventional to regard publishing in distinct sectors: *academic* or *scholarly*, publishing materials for research, scholarship and tertiary education; *educational*, publishing materials for primary and secondary education; and *trade*, publishing all other materials. In some presentations, educational publishing is subsumed under trade; academic or scholarly publishing is always distinct. News, both newspapers and broadcast media, are generally treated separately. There is debate as to whether operators of social media platforms should be regarded as publishers. Hitherto they have not been viewed in this way, despite the fact that they are the main source of news and general information for many people; if they

were treated as publishers, it would have consequences for the way their content is regulated.

It is beyond our scope to deal with publishing and the publishing industry in any detail; rather, we will focus on aspects directly affecting the changing information chain and the work of the information professional. For a comprehensive overview of all aspects and sectors of publishing, see Phillips and Bhaskar (2019), and for a scholarly analysis of economic pressures and technological changes to publishing and bookselling practices which have endured for centuries, see Thompson (2012). For overviews of the publishing of books, journals and magazines, see respectively Clark and Phillips (2019), Cope and Phillips (2014) and Das, Stam and Blake (2021).

The communication of news has changed dramatically, with newspapers largely migrating to digital forms and broadcast media being supplemented, and to a degree supplanted, by niche digital sources and by social media. News is increasingly created with the aid of data, algorithms and social media. One casualty has been the newspaper libraries and archives which previously supported the fact-checking whose importance has since been rediscovered by the more responsible news media. This large topic is beyond our scope; for perspectives, see Brock (2013), Lambie (2016), Diakopoulos (2019) and Marconi (2020).

We will focus on four aspects of the changes being brought about in publishing and information dissemination by digital technologies and economic and social factors: the changing nature of published documents; changing models of publishing; open access; and quality and ethical issues.

The changing nature of published documents

The nature of published documents has changed and continues to change in several respects. Most obviously there is a wider range of document types: to the traditional range of printed documents including books, journals and magazines, reports of various kinds, maps and images are added web resources, blogs and social media, videos and podcasts and many more. Most of these are informal and ephemeral, rather than formally published, overlapping with the older idea of informal *grey literature*. A proportion are more substantial and permanent documents, with identifiers such as DOIs, the digital equivalent of the traditional ISBN identifier for single documents, and ISSN identifier for serials, such as journals and magazines (Swoger, 2013). Furthermore, even straightforward digital equivalents of printed documents may come in multiple formats: web page, tablet such as Kindle or iPad variants and mobile versions for use on smartphones. Nor are the changes simple replacements of printed products by digital equivalents. There are only a few examples of such direct replacement: frequently updated reference tools, such as telephone directories and technical product brochures are such. Even for forms which are generally believed to have 'gone digital', such as academic journals, some still offer a printed version. Also, many of the digital materials are a kind of meta to a main document; journal articles, for example, attract social media mentions of the kind captured in the Altmetric doughnut (Zheng et al., 2019; Holmberg et al., 2020).

The form of documents themselves also changes. In the digital environment they may be updated and edited at will, so that the

stability of a published document is a management choice rather than a technical necessity. The digital realm allows greatly enhanced extension of documents into the infosphere, building on the much more limited facilities of the world of print. Images in printed documents can now be augmented by embedded media, leading to fully multimedia documents. The references and footnotes of the printed document are now hyperlinked, so that new virtual documents can be created by combining material from various sources. Immersive experiences, previously provided by the suspension of belief produced by good writing, are now becoming available through augmented and virtual reality, and the unreality of the multisensory digital document (Robinson, 2015).

With the removal of the costs and storage problems associated with printed journals, there is arguably no longer the same necessity for journal editors to restrict what is published by consideration of how 'interesting' or 'original' it is, or how closely it matches a journal's subject orientation. This has led to a new form of journal, the digital *megajournal*, which publishes material over a very wide subject range and accepts material on a basis of 'soundness only'; if there is nothing evidently incorrect or inappropriate in an article, it will be published, regardless of how interesting, original or relevant it may be considered to be. Current examples are *PLOS One* and *Sci*. How popular such journals prove to be, and whether this is the future of journal publishing, remains to be seen. Other suggestions have included publication before peer review, with articles subsequently reviewed, amended and feedback incorporated, and selected articles recommended for particular audiences; see, for

example, Stern and O'Shea (2019). Such ideas have, as yet, been tried out only on a small scale; see, for example, Thelwall et al. (2020) and Tennant et al. (2021).

Finally, the digital transition is amplifying the trend towards 'bitesize' rather than 'long form' reading; see, for example, Tenopir et al. (2015). The tendency, stemming from information overload even in the days of print, to rely on an abstract rather than an article, or a review rather than a book, is amplified by technologies which provide what are effectively new document substitutes. An increasing proportion of web searches are completed without the user ever leaving the search engine, relying on the document 'snippets' provided in the results; for example, Google's presentation of brief sections of a Wikipedia entry.

Models of publishing

Models of publishing are changing drastically – both the economic aspects and issues of how, and by whom, publishing is done. With documents in digital form, many of the former costs of publishing, printing, warehousing and transporting of printed items disappear. Even for printed books, digital software and printing reduce costs, and remove the cost reductions of large print runs. Short print runs of niche titles, print-on-demand and self-publishing all become economically feasible. The traditional purchasing by libraries of individual journals has been largely replaced by journal 'bundles' or 'Big Deals', by which a publisher supplies a collection of journals; for a critical analysis, see Crawford (2014). This has not stopped continuing controversy over the increasing cost of journals, the so-called 'journals crisis', particularly when pricing is less transparent

in the new environment and when many journals included in bundles may never be used (Mongeon et al., 2021). There has been a trend towards libraries attempting to reduce costs by ‘unbundling’ and returning to ordering individual titles (Ivanov, Johnson and Cassady, 2020).

It should be said that the end of the academic journal has been predicted for some time, now that articles are typically found by using search engines on the web or on publishers’ journal bundles, rather than by seeking material in a particular journal. This is not quite as sharp a division from previous practice as might be thought, since, prior to the internet, journal articles were often found through secondary abstracting and indexing services rather than by going directly to a particular journal or publisher. Nonetheless, it does mark a decoupling of article and journal, emphasising the increasing irrelevance of the journal issue and volume in the digital environment. It has been suggested that journals may cease to exist in the future, or may exist simply as collections of material of a particular kind. The popularity within some disciplines of the use of repositories, such as arXiv for the physical sciences, where articles may be deposited ahead of, or instead of, journal publication also suggests that journals might lose their importance. This does not seem to be the case, for the moment at least, and journals are retaining their place as a marker of quality and significance of publication within disciplines and specialisms, albeit sometimes with enhanced functionality and new models of publication; see, for example, Herman et al. (2020) and Kim et al. (2020). Journals which are strongly selective about the quality and subject-relatedness of

their material may also be attractive to someone seeking information outside their own area of expertise as a guide to what is worth reading.

Open access

Open access (OA) in its various forms has caused some disruption to publishing, though not as much (at least as yet) as was envisaged by its early proponents. Largely limited to academic journals, it has led to a mixed environment: subscription journals, typically purchased by libraries; hybrid journals for which a subscription is charged but authors may pay extra to have an article made OA; fully OA journals funded by author payments; and fully OA journals free to authors, produced by volunteer effort and institutional support. There are other OA solutions: through the deposit of articles in institutional repositories, typically operated by university libraries, and in subject repositories, such as arXiv and Humanities Commons; through deposit in commercial repositories, such as [Academia.edu](https://www.academia.edu/) and Researchgate; and through unethical, and arguably illegal, sites such as SciHub. For reviews of the complex and developing OA environment, see Suber (2012), Eve and Gray (2020) and Pinfield et al. (2021).

Attempts have been made to systematise this complexity by denoting various forms of OA by various colour indicators, following the initial proposal of the Budapest Open Access Initiative (BOAI) in 2002:

- *green*: a version of an article in a subscription journal is self-archived in a repository

- gold*: an article is published OA, in a fully OA or hybrid journal,
- with payment by authors or other parties
 - *bronze*: articles may be read freely on a publisher's web page, but may not be reused
 - *platinum or diamond*: articles are free to use without payment by either subscriber or author
 - *black*: articles are provided freely through unauthorised, and possibly illegal, copying.

Though helpful to an extent, this colour designation is not applied consistently and does not cover all possibilities.

The BOAI also made a distinction, still in use, between *gratis* OA, in which an article may be read freely by an individual, and *libre* OA, by which an article may be read freely and also reused in various ways, including by modifying and text mining.

Another example of OA is an extension of self-publishing, which has been made much more feasible by digital media. This is community or crowdsourced publishing, to produce *wiki* resources of all kinds, of which Wikipedia is the best known and has had most impact. Usually without an association with any formal publisher, these sources vary greatly in their activity, quality and reach, but they provide an important form of published material unknown before use of the web became widespread.

Quality and ethical issues

These issues have always been a concern in publishing, with a desire to ensure, so far as possible, that what is published is correct and that plagiarism, wrong attribution, etc. are avoided. Publishers of all

kinds have developed processes to deal with these problems, which are felt most acutely in the publication of scientific research and medical material. The digital transition, on the one hand, makes it easy to identify and to correct problems and on the other hand, introduces new versions of such problems.

Two methods traditionally used by publishers to ensure that quality is maintained are peer review and the facility for retraction and correction. *Peer review* involves the reading and checking of submitted material by independent experts. For journal publication, it is usual for all articles to be peer reviewed before publication. For books, at least a synopsis and sample chapter would be peer reviewed, and in some cases the whole text of a book. Peer review may be *blind*, where authors and reviewers do not know each other's identity, or *open*, where they do. Although universally applied, the process is not without its problems. On the one hand, there are examples of serious errors escaping detection by the reviewers and being published; on the other hand, there are examples of innovative and original work being denied publication by overly conservative reviewers. The digital environment allows new ways of peer reviewing, for example *post-publication review*, whereby the article is published and comments are made subsequently by reviewers for incorporation into a revised version. Digital tools can also assist reviewers by, for example, checking for plagiarism of previously published material. There is also a new need for ways of checking and validating computational research, which is becoming common in many disciplines (Willis and Stodden, 2020). For reviews of the

peer review process and the ways it is changing, see Horbach and Hoffman (2018) and Tennant et al. (2021).

The ease of publishing academic journals in digital form has led to the emergence of so-called *predatory journals*, which promise rapid OA publication for a fee, and with seemingly little review or quality control. This is a controversial issue, since there are some rapid-publication OA journals with perfectly credible review and quality processes. It has also been suggested that it is wrong to label any journal as ‘predatory’, since they may offer a valuable publication route for some authors, especially those from the developing world or who are not associated with an institution.

Retraction of published papers found to be in serious error, and *correction* of those where less serious errors have been identified, has always been possible. In the days of print, this was achieved by sending out errata slips, to be pasted into printed documents. Digital documents can easily be deleted or amended, making quality control easier but, at the same time, causing concern about the stability of the published record. Human attitudes have not caught up with technology in this respect; many authors still view a notification of an error in a published document, and even more so a retraction, as so undesirable that they will seek to avoid it by whatever means they can. For analysis and discussion, see Vuong (2020) and Ajiferuke and Adekannbi (2020).

The digital environment also enables publication fraud, by providing algorithmic means to generate fake datasets and images and even, to an extent, text; an example of large-scale fraudulent publication of this kind is given by Bilk (2020). These problems are

likely to increase in the future, as AI makes production of credible fake publishable content much easier, through systems such as GPT-3, which automatically produce text appearing to have been written by a person (Floridi and Chiriatti, 2020).

A further concern for scholarly publishing is the distribution of languages in which material is published. Publication in a variety of languages, particularly lesser-known languages, has been held to cause a 'language barrier', hindering access to knowledge. However, recent trends towards the dominance of a few languages, and particularly of English, in the publication record is also problematic, particularly for access to locally significant material unavailable in local languages. For perspectives on these issues, see Amano, González-Varo and Sutherland (2016), Kulczycki et al. (2020) and Koller and Thompson (2021).

So far, the ethical issues discussed have been to do with formal publication of books and journals. There are also considerable issues with less formal publication and with social media, largely to do with the detection and removal of false information, fake news and the like; for a perspective on the public role of social media companies and how they may be ethically regulated, see Floridi (2021). The content moderation of social media has itself been called into question as involving the use of low-paid workers, spending many hours each day dealing with the most unpleasant of materials (Roberts, 2019).

Information places and spaces

The digital transition has caused an existential crisis for the 'information places' of our society. This includes both those which

existed to supply information-bearing documents – libraries, information centres, reading rooms, archives and record centres – and also, to a lesser extent, those which dealt with other forms of document – galleries, museums, etc. When documents began to exist to a considerable extent in digital form, there was not the same need for a physical place where such documents might be stored and accessed. What, then, was the future for the information places? The COVID-19 pandemic of 2020–21 focused minds on these issues, as many libraries and other information institutions had to close their physical spaces and pivot to a largely digital service.

One early response was to speak of a *hybrid library*, involving two essentially separate parts, the physical and the digital. While this was a reasonable initial response, it was never going to be stable in the longer term and we can now see that it does not reflect, or accommodate, the onlife merging of the physical and the digital. Another, less sensible, early response was to reduce or even eliminate the physical space of information institutions, on the grounds that if information is digital there is no need for an information place. Related to this, and perhaps more understandable, has been the move to merge information spaces with other activities of an institution. Some university libraries, for example, have merged their space with general IT provision, or with spaces for individual and group learning activities. The latter may be termed *information commons* or *learning commons*, depending on the emphasis on information transmission and on learning more generally (Allison et al., 2019). Some local government authorities have merged their public library spaces with those offering more

general advice and transactions. While these responses may be understandable, and viable in the short term, they avoid the main question: what is the future for the information place, and where does it fit into the onlife infrastructure? One of the effects of digital's cleaving power is to decouple location and presence; in the example given by Floridi (2017) one may be physically located in a coffee shop but interactively present in a social media site. Information institutions need to consider how to address both the location and the presence of their users, and to consider to what extent they should try to replicate the information experience of the situation when location and presence are synonymous.

One valid response is that there will always be a need for some physical places to hold physical documents. Archives, library collections of old and rare books, museums and galleries will all continue to exist; and experience has shown that the printed book is retaining its appeal to a much greater extent than advocates of e-books originally believed. For this reason alone, there will be a need for the physical information place, with a physical collection, for the foreseeable future, however much it may be complemented by digital resources. The question is how to develop an onlife presence, so that the physical and digital naturally overlap rather than compete.

It is highly unlikely that any single way to redesign information places and spaces will be effective in meeting the needs of the new environment. Rather, as Shannon Mattern (2014) points out, there needs to be an understanding of how developing physical, technical

and social infrastructures and architectures of information interact, and how they fit into the information infrastructures of society.

Of course, the discussion above relates to formal or official information settings. There are many and varied unofficial settings for information provision and exchange – sometimes termed information grounds – from bus queues to coffee shops and from doctors' waiting rooms to office kitchens. These are also information spaces, albeit unplanned and unregulated, and the information activities which take place there are also likely to be affected by the move to onlife.

Summary

The communication of information, in all the forms discussed here, is undergoing profound change. In some ways, this is not yet fully evident. While some changes have been clear and definite – the move of academic journals to digital formats and the disappearance of formerly ubiquitous resources, such as the printed telephone directory – the rate of change elsewhere has been slow. Printed books are still very much with us, OA has not driven publishers out of business and many initiatives, such as the new forms of peer review, remain small scale and experimental. It would be a mistake, nonetheless, to minimise the magnitude of the change that is occurring as information communication takes on new forms in the emerging infosphere and new information infrastructures emerge.

- Information chains and lifecycles are useful, simple models for understanding the nature of information communication and the ways in which it is changing.

- The move to online and the infosphere is changing the ways in which all forms of information and knowledge, data and news, are communicated.
- The nature of the new infrastructures that will be needed is not yet clear.
- New ethical problems, and new solutions, are emerging, especially in scientific and medical publishing.

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14 Information Management and Policy

Information policy is a dry and abstract term; it seems remote from everyday concerns and probably outside our control . . . But actually we experience information policy daily, and it affects our life styles, our political choices, our shopping, as well as other issues that we negotiate or navigate our way through, like health or education.

Ian Cornelius (2010, 3)

Knowledge and wisdom, far from being one
Have oft-times no connection. Knowledge dwells
In heads replete with thoughts of other men
Wisdom in minds attentive to their own.
Knowledge is proud that he has learned so much,
Wisdom is humble that he knows no more.

William Cowper, *The Winter Walk at Noon*, Book 6 of *The Task*
(1785)

Introduction

Information management and policy is a multifaceted concept and can be understood in different ways. Sometimes it is understood as a wide and all-embracing concept including records management, knowledge management, library management and so on, and sometimes with a much narrower focus. It ranges in scope from personal through institutional to national and international. It may or may not be taken to include general management processes of budgeting, managing people, etc. There is a close relation with IT

management, since the great majority of information management (IM) uses digital systems; and a similarly close relation with the legal and regulatory framework for proper information use, the area of *information governance*. For overviews, see Detlor (2017), Schopflin and Walsh (2019), Evans and Price (2017) and Byström, Heinström and Ruthven (2019), and for general management issues, see Evans and Greenwell (2020). For insights into the historical development of IM, see Blair (2010), Black and Brunt (1999), Black, Muddiman and Plant (2007), Bawden & Robinson (2010) and Schopflin and Walsh (2019).

In this chapter we consider first some fundamental issues and models within IM and then some important contexts, including the management of collections, libraries, repositories, records, archives, documents, and knowledge. We then review information policies and strategies and ways of auditing and evaluation information services, ending with a brief consideration of some general management topics.

Information management: fundamentals and models

There are no widely accepted general models for information management (IM), although attempts have been made towards developing such models (Macevičiūtė and Thivant, 2019). There are a variety of conceptual models, generally taking the form of block or process diagrams, which have had considerable influence on practice. Some show the relation between data, information, knowledge and similar entities; for example, the data–information–knowledge–wisdom hierarchy, Popper’s concept of Worlds 2 (personal, subjective knowledge) and 3 (objective, communicable

knowledge) and Floridi's Philosophy of Information; others relate to the value of information.

Fundamental to IM is the idea that information and data, and the documents in which they are carried, have a chain or lifecycle of clearly defined stages. Use of such models is termed a *process orientation*, since it involves analysis of the various processes of the lifecycle. There are a variety of such models, and they will vary in detail according to their context and will change in detail with a changing information environment; they are examined further in [Chapter 13](#).

Information management also assumes that information may be regarded, to some extent, as a resource; hence the term *information resource management*, which emphasises this aspect. This assumes that information resources may be identified, counted and their cost, use (actual and potential) and value assessed. Information, considered as a resource, has some unusual and indeed paradoxical qualities. Information has multiple and unpredictable lifecycles, so that its use and value does not change with time in a regular manner; it is 'diffusive', tending to leak from any system of control, and the more it leaks the more there is; it is sharable, not exchangeable, it can be given away and retained at the same time; its value is subjective, depending upon its context and intended use by particular persons on particular occasions; and, rather than being a static entity, it is a dynamic force. For more discussion of these issues, see Eaton and Bawden (1991), Yates-Mercer and Bawden (2002) and Budd (2011).

Although the objective value of information may be difficult to determine, any approach to IM must make some attempt to do so, or there will be little rationality behind management decisions. It is also assumed that it is possible to add value to information, by appropriate process. An influential model for understanding *value-added processes* in information lifecycles was introduced by Robert Taylor (1986); see Foster and Clough (2018) and Macevičiūtė and Thivant (2019) for recent adaptations. It provides a framework within which value-adding processes – such as classifying and relating, evaluating and interpreting, presenting options, and making choices – are carried out within information systems of all kinds. Information management is, to a large degree, a matter of enabling these processes to occur in an efficient manner.

Information management is usually regarded as an organisational issue, but there is a well-established concept of *personal information management* (PIM) (Whittaker, 2011; Hajibayova, 2019; Feng and Agosto, 2019). It is of relevance here because effective organisational IM relies on a knowledge of the personal information practices of the individuals it serves, and can also support good PIM in its users.

Data management

Involving the management of structured datasets, this is usually taken to encompass database systems, data modelling, metadata, data quality and security issues and the maintenance of data warehouses, archives and repositories; see, for example, Gordon (2013). The management of large datasets has provided a new role for library and information specialists (Buckland, 2011; Cox et al., 2020) and has given rise to the new specialism of *research data*

management (RDM) (Cox and Verbaan, 2018). Before this development, however, data management was seen as lying on the boundaries between LIS and computer science and information systems, and perhaps more naturally belonging to the latter. This led to a split, with documents and data typically being managed through different processes and with different systems. In some sectors, however, information services have a long involvement with data management, the pharmaceutical industry being an evident example (Bawden and Robinson, 2010). The split between data management and other forms of IM may be expected to diminish or disappear in the near future, particularly as technologies close the rather artificial gap between the handling of data and of information. The processes of data management, *data curation* and *data stewardship*, and the *data lifecycle*, are discussed in [Chapter 13](#).

Document management and content management

These are terms which are often used synonymously, with content management sometimes implying a wider scope (van der Meer, 2017; Svärd, 2017). They refer to processes of creating, revising, tracking, disseminating, and storing electronic documents, which may sometimes be images of paper documents. This is usually in the context of publishing, in a very broad sense. It may be ‘traditional’ publishing, creating e-books, e-journals and so on; or it may be the generation of internal documentation, such as procedures or handbooks; or of documents for specific external dissemination, such as regulatory documents in the pharmaceutical industry. *Digital resource management* deals specifically with ‘born digital’ materials; *web content management* is a specific example of this. There is a

relation with the forms of IM more familiar to information practitioners, in that content management systems both draw material from databases and store their products in repositories and institutional intranets (Lustigman, 2015).

Library and repository management

These are concerned with the acquisition, organisation and storage of a collection of documents and with services based on such collections.

The term *library management* is often used to refer to the ‘general management’ issues involved in running a library, perhaps with some sector-specific issues included. Evans and Greenwell (2020) distinguish three main groups of topics: managerial functions (accountability and power, planning, quality control, decision making, change management, communication, and advocacy and marketing); managing people (staffing, enhancing performance, managing diversity and motivating staff); and managing things (managing money, generating income, managing technology, and managing facilities).

A somewhat different meaning is that encapsulated by Hayes’ (2001) classic text on library management and planning: quantitative models for optimising workloads and staffing levels and for assisting management decisions generally. The phrase has also been used in the context of the technology of *library management systems*.

Related to the idea of IM as based on an information chain or lifecycle is the well-established ‘collection-centred’ idea of library services; see, for example, Chowdhury et al. (2008). These can be

summarised as: select and acquire; organise, classify, index; provide access and circulation; reserve and conserve; assist and advise users; weed and dispose. Traditionally, these kinds of library management activities were divided into technical services and public services. These distinctions are increasingly blurring, because of changes to information lifecycles and chains brought about by changing technologies, as discussed in [Chapter 13](#); in particular, a move to the ‘inside-out’ collection, discussed in [Chapter 10](#).

The management of largely or fully digital libraries follows this same pattern, adapted to the digital environment. *Repository management*, dealing both with institutional repositories, typically operated by universities, and with disciplinary repositories, such as arXiv and Humanities Commons, is similar (Arlitsch and Grant, 2018; Digby and Phillips, 2019). The process of selection at the start is replaced by a task of encouraging the deposit of relevant materials, and, as a repository is a de facto archive, weeding and disposal are not usually necessary.

Memory institutions’ information management

Items in collections of museums, galleries, and other heritage institutions may be regarded as documents in their own right. This brings the necessity for their description, categorisation and indexing in a way analogous to that for other forms of documents in libraries, archives, etc. More than four decades ago Roberts and Light (1980, 42 and 44) were stating that ‘museums throughout the world have an overwhelming documentation problem ... every museum needs a documentation system for collections management purposes’. The processes involved have usually been described as

museum documentation, as exemplified by the work of the UK Museum Documentation Association, founded in 1977. A particular issue is the need for specialised vocabularies for describing this kind of material. Otherwise, the processes are much the same as in other contexts, especially archives, with weeding and disposal of less relevance. An account of the principles of museum IM, though dated in detail, is given by Orna and Pettitt (1998); see also Marty (2007), Engel and Wharton (2017) and Jones (2018).

Archives and records management

These aspects of IM are often treated together, although their emphasis is rather different. Records management deals with relatively modern documents created within an organisation for its own purposes and with their use, storage and disposal. Archival management deals with the selection and care of documents to be stored for a long period, typically for legal or historical purposes.

For overviews, see Yeo (2018) and Shepherd (2017); for records management, see Franks (2018) and Oliver and Foscarini (2020) and for archive management see Millar (2017) and Hunter (2020). For reflection on the significance and future of archives, see the volumes edited by Thomas et al. (2017) and by Brown (2018).

Management of records and archives typically happens within an organisation, whether governmental, commercial, educational or non-profit. At national level, national archives may be said to keep the national memory and may also act as a source of expertise for records and archive management.

Documents are regarded as *records* when they are created or acquired by an organisation as part of a business process, so that their existence provides evidence of the fact that the process took place. Most records are created internally, but incoming messages and other documents, if they are used as part of a business process, are also records. They are distinguished from *reference material*, other documents which are used by the business but which do not qualify as records according to the above understanding. These are not kept with the records collection.

Records management

A record therefore has two main purposes. While it is active it supports the business process for which it was created or acquired; later it serves as a memory of the past with respect to that process. In some cases this latter function is so important that the record is kept indefinitely; this is the function of archives.

Records management is usually understood to be the systematic and consistent control of records throughout their lifetime, as specified in international standards (ISO 2016; 2020). More than most other kinds of document, records have a distinct and predictable lifecycle, encompassing the stages from creation or acquisition to circulation, use, storage and final destruction or preservation.

It is usual to categorise records in three stages over their lifetime: *active records* are those which are in use, needed for the purposes of the organisation; *semi-active records* are those which are no longer in use but which must be retained and kept accessible for a defined period of time; *inactive records* are those which are no longer required, and need not be kept. When a record becomes inactive, the

decision is made to destroy it or to preserve it. Those records which are preserved permanently – for historical, cultural or reference purposes – become *archival records* and are dealt with by the methods of *archive management*.

The basic processes of records management are therefore: categorisation, organisation and indexing of records; storage of records appropriately, so that they are secure and can be retrieved when needed; providing access to records; retaining records for so long as they are needed and then destroying them or sending them for archival preservation. The processes of archive management are similar, except that the last of the four points becomes: retaining archival records indefinitely or until archive policy with respect to particular records changes.

Records rarely if ever occur singly, and the basic unit to be managed is the *record series*, a collection of records of similar nature. This may in turn be divided into *files*, usually the unit which is stored and retrieved, and within files are individual documents. Records which are important to the continuance of the business are known as *vital records*. An issue of particular importance in records and archives management is that of *provenance*; by whom was the record created or acquired, and in what context. This will determine how it is treated. The *principle of original order* states that records should be maintained in the order in which they were first created and arranged. Archival management relies on the idea of *fonds*, or archive groups: the whole set of records, in whatever form or medium, created, collected or used by a person, family or organisation, which should be kept together.

Both physical and digital records are generally arranged according to a *file plan* or *file classification*, based on the functions and processes within the organisation which the records support and reflect. Accompanying the file plan is the *retention schedule*, which specifies the lifecycle of each record series, in particular, how long it is to be kept at each stage and how it is finally to be disposed of, i.e. destroyed or archived. This is usually accompanied by a *records retention authority*, a formal policy document which specifies the retention schedule and the reasons for it.

Secure preservation of records is of obvious importance, since their loss can cause the failure of the business on which they rely. The move to digital records and archives has brought advantages and problems. Issues of physical storage and the problems of records which may fit into two or more places in a file plan become less pressing. There is no longer the issue of keeping current records separate and more immediately accessible than non-current, or making particular provision for vital records. Nor is there any need to make duplicate copies, for whatever purpose. Conversely, issues of records held on obsolescent storage media, 'format obsolescence', can cause problems, particularly if it is held that transferring the records to newer media means that they are no longer the originals.

Knowledge management

The idea of knowledge management (KM) originated in the 1980s and came into prominence in the 1990s as a widely promoted business concept, enabling the capturing and sharing of the 'know how' of individuals within an organisation; Koenig and Neveroski (2010) give an account of its origins. It is associated with

organisational learning and knowledge mobilisation. Guidelines and requirements have been encapsulated in an international standard (ISO, 2018).

It is, in essence, the management of ‘tacit knowledge’, the subjective knowledge in people’s heads which it may be difficult to explain to others and which is often not recorded in documents. This is sometimes categorised as personal, rather than propositional, knowledge: ‘knowing how’ rather than ‘knowing that’. It is the kind of knowledge gained by experience rather than by reading and formal education. It is what Popper would claim belongs to his ‘World 2’, although KM theorists usually prefer to refer back to Michael Polanyi’s (1962) theory of knowledge (mentioned in [Chapter 3](#)) encapsulated in his dictum ‘we may know more than we can tell’; see Day (2005) and Baumard (2010) for the significance of Polanyi’s views for KM. The ‘bridging epistemologies’ idea of Cook and Brown (1999) has also been influential.

There are usually three elements to a KM programme, though one often predominates, according to context: content, the type of knowledge to be shared, and the way in which it is represented; technology, the means by which it is shared; culture, promoting the sharing of knowledge.

Two main approaches have been adopted. One, seeing KM as an extension of conventional IM, focuses on creating ‘know-how databases’, to which members of an organisation contribute their knowledge. This shares many of the functions of IM, such as knowledge resource audits by analogy with the information audit.

The second, considering that personal and tacit knowledge, by its nature, cannot be captured in this objective way, focuses on putting people in touch with one another and enabling discussion through tools such as *indexes of expertise* or *corporate yellow pages*. Popular methods are the *community of practice*, a group of practitioners who come together informally to share knowledge and experience, and formats for encouraging learning and knowledge exchange, such as *knowledge cafés*.

KM has generated a large literature of its own; for recent texts, see Dalkir and Beaulieu (2018), Hislop, Bosua and Helms (2018), Massingham (2019) and Milton and Lambe (2019). For an integrated treatment of information and knowledge management, see Schopflin and Walsh (2019) and for examples of KM in a variety of organisations, see Collison, Corney and Eng (2019).

This is, of course, largely focused on Western concepts of knowledge: the indigenous knowledge of the developing world may take very different forms. Nonetheless, there has been concern for using ideas of KM to capture and preserve such knowledge and ensure that it is made, and remains, accessible; see Lwoga (2011) and Maina (2012).

Business intelligence and environment scanning

Rather than a form of IM, these are distinct activities which are often associated with IM and KM (Macevičiūtė and Thivant, 2019). Business intelligence is somewhat broader than the *competitor/competitive intelligence* (CI) activity, from which it has grown. It involves a systematic analysis, from sources both formal and informal, and both internal and external, of changes in the

environment in which a business operates; certainly including the activities of competitors, but also including legal and regulatory issues, economic, demographic, social and political issues, etc. Usually associated with the commercial world, it is increasingly practised by educational and other non-profit organisations. An information service may be asked to undertake this work, or its results may be integrated with an IM or KM service.

Environmental scanning is the acquisition, organisation and dissemination of information about events, issues and trends in an organisation's external environment, used to help plan the organisation's future course of action. It may be closely related to knowledge management and other information functions (Holland, 2015).

Collection management

This involves the management of materials in a variety of *collection institutions* or *memory institutions*, including information centres, libraries, repositories, archives, record centres, museums and galleries. These collections will often comprise digital or printed documents, but may also include items such as historical artefacts, art works, botanical, anatomical or geological specimens, etc.; such items will have documentation associated with them, as well as being documents in their own right. There is a continuing debate as to whether collections are the defining and central feature of such institutions or whether equal or greater priority should be accorded to services, communication and knowledge creation; see, for example, Given and McTavish (2010) and Lankes (2016). Regardless

of this, collection management remains central to information institutions.

Collection management comprises ‘all the activities involved in building and managing . . . collections in all formats and genres, both locally held and remotely accessed’ (Johnson, 2018, 1). A distinction may be made between *collection development*, the building of a collection, and *collection management*, its ongoing maintenance. Four fundamental questions underlie all collection development and management processes: what purpose/s does the collection serve; for whom is the collection being maintained; what type of resources or items will be collected; and how best can the collection be organised?

The processes of collection management will differ in nature and emphasis according to the context, but usually include: selection, acquisition and processing of material so that it is integrated into the collection; preservation and provision of access; assessment of the quality and adequacy of the collection and evaluation of its use; and weeding and disposal. These general processes have been the basis of collection management for many years, although changed in detail, in response to changes in the technological, economic, social and legal environment; see, for example, Curley and Broderick (1979), Clayton and Gorman (2001), Fieldhouse and Marshall (2012) and Johnson (2018).

Changes in the environment have affected collection management most in the areas of selection and acquisition. *Selection* is greatly affected by the wider range of documents, media and formats available. This is not merely a matter of e-books and e-journals complementing or replacing their printed equivalents, but of entirely

new document forms being eligible for collection; examples are art documentation (Engel and Wharton, 2017), and e-mail archives (Dinneen and Krtalić, 2020). It is also influenced by the ready availability of data from digital access systems, allowing *patron-driven*, *demand-driven* or *data-driven* selection, amounting to essentially the same thing, the semi-automated selection of materials in high demand (Blume, 2019). The social justice movement, a change in the ethical environment far wider than the information sector alone, has led to initiatives to *decolonise* the collection; this affects not merely what is selected and weeded but how it is presented and described; see, for example, Blume and Roylance (2020).

The digital transition has moved the *acquisitions* process towards licensing for access rather than purchase for ownership. This has created a complex and specialised area of IM; see, for example, 'Vendor relations, negotiations, and contracts', chapter 5 of Johnson (2018). Other changes have included the controversial 'big deal' or 'journal bundle' subscription model and the similar use of 'aggregators' (Strieb and Blixtud, 2014), a greater interest in collaborative and co-operative collection development (Levenson and Nichols Hess, 2020), electronic legal deposit (Gooding and Terras, 2020) and the impact of OA resources, which have had an uncertain relation with institutional collections (Ball, Stone and Thompson, 2021).

Information governance

Information governance is an umbrella concept, of relatively recent origin, referring to the sets of policies, processes, standards and

guidance on good practice which are applied within an organisation to ensure that all stages of the information lifecycle are managed correctly. They ensure that the organisation observes all relevant laws and regulations, avoids risk and uses its information in a way which is effective, while being legally and ethically correct (MacLennan, 2014; Saffady, 2017; Smallwood, 2019). Information governance and IM are closely connected.

The move to a digital information environment, with its consequences for better use and, equally, more misuse of information, has pointed up the need for effective governance. Floridi (2018) denotes *digital governance* as involving policies, procedures and standards for the proper development, use and management of the infosphere; it is distinct from regulation and ethical behaviour, though it overlaps and complements both.

The concept of information governance is interpreted differently according to the organisation's context. Usually there is a focus on the legal framework for IM; legislation on matters like data protection, freedom of information, privacy, copyright, intellectual property and so on, together with any sector-specific regulations or issues affecting the way information is managed. In health services, for example, information governance is particularly concerned with safeguarding information about patients and ensuring that it is used to best effect in planning treatment.

Correct dealing with information often involves a formal assessment of the quality of procedures; the information governance concept largely subsumes concerns which were previously expressed as quality control and application of standards in IM; see

Brooks (2019) for the role of records management principles in information governance and Haynes (2018) for the applications of metadata in IM and information governance.

Information risk is a concept closely related to information governance, stemming from the realisation that information problems can become serious business problems, can result in financial penalties and other sanctions and can – in extreme cases – threaten the survival of a commercial organisation or the credibility and ability to function of a government department. These typically arise from the release of information which should remain confidential, or from the loss of vital information (Saffady, 2020).

Information policies and strategies

Policies and strategies, plans and objectives, mission and visions form the framework, albeit one with a very inconsistent use of terminology, for IM; for overviews from different perspectives, see Orna (2004), Cornelius (2010) and Evans and Greenwell (2020, chapter 7).

Information policies may be applied at widely different levels: from a small department or function within an organisation to a large organisation, a grouping of organisations within the public or private sector (e.g. 'UK higher education'), a region, a nation, a grouping of nations or even the whole world.

Organisations began formally to state information policies in the 1980s, coinciding with the widespread adoption of what were then termed 'new ICTs' (information and communication technologies). Governments had begun to state national information policies in the

previous decade, though elements of what would later be termed information policy can be traced back to at least the start of the 20th century. All countries have de facto some information policy provision, but only a minority have a clear and fully explicit statement of such; the UK has never had one. National policies tend to focus first on technological infrastructures, second on skills (often in a rather limited 'computer literacy' context) and third on the legal framework, covering copyright, access rights, etc. See Hernon and Relyea (2017) for an account of the historical development of information policy in the United States and Rabina and Johnson (2017) for a similar account for the European Union.

At the outset of any policy statement there may be an explicit statement of *mission*. This is a summary of what the information service is and does, from which everything else can be derived. Mission statements are typically very brief and are often accompanied by a statement of *vision*: where the service would like to be and what it aspires to.

The next level down is the *policy*, or set of *purposes* or *principles*. An information policy is usually a tool for developing information strategies. It is a statement of general ideas rather than details, typically fairly short, and is 'meant to last'; a policy will not normally be changed frequently. Policies may also be devised to cover particular aspects of the overall activities of library and information services; collection policies, IT policies and privacy policies are common.

Below the level of policy, typically come a number of *strategies* or *priorities*. These state in some detail how the policy and purposes

will be achieved, and provide a detailed framework within which IM is carried out and are accompanied by specific *objectives* – verifiable and achievable aims. A strategy will often be written to cover a defined period of time, typically five years, and may be reviewed and modified frequently as circumstances change.

The term *plan* is used inconsistently. Sometimes, typically in the form of a *strategic plan*, it will be a high-level concept, similar to what is more often called policy, though with more concrete objectives. In the form of a *business plan* it will focus on financial and budgetary predictions and objectives, though it may include higher-level issues, such as overall purpose and vision. Fox (2018) gives a guide to the implementation of business plans for library and information services, including the use of analysis tools such as SWOT (Strengths Weaknesses Opportunities Threats), PESTLE (Political Economic Social Technological Legal Environment) and the Boston Matrix in developing such plans.

The British Library's *Living Knowledge 2015–2023* document is a good example:

To make our intellectual heritage accessible to everyone, for research, inspiration and enjoyment. [mission]

A vision of the British Library in 2023 as the most open, creative and innovative institution of its kind in the world. [vision]
[policies]

1. Custodianship. We build, curate and preserve the UK's national collection of published, written and digital content.
2. Research. We support and stimulate research of all kinds.

3. Business. We help businesses to innovate and grow.
4. Culture. We engage everyone with memorable cultural experiences.
5. Learning. We inspire young people and learners of all ages.
6. International. We work with partners around the world to advance knowledge and mutual understanding.

[example of priorities]

Our priorities for 2015–2023 to support our Business purpose are:

Work with partners to secure funding to grow the network of regional & IP Centres to a total of 20 UK city libraries.

Develop and open up our St Pancras campus to maximise its potential for knowledge exchange and innovation at the heart of the Knowledge Quarter.

User needs analysis and information audit

For effective IM it is necessary that the needs of the users of an information service are understood and that these are mapped against a full knowledge of the information resources available to meet them. These are the processes of user needs analysis and of information auditing and mapping, respectively.

Information needs may be those of an individual, a team, an organisation or a profession. They may be expressed explicitly (*what I want to know is . . .*) or implicitly (*I'm a doctor [so obviously I need to know . . .]*). Without a clear understanding of the needs of the users of a service, it is impossible to rationally plan the service's activities or to evaluate its effectiveness. Information needs analysis is treated in

detail by Dorner, Gorman and Calvert (2015). There has been debate about the difference, if any, between *information needs*, *information demands* and *information wants*. Case and Given (2016) discuss the idea of information need, and its complications and the various definitions and models derived to explain it, and Ruthven (2019) gives a detailed analysis of various levels of information need.

A number of frameworks have been devised for information needs analysis. One long-standing and influential example, originally due to Nicholas (2000), examines the required information by a series of characteristics: subject; function; nature; level; viewpoint; quantity; quality/authority; date/currency; speed of delivery; spatial coverage; and processing/packaging. An example of a use of this framework is a study of users of a music library (Inskip, Butterworth and MacFarlane, 2008).

One of the criteria of a resource of any kind is that it may be audited within an organisation; the financial audit is most familiar, but other forms of audit are common. Information resources can be audited, and the *information audit* is a staple tool of IM. At its simplest, an information audit may be a listing, as complete as possible, of the information resources available, with a note of who is responsible for maintaining them. Their cost will also usually be included. More ambitious audits will attempt to establish the use and value of the resources and the effectiveness with which information is brought into the organisation and communicated within it.

Details and examples of information audits are given by Orna (2004), Buchanan and Gibb (2007; 2008) and Frost and Choo (2017).

A variety of methods may be used, including examination of documents and systems, observations, interviews, focus groups and questionnaires and measures of user satisfaction and impact.

Information mapping or *information flow modelling* is a form of information audit which focuses on information flows within an organisation and identifies where information is missing or difficult to access. The same methods are used to identify these factors, as with information audits. Graphical means are typically used to visualise the results, usually in the form either of *rich pictures* or of more austere *data flow diagrams*; examples are given by Durugbo, Tiwari and Alcock (2013).

Valuing information and evaluating information services

The demonstration of value and impact of information services is an integral part of IM and information governance (Goulding, 2020; Streatfield, Markless and Paley, 2020). A variety of quantitative and qualitative measures may be used; these will be most effective if they are tied in with key performance indicators, comparable across institutions, and with quality standards. For overviews of these issues see Hernon, Altman and Dugan (2015), Appleton (2017), Matthews (2017) and Moss and Thomas (2019); an extensive bibliography is given by Poll (2018).

The value of information

While the value of information can be simply stated to be the amount that someone would pay to have the information at a particular time, for example to make a decision, there is no generally applicable way to assign a reliable monetary value to information,

although the question has been extensively analysed over a long period; see, for example, Braman (2006), Leppälä (2015) and Stehr and Adolf (2016). This applies to specific pieces of information, such as facts provided in answer to a reference query or documents retrieved from a database, to collections of information and to the totality of an information service.

This is due to what Koenig and Manzari (2010, 4305) identified as the ‘elastic and amorphous nature of information and information services and the difficulty of quantifying them and in turn measuring their impact’. Although the *cost* of providing a particular item or collection of information can be known accurately, its *value* will depend upon the circumstances of its use and their consequences, which cannot usually be known in advance. Furthermore, it is difficult to determine exactly what results may be said to be due to some particular information being available.

This has led in some cases to the abandonment of any attempt to provide a quantitative value for information, and reliance on an assessment of the information’s correctness, completeness and relevance to an organisation’s purpose or on asserting a broad, qualitative educational or moral value. Value may also be associated with the use of information, or its fitness for use, and assessed by the extent and adequacy of processes which increase its potential value to a user. This was first put forward systematically in Taylor’s well-known ‘value-added’ model (Taylor, 1986), which may be updated in the new networked information environment (Foster and Clough, 2018). Alternatively, some surrogate measure may be used, such as time saved, which may be related to cost or to expressed user

satisfaction. These approaches, however, are not fully convincing, and ways of assessing numerical and cost values are constantly being sought, without resorting to simplistic counting of what is easy to count: 'social value is all about getting the big picture – do not limit the estimate only to the easier financial estimates . . . the emphasis is on social value, and the big picture of how local library services contribute to the outcomes of importance to their employing organisations' (Urquhart, 2020, 3).

Reviews of the ways in which library and information services have attempted to show return on investment in financial terms are given by Aabø (2009), Madden et al. (2016), Urquhart and Turner (2016) and Urquhart (2020).

An approach increasingly widely adopted is the idea of *contingent valuation*, which takes the view that the value of information is what someone would be willing to pay for it, and surveying users as to what they perceive as the value to them for services provided. These methods have been used increasingly widely, as can be seen by examples from various library sectors – national (Pung, Clarke and Patten, 2004), academic (Kingma and McClure, 2015), public (Stejskal, Hájek and Reháč, 2019) and special (Satterley and Woelhaf, 2020) – and archives (Moss and Thomas, 2019). Despite some acknowledged shortcomings, contingent valuation methods seem to be among the most acceptable means at present of evaluating the cost-benefit of information services.

Away from the library sector, there has been a movement towards treating information more generally as a quantifiable asset in

monetary terms, for example in company accounts; as an example, see Cummins and Bawden (2010).

Effectiveness and impact

Evaluation of the effectiveness of information provision is an intrinsic part of IM. There is a long history of such evaluation; see Lancaster (1993) for an influential summary of practice in earlier years, Rapp (2008) for an account of early evaluations at the US National Library of Medicine, a leader in this area, and Tenopir (2012), Urquhart and Turner (2016), Matthews (2017) and Urquhart (2018) for surveys of more recent practice.

All evaluations must 'measure' something, either by quantitative or qualitative assessment or by a hybrid approach. A useful six-way typology of what may be measured, originally suggested by the American library science professor Wilfred Lancaster, is as follows:

- *Cost*: measures what a service costs to provide, a collection to purchase, etc. This can usually be known with accuracy, and replacement costs can be assessed by standard financial techniques. It will be included in an information audit.
- *Effectiveness*: shows how well the system is working, compared to what it is expected to do. It may often be measured quantitatively, using, for example, performance metrics or expressed user satisfaction.
- *Benefit*: implies a knowledge of the 'true' value of a system or service and is notoriously difficult to measure. It is usually approached by qualitative or semi-qualitative measures.

- *Cost-effectiveness*: attempts to relate the measured effectiveness of a system or service with its known costs. Usually this approach is used on an isolated component of a total information service, and usually in a comparison between two alternatives, e.g. whether to subscribe to information from provider A or B, or whether to outsource some function.
- *Cost-benefit*: attempts to relate the cost of providing a service to its 'real' benefits, through methods such as impact studies and contingent valuation.
- *Cost-performance-benefit*: aims at the investigation of the whole set of relationships between the costs, performance and benefits of an information service. Few convincing studies of this type have been carried out.

Performance metrics

These metrics give quantitative measures of effectiveness, which allow comparison within one information service over time, or between information services if standard sets of performance indicators are used. An international standard lists indicators meant for various types of library (ISO, 2014a), while another lists indicators specifically for the national library sector (ISO, 2019). Many national sets of indicators have also been created in past years, though there is an increasing tendency to rely on the international standards. For a detailed discussion of information performance metrics, see Appleton (2017).

These indicators are quantitative measures of service effectiveness, such as: number of loans; occupancy of seats in reading rooms;

number of accesses to e-resources; availability of required items; speed of retrieval from closed stacks; cost per loan; and rate of satisfactorily answered reference requests. They have the advantage that they should be available from a library's management information with a minimum of extra work. Drawbacks are that they are oriented towards traditional paper-based library and information services and that they can give little qualitative insight into the reasons for the adequacy or otherwise of the service. The data also needs to be interpreted with sensitivity to user needs: the relative importance of the speed of an interlibrary loan and of its cost, for example, can be very different for a medical library serving a critical care department and for an academic departmental library serving mainly humanities research students.

User satisfaction, in a necessarily rather limited quantitative fashion, is often included in such metrics. In some sectors, particularly academic libraries, a standard format for assessing this factor is used for comparability. The SERVQUAL protocol, a standard service industry instrument, has been widely used for this purpose, as has LibQUAL+, a protocol derived from SERVQUAL, specifically devised for assessing user satisfaction in library and information services.

In some cases, performance metrics for information services have been explicitly linked with external quality measures, for example the NSS (National Student Survey) or PTES (Postgraduate Taught Experience Survey) student satisfaction measures in the case of UK university libraries. They have also been incorporated into general quality management approaches such as the 'balanced scorecard',

value scorecard and Holbrook's value typology (Urquhart, 2015; 2018; Urquhart and Tbaishat, 2016), and into mathematical modelling techniques such as structural equation modelling (Gómez-Cruz, Harari-Betancourt and Vergara-Mendoza, 2020).

Impact analysis

Impact studies are particularly desirable for service evaluation, since they aim to show the actual impacts made on the users of information services by the information provided, and on the business of the organisation of which the services are a part; see Streatfield and Markless (2012) for an overview of the issues. In an academic library, for example, we might want to show that use of the library and information services results in students getting better grades in assessments, fewer students dropping out, researchers obtaining more grants, lecturers writing more papers, and so on. In a healthcare setting we would like to show that information services result in fewer patient deaths, faster recovery, less hospital-acquired infection, etc. In a commercial context we might look for more innovations, and products getting to market more quickly.

This issue is addressed by two international standards, ISO 16439 for methods for assessing the impact of libraries in general (ISO, 2014b) and ISO 21248 for national libraries (ISO, 2019). ISO 16439 is analysed by Creaser (2017) and de Jager (2017). This standard notes that 'impact' may be a variety of outcomes, including improved skills or competencies, changes in attitudes or behaviours, greater academic or career success, improved well-being, and social inclusion. Impact may be negative as well as positive, including demotivating people from using an information service. Impact may

be: immediate (for example, a student finding a reference for an assignment) or long term (for example, an improvement in overall information literacy); limited (such as small improvement in a skill) or far-reaching (even changing someone's life). The population who may be impacted ranges from individual users, through institutions such as universities, to the whole of a society. ISO 16439 recommends three general ways of gaining evidence for impact: inferred (statistics and performance indicators); solicited (user surveys looking specifically at impact); and observation (direct or via diaries, weblog analysis, etc.).

This style of evaluation has been practised most commonly in the health information sector; see, for example, Ayre et al. (2018). For an example of the development of a comprehensive approach to evaluation of quality, value and impact in the public library sector, see Reid (2020).

General management issues

The management of information, knowledge and documents has a natural overlap with the management of information institutions and departments. A thorough treatment of the latter is given by Evans and Greenwell (2020). We mention here three aspects which have a significant interaction with IM per se.

Promotion and marketing, to ensure that all users are aware of them, and of their value and best use. It is generally conceded that information services have often been rather poor at marketing and promotion, with the result that their contributions have been overlooked or underrated. A number of distinct, though interrelated, activities are involved. It is usual to distinguish *public relations*

(telling a story), *marketing* ('selling' a product or service) and *advocacy* (creating partnerships); the last may include *lobbying* (influencing decision makers). *Branding* is the development of a symbol to encapsulate the merits of a product or service. Best practice is discussed in the chapter on 'Advocacy and marketing' in Evans and Greenwell (2020), and also by Goodman (2020) and Kendrick (2021).

Equality, inclusion and diversity issues are of importance for the same reasons as for any institutions and departments, but particularly because they complement and facilitate diversity and inclusivity in services and collections; see the chapter on 'Managing diversity' in Evans and Greenwell (2020), Jones and Murphy (2019) and Adkins, Buchanan and Alston (2020).

Leadership, managing projects and managing change are of evident importance for the information sector in times of unprecedented change; see the relevant chapter of Evans and Greenwell (2020), and see also Weaver and Appleton (2020), McAllister (2019) and Daugherty and Hines (2018).

Summary

Information management and policy is a complex area, encompassing a variety of entities, ranging from archival materials to tacit knowledge, applied in contexts from the personal to the international. The digital transition presents issues and problems for all aspects, making it all the more important to identify underlying concepts and models to allow a unified, rational approach across the spectrum.

- IM is a complicated field, encompassing different forms of
- information and types of documents in diverse environments.
 - A variety of models and frameworks can be used to understand IM, based on ideas of the communication chain and information lifecycle, and value-added processes.
 - Demonstrating the value and impact of information and information provision continues to be a challenge for IM.

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15 Information Law and Ethics

The task is to formulate an ethical framework that can treat the infosphere as a new environment worthy of the moral attention and care of the human inforgs inhabiting it. Such an ethical framework must address and solve the unprecedented challenges arising in the new environment.

Luciano Floridi (2014, 219)

Ethical issues involving data may be more challenging than the ethical challenges of some other advanced technologies, partly because data and data science are so ubiquitous, having the potential to impact all aspects of life, and partly because of their intrinsic complexity . . . We must also recognize that we cannot expect to give simple answers to complex moral problems involving data. This will often be impossible, not least because the data environment is changing so rapidly. Instead, the aim must be to help focus on the issues and attempt to remove confusion and ambiguity, so as to provide *principles* that can help people come to a conclusion about what is the right way to act and the right thing to do in the circumstances in which they find themselves.

David J. Hand (2018, 176)

Information law issues dovetail together, creating a sophisticated, complex and delicate balance between facilitating open and free access to information – thus supporting open and accountable government, democratic freedom, and an uncensored press – simultaneously with the necessity for control and privacy. This balance is not always successfully achieved . . .

Charles Oppenheim, Adrienne Muir and Naomi Korn (2020, xv)

Introduction

In this chapter we examine the law and ethics of information. This is a broad topic and we can give only an overview with some relevant examples. In particular, the sections on information law are not in any way intended to provide legal advice. Examples are largely chosen from UK law as of 2021 and may not be relevant at other times, or in other jurisdictions. Ethical issues for specific aspects of information science are dealt with in the appropriate chapters.

The chapter is organised into four sections. After a discussion of the context, and how law, ethics, governance, regulation, standards, etc. interact, the following two sections deal with the generalities of information law and of information ethics. The fourth section examines the legal and ethical aspects of two topics, privacy and artificial intelligence, to illustrate general principles with specific examples.

The legal and ethical context

In discussing information law and information ethics we must be aware that there are several related but distinct factors. Law, ethics, regulation, standards and good practice, and governance all affect the right handling of information. The distinction between them matters, in deciding what is done, how, and by whom. All of these factors, and their interrelations, have been greatly affected by the move to a digital information environment.

Law is a set of formal, codified rules of behaviour imposed on individuals and organisations through social institutions, and

enforced by states through legislation. It is applicable within a *jurisdiction*, either within a nation-state or internationally, through *legislation*.

Ethics is a system of morals or rules of behaviour, objective and codified, accepted by a social group. Ethics help to determine right behaviour in those cases where there is no law to say what to do, or when it may be justifiable to go against the law. Ethics shapes regulation and governance by allowing a consideration of what is socially acceptable or preferable, regardless of what is mandated by law. Floridi (2018) helpfully distinguishes between *hard ethics*, which come before legislation is in place and shape changes in the law, and *soft ethics*, which apply after the law has been observed, because the law cannot stipulate what is the best thing to do in all cases.

Regulation, rules applicable to specific issues and contexts, derives its coercive force from law and is associated with specific legislation. Compliance with regulation generally requires extensive record keeping, and hence has special relevance for information professionals, and also implications for privacy.

Standards and *codes of good practice* may be local, national or international, and may be institutional or industry wide. They are not generally enforced by law, and most are voluntary, though some may be incorporated into regulations.

Information *governance*, as noted in Chapter 14, is a set of policies, processes, standards, and guidance on good practice which are applied within an organisation to ensure that information is managed correctly, including adherence to all relevant laws and regulations. While there is clearly a strong interrelation between

these processes, as Floridi (2018) emphasises (see [Figure 15.1](#)), governance, ethics and regulation, while complementary, must be clearly distinguished.

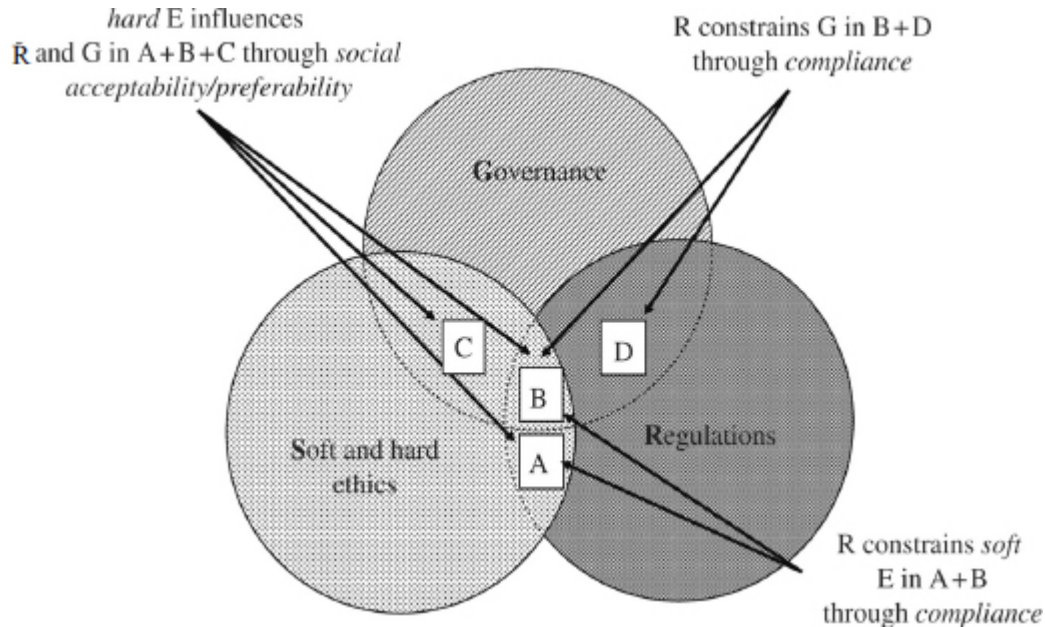


Figure 15.1 *Relations between governance, regulation and ethics* (Floridi, 2018, DOI: 10.1098/rsta.2018.0081)

Finally, at the national and international levels the concept of *digital sovereignty* addresses the issues of how, and by whom, the digital environment of data, software, apps, networks and protocols is controlled, and potentially pits nations-states and international organisations against multinational technology providers (Pohle and Thiel, 2020; Floridi, 2020a). This takes us well into the realm of *information politics* and the governing of the information society, as discussed in [Chapter 16](#).

Information law

The law is complex, and ensuring adherence to it is a task for the specialist, in the information context as much as in any other. In this

section we will give an overview of information law from a lay perspective; for more detailed coverage of these topics, see Oppenheim, Muir and Korn (2020) and Maguire (2018). Legislation can have a very considerable impact, for good or ill, on the development of the information environment. As Floridi (2018) points out, the wrong legislation may kill digital innovation entirely, or destroy whole technological sectors and developments, and simply reacting to technical innovations is a poor approach. There needs to be a proactive decision as to what kind of information environment is needed, and then the construction of legislation to enable it.

There are three legal issues of particular importance to information science, because they are focused on data, information and documents: intellectual property, freedom of information and data protection. These will be considered first, and then other laws relating to information provision and access will be considered.

Intellectual property

Intellectual property (IP) law is fundamental to the work of the information professions and is arguably the most important aspect of information law. It subsumes the laws of copyright, database rights, patents, design rights, trademarks, and trade secrets. For an overview with an international perspective, see Sinnreich (2019). All IP law is intended to strike a balance between the rights of creators to profit from their creations, including moral rights to be recognised as the creators, and the rights of society to benefit from them. *Copyright* is the right to prevent unauthorised copying of any work created by intellectual effort, for a defined time period; an analogous

database right prevents unauthorised copying of substantial parts of a database. There are exceptions for *fair dealing*, i.e. allowing a reasonable extent of copying for specific appropriate purposes. There are problems with *orphan works*, where the copyright owner is unknown, and complexities due to new forms of licences allowing various degrees of openness of digital materials, most notably Creative Commons licences (Creative Commons, 2020). IP is governed by a variety of international treaties, conventions and directives, typically incorporated into national laws and regulations. IP is a controversial subject with considerable ethical implications, including for information professionals, because of the barriers it presents to accessing information, which may be affected by cultural norms. It is a complex area for the practitioner, requiring a specialist understanding; see Padfield (2019), Cornish (2019) and Oppenheim, Muir and Korn (2020, chapters 1 and 6) and, for a US perspective, Crews (2020).

Freedom of information

Freedom of information (FOI) may be taken to be part of the general movement toward openness of information, including OA, open data, open source and open science. However, FOI generally refers to specific legislation and regulations. It refers to an obligation placed on public bodies to provide information held by them, or about their activities, on request, with some exemptions. This was first recognised as an integral part of a right of freedom of expression by the United Nations in 1946 (Resolution 59). Information professionals in public bodies may be required to provide information on their activities, and information specialists in

general may be well placed to promote and facilitate FOI. For more detail, see Gibbons (2019), Maguire (2018) and Oppenheim, Muir and Korn (2020, chapters 3 and 8).

Data protection

Data protection, based on a human right to private and family life, home and correspondence, is instantiated in laws and regulations governing the collection, storage, processing and use of personal data of any kind; personal data being, loosely speaking, any data which may identify an individual. A well-known example is the European General Data Protection Regulation (GDPR). Such regulations give individuals the rights to access their data, to correct errors and, to an extent, to erasure, to a 'right to be forgotten'. A set of basic principles apply to collection and processing of personal data:

- it must be fair, lawful and transparent
- the purpose must be specific, explicit and legitimate
- only the minimum data necessary for the purpose must be collected; it must be adequate, relevant and not excessive
- stored data must be accurate and kept up to date
- data must be kept for no longer than necessary
- security, integrity and confidentiality of data must be maintained
- there must be responsibility and accountability, and proper records must be kept.

Data protection issues affect many contexts of information provision and information management, not merely where data is collected

and managed as part of an IM function but also in library services, with respect to user details, borrowing records, browsing histories, etc., and in information research with human subjects. For more detail, see Maguire (2018) and Oppenheim, Muir and Korn (2020, chapters 2 and 8).

Other legal matters may affect information providers. Some jurisdictions, including several countries of continental Europe, have a 'library law', specifying library provision in detail. The UK has not such comprehensive legislation. Rather, it has a series of laws regarding such things as: the provision of a public library service by local authorities; copyright deposit, a duty on publishers to deposit a copy of everything published in the UK, including digital material, in major 'copyright libraries' (Gooding and Terras, 2020); and public lending right, which establishes a right for authors to be paid for the borrowing of their works from public libraries. UK legislation is far from consistent; notoriously, there is a legal requirement to provide a library in a prison, but not in a school. Libraries and other information institutions are also affected by legislation on obscene materials, on prevention of crime and terrorism, on cybercrime and computer misuse, on defamation and libel (for which they may potentially be held responsible for resources which they provide) and on the reuse of government and public sector information. They are also strongly influenced by legislation on contracts and licences, particularly as the model of purchasing and owning printed materials moves to one of licensing digital resources (Oppenheim, Muir and Korn, 2020, chapter 7). Finally, like other institutions, they

must comply with general legislation on matters such as human rights, equalities and health and safety.

Information ethics

Ethical issues arise in information research and practice when applicable laws and regulations do not give sufficient guidance. Some of the main areas of concern within information ethics include: censorship and intellectual freedom; privacy, confidentiality and data protection; ownership of information, and commercial use of public information; balance in collections, and appropriate metadata; universal access, information poverty and the digital divide; respect for IP, combined with fair use; and issues of balance and bias in information provision, collection development and metadata creation. For a detailed review of principles and applications of information ethics, see Burgess and Knox (2019) and Graff et al. (2020). Specific ethical issues for information organisation are discussed in [Chapter 8](#), for publishing and social media in [Chapter 13](#), for collection development in [Chapter 14](#) and for information research in [Chapter 18](#).

We use the general term ‘information ethics’ here, though much of what follows might equally be denoted ‘data ethics’. The move to ‘datafication’ and to a ‘Big Data Society’ puts data collection, analysis and search at the heart of many of society’s activities, evoking a need for specific ethical consideration; for fuller discussion, see Zwitter (2014), Hand (2018) and Hasselbalch (2019). Floridi and Taddeo (2016) discuss data ethics as distinct from information ethics, and point out that the ethics of data, algorithms and practices are generally intertwined. Internet search engines are

also a source of ethical concern, due to biases in search and ranking algorithms, in autocomplete functions, etc. (Noble, 2018).

While the ethical dilemmas of information science are often dealt with pragmatically as they occur, a more structured and systematic way of deciding such questions is desirable. This is done by drawing either on one of a number of general ethical approaches or on a set of values or code of ethics specifically designed for the information sector.

Types of ethical theory

It is usually argued that there are four general types of ethical theory which may be useful. Blackburn (2021) sets them out succinctly, and Fallis (2007) gives examples of their application in information contexts. Existentialism, which emphasises individual freedom and responsibility for whatever we choose to do is sometimes mentioned as a fifth ethical theory, though one which offers little concrete guidance. The four main types of ethics are:

- *Consequence-based theories*: these focus on the outcome, rather than the action itself, arguing that the right action is that which brings about the best outcome for the greatest number. The best-known example is the philosophy of *utilitarianism*, as originally advocated by John Stuart Mill. This approach may be used in information ethics to argue, for example, that we should not censor potentially offensive materials, even though we know that not doing so will cause distress to some people, since it will be of overall benefit for them to be available.

- *Duty-based, deontological theories*: these argue that there are certain ethical duties which must always be obeyed, regardless of their consequences, that there are some actions or values which are always inherently good or bad. These are justified by our moral intuition; the best-known example is Kant's idea of the *categorical imperative*. In an information context we could, for example, argue that we always have a duty to provide full and correct information, which overrides any other consideration. This approach can be reactive to circumstances, giving guidance on what to do in a particular circumstance.
- *Rights-based theories*: these suggest that the correct thing to do in any situation is determined by the rights we assign to others. This approach was first propounded by John Locke, and recently developed by John Rawls for the rights of members of society. This approach, which is the justification for all human rights legislation, may be particularly useful for 'code-based' information ethics, since these are often expressed as the rights which people have: to have access to information, to be treated respectfully and equally by information providers, to have privacy in their information dealings, and so on.
- *Virtue-based theories*: these suggest that the right thing to do in any situation is what a virtuous person would do; to live a good life is to become a particular kind of person. It is a proactive and constructionist approach; we can all aim to become better people in all circumstances. These theories, which can be traced back to Aristotle and Confucius, ascribe to the virtuous person traits such as courage, calmness and friendliness, and a quest for a balance

between excess and deficit in all things. The recent approach of *communitarianism* is usually understood as a form of virtue ethics, focusing on the idea of the virtuous citizen as part of a strong and functioning community, often associated with promotion of social justice, and hence arguably a particularly suitable approach for the information sciences (McMenemy, 2021).

While all of these approaches have some value for the information context, none is a panacea and none is without problems in application to specific cases. Perhaps the main value of such ethical theories is that they force us to be clear about how we approach specific ethical dilemmas and to be explicit about how we try to solve them. For an example of how the ethical approach taken may influence decisions about information provision in practice, see comparisons of consequentialist, rights-based, virtue, and communitarian stances to the UK's free public library service (McMenemy, 2020) and of consequentialist and deontological approaches to the ethics of recommender systems (Milano, Taddeo and Floridi, 2021).

Regardless of the ethical system adopted, it must be used conscientiously. Floridi (2019) identifies five types of ethical risks, all of which can be identified with respect to information ethics:

- *ethics shopping*: choosing ethical principles to justify existing behaviour
- *ethics bluwashing*: taking limited and superficial steps to make practices seem more ethically sound than is the case

- *ethics lobbying*: using an ethical stance as an argument against legislation or regulation
- *ethics dumping*: carrying out unethical activities in countries with a low level of legislation and regulation
- *ethics shirking*: avoiding ethical responsibilities when this can be done without consequences.

A new form of information ethics?

Some have argued that we need an entirely new form of ethics to deal with information issues, based on the centrality of information as a universal entity. This was first suggested by Wiener (1954) and was put forward most fully by Floridi (2013) as a part of his Philosophy of Information. Such an information ethics focuses on information itself, rather than on any particular technologies, and on the idea that a person may use information as a *resource*, to generate other information as a *product*, which in turn may transform the informational environment as a *target*. Floridi argues that all things which exist should be regarded as 'informational entities': this certainly includes people, but also animals, plants, paintings, books, stars and stones. Information ethics is that set of principles which enable all information entities to flourish; specifically, by minimising *entropy*. Floridi here uses entropy not in the sense of physical, thermodynamic entropy, nor in the sense in which the word is used in Shannon's theory, but, rather, to mean any destruction or corruption of an informational entity. All entities have some informational value, but not all are equal. Those which contribute most to the well-being of the infosphere have most value, so that we

may reasonably destroy bacteria which threaten human lives, even though the bacteria have informational value in themselves.

Floridi derives four main principles of information ethics for the 'infosphere' – the universe understood in informational terms:

- entropy ought not to be caused
- entropy ought to be prevented
- entropy ought to be removed
- the flourishing of informational entities as well as of the whole infosphere ought to be promoted by preserving, cultivating and enriching their well-being.

This is an ambitious attempt to set up a whole system of ethics with information as its basic concept; not, as Floridi (2013) insists, to replace other ethical systems and principles, but to augment them; it is a 'nonstandard' form of ethics. It is related to duty-based ethics, inasmuch as it evaluates the duty of any rational being in terms of a contribution to the well-being and flourishing of the infosphere, but is also related to virtue ethics, as it sees humans, as informational entities, as works in progress, continually seeking to develop ourselves.

In Floridi's scheme, documents have a particular value. He instances both the example of the Italian Library Association's code, which includes a duty towards information and documents as well as to users and to the profession, and also UNESCO's recognition that documents can have value in their own right (Floridi, 2013).

It remains to be seen whether Floridian ethics will lead to a more coherent and practically useful way of dealing with the ethical

problems of the information disciplines and professions than earlier frameworks. An example of how this might be done is an analysis of the anti-counterfeiting trade agreement (ACTA) in terms of information ethics (Floridi, 2015).

Professional values and principles

In the specific context of the information sciences, one approach to ethical decision making is to rely on *professional values* as a guide. Michael Gorman (2000), for example, lists eight 'enduring values' of librarianship:

- stewardship
- service
- intellectual freedom
- rationalism
- literacy and learning
- equity of access to recorded knowledge and information
- privacy
- democracy.

Rubin and Rubin (2020) set out seven values of LIS:

- service
- the importance of reading
- respect for the search for truth
- tolerance
- serving the public good
- justice

- cultural excellence.

While such value lists may be valuable as a basis for discussion, they are – like the rather similar codes and guidelines mentioned above – of limited value in specific situations.

A more concrete alternative is given by *codes of conduct*, a number of which have been produced for the information professions since the first was introduced by the American Library Association in 1938. There are a variety of these, some of which are reviewed and compared by Rubin and Rubin (2020) and Foster and McMenemy (2012). These typically emphasise that information specialists have an ethical duty to their employers, clients and patrons, to the wider society and to their profession. They may be criticised on the grounds that they provide some general principles, but little guidance on how they are to be applied in specific cases; see Ferguson, Thornley and Gibb (2016) for discussion of how well such codes deal with ethical issues.

In the UK, the first library/information code of practice was developed in 1983; the latest development, announced by CILIP in 2018, is a set of seven ethical principles to be promoted:

- human rights, equalities and diversity, and the equitable treatment of users and colleagues
- the public benefit and the advancement of the wider good of our profession to society
- preservation and continuity of access to knowledge
- intellectual freedom, including freedom from censorship
- impartiality and the avoidance of inappropriate bias

- the confidentiality of information provided by clients or users and the right of all individuals to privacy
- the development of information skills and information literacy.

Like most such codes, it is a composite of general and specific ethical issues with pragmatic professional concerns.

To help set the general principles in specific context, we now look in more detail at the legal and ethical implications for information science of one issue, privacy, and of one technology, AI.

Ethics of informational privacy

We [. . .] have a private life [and] hold it infinitely the dearest of our possessions Virginia Woolf, *Montaigne*, in *The Common Reader* (1925)

Privacy has been thought important for many years, as the quotation from Virginia Woolf illustrates, but has gained increased ethical significance in a digital information environment, being denoted as ‘a central issue of the information age’ (Wu, Vitak and Zimmer, 2020, 485) and ‘one of the defining issues of our hyperhistorical time’ (Floridi, 2014, 102). It is an issue of particular significance for information science, for a number of reasons, including the need to protect the privacy of users of information services, and also more generally the privacy of anyone whose data may be stored and processed.

Privacy is defined by the Oxford English Dictionary (2007) as ‘The state or condition of being alone, undisturbed, or free from public attention, as a matter of choice or right; seclusion; freedom from interference or intrusion’. It has been described as a right to be let

alone, or the right to be forgotten, and hence is generally treated in the context of rights-based ethics. Privacy is a deceptively simple idea whose nature seems obvious at first sight, and the protection of privacy an obvious ethical good, but the issues become complex on investigation (Solove, 2008; Wacks, 2015; DeCew, 2018).

It is usually believed, rightly, that privacy is increasingly under threat, due to digital collection and processing of personal information. We may distinguish between *anonymity* (information about someone has never been recorded) and *obscurity* (the information has been recorded, but it is too difficult and time consuming to find). Both anonymity and obscurity are easily removed in a digital information environment; hence the need for explicit ethical concerns for informational privacy.

There have been numerous analyses of the nature of privacy and how it is changing in a digital age. There is something of a paradox here, in that there is an increasing concern about privacy in the digital realm at the same time as many people seem happy to make much of their life digitally public, particularly through social media and the more extreme phenomenon of ‘lifelogging’ (Mai, 2016; Gupta, Crane and Gurrin, 2020). Studies of the concept in a variety of disciplines have produced many definitions, concepts, frameworks, and models, some of which have clear relevance for information science; for reviews, see Mulligan, Koopman and Doty (2016), Bawden and Robinson (2019), Mai (2019), Rønn and Søre (2019) and Wu, Vitak and Zimmer (2020).

There are several typologies of different kinds of privacy. Tavani (2008), for example, distinguishes four types of privacy:

- physical
- decisional
- psychological
- informational.

Rønn and Sør (2019) apply this model to issues of social media privacy.

Koops et al. (2017) distinguish nine types:

- bodily
- intellectual
- spatial
- decisional
- communicational
- associational
- proprietary
- behavioural
- informational.

Nissenbaum's (2011) 'framework of contextual integrity' provides a set of conceptual entities relevant to online privacy, including actors (subjects, senders and recipients), attributes (types of information) and transmission principles (constraints on flow of information), enabling the derivation of a model directly comparable to, or able to be integrated into, established information behaviour models. It has been applied by McMenemy (2017) to privacy issues affecting information professionals. Mai (2019) develops a similar approach,

including the contextual nature of the information itself, in addition to the situation and the relations of the persons with the information, in a model designed for a digital environment.

Floridi takes a different perspective, seeing privacy as essentially informational, rather than informational privacy as merely of one type. In this view, because personal information plays a crucial constitutive role in who an individual is and who they may become, protection of privacy should be identified as protection of personal identity, and a breach of informational privacy as an aggression against personal identity and self-development. Defence of privacy should be based directly on the protection of human dignity, rather than on secondary considerations such as a right to property, to freedom of expression or to privacy per se. In this view, 'my privacy' is to be regarded as more like 'my arm' than 'my bicycle', and loss of privacy to be more akin to 'kidnapping' than 'theft'. For Floridi's conception of privacy, see Floridi (2005; 2006; 2016; 2017), and for commentary and discussion see Bawden and Robinson (2019; 2020).

Privacy is usually taken to be an individual matter, but group privacy is now becoming regarded as equally important, particularly now that groups may be formed algorithmically from data analysis. Such groups need not conform to intuitively understood groupings, and their members will not be aware that they are being treated as belonging to a group which they did not ask to join, and with which they may not necessarily wish to be associated (Taylor, Floridi and van der Sloot, 2017; Mittelstadt, 2017). There is also a concern for particular demographic groups, such as children (Stoilova, Nandagiri and Livingstone, 2021).

Informational friction is an important concept in considering the ethics of privacy. Floridi (2005) argues that privacy itself is a function of informational, or ontological, frictions in the infosphere. An information friction is anything which interrupts the flow of and access to information, including such diverse factors as noise, limited bandwidth, paywalls, copyright, language barriers, access rights, and data protection regulations. ICTs can both increase and decrease frictions, and can therefore both support and damage privacy. It is usually held that the task of information science is always to reduce such frictions and improve information flow and access, but the more frictions are reduced, the more likely it is that privacy will be damaged. Control, not elimination, of frictions is the key to allowing the maximum information flow and access, while maintaining a proper degree of privacy. This is likely to be achieved by a mix of legislation, regulation, codes of conduct for particular settings, and an ethical awareness of the right to privacy. These will vary according to the context. For example, some jurisdictions have specific laws related to privacy, while others do not; the UK, for example, has no specific privacy legislation, privacy issues usually being addressed by data protection law. Privacy in library/information settings is usually dealt with by codes of practice rather than by law or regulation, with the exception of data protection legislation. The means for protecting privacy will also need adaption to changing technological capabilities, but ideally not in a purely reactive way, but based on clear and lasting ethical principles.

While maintenance of privacy is certainly a matter for legislation and regulation, it is also important that individuals are aware of the issues and able to take steps to protect their own privacy. *Privacy literacy* has been suggested as a form of information and digital literacy (Wissinger, 2017).

Legal and ethical implications of artificial intelligence

Of all new digital technologies, AI has generated the most controversy about the legal and ethical aspects of its implementation. In part this is because of concerns, which would better remain in the realm of fiction, about super-intelligences, singularities and killer robots. More realistically, it is because its reach and influence, and consequent ethical implications, are greater than those of other technologies. As Taddeo and Floridi (2018, 751) put it:

[AI] is not just a new technology that requires regulation. It is a powerful force that is reshaping daily practices, personal and professional interactions, and environments. For the well-being of humanity, it is crucial that this power is used as a force of good.

Ethics plays a key role in this process by ensuring that the regulations of AI harness its potential while mitigating its risks.

Ethical issues arising from current AI systems include: discrimination and inequalities, due to bias in the datasets from which AI systems learn interacting, and to unanticipated algorithmic judgements; an inability to understand, and hence correct, the behaviour of complex AI systems which are unable to 'explain themselves'; the loss of privacy, due to extensive data collection and integration; surveillance, both directly through cameras with facial

recognition and indirectly through tracking of digital presence; potential job losses, due to the automation of intellectual as well as manual work; and the difficulty of assigning responsibility for harms arising from the action of intelligent agents. For overviews of these issues, see Bruneault and Laflamme (2020), Coeckelbergh (2020) and Dubber, Pasquale and Das (2020). Ethical issues with AI are often complex and multilevel, and not amenable to simple analysis and solution; as an example, see a review of ethics of AI in healthcare which identified ethical issues arising on individual, interpersonal, group, institutional and societal levels (Morley et al., 2020).

A number of sets of principles have been put forward to ensure that AI is used in a socially beneficial manner, sometimes described as Artificial Intelligence for Social Good, which may be aligned with the United Nations' Sustainable Development Goals (Floridi et al., 2020; Cows et al., 2021). A framework of five core principles for ethical AI is presented by Floridi and Cows (2019). The first four principles are commonly used in bioethics, showing the value of foundational ethical ideas in this area; the fifth is developed for the AI context. The principles are:

- *beneficence*: AI should be used to promote well-being, preserve dignity and sustain the planet
- *non-maleficence*: AI should not do harm, for example by infringing privacy
- *autonomy*: a balance must be struck between how much autonomy is given to machines and how much is retained by humans

- *justice*: AI should be used to promote prosperity, preserve solidarity and avoid unfairness
- *explicability*: there should be clarity as to how any AI works and who is responsible for the way it works.

There is inevitably a gap between such typically high-level statements of principle and practical guidance on incorporating ethical considerations in the design of systems. This problem may be overcome by the idea of ‘ethics as a service’, including codes of practice, ethics advisory boards, ethics-based system audits and full documentation (Morley et al., 2021; Mōkander and Floridi, 2021). Following such recommendations, AI systems should become just another tool, to be used in a transparent, responsible and trusted way, as in the intended use of AI in public services by the cities of Amsterdam and Helsinki (Floridi, 2020b).

Summary

The new digital information environment has raised a host of new issues requiring careful governance if new technologies are to be beneficial to humanity. These issues involve information, data and algorithms in a complex and interrelated landscape, and require an equally complex and interrelated set of laws, regulations, standards, governance, and ethics. A variety of laws relate to information and information provision, varying between jurisdictions; the most significant bodies of information law relate to IP and the balance between the rights of creators and users, to access to information and to the privacy of the information and data of groups and individuals. There is no single accepted form of information ethics; rather, several traditional ethical approaches have been adopted, augmented by

specific systems of ethics designed to deal with information issues. Of all information-related ethical issues, privacy has received the most attention, and AI the most attention of all technologies.

- Law and ethics have not kept up with technology in the new infosphere.
- To ensure the proper treatment of information and data requires a mix of law, ethics, regulation, standards and good practice and governance.
- Currently the most important information laws relate to IP, especially copyright, freedom of information and privacy and data protection.
- All the main ethical systems have been applied to information issues; Floridi's information ethics is the most fully developed of systems specifically for the information environment.

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16 Information Society

We can see now that information is what our world runs on: the blood and the fuel, the vital principle.

James Gleick (2011, 8)

We are so familiar with the talk of '*the* information society' that we sometimes forget there is no such thing, but rather a multitude of societies, unlike from each other, some of which may qualify as information ones in different ways and degrees.

Luciano Floridi (2016, 1)

Introduction

Since the later years of the 20th century it has become commonplace for commentators to declare that we are living in an 'information society'. In this chapter we will first consider the nature of, and criteria for, an information society. We will then look at some of the frameworks and infrastructures which support it, and some of the problems and issues which it brings with it.

Understanding information society

Information society is not a recent phenomenon. As Floridi (2014a; 2016) reminds us, it would be reasonable to say that information society emerged in the Bronze Age with the introduction of written communication and records. The contributors to Rayward (2008) identify origins of our present information society in the early 20th century. However, we tend to think of the development of information society as synonymous with the widespread use of digital information systems (Gleick, 2011, chapters 14 and 15; Floridi, 2014a; Lindgren, 2017). While information use has been a feature of

all historic societies, in our current situation of *hyperhistory* (Floridi, 2014a), our use of information and communication technologies defines our society as one rooted in the infosphere. Put simply, as Gleick points out in our opening quotation, information is what our world runs on. There are different kinds of information societies, but a mature information society is one 'whose members assume the digital as a foregone phenomenon' (Floridi, 2019, 183).

Considered in more detail, however, there are a number of different criteria by which we may recognise and define information society. We will consider these in outline; for detailed analysis, see Feather (2017), Webster (2014), Floridi (2016) and Beaulieu and Leonelli (2021).

Types of information society

An *economic information society* is one in which most of the economic activity and wealth of a society is based on intangible, information-based goods, rather than on manufacture and agriculture. This viewpoint builds on the studies of scholars such as Fritz Machlup, Peter Drucker and, particularly, Marc Porat, who pioneered the idea of the 'information economy' in the 1970s (Machlup, 1972; Porat, 1977). Since, for example, more than 70% of the economic activity of the G7 group of developed countries is associated with intangible, i.e. information-based, goods, in this view these may be counted as information societies.

An *occupational information society* is one in which most people are employed as 'information workers', however this may be defined. This viewpoint was pioneered in the 1960s by the American sociologist Daniel Bell, with his idea of the 'post-industrial society'

(Bell, 1976), and developed in the 1990s in the 'knowledge society' idea of the German sociologist Nico Stehr, who examined the extent to which non-information occupations were increasingly knowledge based (Stehr, 1994).

A *technological information society* is one built around, and dependent upon, widespread ICTs. The problem here is to determine what counts as an 'information technology' and how much of it there must be to create an information society. The presence of a digital infrastructure, particularly access to broadband and to digital devices, can readily be measured, but their influence on society may be varied. Three key aspects are social media, e-commerce (both online retail and business-to-business transactions) and e-government; widespread adoption of these is taken as indicative of an information society.

A *political information society* is one in which information and ICT issues become the subject of political debate, and of action at local, national and international levels. Here there is the problem of how much debate is necessary, and also whether the political attention needs to be positive. Governments, and powerful individuals, have engaged in censorship and propaganda since written communication became widespread. Should these be counted as progress towards an information society? For thoughts on the politics of mature information societies, see Floridi (2019).

Sociocultural factors have also been proposed as the basis for an information society. Most notable is the idea of the *network society* put forward in the 1990s by the Spanish sociologist Manuel Castells (1996). Such concepts point to the power of information, and

specifically of information networks, to change our perceptions of time and distance and to change the way we behave and interact in all aspects of our social dealings (Graham and Dutton, 2019). This was anticipated by Karl Popper, who in 1945 foresaw an *abstract society*, in which people practically never meet face to face, with business conducted by individuals in isolation, who communicate by typed letters or by telegrams (Popper, 2020, 166) – an idea which has resonance in the COVID pandemic of 2020–21 (Chmielewski, 2020). The rise of social media since such ideas were first advanced has only served to further promote this idea. It seems undeniable that society, in those parts of the world affected by these trends, has changed greatly; but it is problematic to say exactly when, and why, it became an information society. See Graham and Dutton (2019) for analyses of the influence of the internet on society.

Finally, the British sociologist Frank Webster has argued for the idea of a *theoretical information society*, one in which the main guide to practical action is abstract theoretical knowledge, recorded in documents, rather than tradition, experience, and tacit knowledge (Webster, 2014). This is similar to Michael Buckland's suggestion that it would be better to speak of a *document society* rather than an information society, as it is increase in use of recorded information which demarcates our society from others (Buckland, 2017, 17).

All of these conceptions of an information society offer some insight, but none in itself is fully satisfactory as a definition. It is reasonable to say, following Floridi, that there are many kinds of information society, but the defining feature of any mature information society is the unreflective and implicit expectations of its

citizens that they will be able to access, communicate and process whatever information they want at any time.

We can mention one other concept which, though not a theory of the information society, has some resonance with it. This is Karl Popper's idea of *open society*, which he initially proposed in the 1940s as an alternative to the closed societies of fascism and communism. Popper's open society is a society based on the rule of law, in which everyone knows the law and is equal before it, in which those who are responsible for making the laws can be changed without the use of force, and in which society is advanced by rational argument and open criticism of the status quo. It is therefore a form of society dependent on the free flow of information, and Popper's principles can be used as a guide for information providers (Robinson and Bawden, 2001). Popper's protégé, the financier and philanthropist George Soros, used these ideas as the basis for his Open Society Institute (OSI) (Tamkin, 2020). OSI has always supported library and information programmes as an important strand of its work; for one example, support for the professional development of information specialists in countries in transition, see Robinson and Glosienė (2007).

With these ideas of the nature of an information society in mind, we look at some of the infrastructures which govern information within societies.

Information society infrastructures

National and international policies have a major effect in determining the form of information society in parts of the world affected by them. National governments and international bodies

have the power to enact the laws and regulations which provide the framework for information societies. These control the nature of IP and copyright, data protection and privacy, freedom of information and censorship in any region. These aspects of information society are covered in [Chapters 14](#) and [15](#). Governments and international bodies may also choose to fund the development of the physical and digital information infrastructure, promote education and training for digital literacy, advocate e-procurement and e-commerce and so on. They are able to lead by example, since governments are among the largest producers, collectors, preservers and users of information, and can promote e-government, giving citizens digital access to government information and services. Public libraries, established in most of the developed world in the 19th century and described as ‘the prototypical spaces of the knowledge society’ (Mainka et al., 2013, 295), have the potential to be an effective support for many aspects of information society, if given the opportunity to do so (McMenemy, 2008; Leorke and Wyatt, 2018). A recent marker of the development of information society is the *smart city* or, as Castells termed it, the *informational city*, whose physical and digital infrastructure is designed to promote an onlife existence, as well as managing the city’s resources to best effect (Mainka, 2018). Such a city may act as an information repository and a knowledge hub, though with consequent concerns about surveillance and privacy (Simeone, 2020; Israilidis, Odusanya and Mazhar, 2021; Mattern, 2021).

The nature of information policies varies greatly between countries and regions. The UK and the USA, for example, have had

rather partial and reactive policy provisions, while the countries of continental Europe have adopted a more co-ordinated approach (Hernon and Relyea, 2017; Rabina and Johnson, 2017). Non-governmental agencies and philanthropic foundations, such as OSI, the Bill and Melinda Gates Foundation, the United Nations through its annual World Summit on the Information Society (WSIS) Forum and the World Bank, have played a significant part in promoting the development of information societies.

The technical infrastructure for an information society comprises the ICTs outlined in Chapter 9. A variety of standard metrics are available for international comparison of progress. A well-known example is the ICT Development Index (IDI) of the International Telecommunication Union (ITU), which is designed to measure the *level and progress over time* of ICT developments in both developed and developing countries, the *digital divide* between countries in terms of their levels of ICT development and the *development potential* of ICTs and the extent to which countries can make use of them to enhance growth and development (Gerpott and Ahmadi, 2015). The IDI uses indicators under three headings: *ICT readiness* (infrastructure and access), such as proportion of households with a computer and with internet access, and reach of broadband and mobile networks; *ICT use*, such as proportion of individuals using a mobile phone and using the internet; and *ICT capability* (skills), such as duration of formal education and percentage of individuals with specified digital skills.

The European Union's Digital Economy and Society Index (DESI) is similar, with metrics for connectivity, social capital/digital skills,

use of internet services, integration of digital technology and digital public services (European Commission, 2020). With all such indicators, there are disagreements as to the most appropriate metrics and ways of measurement.

Social informatics

Metrics such as the IDI give a simple quantitative assessment of the extent and, to a degree, the nature, of information society. A more nuanced assessment is given by the discipline of *social informatics*, most simply understood as ‘the systematic study of social aspects of computerization’ (Kling, Rosenbaum and Sawyer, 2005, 3). It uses an eclectic and flexible mix of methods, from the analytical and technical to the ethnographic, and also includes a strong element of action research (Fichman and Rosenbaum, 2014; Smutny and Vehovar, 2020). Focusing on tasks, practices, communities, tacit knowledge, and the ways in which these are affected by ICTs and, in turn, affect the way ICTs are used, social informatics has an overlap with ideas of knowledge management (discussed in [Chapter 14](#)).

Pathologies of the information society

As mature information societies develop they are afflicted by a number of information problems or pathologies (Bawden and Robinson, 2009). We will mention two of the most significant general issues: information poverty and the digital divide, and the divide by generation; and information overload and information anxiety. A third issue, that of the prevalence of misinformation, disinformation and malinformation, particularly on social media and digital news sources, is dealt with in Chapter 17; for its direct relevance to information society, see Goldstein (2020).

Information poverty and digital divide

These are closely related ideas which denote the differential levels of engagement of individuals and groups with the information society. Both of these are contested concepts, with lack of agreement as to their exact scope and meaning.

‘Information poverty’ has been, and is, understood in various ways: as lacking money to pay for information; as living in a region where access to information sources is limited; as living under a government which deliberately restricts access to information; or as lacking the basic education or the information literacy necessary to use a range of information resources. See Feather (2017, chapter 5), Haider and Bawden (2007), Duff (2015) and Marcella and Chowdhury (2020) for analyses of the concepts of information rich and poor. Duff regards information poverty as a deficiency in ‘normal democratic information’, and hence an issue affecting full membership of society; Marcella and Chowdhury consider it to be a lack of access to the information necessary for survival, self-sufficiency, sustainability or development. Note that the term is sometimes also used in respect of those who, though not fitting any of the categories above, have problems in finding necessary information for some tasks, as in a study of healthcare information managers (MacDonald, Bath and Booth, 2011).

Where information poverty is a term mainly used in respect of individuals and distinct groups, referring to all forms of information, digital divide is typically used on a larger scale, and specifically in respect of digital information, although the implications of the terms are much the same. *Digital inclusion* and *exclusion* are also used for

the same idea. Divides have been identified between the developed and developing worlds, between continents, regions and countries, and between rural and urban areas. The divide is generally taken to be between those who have access to digital resources, the money to pay for them and the ability to use them effectively and those who lack one or more of the requisites. See van Dijk (2020) and Flanagan (2018) for reviews of the issues, and Stevenson (2009) for a critique of the whole concept. Manžuch and Macevičiūtė (2020) give an analysis of how the divide may be addressed by library services.

The advent of ICTs is often held to have worsened the divide between those who have ready access to information and those who do not. On the other hand, it has been argued that a range of new ICTs, from open access e-journals to smartphones, will enable developing countries to catch up with the developed world in information terms, in a way that would be impossible without such technologies. For an evaluation of these issues, see Mabweazara (2021).

Generational divides

Another form of divide in the information society has been suggested; that between *generations*. It is argued that people of different ages form distinct generations with different attitudes and behaviours (Edmunds and Turner, 2005; Wacheider, 2016). This might be expected to include information behaviour, due to their growing up in very different information environments. The baby boomer generation, brought up with the novelties of television, vinyl records and paperback books, will be very different from the

Millennials, brought up with the web, the mobile phone and the computer game, from the Google Generation and so on.

Inasmuch as this is the case, it may be a problem in various ways: different generations may need different forms of information provision; older generations, unfamiliar with new ICTs, may form a group of 'information poor' in their own right; and the younger generations, while very confident with new ICTs, may be unable to engage with information resources in a deep and sustained way.

It has to be said that many concerns are based on anecdote and opinion, rather than on research. Studies have not shown any consistent generational differences in attitudes and behaviour fixed over a lifetime; indeed, digital technologies may be influencing the concept of generation itself (Au, 2020). There may well be typifying generational attitudes and preferences in information behaviour; see, for example, Benselin and Ragsdell (2016), Herrando, Jimenez-Matinez and Martin-De Hoyos (2019) and Deyrup, Bekbalaeva and Tammaro (2020). It is not viable to categorise people by age in this respect; other factors may be of more importance, there are always outliers and people change over time. In particular, the idea of *digital natives*, the newer generations brought up in a world of ubiquitous digital technology and therefore supposedly intuitively capable in the use of ICTs, has been largely discredited.

Information overload and anxiety

The phenomenon of information overload has been known by many different names, including *infoglut*, *data smog*, *information fatigue* and *cognitive overload*. There is no single, generally accepted definition, but it is usually understood as that situation which arises when there

is so much relevant and potentially useful information available that it becomes a hindrance rather than a help. Overload has been claimed both to be the major issue of our time and to be a complete non-issue. Its nature has not changed with changing technology, though its causes and proposed solutions have changed much.

It is not a new problem, although, as Gleick (2011) points out, it has always *felt* new. For almost as long as there has been recorded information, there has been a perception that humans have been overloaded by it, and concerns about ‘too much to read’ have been expressed for many centuries (Blair, 2010). These have become more urgent since the arrival of ubiquitous digital information in the late 20th century. Originally seen as a problem primarily affecting scholars and academia, it was then claimed to afflict business and the information-intensive professions such as medicine, and has been more recently seen as affecting the whole of society, including education, government, home life and leisure, and citizenship, and has even influenced literature. For reviews, see Bawden and Robinson (2020), Schmitt, Debbait and Schneider (2018), Koltay (2017), Groes (2017) and Case and Given (2016, 122–7).

Information anxiety is a closely related issue, described by Wurman (1989, 34) as ‘the ever-widening gap between what we understand and what we think we should understand . . . the black hole between data and knowledge’. It has remained an issue of concern: see, for example Wurman (2001), Hartog (2017) and Naveed and Anwar (2020). The irony is that increasing volumes of data and information may cause a lack of knowledge and understanding; overload and anxiety are linked.

The causes of overload are primarily associated with having 'too much information' (TMI), particularly when it comes through a diversity of media and when it appears interesting and potentially useful. Lack of time to deal properly with information is often cited as a cause; see, for example, Savolainen (2006). So is the 'paradox of choice', the paralysis of action caused by too many alternatives, for information as much as for anything else. It has been blamed for many consequences, including inefficient working, poor decision making and ill-health. On the other hand, it is suggested that overload, for many people at least, does not exist; it is a 'phantom', a 'myth' and something which may be avoided simply by ignoring most available information; Bawden and Robinson (2020) give examples of this viewpoint. Floridi suggests that we should no more complain about there being too much available information and data than we would complain about there being more food provided at a banquet than we can eat; in both cases, it is a matter of thoughtfully deciding what we should consume (Floridi, 2014b).

There are many suggested solutions to information overload and anxiety (Bawden and Robinson, 2020). The best ways of avoiding overload, individually and socially, appear to lie in a variety of coping strategies, such as filtering, withdrawing, avoiding and 'satisficing' (Savolainen, 2007; Lauri, Virkus and Heidmets, 2020). Better design of information systems, effective personal information management (PIM) and the promotion of digital and media literacies also have a part to play. Overload may perhaps best be overcome by seeking a mindful balance in consuming information.

Summary

It is generally agreed that we live in an information society, but there is no general agreement about exactly what this means, other than that information is somehow central to the way in which the society works. A variety of theories invoke technology, politics, economics, occupations and roles, networks and communication, knowledge and abstract theory, and other factors, as the basis for the idea. It is probably best understood in terms of a combination of these.

Despite the wealth of information available in the digital environment of the infosphere, the pressure of 'too much information' and the inequitable nature of access have led to concerns about information overload, information anxiety, the digital divide and the rest. Whether these will fade away into insignificance as societies adapt to their new nature, or whether they will continue to require active treatment through IM and digital literacy remains to be seen.

- Information society is a concept with no single agreed definition; it may be understood in a number of ways, expressed as theories or frameworks.
- It can be said that an information society has existed for as long as there has been recorded information, but modern information society has been catalysed by digital technologies, and by the move into hyperhistory and onlife.
- The idea of information society is closely related to other concepts: network society, knowledge society, information economy and post-industrial society.

- Perceived problems of information societies include information overload, information poverty and the digital divide.

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17 Digital (Onlife) Literacies

Any attempt to constitute an umbrella definition or overarching frame of digital literacy will necessarily involve reconciling the claims of myriad concepts of digital literacy, a veritable legion of digital literacies.

Colin Lankshear and Michele Knobel (2008, 4)

There's no doubt that it will be difficult to shift the information literacy narrative from emphasizing finding, evaluating, and using information in an academic setting to something that addresses a broader understanding of how information 'works' these days.

Barbara Fister (2021, 5)

Introduction

The ability to use information effectively, in all formats, in a largely digital information environment is a vital life skill for everyone. This ability is variously termed *information literacy*, *digital literacy* and *media literacy*, with numerous other literacies having been mentioned. We will use *digital literacies* as an overall term, linked to the concept of *onlife*, for reasons explained later, using the other terms where appropriate. What constitutes each of these literacies, how they relate to each other, and how they can best be taught and promoted, has caused much debate.

The promotion of digital literacies has been seen as an important task for information professionals in all sectors. Indeed, CILIP includes, as one of its seven principles for ethical information professionals, a commitment to uphold the development of information skills and information literacy. Identified as an

important issue since the 1990s, it has gained in importance as a claimed solution to the pathologies of the current information environment (Bawden and Robinson, 2009; 2011; 2020): information overload, misinformation, disinformation, information anxiety, etc. While information literacy training has largely been centred in libraries in secondary and tertiary education, as a replacement for traditional bibliographical instruction, there is an increased interest in its promotion in the workplace, and in society generally. The question of whether this can be a generic ability or whether it can be promoted only in the context of a specific subject is still debated, while the issue of multicultural information literacy has gained attention.

In this chapter, after explaining why we prefer the ‘digital literacies’ term, we will look at the variety of literacies of information. We will outline some of the more important concepts and models, consider the contexts in which they may be applied and the relation of generic and subject-specific digital literacies and how it may be affected by linguistic and cultural factors. We will then assess the contribution of digital literacies to the evaluation of information, and to the identification of misinformation and disinformation. Finally, we will look at the ways in which these abilities have been promoted and taught.

These topics have generated a very large literature, and our coverage here is necessarily selective and the examples only illustrative.

Why ‘digital literacies’?

Paul Gilster (1997), introducing the concept of digital literacy as the term is now generally used, explained it as an ability to understand and to use information from a variety of digital sources, and regarded it simply as literacy in the digital age. He emphasised that digital literacy is about mastering ideas and concepts, not keystrokes, thus distinguishing his conception from the more limited 'technical skills' view of digital literacy. It is 'the ability to understand and use information in multiple formats from a wide variety of sources when it is presented via computers' (Gilster, 1997, 1). Gilster specifically notes that digital literacy involves an understanding of how to complement digital resources with such things as reference works in libraries, printed newspapers and magazines, radio and television, and printed works of literature. This is particularly so since printed resources have probably been created from digital originals, if modern, or are liable to be digitised, if older.

There is a clear analogy with Floridi's conception of *onlife* (Floridi, 2014), with its seamless blending of physical and digital. Indeed, there is an analogy between Gilster's digital literacy as literacy for the digital age with Floridi's PI as philosophy for the information age. For this reason, we have chosen this term as most appropriate, but make it plural to distinguish it from some narrower conceptions called digital literacy. [We were tempted to write of *onlife literacy* or *infosphere literacy*, which would best fit our perspective, but the area is bedevilled by too many terms as it is.]

A variety of literacies

The terminology here is unclear, with a substantial and potentially confusing variety of 'literacies' of information, and of changing understandings of the same literacy (Bawden, 2001; 2008; Pinto, Cordon and Diaz, 2010; Koltay, 2011; Julien, 2016; Carrington, 2018; Sample, 2020). A bibliometric study of the information literacy literature identified over 70 different literacies associated with information between 1975 and 2018 (Onyancha, 2020). Literacies are sometimes combined, as for example 'media and information literacy' or 'information and data literacy', and the interrelationships are far from clear; for example, in some formulations digital literacy is a part of information literacy, in others vice versa. Sometimes they are combined in meta literacies or fluencies. One clear trend has been a move away from seeing these literacies as primarily skills based, in essence lists of things people should be able to do, to more broad and holistic conceptions, including attitudes and values, and social and ethical factors. Walsh (2015) describes them as 'competence' and 'relational' models, respectively, suggesting that the second type begins with the work of Christine Bruce (1997). Although the relational models are generally regarded as more realistic and comprehensive, the competence models are still of pragmatic value for structuring resources and instruction.

We will outline briefly some of the more significant definitions of, and models for, the various literacies, particularly showing their development over time. They are of importance because they provide the basis for all training and promotion in this area. Although this is usually designed around one model, it is possible to combine models as a basis; see, for example, a combination of

ANCIL (A New Curriculum in Information Literacy) and the ACRL (Association of College and Research Libraries) Framework (Dodd, 2017).

Information literacy

Information literacy is generally agreed to be a successor to traditional *bibliographic instruction* or *library orientation*. The term *library literacy* has also been used.

The concept of information literacy was introduced by Paul Zurkowski in 1974, in a report on the likely impact of the digital computer on information work, in which he wrote 'People trained in the application of information resources to their work can be called information literates'. It was first given a definite content in the influential American Library Association (ALA) definition of 1989: 'To be information literate, a person must be able to recognize when information is needed and have the ability to locate, evaluate, and use effectively the needed information.' This set a trend for information literacy to be expressed as a series of linear steps to be carried out, somewhat reminiscent of the process models for information behaviour considered in [Chapter 12](#). It may be noted that there has been surprisingly little explicit linkage between the conceptions and models of information literacy and of information behaviour, although the two clearly have much in common (Julien, 2016).

Similar was the 'Seven Pillars' model produced in 1999 by the UK Society of College, National and University Libraries (SCONUL). This saw information literacy as consisting of understanding and competencies in each of seven 'pillars' or 'headline skills': recognise

information need; find ways of meeting the need; construct strategies for locating information; locate and access; compare and evaluate; organise, apply, and communicate; synthesise and create.

In 2011, SCONUL revised their approach, making their pillars concept broader and more general: 'Information literacy is an umbrella term which encompasses concepts such as digital, visual and media literacies, academic literacy, information handling, information skills, data curation and data management. Information literate people will demonstrate an awareness of how they gather, use, manage, synthesise and create information and data in an ethical manner and will have the information skills to do so effectively.' This was a non-linear model, without the prescriptive sequential process and with 'lenses' for context; for example, a research lens, and a graduate employability lens.

In the United States ACRL produced a similar formulation in their Standards of 2000. Their six steps were: determine the extent of information needed; access the needed information effectively and efficiently; evaluate information and its sources critically; incorporate selected information into one's knowledge base; use information effectively to accomplish a specific purpose; understand the economic, legal, and social issues surrounding the use of information; and access and use information ethically and legally.

In 2015, ACRL introduced a new approach, their Framework, based on the idea of information literacy as a social practice rather than a set of competencies (Foasberg, 2015). The definition, 'Information literacy is the set of integrated abilities encompassing the reflective discovery of information, the understanding of how

information is produced and valued, and the use of information in creating new knowledge and participating ethically in communities of learning', is non-linear and inclusive, and accompanied by a series of six 'frames' or threshold concepts (Oberlies and Mattison, 2018). These are:

- authority is constructed and contextual
- information creation as a process
- information has value
- research as inquiry
- scholarship as conversation
- searching as strategic exploration.

Like the preceding Standards, and despite some criticisms of a lack of structure and specificity, the Framework has been influential in the design of instructional courses and materials (Hsieh, Dawson and Yang, 2021).

In the UK, CILIP produced their guidance in 2004. This defined information literacy as 'knowing when and why you need information, where to find it, and how to evaluate, use and communicate it in an ethical manner', calling for an understanding of: the need for information; the resources available; how to find information; the need to evaluate results; how to work with or exploit results; ethics and responsibility of use; how to communicate or share your findings; how to manage your findings. This was revised in 2018 along the same lines as SCONUL and ACRL: information literacy in this formulation is 'the ability to think critically and make balanced judgements about any information we

find and use. It empowers us as citizens to reach and express informed views and to engage fully with society.'

International organisations also became interested in information literacy, with UNESCO holding meetings in Prague (2003), Alexandria (2005) and Moscow (2012). The ensuing Prague Declaration followed the skills-based approach: 'Information literacy is a prerequisite for participating effectively in the information society ... Information literacy encompasses knowledge of one's information concerns and needs, and the ability to identify, locate, evaluate, organise and effectively create, use and communicate information to address issues or problems at hand'. The Alexandria Proclamation (2005) emphasised information literacy as a universal social good: 'Information literacy empowers people in all walks of life to seek, evaluate, use and create information effectively to achieve their personal, social, occupational and educational goals. It is a basic human right in a digital world and promotes social inclusion in all nations.'

The Moscow conference conjoined information and media literacies, as noted below.

An example of an information literacy model aimed at a specific context is A New Curriculum in Information Literacy (ANCIL), aimed at new entrants to the British university system (Secker and Coonan, 2012). This defines information literacy as 'a continuum of skills, behaviours, approaches and values that is so deeply entwined with the use of information as to be a fundamental part of learning, scholarship and research. It is the defining characteristic of the discerning scholar, the informed and judicious citizen and the

autonomous learner.’ It has ten ‘strands’: transition from school to higher education; becoming an independent learner; developing academic literacies; mapping and evaluating the information landscape; resource discovery in your discipline; managing information; the ethical dimension of information; presenting and communicating knowledge; synthesising information and creating new knowledge; the social dimension of information [transferring information literacy skills to the workplace, everyday life, lifelong learning]. The strands are not intended to form ten discrete modules or classes; rather, they are to be incorporated into subject-specific curricula as required by the nature of the discipline. ANCIL is based on the idea that separating basic skills and competencies from higher-order abilities disadvantages students, and regards information literacy as a continuum of skills, competencies, behaviours and values around information.

Another model for a specific context is that produced as the basis for development of an information literacy MOOC (Massive Open Online Course) (Dreisiebner et al., 2021). This defines information literacy as ‘a set of competencies that empowers citizens to access, retrieve, understand, evaluate and use, create, as well as share information and media content in all formats, using various tools, in a critical, ethical and effective way, in order to participate and engage in personal, professional and societal activities’, and focuses on five areas: understand and engage in digital practices; find information; critically evaluate information, online interactions and online tools; manage and communicate information; collaborate and share digital content.

Finally, we should mention *critical* or *radical information literacy*, not a model so much as a general approach to the relation between people and information through the perspectives of critical theory and critical literacy, and an understanding of the social construction of information, often with an explicitly political or social justice emphasis; see, for example, Whitworth (2014), Downey (2016), Špiranec, Banak Zorica and Kos (2016), Tewell (2018), Špiranec, Kos and George (2019), Pangrazio and Selwyn (2019) and Flierl and Maybee (2020).

Media literacy

Media literacy, also termed *mediacy*, and subsuming *social media literacy* and *news literacy*, is generally regarded as the skills needed to access and evaluate information from a variety of media and to understand the nature of different media (Livingstone, 2004; Potter, 2021). Creating media is also commonly included, and the concept has been associated with *visual literacy*, the ability to critically and ethically view, understand, use and create images and visual media (Brown et al., 2016; Michelson, 2017; Thompson, 2020), with *critical thinking* or *critical skills* (Dodd, 2017; Goodsett, 2020) and with *civic literacy*, *data citizenship* and *digital citizenship* (Carmi et al., 2020; Hintz, Dencik and Wahl-Jorgensen, 2019).

Media literacy and information literacy, though clearly having much in common, developed independently and have been combined only relatively recently (Lee and So, 2014). The UNESCO Moscow Declaration on media and information literacy (2012) noted that ‘media and information literacy (MIL) is a prerequisite for the sustainable development of open, plural, inclusive and participatory

knowledge societies . . . MIL is defined as a combination of the knowledge, attitudes, skills and practices required to access, analyse, evaluate, use, produce and communicate information and knowledge'. In July 2021, the Media and Information Literacy Alliance (MILA) was established by CILIP and its Information Literacy Group to promote MIL in the United Kingdom as a fundamental building block for a healthy, inclusive and democratic society (<http://infoilit.org.uk/mila/>). MILA endorses the definitions and concepts set out by UNESCO in their Media and Information Literacy programme.

Digital literacy

Digital literacy is generally taken as an all-embracing successor to a variety of technical literacies, including *computer literacy*, *IT literacy*, *internet literacy* and *network literacy*, together with the more recent idea of *algorithmic literacy*, understanding the nature of algorithmic decision making and its potential for inaccuracy and bias (Carrington, 2018; Lloyd, 2019; Ridley and Pawlick-Potts, 2021).

These earlier concepts typically related to a basic ability to use computing hardware and software. More recently they have come to mean an ability to cope in a digital environment, including using search engines effectively, evaluating digital information, creating and sharing digital information, using social media appropriately and avoiding online risks, and behaving ethically online. Gilster's conception of digital literacy is of this kind, as are others (Gilster, 1997; Lankshear and Knobel, 2008; Reedy and Parker, 2018). The term *information fluency* has also been used for this idea (Bawden, 2014; Emmett, 2016).

IFLA's 2017 statement on digital literacy is of this nature: 'To be digitally literate means one can use technology to its fullest effect – efficiently, effectively and ethically – to meet information needs in personal, civic and professional lives . . . the definition of digital literacy remains fluid . . . digital literacy overlaps with other areas of competence, notably media and information literacy . . . and finally digital literacy is a lifelong learning process'.

Marcum and O'Shea (2017) use the term *information technology literature* to describe this kind of broad, multidimensional set of skills and knowledge, regarding it as more complex than information literature and heavily dependent on the context.

Digital literacy also typically includes *data literacy*, the ability to find, understand, use and communicate data in all its forms, including visualisations, and incorporating an understanding of how to create, control and protect our personal data (Pangrazio and Selwyn, 2019). Data literacy, sometimes conflated with information or media literacy, is of increasing importance in view of the greater availability and usage of datasets (Carmi et al., 2020; Špiranec, Kos and George, 2019; Carrington, 2018; Usova and Laws, 2021).

Metaliteracies

Finally, we may mention *metaliteracies* of information. The metaliteracy model (Mackey and Jacobson, 2014; Jacobson and Mackey, 2016) is an open, non-linear and flexible framework that unifies information literacy with related literacies, in particular with media literacy and with computer literacy; it places great importance on collaboration and reflection and on an ability to critically self-assess different competences. It regards skills-based models as

inadequate for the present digital information environment, especially its emphasis on open learning, social media and participation/creation/collaboration. Foregrounding the importance of information literacy for lifelong learning, it also recognises that, since the information environment is constantly changing, courses and learning materials based on the metaliteracy model must also constantly be revised.

Contexts for digital literacies

Information, digital and media literacies have been promoted in higher education, much more than in other settings. This is slowly changing, as they are increasingly seen as worth promotion in schools (Laubersheimer, Ryan and Champaign, 2016; Walton, Pickard and Dodd, 2018), in workplaces (Forster, 2017) and in society generally (Goldstein, 2020). To be fully effective, this is likely to require different concepts of digital literacies and different methods of promotion.

Clearly, much information literacy is discipline and context specific: as examples, see Bawden and Robinson (2017) for chemistry and Houlihan, Click and Wiley (2020) for business and, for a comparison over several disciplines, Dreisiebner and Schlögl (2019). Indeed, there has been a long-lasting debate whether there is such a thing as generic information, digital or media literacy, or whether it must always be context bound. The evidence suggests that there is a core of basic principles, but these are best presented in the relevant context, as is current good practice.

Most conceptual and pedagogic development of digital literacies has been carried out in the developed Anglophone world and there

has been relatively little consideration of what adaptations may be necessary for different cultural settings. This issue has been recognised for many years (Bruce, 1997). As Johnson and Webber (2003, 112) put it: 'in terms of local and national culture, the information literate person is a self- and socially-conscious being, rather than a simple repository of skills and knowledge. This is underlined by cross-cultural difference, where issues of behaviour and acceptability of kinds of information become sensitive'. See Zimmerman and Chaoqun (2021) for an analysis of the context in which information literacies are discussed in academic papers from different parts of the world. There have been many descriptions of information literacy education in different settings, but little analysis of the needs of different cultural groups, and little agreement on good practice. Suggestions have included using locally relevant examples, considering whether individual or group working is more acceptable and using models such as Hofstede's cultural dimensions (ACRL, 2017; Nowrin, Robinson and Bawden, 2019).

An extensive set of studies are those of Dorner and Gorman, drawing on analyses of the contexts of Asia and Oceania (Dorner, 2012; 2017, Dorner and Gorman, 2011). They argue for explicit consideration of cultural factors, using Hofstede's dimensions, and in favour of models using a critical form of information literacy, and suggest, for example, that student-centred learning may not be appropriate and that collaborative group working will be better accepted than individual work. Špiranec (2017) argues that critical information literacy, because of its support for multiple perspectives and for societal as well as personal development, offers the best

framework for instruction in transitional and post-conflict societies, and that this is better supported by approaches such as the ACRL Framework, rather than more prescriptive approaches such as the ACRL Standards.

Evaluating information and avoiding mis- and disinformation

Evaluation of information has been a part of all formulations of digital literacies. The need for explicit evaluation of information has always been important but it has become more pressing in a largely digital information environment, for several reasons:

- more information is available, so a greater degree of selectivity is needed
- a greater proportion of information is generated outside the formal 'publication chain' and therefore may have little or no initial quality control
- web browsers have a homogenising effect. All information has a similar 'look and feel', whereas previously the differences between, say, a textbook, a legal case report, a newspaper article and a Post-It note were immediately obvious
- the origin of information, the motives of its originator(s) and its trustworthiness may be unclear
- many people rely on social media sources, rather than on formal media, for news, and even for information on topics such as health
- technology increasingly allows the easy creation of 'deep fakes', false images and videos.

The result has been a concern for the proliferation of *misinformation* (false information), *disinformation* (deliberately false information) and *malinformation* (true information presented for malicious motives) and for the phenomena of *fake news* (Buschman, 2019; Rubin, 2019; O'Connor and Weatherall, 2019; Carmi et al., 2020; Zimdars and Kembrew, 2020) and of *post-truth* and *alternative facts* (Cooke, 2017). This has led to a revival of the long-established practice of 'fact checking' and, of particular relevance here, to initiatives for training in the evaluation of information and information sources, often within a framework of information or digital literacy (Revez and Corujo, 2021; Agosto, 2018; Cooke, 2018). A somewhat naïve view was held for a long time that all that was necessary to defeat these problems was more training in information, media and digital literacies. This optimistic view has been dispelled by the realisation that the reasons why such problems exist are far more complex than a simple lack of digital literacy. The commonly used checklist methods of information evaluation, such as the widely used CRAAP (Currency, Relevance, Authority, Accuracy, Purpose) list, applied within a source, are of limited value beyond a basic appreciation of evaluation criteria. An approach of *lateral reading*, leaving the source being evaluated in order to see its context and what other sources say about it, is needed in many cases (Wineburg and McGrew, 2019). It is also clear that many of those who provide false information, for whatever motives, have a very sophisticated understanding of information handling. Improvement in levels of digital literacies will therefore be a help, but it will require more sophisticated approaches than heretofore, involving a

variety of actors (Bontcheva and Posetti, 2020) and focusing on promoting a broader understanding of 'how information works' (Polizzi, 2020; Fister, 2021).

Promoting digital literacies

How to teach and promote information literacy is a continuing debate, and many different approaches have been tried and are in use (Robinson and Bawden, 2018; Saunders and Wong, 2020). There has been a general reliance on traditional classroom settings, supported by online tutorials and LibGuides (Courtney and Wilhoite-Mathews, 2015; Miller, 2015; Burkhardt, 2016; Saunders, 2018). Other methods have been considered, including MOOCs (Dreisiebner, 2019; Dreisiebner et al., 2021), games (Urban, 2019; Jerrett, Bothma and de Beer, 2017; Walsh, 2015), peer-led learning (Aylward, Sbaffi and Weist, 2020) and activities such as source evaluation, creation of resource lists, reflective writing, and flipped classrooms (Rodriguez, 2016; Johnson and Barrett, 2017; Humrickhouse, 2021). Learning materials may be in the form of text and audio or, increasingly, video and animation. Walsh (2015; 2020) notes that the use of instructional games may be extended to a wider approach of *playful learning*, including elements such as storytelling and performance.

No single pedagogical framework has been agreed as optimal. Rather, a variety of frameworks have been used, seemingly ad hoc, for both the creation and the application of instructional materials. These include: Biggs constructive alignment pedagogy; Bloom taxonomy of educational objectives; Honey Mumford learning

styles; Kolb experiential learning style theory; Schön reflective practice; and Knowles androgogy.

A range of forms of pedagogic interaction have been applied in digital literacies instruction, including:

- traditional instruction (instructor led, didactic, passive)
- online tutorial/workbook instruction (didactic, passive)
- active learning (instructor takes a facilitator role)
- learner-centred (focus on each student's individual needs)
- self-directed learning (independent, student takes responsibility)
- interaction and collaboration between learners.

Where these have been evaluated and compared, little demonstrable difference in effectiveness has been demonstrated between particular instructional methods, or between face-to-face and online delivery, or between online instruction with different degrees of interaction. As Reichart and Elvidge (2015, 146) summarise the situation: 'The variety of methods employed illustrates that there is no magic bullet approach to IL.'

Summary

Many different literacies of information, and components and models for them, have been proposed. Literacies for onlife, which we think are best termed digital literacies, must include specific competences, as in the older conceptions of information, media and computer literacy, but must go beyond these to broader capabilities and attitudes, including a full understanding of the new information environment, the infosphere. More interaction between the

understanding of information behaviours and practices, and promotion of digital literacies, is desirable. A variety of methods have been used for the promotion of digital literacies. Imaginative approaches will be needed in the future, particularly for developing means of combatting misinformation and disinformation.

- There are many names, definitions, and models for the literacies of information, and equally many means for promoting them.
- Capabilities, attitudes and values are as important as skills and competencies.
- An appreciation of the new information environment is essential.

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18 Research in the Information Sciences

Can it be that there are no problems in library service that call for scientific research? Nothing more to learn? No unsolved problems? Is it to be classed with – what shall I say? – the activities of the street-car conductor, the waiter, the clerk?

Charles Williamson (address delivered at the Founder's Day Exercises, School of Library Science, Western Reserve University, 10 June 1930)

Research is to see what everybody else has seen, and to think what nobody else has thought.

Albert Szent-Györgi, Hungarian biochemist

Research is formalized curiosity. Poking and prying with a purpose.

Zora Neale Hurston, American novelist and anthropologist

Introduction

Research is generally regarded as the creation of new knowledge; more specifically, as the creation of knowledge which is new to everyone, not just to the researcher, in a form which can be communicated through objective information, and which builds on existing knowledge. Communication to the wider community is the vital final stage of research, through the processes of scholarly communication and further dissemination discussed in [Chapter 13](#).

Research is often described in terms of major discoveries by brilliant individuals. But it is usually more realistic to think of it as the creation of new knowledge within a discipline by an evolutionary and impersonal process, with contributions by many

individuals, most of whom will never be credited with a major advance (Scerri, 2016).

For a discipline with an associated professional activity, such as information science, research has two purposes: academic, to improve the discipline's knowledge base; and professional, to improve practice. Information research may be carried out by university researchers, advanced students, research institutes and consultancies, government agencies, system and service designers and suppliers and practitioners. Research has been an important aspect of information science since the discipline's beginnings.

In this chapter we will consider first the somewhat troubled research/practice interaction, then consider information research and methods and finally mention research ethics. There are several textbooks for this area, and the coverage here will not be detailed.

Research and the practitioner

Interaction between academics and researchers on the one hand, and practitioners on the other, is as desirable in the information disciplines as in any other academic area with an associated professional discipline. Practitioners may be in a position to carry out research themselves, and should certainly be aware of, and make use of, research results. It is beneficial for both if practitioners carry out their own research and use research results and if researchers are aware of practice issues in planning and carrying out studies, as has been recognised for many years; see, for example, Williamson (1931).

In the information sciences there has in the past, as with other disciplines, been a divide between academics/researchers and

practitioners. The latter have been held to be uninterested in research results, still less in carrying out research themselves; the counter prejudice has been that information researchers undertake 'useless' research, ignoring the needs of practice. As long ago as 1983, Blick presented this as a kind of contest or conflict between practitioners and researchers in information science, and the nature of the divide has been investigated, and solutions proposed, since then: see, for example, Haddow and Klobas (2004), Hall (2010), Finlay et al. (2013), Partridge et al. (2014), Pilerot (2016) and Mizrachi (2018). There is an analogy with the use, or non-use, of theory by practitioners; see, for example, Pinfield, et al. (2020, chapter 4).

The gap may be bridged by research embedded in a practice context, where the aim of the research is the improvement of practice as much as the extension of the disciplinary knowledge base, typically termed *action research*. The description *use-inspired research* has been applied to research which is an academic study of real-world problems, producing fundamental understanding but also benefitting practice. Application of the principles of *knowledge translation* might assist this form of gap bridging (Garnett, 2011). There is also the rather different issue of academic librarians becoming involved with research in disciplines other than LIS, typically assisting researchers in their institutions with the communication and data curation aspects of research (Borrego and Pinfield, 2020).

Perhaps not surprisingly, practitioner research has been concentrated in the library/information services of institutions which are themselves research based, notably university libraries and

industrial research centres, where a research culture is naturally developed (Picton, 2016; Walkley Hall, 2016). Other initiatives have set up networks for researcher–practitioner interaction and potential collaboration; see, for example, Hall, Cruickshank and Ryan (2019) and Hall and Ryan (2020).

There is now greater pressure for practitioners to be involved in research and in using research results and insights, and for academics and other researchers to pay greater attention to the dissemination of results in a way useful for practice: funders of research are increasingly requiring attention to be given to social and economic impact. Communication between the two groups is of obvious importance: scholarly publishers increasingly ask for statements of implications for practice to be made in reports of information research, while practitioner publications carry evidence summaries and research reports.

Many practitioners carry out research as an integral part of reflective practice, usually aimed at service evaluation and improvement and not necessarily intended for publication (Mizrachi, 2018). This has led to the increasingly accepted idea that library and information service practice should be *evidence based*, with much evidence provided by research (Koufogiannakis and Brett, 2016). The usual idea of evidence-based practice has been criticised as narrow-minded and mechanical, by those who argue that practitioners need to have a fuller understanding of research and theory; see, for example, Hjørland (2011).

Information research approaches

All research is based in a philosophical viewpoint or paradigm, although this is not always explicitly recognised by researchers, and the viewpoint determines the research approach (Creswell and Creswell, 2018). A rough distinction can be made between studies conducted from a *positivist* viewpoint, where it is assumed that an objective reality exists, which the researcher may identify and study, and those conducted from a *constructivist* or *interpretivist* viewpoint, which assumes that reality is subjective and must be constructed or interpreted by the researcher. For information research, the former assumes, for example, that concepts such as the meaning or relevance of a piece of information to a user, or the proper role of an information service, are objective facts which may be determined correctly; the latter assumes that they are subjective, and created and modified by those involved. The former assumes that the researcher is detached from the context studied; the latter allows for them to be involved, and may be termed *participatory research*. There is also a *pragmatic viewpoint*, which assumes no philosophical position. This should be a deliberately chosen stance, not a thoughtless, seemingly easy default.

While information researchers need not be philosophers, it is important to recognise what assumptions are implicit in the methods used, and what assumptions or viewpoints the researchers themselves hold which may affect their approach. A detailed discussion of the relation between research, philosophies, approaches and methods is given by Creswell and Creswell (2018); see Berryman (2019) and Polkinghorne and Given (2021).

A common distinction is between *quantitative* and *qualitative* research, approximately corresponding to positivist and interpretivist viewpoints. These describe a difference in general approach: whether the focus is on measuring, counting and testing formally stated hypotheses or on an interpretation of the meaning of events, issues and opinions, in an attempt to gain understanding. In practice there is overlap: few studies are solely one or the other. Each has its own ways of analysing the data generated in the research process, and qualitative data collection and analysis are no less rigorous than quantitative methods.

Mixed methods studies, corresponding to a pragmatic viewpoint and incorporating both qualitative and quantitative methods, have become increasingly common in information research. They have advantages in the richness of data collected, but require careful analysis and synthesis of the results (Tashakkori, Johnson and Teddlie, 2021; Granikov et al., 2020).

Information research methods: general aspects

A wide range of research methods and techniques are used in the information disciplines (Ullah and Ammen, 2018). These methods are explained in detail in textbooks such as Pickard (2013), Connaway and Radford (2021), Williamson and Johanson (2017), Denscombe (2017) and Billups (2020).

We consider methods in four categories: document-based research; experiment and observation; evaluation and design; and surveys. These are not mutually exclusive, and mixed method studies are increasingly common. The term *triangulation* is used to describe studies which mainly rely on one method but then use a

second method to check the validity of the results; for example, questionnaire surveys which include some interviews.

Evaluation is such an important part of research in many subjects, including the information sciences, that it is often treated separately. Evaluation may be carried out on a micro level, evaluating, for example, the content of a small set of literature, or a macro level, evaluating systems and services. Research embedded in a practice context, where the aim of the research is the improvement of practice rather than an objective study of the situation, may be termed *action research*.

All research of whatever kind will involve an element of *document-based research*, in the form of a literature review for context setting and to establish if similar studies have already been done. For document research proper, this will form an introduction; for example, a literature review of subject X will begin with a description and analysis of any previous reviews of that subject. 'Literature' here is defined very broadly, as it may include information in sources other than academic and professional literature.

All research which has any element of data collection will need to consider the question of sampling: how to know that an adequate number and range of cases have been included for the results to be valid, and how to know how generalisable to other situations the results will be. The sampling issue can be avoided only when all relevant cases are considered; for example, in studying some aspect of national libraries in the British Isles it is feasible to study all of

them. The extreme example is the case study, where a single example is studied in great detail.

Advances in information and communication technologies, allowing greatly increased options for communication between researchers and for collection, integration and analysis of large amounts of data, are leading to the new style of research, termed *e-research* or *digital research* (Costa and Condie, 2020). This may affect information research directly, and information specialists may have a role in promoting and facilitating it. Research using artistic and visual methods is also a relatively new departure (Ward and Shortt, 2020; Bedi and Webb, 2020).

It is essential to plan a study by starting with research questions and working through the kind of results needed to answer them and then to the data collection and analysis methods which will give those results. All research, even small-scale and relatively informal studies, requires careful planning to ensure that all the resources needed, including access to people and systems to be studied, are in place and that everything happens in the right order. Ethical issues, management of the data collected and dissemination of the results must also be planned in advance.

Research methods (1): document-based research

Document-based research, commonly termed *desk research*, involves research carried out by any form of analysis of documents of any kind. This is a part of all research, in the form of a literature search to establish the context and to identify previous relevant work. It may provide the methods for studies in their own right, which are entirely valid research, inasmuch as they have the potential to

provide new knowledge. Document research may also be referred to as *literature research* or *internet research*, although both these terms are too narrow to cover all forms of document research. Several styles can be distinguished, although the distinctions are not sharp, and the terms are often loosely used; some studies cross the boundaries between styles.

Literature reviews are the most common form, and may form a study in their own right as well being a precursor to other research methods (Fink, 2014; Hart, 2018). They may be designated as *comprehensive* or *selective*, according to whether an attempt is made to cover all relevant material or only a subset which the reviewer finds significant. A *systematic review* carefully defines and justifies the research question, the sources and search strategies and the relevance criteria to be used, before material is identified. A *scoping review* or *mapping review* or *narrative review* is a comprehensive examination of the scope, extent and content of the literature in a particular topic, without the very specific criteria of a systematic review. For a comparison of different forms of comprehensive review that may be done, see the reviews of eye-tracking research on search engine use by Lewandowski and Kammerer (2021) and by Strzelecki (2020).

A review may be objective, simply reporting what is in the literature, or subjective, in that the reviewer gives a judgement on the quality of the material and its content; the latter may be called a *critical* review. Reviews focused on an emergent or developing technology and its prospects may be termed *technology assessments*.

Meta-analysis is a specific kind of literature review, in which quantitative data from a number of sources are combined and analysed together; necessarily it requires comparable numeric datasets. *Meta-synthesis* is the equivalent term for systematic analysis of multiple qualitative studies (or sometimes a mix of qualitative and quantitative studies), using interpretivist methods such as thematic analysis and grounded theory, rather than statistical analysis, to combine the findings of the studies (Braun and Clarke, 2006; Bawden, 2012; Kie et al., 2020). Meta-synthesis is likely to be of more value than meta-analysis in information science, because of the limited number of comparable quantitative studies for inclusion.

Conceptual analysis and *philosophical analysis* are used to analyse and clarify terms, concepts and issues (Furner, 2004). Conceptual analysis relies on examining usage of concepts in documents; philosophical analysis may be more abstract.

Historical research analyses developments over time, using documentary and archival evidence, personal recollections, etc. For the information sciences, this has typically been focused on the development of libraries and other information institutions, or information systems, services, processes and concepts, and of information phenomena in society.

Content analysis is a varied form of research, involving some kind of quantitative assessment of the content of a set of documents to assess the extent to which concepts and issues are mentioned, or not (White and Marsh, 2006; Armann-Keown and Patterson, 2020). It may include a qualitative dimension, to record *how* concepts and issues are described.

Discourse analysis is a form of content analysis, focusing on the way in which spoken or written language is used. It is used to analyse, often in detail, the way in which concepts and issues are mentioned and what this shows about how they are understood (Budd, 2006).

Bibliometrics and *webometrics* are quantitative methods for describing and analysing patterns of recorded communication. They are used to carry out research into such topics as: the size and growth of information within disciplines; significant sources, authors, institutions and countries; linkages and influence between information producers; and changes in communication patterns (Thelwall, 2008; Sugimoto and Larivière, 2018). A related method is *social network analysis*, for the study, primarily quantitative, of interacting and communicating individuals and groups (Hicks, Cavanagh and VanScoy, 2020).

Research methods (2): experiment and observation

This category of research covers a variety of methods. At one end of the spectrum are those described as *experimental*, based in a positivist, scientific worldview. *Observation* research was originally located within this approach as a largely quantitative technique counting how many people did what and how often; it now embraces methods within an *interpretivist* approach, based in the traditions of ethnography and similar disciplines. These methods are also often subsumed into mixed method approaches. They are typically used to understand information behaviour and to understand in detail how information systems and services are used

and how well they work, so as to provide a basis for the improvement of information systems and services.

The experimental style of research in library/information science derives from the approach of the physical sciences. The topic to be studied is isolated, so far as possible, from the complexities of the real world so that all the variable factors in the situation can be held constant, apart from the one being studied. It is therefore particularly suitable for comparisons of algorithms and interfaces. Its opposite is *operational evaluation*, analysing the totality of an information service and including all the messiness of the real-world context. The experimental approach was pioneered in information retrieval research (Sparck Jones, 1981; Robertson, 2008) and is still most commonly used there, and in HCI studies, using methods such as eye-tracking (Lund, 2016). Where users are involved, data on their success in using the system may be augmented by data gathered from special equipment, such as eye-tracking devices to see where they look on the screen, and by user opinions given by conventional interview or by 'talk/think aloud' or 'talk/think after' features of the experiment.

At the other extreme, *observation* is usually regarded as a technique for understanding a real-world situation without affecting or influencing it at all; hence a phrase commonly used in the past, *unobtrusive observation*, when the subjects are not aware that they are being studied. The use of the so-called *mystery shopper* or *mystery visitor* is a variant for examining user-facing services. Observation is increasingly no longer carried out visually; rather, it is done by means such as asking participants to keep detailed diaries (including

photo and video diaries) or by recording their activities by text messaging. There are obvious ethical issues involved in unobtrusive approaches, but observation may be a participatory technique, if carried out openly, with the observer as part of the context.

The current trend is for observation, often used in conjunction with other data-gathering methods, to follow approaches such as anthropology, ethnography and phenomenology/phenomenography, which emphasise an interpretation of the situation from the viewpoint of those observed (Forster, 2016; Bamkin, Maynard and Goulding, 2016). Phenomenology focuses on the subjective first-person experience, and phenomenography on observation of people's experiences, though the terms are used inconsistently. The term *information experience* is increasingly used to describe this approach (Bruce et al., 2014). Results are increasingly expressed in terms of *personas*, archetypal or representative imaginary persons. This unstructured way of proceeding, typically based on observation, is sometimes referred to as *grounded theory* – often incorrectly, since this is a quite specific theory and way of proceeding (Charmaz, 2014; Urquhart, 2014).

A new form of observation became possible with the move to a largely digital information environment: the ability to analyse the logs of websites and search engines so as to establish exactly what a very large number of users are doing, in a way not possible with printed information. However, such log analysis cannot give any explanation for why users are doing what they do, nor how satisfied they are. It is therefore often combined with other methods, typically

asking the subject to say what they are doing at the time (*talk aloud*) or after the session (*talk after*).

Research methods (3): evaluation and design

This category of research covers methods to evaluate the performance of information systems and services and to design new ones; design research is now accepted as a research method in its own right.

Research studies of many kinds, including the experiment and observation methods noted earlier, usability testing in particular, can be used for the practical purpose of the *evaluation* of the performance of information systems and services. Here, research overlaps very much with practice, since the monitoring of performance should be a routine activity for information providers, using tools such as standard sets of performance indicators, user-satisfaction surveys and cost-benefit analyses. Research in the context of the evaluation of operational systems tends to mean particular initiatives or projects. Examples are: an *information audit*, whose purpose is to enumerate and evaluate all the available information resources (Frost and Choo, 2017); an *impact study*, aiming to show the ‘real benefit’ to the users (for example, Rutter and Rutter, 2019, on the impact of medicines information services); or a financial *return on investment* (ROI) study (Madden et al., 2016; Urquhart, 2020). In keeping with our understanding of research, they should be communicated outside the organisation in which they are conducted, and should add to the knowledge base of the discipline.

Research may involve the design and implementation of new systems and services, or new features within existing systems

(Clarke, 2018; Vermeeren et al., 2016); by definition, this is an example of *action research*, since one of the results of the research should be improvements to practice. *Participatory design* is a user-centred approach increasingly used in developing information systems and services (Yip et al., 2020). Again, there is an overlap with practice, as these are also normal tasks for the practitioner; they are typically regarded as research when there is some novelty in the new system or feature or when an established feature is introduced into a new context. Research into information needs and behaviour is also needed in order to develop or extend systems to go beyond incremental improvement of what is already available.

Research methods (4): surveys, interviews and Delphi

Methods of this kind, and particularly questionnaires and interviews, are the most widely used information science studies and are commonly used in mixed-methods research, typically combining a survey with some form of observation or document-based research. *Longitudinal* studies are carried out over a period of years so as to assess changes over time; as examples see Tenopir et al. (2009) and Borgman et al. (2021).

Most library/information surveys are small scale, rarely involving more than one hundred participants, and are usually aimed at a closely defined group of people. They do not therefore usually adopt the rigorous methods of sampling and analysis used for purposes such as public opinion polling or large-scale market research.

Questionnaires are usually regarded as being in the positivist/objective style of research, since they assume that researcher and participants share a common perspective of the

situation. A series of short, structured questions with prescribed answers (yes/no, Lickert scale, multiple choice) are generally used, with limited opportunity for expressing extra information, giving reasons or raising other topics. This approach is appropriate for getting large numbers of responses quickly and simply and for giving results which can be analysed quantitatively. However, the value of the results relies in the researcher's having understood all the relevant factors and expressed them in a way which the participants can understand. For a basic introduction to questionnaire design, see Kara (2019).

Questionnaires are now usually administered electronically rather than in printed form. This is convenient and gives greater reach, but may mean that the population sampled, the response rate and any bias in response are unknown. If such a survey is placed on a website, for example, all that can be known is those who replied to it; the number who saw it and did not respond cannot usually be established, nor whether there is anything different about those who responded and those who did not.

Interviews are regarded as appropriate for getting a greater, and richer, amount of information from a smaller number of participants; they may be carried out in person or by telephone, video, or e-mail. There are several forms of interview. *Structured interviews* resemble questionnaires, in that a predefined set of questions are asked, often requiring a choice of answer from a list. At the opposite extreme *unstructured interviews* have no predefined structure and resemble a free-flowing conversation; they are most appropriately used in an exploratory way, in situations where the interviewer does not have a

good understanding of the issues or context. An intermediate form, and often regarded as most appropriate for most library/information research, is the *semi-structured interview*, where a number of predefined questions are always asked, but subsequent questioning will depend on the responses given.

Interviews are usually carried out one to one. *Group interviews* may be used on grounds of practicality and time saving. They may also be useful in allowing interviewees to discuss the questions asked, reach consensus and spark ideas off each other; but equally they run the risk of being dominated by a small number of participants, such that other views are not heard. The term *focus group* is often used synonymously with group interview, but may imply a preference for discussion of an issue, rather than answering a set of questions.

A tool specifically designed for reaching consensus within a group is a *Delphi study* (Lund, 2020). This brings a virtual group of participants together, typically by e-mail. They are asked individually to state their views on some topic. The researcher writes a summary of their responses and sends it out to them for further comment; this process is repeated as many times as necessary. Usually two or three 'rounds' are enough to reach consensus or to establish that there are two or more incompatible viewpoints.

An approach which may be used with any of the above methods is the *critical incident* technique. This asks participants to specify one particular occurrence of what is under investigation; for example, one occasion when use of a particular information source made a difference to their work. This approach avoids very general and

bland responses and is particularly useful in assessing the impact of information systems and services.

When a survey is begun, a *pilot study* may be carried out with a small number of participants to check the validity of the survey questions and allow for changes to be made for the main study. In a large-scale study, particularly if it is to be analysed quantitatively, the results of the pilot study would not be included, though in smaller surveys they may be.

Sampling, data analysis and statistics

A full treatment of statistics and data analysis is beyond the scope of this book. There are many good resources available for these topics, so here we reiterate some central points, making some remarks about sampling and about the question of ‘how much is enough’. For basic introductions, see Frey (2015), Campbell and Swinscow (2009), Rowntree (2018) and Spiegelhalter (2020) and from an LIS perspective Walters (2021).

A fundamental distinction is between *quantitative* and *qualitative* data analysis. Qualitative analysis, not primarily involving numbers, has its own rigorous methods, typically involving repeated coding, categorising and comparing of data. Quantitative analysis is typically divided into *descriptive* and *inferential* statistics. The former seeks to describe and summarise a set of collected data through numeric calculations and through diagrams and visualisations. The latter seeks to decide what may be reliably concluded and generalised from a set of data; this is the realm of *statistical significance* and *hypothesis testing*. What kind of statistical tests are appropriate will depend upon the nature of the data available:

nominal (categories or groups); ordinal (ordered); ratio (positive numbers); interval (numbers). The ready availability of easy-to-use statistical analysis packages makes it important that we are aware of which tests are appropriate to use, in view of the kind of data we have, and how to interpret the results.

Sampling

‘Sampling’ is the procedure by which a selection is made of entities (people, organisations, documents, etc.) to study, when there are too many for all to be studied. The aim of sampling is to choose a sample representative of the larger population from which they are drawn, so that the results obtained have a more general validity beyond the particular entities which were studied.

The issue of sampling very often arises in information research, regardless of the method used. It is often thought of in the context of surveys – how should we choose our participants? – but is equally applicable in, for example, the choice of documents for content or discourse analysis, or the choice of place and time for observations. It raises two questions: what kind of sample is to be used; and how to know when it is enough. This is a complex subject, and only some simple points are made here. Texts on research methods and on statistics cover the topic more thoroughly.

Four general kinds of sample may be taken.

- A *complete sample*, where we include all examples of the population. For example, in a study of archiving policies in local government in London, an archivist in each of the 33 government units could be interviewed.

- A *random sample* of our population. This has to be done formally, typically using tables of pseudo-random numbers. Many samples described as 'random' are no such thing, having simply been picked arbitrarily.
- A *purposive sample*, chosen to include examples of different types within our population. For example, to get a sample of users of a website, we might choose to select by age, gender, nationality, education level, place of residence, occupation, etc. Having chosen these groups, we might then select individuals within them randomly. This is a complex process and must be done carefully, according to the nature of the research. This may be referred to as a *systematic, clustered, quota* or *stratified* sample, according to the exact procedure used. If the participants are individually chosen by the researchers, it may be termed a *judgement* sample.
- A *convenience sample*, where a set of appropriate participants are selected because they are available: family, friends, fellow students, regular users of an information service, etc. This may be an appropriate method for small-scale and exploratory studies, provided that (1) it is declared as such and (2) no attempt is made to generalise the results or to claim statistical validity; both of the latter would need a more formal sampling process. A *snowball* sample is a convenience sample where participants are asked to select others.

The question of how large a sample is needed is often asked, but is difficult to answer exactly. A numerical answer can be given only in those cases when the magnitude of the difference looked for, the

level of statistical confidence required in the results and the underlying distributions of the variables are all known. In much information research little can be assumed about the underlying distributions, and the other parameters are uncertain. What is usually done is to judge whether the sample allows a reasonably confident answer to the research questions. This usually this means two things: does the sample include all groups of the population studied; and are there enough cases in each group. There is no formula to give the answer, and it must be decided by common sense in each case – usually, in small-scale studies, by doing as many cases as is feasible. In the case of qualitative studies, the usual process is to continue to *saturation*, the point when no new information is being found.

Research ethics

Ethics are of relevance to all research, if only because of the idea that it is unethical to waste resources by doing poor research and producing potentially misleading results. The principles of research ethics are the same as those for ethics in general, as discussed in [Chapter 15](#), but there are certain ethical issues specific to the research context; for a basic introduction, see Poth (2021). For information research, some of the main issues are:

- recognising potential preconceptions and prejudices on the part of the researcher
- transparency: making clear issues such as the nature and purpose of the research, who may benefit from it, who is funding it and how the results will be disseminated

informed consent: making sure that all research participants are

- fully aware of the nature of the research and have the chance to decline to take part and to withdraw from the study at any time
- ensuring anonymity and confidentiality as necessary
- protecting participants' rights, particularly for vulnerable groups such as children or those in ill-health
- dealing correctly with documents, both in the study and in its dissemination, including observing copyright and avoiding plagiarism
- not collecting more data than necessary, and ensuring that data is stored and disposed of correctly
- choosing samples, analysing data and presenting results in ways which, so far as possible, avoid bias and distortion.

Digital research brings its own particular ethical issues, such as those around the collection of data or observation of behaviour without those affected being aware that they are being investigated. Care needs to be taken to observe the rules of digital environments, particularly social media platforms, regarding data collection.

Researchers will often be required by their institutions or funders to observe processes of *research integrity*, for example by observing codes of conduct, by collecting only that data which is strictly necessary for the research, by observing norms of data protection and so on.

Summary

Research is fundamental to the information disciplines and professions. Without it, the academic knowledge base will stagnate

and improvements in practice will be made more difficult. Academics and practitioners have a joint duty to carry out appropriate research, to be aware of and make use of existing research findings and to communicate across the academic-practitioner divide on these issues.

The range of methods available for and used in information research has expanded greatly, and novel methods still arise. Research is likely to be even more important in the future, to help deal with dramatic changes in the information environment.

- Research is essential for the future of the information disciplines and professions.
- Practitioners and academic researchers have a joint responsibility to carry out appropriate research and to make use of existing research.
- A wide variety of methods are available for information research, but these must be chosen and applied correctly in each study.

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19 The Future of the Information Sciences

And little he knew of the things that ink may do, how it can mark a dead man's thoughts for the wonder of later years, and tell of happenings that are gone clear away, and be a voice for us out of the dark of time, and save many a fragile thing from the pounding of heavy ages; or carry to us, over the rolling centuries, even a song from lips long dead on forgotten hills.

Lord Dunsany, *The King of Elfland's Daughter* (1924)

Prediction is very difficult, especially about the future.

Robert Storm Petersen, Danish poet and philosopher – also attributed to the physicist Niels Bohr

Introduction

In this final chapter we give an overview of some ideas about the future of the discipline and profession of information science. This has been a source of debate almost since the beginnings of the discipline, and of its sibling, library science. Much of the debate has been negative, with a narrative around disciplines and professions permanently in crisis. And, indeed, if one worries about the structure of academic departments and the names of professional associations one might find cause for concern. If one takes a longer view and thinks, with Lord Dunsany, about the power of recorded information and the continuing need for disciplines and professions to deal with it, the concerns come into better focus.

Predicting and prophesying

Karl Popper distinguished between *prediction*, when sufficient knowledge is available to make reasonable conclusions about future

events, and *prophecy*, when statements are made about the future without the necessary knowledge being available for them to be rational or realistic. We have seen examples of both of these relating to information science, but, because of the great and rapid changes in the information environment, most fall under the heading of prophesy, and much of it bad prophesy. David Deutsch (2011) builds on Popper's view to argue that the continuous development of knowledge of all kinds reduces all speculation about the future to prophesy, and almost inevitably in error.

Arthur C. Clarke, writer of popular science books and science fiction novels and originator of the concept of the communications satellite, identified two general problems in predicting the future. A 'failure of imagination' occurs when future developments are unforeseen because some necessary piece of knowledge was as yet unavailable; more common is the 'failure of nerve', in which all the relevant facts are known, yet they are not followed to their logical conclusion (Clarke, 1999). Again, we have seen examples of both of these relating to the information sciences.

Speculations about a future likely to be influenced by the growth of knowledge and by the new technologies and capabilities arising from it are likely to be just that: speculations, possibly interesting, but almost always wrong. The future of the information environment and the information disciplines is most certainly of this kind, and it is straightforward, if a little unkind, to reflect on the many poor predications of the value and use of information and communication technologies and to point out that authors of fiction were often more successful; see Bawden (1997) and the equivalent

chapter of the first edition of this book (Bawden and Robinson, 2012, chapter 15) for examples.

Having acknowledged the difficulties of speculating on information futures in detail, we will now consider the changes in the information environment which will influence future developments. In the first edition of this book we expressed these as a series of drivers for change; we are now able to summarise them more coherently as the advent of the infosphere.

Curating the infosphere

Throughout this book we have emphasised the significance of Luciano Floridi's ideas of the fourth revolution, the infosphere, and onlife (Floridi, 2014). Humankind has passed through a once-only transition into a new information environment and a new form of society, with far-reaching implications. It is hardly surprising that this has caused disruption to the disciplines and professions focused on the handling of recorded information. To give just one example, from the 1970s to the 1990s one of the most important roles for information professionals was online searching, carrying out searches in digital sources on behalf of clients. This was necessary because the nature of the digital sources and their interfaces, and the cost of searching, required specialist expertise and equipment. Since the advent of the internet and web search engines, this role has largely disappeared and online searching is the remit of everyone with a smartphone, including primary school children. It is true that much could be said about the quality of most search and the continuance of the intermediary searcher role in some environments, but that is beside the point; this typifies the way in which the move

to the infosphere has disrupted much of what was assumed to be the *raison d'être* of the information disciplines and professions.

In our view, this is not a cause for lamentation but, rather, offers new opportunities. The importance of documents in the keeping of the human record is not diminished in the infosphere, it is just reframed. We have argued that the information professions may gain a new role as 'curators of the infosphere' (Bawden and Robinson, 2018). This involves agreement with the view expressed by Floridi at the 2017 CILIP annual conference that the library and information community are experiencing a profound evolution of their role from keepers to curators of the infosphere. It involves an active role of curation and ontic stewardship of the new information environment (Fyffe, 2015; Van der Veer Martens, 2017), a novel, and we think appropriate, orientation of the information professions for the infosphere and onlife.

The future of a multidiscipline

If the information professions are to have a future as curators of the infosphere, what of the academic disciplines which support them? We suggested in [Chapter 1](#) that information science is a multidisciplinary field of study focused on recorded information. As information and data become more significant in many, if not all, academic disciplines, it is not surprising that we see even more interdisciplinary and multidisciplinary activity, not all of it necessarily under the heading of 'information science', 'information studies' or 'library and information science'. At a time when many academic disciplines are experiencing a change in their boundaries and interactions, it would be surprising if information science,

arguably the most multidisciplinary of the multidisciplines, kept a tidy and exclusive place. Rather, we are likely to see research, scholarship and teaching in the subject developing, as they always have, in different academic sectors, and partnering with a variety of related areas. We cannot, and should not try to, claim all information science issues as belonging to that discipline; rather, we should seek to ensure that their importance is recognised in the wider world.

We can see three different, though not incompatible, places for information science in the changing academic firmament. One is a closer alignment with the informatics and data science areas; this is essentially the route taken by the iSchools movement. Another is an alignment with media, communications, journalism, publishing and similar subjects. A third is an alignment with cultural studies and the digital humanities. Examples of all three can be seen, and all seem to be viable. It is important, however, not to choose one of these and declare that this is the future of the information sciences. They, and others which will emerge in the future, are simply aspects of the multidiscipline, emphasising different perspectives on the central focus of documents and the human record.

The big issues

In the first edition of this book we proposed a series of seven rather general, research questions which, we argued, would, and should, influence the future of information science. They are:

- What is the best theoretical basis for the information sciences (if indeed there is a single such basis)?

- How is the concept of information in the information sciences related to the ideas of information in other domains?
- How can principles of information science be applied to all sectors where documents and information resources (widely understood) are used?
- How can lessons from the past be applied to current and future information environments?
- How can we best operate in a world where an information glut, previously unimaginable, will be the norm?
- How can we best show the impact and value of knowledge and information resources and services in changing environments?
- How can information research and information practice best be linked, so that each informs the other?

We believe these still to be valid, and would not change any of them. We would, however, subsume them under a wider question:

- How can individuals and societies best operate, in information terms, so as to best adapt to the new infosphere and pursue what Floridi terms ‘the human project’?

This implies investigation of the ontology, epistemology and ethics of the section of the infosphere of particular relevance to LIS: essentially, that involved with documenting the human record, particularly in light of developments with big data and new forms of document.

We believe that it is important to keep these big questions and their wider context in mind, so as not to be overly focused on issues of the moment. Currently important general issues for the

information sciences are such things as sustainability and climate change, global and local inequalities, democracy and social justice, OA and open scholarship, the ethics of algorithms and AI, privacy and big data, and more. These are, of course, of great importance and should be priorities. Some will no doubt remain significant issues in the long term; for example, issues of sustainability of information provision (Chowdhury, 2016; Fedorowicz-Kruszewska, 2021). But we should not be misled into thinking that any one of them is what information science is, or should be, about. Our subject should, perhaps more than most, take a long view and fit the concerns of the moment into this.

Summary

No one knows in detail what the future will be like. This applies particularly to a discipline such as information science, which will be greatly changed by new knowledge, in ways which it is not possible to imagine.

Given the rapid and dramatic changes in the information infrastructure, due to the move to online and the infosphere, we should not be surprised that the professions and disciplines centred on the communication of information should feel a sense of disorientation, and even crisis. This does not mean that they have become irrelevant or outmoded, but it does mean that it is necessary to focus on the big issues, rather than try to defend particular roles. It is equally necessary to align with other information-related disciplines and professions, some of which are experiencing equally challenging changes.

We believe that information science has a bright future, though its details cannot be known. Rather than attempting to defend their present disciplinary niche, academics and practitioners alike should concentrate on ensuring that the insights of information science are recognised and applied in wider contexts.

- The detailed future of information science is inherently unpredictable, since it will depend on the growth of new knowledge.
- The future role of the information professions may be as curators of the infosphere, and that of the information disciplines as providers of supporting scholarship and research.
- Information science has a bright future; but it will change in ways we cannot now imagine.

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Additional Resources

Although this book is intended to be self-contained, with the provision of extensive literature references for readers interested in any particular aspects, we should mention some important resources for the study of information science.

Other textbooks

Two general texts are useful complements to this one. Rubin and Rubin cover the topics from a library science and US perspective, and are particularly good on ethics and professional issues. Stock and Stock emphasise the information science end of the spectrum, particularly strong on information retrieval and knowledge representation.

- Rubin, R. E. and Rubin, R. G. (2020) *Foundations of Library and Information Science* (5th edn), London: Facet Publishing.
- Stock, W. and Stock, M. (2013) *Handbook of Information Science*, Berlin: de Gruyter.

Journals

Relevant papers will be found in the journals of a wide variety of subjects, from computer science to cultural studies, from librarianship to business management, and from information systems to archiving. A selective list of journals particularly relevant to the topics of this book is:

- *Aslib Journal of Information Management*
- *Journal of the Association for Information Science and Technology*
- *Journal of Documentation*

- *Journal of Information Science*
- *Journal of Librarianship and Information Science*
- *Information [MDPI]*
- *Information Processing and Management*
- *Information Research*
- *International Journal of Information Management*
- *Library and Information Research*
- *Library and Information Science Research*
- *Library Trends*
- *Knowledge Organization.*

Abstracting and indexing services

Again, information science literature may be covered by a wide variety of secondary sources, especially if it relates to information in a particular subject area; *Medline/PubMed*, for instance, covers healthcare information. However, the main subject-specific resources are:

- *LISA (Library and Information Science Abstracts)*
- *LISTA (Library and Information Science Abstracts)*
- *SSCI (Social Science Citation Index)* part of the *Web of Knowledge/Web of Knowledge* service.

Reference sources

- The fourth edition of the *Encyclopedia of Library and Information Science* (CRC Press, 2018) is a valuable resource, with short

accessible articles of generally good quality covering a wide area, though not all are up to date.

- The *Epistemological Lifeboat*, <https://web.archive.org/web/20180226161042/http://lifeboat.iva.ku.dk>, a web resource compiled by Birger Hjørland and Jeppe Nicolaisen, has a useful compilation of short authoritative articles on concepts and theories in information science.
- The ISKO online *Encyclopedia of Knowledge Organization*, www.isko.org/cyclo, is a useful developing collection of articles on a wide range of topics.
- The *Stanford Encyclopedia of Philosophy*, <https://plato.stanford.edu>, has numerous articles of relevance to information and knowledge, generally at a scholarly but accessible level.

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